

APPENDIX 2

Part 1 of 3

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First Named Inventor :	Patrizio Vinciarelli	Art Unit :	Unknown
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Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

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PRELIMINARY AMENDMENT

Prior to examination, please amend the application as indicated on the following pages.

Amendments to the Claims:

This listing of claims replaces all prior versions and listings of claims in the application:

Listing of Claims:

1. – 131. (Cancelled)

132. (Original) A method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising:

providing one or more driver circuits for deployment outside the semiconductor package, each having an input for receiving power from the source and a driver output;

providing one or more output circuits for deployment in or near the semiconductor package, being connected to receive power from the driver circuit at a driver voltage;

arranging the output circuits to form a plurality of output rails configured to supply power at a plurality of respective unique rail voltages to the one or more semiconductor chips;

configuring each output rail to provide a rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential;

configuring the output circuits to provide the rail voltages as a function of the driver voltage;

providing one or more rail selection circuits on the one or more semiconductor chips, each having inputs connected to the output rails and an output connected to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches connected to provide a selectable path for current flow from a selected one of the rails to the output;

providing a power bus for carrying power from the driver output to the input of each of the one or more output circuits; and

deploying the driver circuit at a first location outside of the semiconductor package;
wherein each rail selection circuit is operative to enable a respective rail switch to
conduct current from selected output rail to its respective output.

133. (Original) The method of claim 132, further comprising:
providing power management circuitry on the one or more semiconductor chips, the
power management circuitry having one or more outputs for sending commands to the one or
more rail selection circuits, the commands including a voltage command for setting a desired
output voltage of respective rail selection circuits;
wherein each rail selection circuit is further adapted to receive the voltage command and
enable the respective rail switch according to predetermined criteria, the predetermined criteria
including selection of the rail voltage that is closest to the desired output voltage.

134. (Original) The method of claim 133 further comprising:
providing regulation circuitry adapted to regulate power conducted from the selected rail
to the respective output; and
wherein the predetermined criteria further include selection of the rail voltage that is
greater than the desired output voltage by a minimum voltage drop required for the regulation.

135. (Original) The method of claim 134 wherein the regulation circuitry is further
adapted to adjust a drive signal to the selected rail selection switch as a function of an error
signal derived from a difference between the respective output voltage and a reference signal to
vary a voltage drop across the selected rail selection switch.

136. (Original) The method of claim 135 further comprising providing circuitry
adapted to adjust the reference signal as a function of the respective voltage command.

137. (Original) The method of claim 132 further comprising providing one or more
regulation stages for regulating the driver voltage delivered to the one or more driver circuits.

138. (Original) The method of claim 137 wherein the one or more driver circuits are further adapted to adjust the driver voltage as a function of commands received from the one or more semiconductor chips.

139. (Original) The method of claim 132 wherein the one or more output circuits further comprise:

one or more transformers, each having a primary winding connected to receive the driver voltage and one or more secondary windings;

rectification circuitry connected to convert AC signals received from the one or more secondary windings to unipolar voltages.

140. (Original) The method of claim 139 wherein the one or more transformers comprise at least one transformer per output rail.

141. (Original) The method of claim 139 wherein the one or more output circuits comprise at least one pair of output circuits, each having at least one transformer.

142. (Original) The method of claim 139 wherein the one or more output circuits comprise at least one pair of output circuits, each having at least one transformer, and at least two of the output rails are connected in parallel.

143. (Original) The method of claim 132 wherein the one or more output circuits are provided in the semiconductor package and each output circuit includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output and an output connected to the second winding for supplying power to respective ones of the plurality of output rails.

144. (Original) The method of claim 143 further comprising:
providing a controller adapted to operate the one or more driver circuits and the one or more output circuits in a series of converter operating cycles, each converter operating cycle

being characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the one or more driver circuits to the plurality of output rails via the transformers.

145. (Original) The method of claim 144 wherein the one or more driver circuits and the one or more output circuits are adapted for power to be supplied to the semiconductor package by the one or more driver circuits at a bus voltage, V_{bus} , that is a multiple, X , times greater than at least one of the rail voltages.

146. (Original) The method of claim 132 wherein the one or more output circuits each includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to conduct AC power received from the driver output to the first winding for driving the power transformer and an output connected to conduct power from the second winding to respective ones of the plurality of output rails and further comprising:

providing a controller adapted to operate the one or more driver circuits and output circuits in a series of converter operating cycles, each converter operating cycle being characterized by two power transfer phases of essentially equal duration during which a magnetizing current flows in a winding of a respective power transformer and power is transferred from the one or more driver circuits to at least one of the plurality of output rails via the respective power transformer.

147. (Original) The method of claim 146 further comprising:

forming a resonant circuit including the respective power transformer and a resonant capacitor connected in series with the first winding, the resonant circuit being characterized by a characteristic resonant period and frequency;

wherein each power transfer phase is further characterized in having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency.

148. (Original) The method of claim 132 further comprising:

providing a controller adapted to operate the one or more driver circuits and output circuits in a series of converter operating cycles;

providing a clamp switch connected across the resonant capacitor;

wherein the one or more output circuits each includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to conduct AC power received from the driver output to the first winding for driving the power transformer and an output connected to conduct power from the second winding to respective ones of the plurality of output rails;

wherein the driver output includes a first terminal driven by a switching node and the driver input includes a source terminal and a common terminal;

and each converter operating cycle is characterized by

(a) two power transfer phases of essentially equal duration during which the switching node is connected to either the source or the common terminal, a magnetizing current flows in a winding of a respective transformer, and power is transferred from the one or more driver circuits to at least one of the plurality of output rails via the respective transformer, each power transfer phase including

(i) first and second resonant intervals, each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

(ii) a clamp interval occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

149. (Original) The method of claim 132:

wherein the driver output includes a first terminal driven by a switching node and a second terminal, the driver input includes a source terminal and a second terminal referenced to a voltage potential within a predetermined range of a common terminal; and further comprising

providing a controller to operate the driver circuit in a series of converter operating cycles;

providing a first capacitor connected to block direct current between the switching node and the primary winding, wherein the driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts;

wherein the driver voltage comprises a maximum peak-to-peak voltage, V_{peak} , that is at least a multiple, X , times greater than at least one of the rail voltages and X is at least six; and

each converter operating cycle is characterized by two power transfer intervals of essentially equal duration during which the switching node remains connected to either the source or the common terminal, a magnetizing current flows in a winding of the transformer, and power is transferred between the driver input and at least one of the multiplicity of output rails via the transformer.

150. (Original) The method of claim 145 wherein X is greater than or equal to 15.

151. (Original) A method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising:

providing one or more driver circuits for deployment outside the semiconductor package, each driver circuit having an input for receiving power from the source, and a driver output having a first terminal driven by a switching node and a second terminal referenced to a voltage potential within a predetermined range of a common terminal;

deploying the driver circuit at a first location outside of the semiconductor package;

providing one or more output circuits for deployment in or near the semiconductor package for supplying power to the one or more semiconductor chips, at least one of the output circuits including a transformer having a primary winding adapted for connection to the first and second terminals of the driver output and a secondary winding;

providing a power bus for carrying power from the first and second terminals of the driver output to the primary winding; and

providing a first capacitor connected to block direct current between the switching node and the primary winding;

wherein driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts;

wherein the one or more output circuits are adapted to receive power from the driver circuit at a peak-to-peak bus voltage, V_{peak} , and supply power to the one or more semiconductor chips at a load voltage, V_{load} , wherein V_{peak} is a multiple, X , of V_{load} and X is at least six.

152. (Original) The method of claim 151 wherein the one or more output circuits are deployed in the semiconductor package.

153. (Original) The method of claim 151 wherein the one or more driver circuits comprise a first semiconductor switch connected between a positive terminal of the driver input and the switching node and a second semiconductor switch connected between the common terminal and the switching node, the first and second semiconductor switches being adapted to alternatively connect the switching node to the positive terminal or the common terminal in a succession of switching cycles.

154. (Original) The method of claim 151 wherein the first capacitor is connected between the switching node and the driver output.

155. (Original) The method of claim 154 further comprising:
forming a resonant circuit including the transformer and a resonant capacitor connected in series with the primary winding, the resonant circuit being characterized by a characteristic resonant period and frequency.

156. (Original) The method of claim 155 wherein the first capacitor comprises the resonant capacitor.

157. (Original) The method of claim 155 wherein the resonant capacitor is connected between the transformer and the second terminal of the driver output and further comprising

providing a clamp switch connected across the resonant capacitor and the first capacitor is at least an order of magnitude greater than the resonant capacitor.

158. (Original) The method of claim 157 further comprising providing a resistance in parallel with the resonant capacitor.

159. (Original) The method of claim 157 further comprising:
providing a controller to operate the driver circuit in a series of converter operating cycles, each converter operating cycle characterized by:

(a) two power transfer phases of essentially equal duration during which the switching node is connected to either the source or the common terminal, a magnetizing current flows in a winding of the transformer, and power is transferred between the driver input and the output circuit via the transformer, each power transfer interval including

(i) first and second resonant intervals, each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

(ii) a clamp interval occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

160. (Original) The method of claim 151 wherein the predetermined range is less than or equal to 10% of the source voltage.

161. (Original) The method of claim 155 further comprising:
providing a controller to operate the driver circuit in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which the switching node remains connected to either the source or the common terminal, a magnetizing current flows in a winding of the transformer, and power is transferred between the driver input and the POL circuit output via the transformer.

162. (Original) The method of claim 161 wherein each power transfer phase is further characterized in having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency.

163. (Original) The method of claim 151 wherein X is greater than or equal to 20.

164. (Original) The method of claim 151 wherein X is greater than or equal to 10.

165. (Original) A method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage, comprising:

performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage;

conducting power from the first output through a portion of the semiconductor package to a point of load (POL) circuit located within the semiconductor package;

performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering power to the semiconductor chip at a chip voltage; and

performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip;

wherein the package voltage is at least, 5, times greater than the load voltage.

166. (Original) The method of claim 165 wherein the third power conversion comprises voltage regulation.

167. (Original) The method of claim 166 wherein the third power conversion comprises linear voltage regulation.

168. (Original) The method of claim 166 wherein the third power conversion comprises switching voltage regulation.

169. (Original) The method of claim 166 wherein the second power conversion comprises voltage regulation.

170. (Original) The method of claim 169 wherein the POL circuit comprises a multi-phase switching regulator.

171. (Original) The method of claim 169 wherein the first power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current, in which V_{in} is the source voltage and V_o is the package voltage.

172. (Original) The method of claim 169 wherein the package voltage is greater than 20 VDC and the chip voltage is less than 3 VDC.

173. (Original) The method of claim 169 wherein the package voltage is greater than 5 VDC and the chip voltage is less than 2 VDC.

174. (Original) The method of claim 165 wherein the second power conversion comprises voltage regulation.

175. (Original) The method of claim 174 wherein the POL circuit comprises a multi-phase switching regulator.

176. (Original) The method of claim 165 wherein the second power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current, in which V_{in} is the package voltage and V_o is the chip voltage.

177. (Original) A method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage, comprising:

performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage;

conducting power from the first output through a portion of the semiconductor package to a point-of-load (POL) circuit located within the semiconductor package and outside of the semiconductor chip;

performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering power to the semiconductor chip at a chip voltage; and

performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip;

wherein the package voltage is at least, 5, times greater than the load voltage.

178. (Original) The method of claim 177 wherein the third power conversion comprises voltage regulation.

179. (Original) The method of claim 178 wherein the third power conversion comprises linear voltage regulation.

180. (Original) The method of claim 178 wherein the third power conversion comprises switching voltage regulation.

181. (Original) The method of claim 178 wherein the second power conversion comprises voltage regulation.

182. (Original) The method of claim 181 wherein the POL circuit comprises a multi-phase switching regulator.

183. (Original) The method of claim 181 wherein the first power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current.

184. (Original) The method of claim 181 wherein the package voltage is greater than 20 VDC and the chip voltage is less than 3 VDC.

185. (Original) The method of claim 181 wherein the package voltage is greater than 5 VDC and the chip voltage is less than 2 VDC.

186. (Original) The method of claim 177 wherein the second power conversion comprises voltage regulation.

187. (Original) The method of claim 186 wherein the second stage comprises a multi-phase switching regulator.

188. (Original) The method of claim 177 wherein the second power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current, in which V_{in} is the source voltage and V_o is the package voltage.

189. (Original) The method of claim 188 wherein the source voltage is greater than 40 VDC.

190. (Original) The method of claim 189 wherein the package voltage is greater than 20 VDC and the chip voltage is less than 3 VDC.

191. (Original) The method of claim 188 wherein the package voltage is greater than 5 VDC and the chip voltage is less than 2 VDC.

192. (Original) The method of claim 177 wherein:
the semiconductor package comprises a substrate and the semiconductor chip is mounted to the substrate; and
the POL circuit further comprises one or more semiconductor switches and at least one inductive component, the POL circuit being substantially contained within an electronic package mounted to the first substrate.

193. (Original) An apparatus comprising:
a semiconductor package comprising one or more semiconductor chips, each semiconductor chip comprising one or more circuit loads;
one or more driver circuits disposed outside of the semiconductor package, each driver circuit having a driver input and a driver output, in which the driver input is configured to receive power from an input source at an input voltage;
one or more output circuits disposed inside or in a vicinity of the semiconductor package and configured to receive power from the driver circuit at a driver voltage, in which the output circuits are configured to form a plurality of output rails to supply power at a plurality of respective unique rail voltages to the one or more semiconductor chips, each output rail is configured to provide a DC rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential, and the output circuits provide the rail voltages as a function of the driver voltage;
one or more rail selection circuits disposed on the one or more semiconductor chips, each having inputs electrically coupled to the output rails and an output electrically coupled to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches electrically coupled to provide a selectable path for current to flow from a selected one of the output rails to the respective output; and

a power bus configured to carry power from the driver output to the input of each of the one or more output circuits.

194. (Original) An apparatus comprising:

a semiconductor package comprising one or more semiconductor chips, each semiconductor chip comprising one or more circuit loads;

a first power conversion stage disposed outside of the semiconductor package and configured to perform a first power conversion, in which the first power conversion stage is configured to receive power from an input source at a source voltage, make a voltage adjustment, and deliver power from a first output to the semiconductor package at a package voltage;

a power bus configured to conduct power from the first output through a portion of the semiconductor package to a point-of-load (POL) circuit located within the semiconductor package;

the POL circuit configured to perform a second power conversion, in which the POL circuit is configured to receive the conducted power, make a second voltage adjustment, and deliver power to at least one of the semiconductor chips mounted in the semiconductor package at a chip voltage; and

a third power conversion stage located on the semiconductor chip and configured to perform a third power conversion, in which the third power conversion stage is configured to receive power from the POL circuit, make a third voltage adjustment, and deliver power at a load voltage to at least one circuit load on the semiconductor chip, and the package voltage is at least 5 times greater than the load voltage.

First Named Inventor : Patrizio Vinciarelli
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REMARKS

The Applicant asks that all claims be examined in view of the amendment to the claims.

Excess claims fees are being paid with this reply on the Electronic Filing System. In addition, please apply any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: June 7, 2017 _____

/Rex I. Huang/ _____

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Title of Invention:		METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		
First Named Inventor/Applicant Name:		Patrizio Vinciarelli		
Filer:		Rex I. Huang/Rita Liston		
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UTILITY APPLICATION FILING	1011	1	280	280
UTILITY SEARCH FEE	1111	1	600	600
UTILITY EXAMINATION FEE	1311	1	720	720
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	44	80	3520
INDEPENDENT CLAIMS IN EXCESS OF 3	1201	3	420	1260
Miscellaneous-Filing:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
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Post-Allowance-and-Post-Issuance:				
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First Named Inventor/Applicant Name:	Patrizio Vinciarelli
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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		00614_0180001_Spec.pdf	338484	yes	87
			2de841b10a8cb1336f94cb27bb205c0b40230c5f		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	46	
	Claims		47	86	
	Abstract		87	87	
Warnings:					
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2	Drawings-only black and white line drawings	00614_0180001_Dwgs.pdf	1064517	no	30
			241b36e792ec3bfac908575fcbca4d8551e9f992		
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3	Foreign Reference	JPH0228066.pdf	2135385	no	13
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4	Foreign Reference	EP2863531.pdf	2958886	no	23
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5	Non Patent Literature	EPOA.pdf	183592	no	2
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6	Non Patent Literature	EESR.pdf	658356	no	8
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7	Non Patent Literature	Okudaira.pdf	847862	no	6
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8	Non Patent Literature	Watson.pdf	2391132	no	32
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11	Transmittal Letter	00614_0180001_PAP.pdf	110125	no	3
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	Document Description		Start	End	
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	Information Disclosure Statement (IDS) Form (SB08)		2	2	
Warnings:					
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13	Application Data Sheet	00614_0180001_ADS.pdf	1822821	no	8
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14	Nonpublication request from applicant	00614_0180001_Non_Pub.pdf	96356	no	1
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	Preliminary Amendment		1	1	
	Claims		2	15	
	Applicant Arguments/Remarks Made in an Amendment		16	16	
Warnings:					
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METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation-in-part of, and claims priority to, U.S. patent application 15/091,346, filed on April 5, 2016, which is incorporated by reference in its entirety.

FIELD OF THE INVENTION

[0002] This invention relates to the field of switching power supplies and more particularly to power converters that supply power to large semiconductor devices such as processors and ASICs.

BACKGROUND

[0003] Many large scale semiconductors require high current, e.g. 100A or more, at low voltages, e.g. 1V, or less, dissipate large amounts of power, challenging packaging technologies to accommodate power, thermal, and signal demands. Certain microprocessors incorporate on-silicon switching regulators which because of design tradeoffs occupy precious silicon real estate and inefficiently convert power. Resonant switching power converters may be used as fixed ratio current multipliers in power conversion systems to provide high current at a low voltage.

SUMMARY

[0004] In one aspect, in general, a method of supplying power received from an input source at an input voltage to circuitry on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage is provided. The method includes providing a driver circuit for deployment outside the semiconductor package having an input for receiving power from the source, circuitry adapted to drive a transformer, and a driver output; providing one or more output circuits for deployment

in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits in the semiconductor package; operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and deploying the driver circuit at a first location outside of the semiconductor package. Power is supplied to the semiconductor package by the driver circuit at a bus voltage that is a multiple, X, times greater than a load voltage required by the one or more semiconductor chips.

[0005] In another aspect, in general, an apparatus includes a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board; one or more output circuits housed in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; and a power bus connected to selected interface connections for carrying AC power at a bus voltage from the system board to the input of each of the one or more output circuits in the semiconductor package. The bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips.

[0006] In another aspect, in general, a method of packaging a semiconductor die includes providing a substrate for mounting to a bottom surface of the semiconductor die; providing a lid to extract heat from a top surface of the semiconductor die; providing one or more output circuits having a bottom surface mounted to the substrate for supplying power to the semiconductor die; and conducting heat from the bottom of the semiconductor die, into the substrate, from the substrate into the bottom of the one or more output circuits, out of the top of the one or more output circuits, into the lid.

[0007] In another aspect, in general, an apparatus includes a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board; and one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, the inductive component having a magnetically permeable core having an effective permeability of at least 25. The ratio of input voltage to output voltage is fixed, subject to a series resistance, and is at least 5.

[0008] In another aspect, in general, an apparatus includes a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board; and one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, wherein the switch controller is configured to turn the at least one switch ON or OFF at essentially zero voltage. The power conversion circuit ratio

of input voltage to output voltage is fixed, subject to a series resistance, and is at least 5.

[0009] In another aspect, in general, an apparatus includes a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board; and one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, wherein the switch controller is configured to turn the at least one switch ON or OFF at essentially zero current. The power conversion circuit ratio of input voltage to output voltage is fixed, subject to a series resistance, and is at least 5.

[0010] In another aspect, in general, an apparatus includes a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board; and one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips. The switch controller is configured to operate the at least one switch to limit slew rates of voltage in the converter to 5 (V_{peak}/Top).

[0011] In another aspect, in general, an apparatus includes a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board; and one or more power conversion circuits housed in the

semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips. The switch controller is configured to operate the at least one switch to limit slew rates of current in the converter to less than or equal to $5 \cdot (I_{\text{peak}}/T_{\text{op}})$.

[0012] In another aspect, in general, a method of supplying power received from an input source at an input voltage to circuitry on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage is provided. The method includes providing one or more output circuits for deployment at one or more locations near or adjacent to the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; providing a driver circuit for deployment at a location spaced apart from the one or more output circuits, the driver circuit having an input for receiving power from the source, circuitry configured to drive a transformer, a driver output for supplying AC power to the one or more output circuits, and circuitry configured to control the power supplied to the one or more output circuits; providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits; operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and deploying the driver circuit at a first location outside of an area immediately near or adjacent to the one or more output circuits. Power is supplied to the one or more output circuits by the driver circuit at a bus voltage that is a multiple, X , times greater than the load voltage required by the one or more semiconductor chips.

[0013] In another aspect, in general, a method of supplying power received from an input source at an input voltage to a load at a DC output voltage is provided. The method includes providing one or more current multiplier modules at one or more locations in close proximity to the load, the one or more current multiplier modules having a transformer and circuitry for supply DC power to the load; and providing a driver module at a location spaced apart from the one or more current multiplier modules, the driver module having circuitry including an input for receiving power from the input source, control circuitry for generating a controlled driver voltage, and driver circuitry for generating AC power for driving the transformer in each of the one or more current multiplier modules.

[0014] In another aspect, in general, a method of making a planar magnetic component is provided. The method includes providing a multilayer printed circuit board (PCB) including conductive features arranged on conductive layers of the PCB to form one or more windings around one or more predetermined axes; forming a hole in the PCB at each of the one or more predetermined axes to accommodate one or more core legs, wherein for each hole, an inner edge of one of the windings overlaps an edge of the hole in a lateral direction after the hole is formed; assembling a first magnetically permeable plate to a first surface of the PCB covering a selected one or more of the holes at the one or more predetermined axes; assembling a second magnetically permeable plate to a second surface of the PCB covering the selected one or more of the holes at the one or more predetermined axes; filling the selected one or more of the holes with at least one of a magnetically permeable fluid or powder; and sealing the selected one or more holes to prevent the at least one of magnetically permeable fluid or powder from escaping.

[0015] In another aspect, in general, an apparatus includes a planar magnetic structure including a multilayer printed circuit board (PCB) having a first surface, a second surface, and conductive features arranged on conductive layers of the PCB to form one or more windings around one or more predetermined axes; a hole in the PCB at each of the one or more predetermined axes, each hole having an inner edge aligned with an inner circumference of one of the windings; a first magnetically permeable

section affixed to the first surface of the PCB covering a selected one or more of the holes; a second magnetically permeable section affixed to the second surface of the PCB covering the selected one or more of the holes; and at least one of a magnetically permeable fluid or powder disposed in the selected one or more of the holes. The at least one magnetically permeable fluid or powder is contained within the selected one or more holes.

[0016] In another aspect, in general, a method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage is provided. The method includes providing one or more driver circuits for deployment outside the semiconductor package, each having an input for receiving power from the source and a driver output; providing one or more output circuits for deployment in or near the semiconductor package, being connected to receive power from the driver circuit at a driver voltage; arranging the output circuits to form a plurality of output rails configured to supply power at a plurality of respective unique rail voltages to the one or more semiconductor chips; configuring each output rail to provide a rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential; configuring the output circuits to provide the rail voltages as a function of the driver voltage; providing one or more rail selection circuits on the one or more semiconductor chips, each having inputs connected to the output rails and an output connected to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches connected to provide a selectable path for current flow from a selected one of the rails to the output; providing a power bus for carrying power from the driver output to the input of each of the one or more output circuits; and deploying the driver circuit at a first location outside of the semiconductor package; wherein each rail selection circuit is operative to enable a respective rail switch to conduct current from selected output rail to its respective output.

[0017] In another aspect, in general, a method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more

semiconductor chips mounted in a semiconductor package at a DC output voltage is provided. The method includes providing one or more driver circuits for deployment outside the semiconductor package, each driver circuit having an input for receiving power from the source, and a driver output having a first terminal driven by a switching node and a second terminal referenced to a voltage potential within a predetermined range of a common terminal; deploying the driver circuit at a first location outside of the semiconductor package; providing one or more output circuits for deployment in or near the semiconductor package for supplying power to the one or more semiconductor chips, at least one of the output circuits including a transformer having a primary winding adapted for connection to the first and second terminals of the driver output and a secondary winding; providing a power bus for carrying power from the first and second terminals of the driver output to the primary winding; and providing a first capacitor connected to block direct current between the switching node and the primary winding; wherein driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts; wherein the one or more output circuits are adapted to receive power from the driver circuit at a peak-to-peak bus voltage, V_{peak} , and supply power to the one or more semiconductor chips at a load voltage, V_{load} , wherein V_{peak} is a multiple, X , of V_{load} and X is at least six.

[0018] In another aspect, in general, a method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage is provided. The method includes performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage; conducting power from the first output through a portion of the semiconductor package to a point of load (POL) circuit located within the semiconductor package; performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering power to the semiconductor chip at a chip voltage; and performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL

circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip; wherein the package voltage is at least, 5, times greater than the load voltage.

[0019] In another aspect, in general, a method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage is provided. The method includes performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage; conducting power from the first output through a portion of the semiconductor package to a point of load (POL) circuit located within the semiconductor package; performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering power to the semiconductor chip at a chip voltage; and performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip; wherein the package voltage is at least, 5, times greater than the load voltage.

[0020] In another aspect, in general, a method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage is provided. The method includes performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage; conducting power from the first output through a portion of the semiconductor package to a point-of-load (POL) circuit located within the semiconductor package and outside of the semiconductor chip; performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering

power to the semiconductor chip at a chip voltage; and performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip; wherein the package voltage is at least, 5, times greater than the load voltage.

[0021] In another aspect, in general, an apparatus includes a semiconductor package having one or more semiconductor chips, each semiconductor chip comprising one or more circuit loads; one or more driver circuits disposed outside of the semiconductor package, each driver circuit having a driver input and a driver output, in which the driver input is configured to receive power from an input source at an input voltage; one or more output circuits disposed inside or in a vicinity of the semiconductor package and configured to receive power from the driver circuit at a driver voltage, in which the output circuits are configured to form a plurality of output rails to supply power at a plurality of respective unique rail voltages to the one or more semiconductor chips, each output rail is configured to provide a DC rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential, and the output circuits provide the rail voltages as a function of the driver voltage; one or more rail selection circuits disposed on the one or more semiconductor chips, each having inputs electrically coupled to the output rails and an output electrically coupled to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches electrically coupled to provide a selectable path for current to flow from a selected one of the output rails to the respective output; and a power bus configured to carry power from the driver output to the input of each of the one or more output circuits.

[0022] In another aspect, in general, an apparatus includes a semiconductor package having one or more semiconductor chips, each semiconductor chip having one or more circuit loads; a first power conversion stage disposed outside of the semiconductor package and configured to perform a first power conversion, in which the first power conversion stage is configured to receive power from an input source at a source voltage, make a voltage adjustment, and deliver power from a first output to

the semiconductor package at a package voltage; a power bus configured to conduct power from the first output through a portion of the semiconductor package to a point-of-load (POL) circuit located within the semiconductor package; the POL circuit configured to perform a second power conversion, in which the POL circuit is configured to receive the conducted power, make a second voltage adjustment, and deliver power to at least one of the semiconductor chips mounted in the semiconductor package at a chip voltage; and a third power conversion stage located on the semiconductor chip and configured to perform a third power conversion, in which the third power conversion stage is configured to receive power from the POL circuit, make a third voltage adjustment, and deliver power at a load voltage to at least one circuit load on the semiconductor chip, and the package voltage is at least 5 times greater than the load voltage.

[0023] The details of one or more embodiments of the invention are set forth in the accompanying drawings and description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF DRAWINGS

[0024] FIG. 1 shows an isometric view of a semiconductor package with integrated power-converter output circuitry.

[0025] FIG. 2 shows a top plan view of the semiconductor package.

[0026] FIG. 3 shows a side view of the semiconductor package.

[0027] FIG. 4 shows a cross-sectional view of a portion of the substrate of the semiconductor package revealing power input connections for the output circuitry.

[0028] FIG. 5 shows a schematic drawing of a first modular power converter based on the SAC topology.

[0029] FIG. 6 shows a schematic drawing of a second modular power converter based on the SAC topology.

[0030] FIG. 7 shows a schematic drawing of a system including a third modular power converter with multiple output chips integrated into a semiconductor package to supply power to a large semiconductor die.

[0031] FIG. 8 shows an isometric view of an output circuit.

[0032] FIG. 9 shows a schematic block diagram of a system including a fourth modular power converter including an improved driver circuit.

[0033] FIG. 10 shows an isometric view of the semiconductor package with the lid exploded from the assembly.

[0034] FIG. 11 shows heat flow in a side view of the semiconductor package with lid.

[0035] FIG. 12 shows an exploded view of a transformer structure for use in an output circuit.

[0036] FIG. 13 shows a schematic block diagram of a system including a fifth modular power converter.

[0037] FIG. 14A and FIG. 14B show exploded and assembled views of an alternate transformer structure for use in an output circuit.

[0038] FIG. 15A and FIG. 15B show exploded bottom and top isometric views of a second semiconductor package.

[0039] FIG. 16A and FIG. 16B show top and bottom isometric views of the second semiconductor package assembled.

[0040] FIG. 17 is a side view of the second semiconductor package.

[0041] FIG. 18 is a bottom isometric view of a third semiconductor package.

[0042] FIG. 19 is a schematic drawing of a system including a sixth modular power converter with an output chip integrated into a semiconductor package to supply power to a large semiconductor die.

[0043] FIG. 20A and FIG. 20B show first and second mounting positions for the second and third semiconductor packages.

[0044] FIG. 21A shows a block diagram of a seventh modular power converter.

[0045] FIG. 21B shows a block diagram of a system including an eighth modular power converter including a multi-rail output point-of-load (POL) circuit co-packaged with a semiconductor die.

[0046] FIG. 22 shows a schematic of a multi-rail output POL circuit.

[0047] FIG. 23 shows a schematic of an alternate embodiment of a multi-rail output POL circuit.

[0048] FIG. 24 shows a generalized block diagram of multiple loads on an ASIC with power management control and individual rail selection and regulation circuitry.

[0049] FIG. 25 shows an example of rail selection and regulation circuitry for a single load.

[0050] FIGS. 26A and 26B show modified driver circuitry.

[0051] FIG. 27 shows a power system including a modified driver circuit.

[0052] FIG. 28 shows a block diagram of a dual integrated driver module.

[0053] Like references symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0054] In contemporary electronic systems, space is at a premium on customer circuit boards, e.g. on a circuit board near a processor. Additionally, thermal

management considerations place limits on the efficiency and power dissipation of power supplies at, or near, the point of load. Many very large scale integrated (“VLSI”) semiconductor dies such as central processing units (“CPU”), graphics processing units (“GPU”), and application specific integrated circuits (“ASIC”) are mounted to a multilayer ceramic substrate which translates the electrical connections from the die to larger connections suitable for interfacing with a customer motherboard. As feature sizes decrease and transistor counts increase, so too do the power supply current requirements for such large chips. Current requirements for a typical CPU can easily exceed 200 amps creating challenges for the package and system designers to efficiently supply such high currents. For example, power connections between the component package (such as a chip carrier or substrate or other package in or on which the semiconductor die is mounted) and the printed circuit board (PCB) on which the package is mounted may demand a large number of connector pins, leads, solder bumps, etc., to carry very high currents challenging package designers to accommodate both power and signal requirements. In many cases the large number and high frequency demands of signals may limit the maximum voltage, e.g. the interlayer breakdown voltage, to which the substrate or package may be subjected, in some cases as low as a few volts, further challenging power connections to and within the package or substrate.

[0055] A Factorized Power Architecture well suited for supplying power to low voltage high current loads is described in Vinciarelli, *Factorized Power with Point of Load Sine Amplitude Converters*, U.S. Patent No. 6,975,098, issued Dec. 13, 2005 (the “Micro FPA Patent”) and U.S. Patent No. 6,984,965, issued Jan. 10, 2006 (the “FPA Patent”) (both assigned to VLT, Inc. of Sunnyvale, CA, and the entire disclosure of each patent is incorporated herein by reference). Power converters which function as DC-to-DC transformers called Voltage Transformation Modules (“VTM”) and Sine Amplitude Converters (“SAC”) which have a transfer function approximating $V_o = K_{VTM} * V_{in} - I_o * R_{VTM}$ are described in Vinciarelli, *Factorized Power with Point of Load Sine Amplitude Converters*, U.S. Patent No. 6,930,893, issued Aug. 16, 2005 (the “SAC Patent”) and in Vinciarelli, *Point of Load Sine Amplitude Converters and Methods*, U.S. Patent No. 7,145,786, issued Dec. 5, 2006 (the “POL SAC Patent”)

(both assigned to VLT, Inc. of Sunnyvale, CA, the entire disclosure of each patent is incorporated herein by reference).

I. In-Package Power Conversion Topologies

A. Fault-Tolerant Topology

[0056] Fig. 5 is a replica of Fig. 14 from U.S. Patent 9,112,422, issued August 18, 2015, which is incorporated herein by reference in its entirety (hereinafter the “FT Patent”). In Fig. 5, a power converter 400 is shown including a driver 420 connected to drive one or more point-of-load (“POL”) current multiplier circuits 430 having inputs 447, 448 connected to the driver outputs 427, 428 via an AC power bus 410. The driver 420 may comprise a full-bridge fault-tolerant input circuit such as input circuit 250 of Fig. 11 in the FT Patent and a switch controller 425 similar to the switch controller described in connection with Fig. 11 of the FT Patent.

[0057] The POL circuit 430 may include a transformer circuit 440 and a rectification circuit 450. The transformer circuit 440 may include none, one, or both, of resonant capacitors 441, 442 shown in Fig. 5 connected to the primary winding 82 of transformer 81. The secondary winding may be connected to a full-bridge fault-tolerant rectification circuit 450 as shown in Fig. 5 and described in the FT Patent. The full-bridge fault-tolerant rectification circuit 450 may use switches, R1, R2, R3, R4, operated as rectifiers in the manner described in the FT Patent in connection with output circuit 100 of Fig. 3 (of the FT Patent) and may preferably employ the common-source synchronous rectifiers described in connection with Figs. 7 and 8 (of the FT Patent). Note that a simplified symbol is used in Fig. 5 for switches S1-S4 and R1-R4 (instead of the enhancement mode MOSFET symbols used, e.g. in Figs. 2-4, 7, 8, 11 of the FT Patent) in which the arrow indicates the direction of current flow through the intrinsic body drain diode when the switch is open.

[0058] As its name implies, the POL circuit 430 (Fig. 5) may be designed to be deployed as close to the point of load, where space and thermal requirements are stringent, as possible. Because the driver circuit 420 does not need to be close to the

point of load, it may be deployed elsewhere, away from the point of load, reducing the space required by the POL circuitry and reducing the power dissipation in proximity to the load. One benefit of removing the driver circuitry from the POL is that a larger transformer structure and array of output switches (R1-R4) may be used in the POL circuit thereby improving overall converter efficiency and further reducing dissipation at the POL. Similarly, larger input switches (S1-S4) may be used in the driver circuit to further improve overall efficiency without impacting space considerations at the POL.

[0059] However counter intuitive separating the driver 420 from the POL circuitry 430 and deploying an AC bus may initially seem, closer inspection refutes such objections. For example, power carried by the AC bus 410 may be spectrally pure (sine wave) and voltage and current slew rates substantially lower than those typically found in other switching power converter topologies, such as buck and multiphase buck converters, and even in the signal paths of computer circuitry, reducing concerns about noise and electromagnetic emissions.

[0060] The POL circuit 430 may be enclosed as a single module, i.e. packaged for deployment as a single self-contained unit (as shown in Figs. 5, 6, and 8), or as a multiplicity, specifically a pair, of modules for deployment as component pairs, e.g. 440 and 450, (as shown in Figs. 1-3, 7). Because switches R1-R4, need only withstand the output voltage, the rectification circuit 450 may be integrated (with or without the control circuitry, e.g. as shown in Figs. 7 and 8 of the FT Patent) onto a die with circuitry to which it supplies power, e.g. a processor core or an ASIC.

B. Alternative POL Topology

[0061] Referring to Fig. 6, an alternate embodiment 401 of the converter topology is shown comprising driver circuit 481 and POL circuit 431. In addition to outputs 427, 428 for driving the AC power bus 410, the driver circuit 481 as shown also includes a bias output 429 for supplying a small amount of power to operate control circuitry in the POL, and a control output 426 for supplying timing and/or control information to the POL circuit 431. A small signal bus 415 may be provided to

connect driver outputs 429, 426, which may be low power and low voltage signals, to the input of the POL circuit 431.

[0062] An example of suitable control circuitry is described in *Digital Control of Resonant Power Converters*, Vinciarelli et al., U.S. Patent no. 9,166,481, issued Oct. 15, 2015, assigned to VLT, Inc. and incorporated here by reference (the “Controller patent”), e.g. in connection with Figs. 15 and 16. The driver may further include clamp and control circuitry to implement the clamped capacitor techniques to increase efficiency as described in *Clamped Capacitor Resonant Power Converter*, Vinciarelli, U.S. Patent Application Ser. no. 14/874,054, filed Oct. 2, 2015, now U.S. Patent No. 9,325,247, issued on April 26, 2016 (the “CSAC” patent application and patent, respectively), assigned to VLT, Inc. and incorporated here by reference. As shown in Fig. 6, the converter 401 may be partitioned with the resonant capacitors 441, 442 located in the driver module 481. Alternatively, a single resonant capacitor may be located within the driver for ease of implementing the clamp circuitry.

[0063] The POL circuit 431 as shown may include a switch driver 460 having inputs 449, 446 for receiving a bias voltage (449) and a control signal (446) from the driver circuit 481. The bias voltage provides power to operate the switch driver and the control signal provides timing information to the switch driver to synchronize operation of the secondary switches 451, 452, 453, 454 as controlled rectifiers. The secondary controller 200B shown in Fig. 15 of the Controller patent may be used for the POL switch driver 460 in Fig. 6. Although the switch driver 460 is shown in Fig. 6 with a dedicated bias supply from the driver, it may, as shown in the Controller patent, derive the power it requires to operate from the control signal or from an independent bias supply. The switch driver 460 may alternatively be co-packaged with, or integrated within, the driver circuit 481 and the secondary switches may be driven directly by the driver circuit 481, in which case the small signal bus may be used to carry gate drive signals for the secondary switches instead of the bias and timing/control signals shown.

[0064] The POL circuit 431 may be enclosed as a single module, i.e. packaged for deployment as a single self-contained unit, or as a plurality of modules for deployment as component parts, e.g. transformer module 440, secondary switches 451-454, and driver circuit 460. Because the secondary switches 451-454 need only withstand the output voltage, the rectification circuit 450 may be integrated with driver circuitry 460 onto a single die or even on the same die as the circuitry to which it supplies power, e.g. a processor core, such as a GPU, CPU, or ASIC.

C. Single-Driver Multi-POL Topology

[0065] Referring to Fig. 7, another embodiment of the topology is shown as converter 402 including a single driver circuit 481 and a plurality of POL circuits, 431-1, 431-2. The driver circuit 481 and POL circuits 431-1, 431-2 may be the same as driver circuit 481 and POL circuit 431 shown in Fig. 6, respectively. The converter 402 may include an AC bus 410 connected to outputs 427, 428 of driver circuit 481 and a small signal bus 415 (low power, low voltage) connected to the bias 429 and control 426 outputs of the driver 481 for establishing the requisite connections to each POL circuit 431-1, 431-2.

[0066] The POL circuits may be connected to operate with their outputs 413, 414 connected in parallel for low voltage loads such as a CPU, GPU, or ASIC 101. Alternatively, inputs to the POL circuits may be connected in series for lower output voltages. To summarize, power may be supplied to the POL circuits 431 by the driver 481 at a bus voltage, V_{bus} , that is a multiple, X , times greater than the voltage, V_{load} , required by the load (e.g., one or more semiconductor chips 101). The multiple X may preferably be an integer (or alternatively a non-integer rational number), preferably at least 5, or greater, e.g., 10, 20, and more preferably 40 or more. Each POL circuit may have a fixed voltage transformation ratio, $K = V_{out}/V_{in}$ at a load current, where K may be equal to or greater than the turns ratio or step-down ratio, N , of the respective transformer in each POL circuit, depending for example on the output circuitry. The voltage transformation ratio, K , of each POL circuit may be less than or equal to the inverse of the multiple, $X = V_{bus} / V_{load}$, depending on the number and configuration

of POL circuits supplying the load. For example, with the inputs and outputs of two or more POL circuits connected in parallel, the bus voltage, V_{bus} , may be set to $X = 1 / K$ times the load voltage, V_{load} : $V_{bus} = V_{load} / K$. Alternatively, it may be preferable for very low output voltages, to arrange a number, M , of POL circuits with their respective inputs connected in series and outputs connected in parallel, in which case the bus voltage V_{bus} may be set to $X = 1/(M * K)$ times the load voltage, V_{load} : $V_{bus} = V_{load} / (M * K)$. The POL circuits 431-1, 431-2 may be deployed as close as possible to the load or preferably co-packaged together with the load as shown schematically in Fig. 7 and mechanically in Figs. 1-3.

D. Integrated Driver Regulator

[0067] Referring to Fig. 9, another power converter system 403 is shown including a driver 490 electrically connected to the semiconductor package 100 via connections formed by a system PCB in which the driver 490 and semiconductor package 100 may be mounted. The bias and control connections (415, Fig. 6) and AC power connections (410, Fig. 6) between the driver 490 and the substrate 102 are not shown in Fig. 9 for clarity, however, it should be understood that the desired connections, e.g. as shown in Figs. 5-7 may be provided in the manner described above. The driver 490 as shown in Fig. 9 may include the transformer driver circuitry 481 (which may be of the type shown in Fig. 5 or 7), a power regulator circuit 482, and a supervisory circuit 483. The power regulator 482 may be used to control the voltage, V_F , input to the driver circuit 481 as a means of controlling the AC voltage supplied to the POL circuits 431-1, 431-2 and in turn the DC output voltage to the semiconductor die 101.

[0068] The supervisory circuit 483 may be connected to communicate with the semiconductor die 101 and optionally the POL circuits 431-1, 431-2 via a digital or analog communication bus 497 as shown in Fig. 9. Although shown as a single bus, the semiconductor die 101 and POL circuits 431 may have one or more separate buses for direct communication on the substrate 102 and the supervisory circuit 483 may have separate busses for communication with the die 101 and with the POL circuits

431, e.g. to accommodate different communication speeds and protocols. The supervisory circuit 483 may be connected to communicate with external system components via a digital or analog communication bus 495 as shown in Fig. 9, e.g. to report on conditions in the semiconductor package or power system, e.g. temperature, voltage, current, power, fault conditions, etc. or to receive commands, e.g. reset, disable, etc. For example, some CPU's require the power system to adjust the voltage supplied to the CPU in response to commands issued by the CPU, e.g. many Intel processors send voltage identification (VID) information to a voltage regulator which in turn adjusts the voltage supplied to the processor. The supervisory circuit 483 may receive such voltage commands from the semiconductor die 101 via bus 497 and issue appropriate commands to regulator 482 via digital or analog communication bus 496 to adjust the output voltage. The regulator may, in response to commands received from the supervisory circuit 483, adjust the DC output voltage of the output circuits (via the control voltage, V_F) to comply with the requirements of the semiconductor die 101.

[0069] An integrated dual driver module 490-4 will be described in connection with the block diagrams of Figs. 27 and 28. In the electrical block diagram of Fig. 28, the driver 490-4 is shown including a power regulator 482 (similar to regulator 482 described in Fig. 9) supplying power to two driver circuits 481-1 and 481-2 (each similar to the driver 481 described in Figs. 6, 7, and 9). The dual driver is similar to the integrated driver 490 described above in connection with Fig. 9, however two driver circuits, 481-1, 481-2 are connected to receive power from the regulation stage 482. As shown in Fig. 28, the regulation stage may be implemented using a single regulator 482 supplying power to both drivers 481-1, 481-2 or optionally as a plurality of regulators, 482A, 482B, as shown by the broken line 482-AB, each supplying power to its respective driver. The dual integrated driver module 490-4 may preferably be arranged with the regulation stage or stages located in the center of the package and the drivers located at each end of the package as best illustrated shown in Fig. 27.

[0070] The dual integrated driver module 490-4 may be configured and arranged to enable efficient routing of power and control signals as shown in the system block

diagram in Fig. 27. For example, the regulator power input terminals 491, 492 and system control signals 495 may be provided in the central portion, and the driver output terminal pairs, 427-1 and 428-1, 427-2 and 428-2, may be situated at the ends of the dual integrated driver module 490-4 to facilitate routing that avoids the footprint of, and the primary signal paths to and from, the die 101. For example, the power system 408 shown in Fig. 27 includes the dual integrated driver 490-4 positioned on a system board 11C along one side of a semiconductor die 101, with two POL driver circuits 431-1, 431-2 positioned along the two sides of the die 101 adjacent to the driver 490-4. The die 101 may be a large semiconductor die such as a CPU, GPU, ASIC or other large scale integrated circuit. The large arrows in Fig. 27 demonstrate a preferred power flow in the system 408 with the input power (preferably a relatively high voltage) being routed from the right side of the board 11C to the driver inputs 491, 492, i.e. to the regulator(s) 482 at the center of the dual driver 490-4 (arrow A); power flows through the dual integrated driver module 490-4 from the center (the regulator(s)) along the longitudinal axis of the module to the opposite ends (the drivers); power from the driver outputs, terminals 427-1, 428-1, 427-2, 428-2, at each of the opposite ends flows a short distance around the respective corners of the die (arrows B) to the input terminals, 447-1, 448-1, 447-2, 448-2 (e.g., see FIG. 7), at the ends of the POL output circuit modules 431-1, 431-2; and power flows out of each of the long sides of the POL output circuit modules 431-1 and 431-2 to the semiconductor die 101 (arrows C). Note that the connections (B, 410-1, 410-2) between the driver outputs and the POL output circuit inputs correspond to the high voltage bus described in connection with previous figures, e.g. 410 in Fig. 6, 410-1, 410-2 in Fig7, etc. The POL circuit module output terminals 413-1, 414-1, 413-2, and 414-2, supply power to the semiconductor die 101. Finally, the low power timing and control signals (bus 415 in Figs. 6, 7, and 19) flowing through, e.g. terminals 446-1, 449-1, 446-2, 449-2, between the driver and the POL output circuits may be routed from the far ends of the POL output circuit modules 431-1, 431-2 to terminals 497 on the central die-facing portion of the module 490-4.

E. Multi-Driver Multi-Rail Topology

[0071] Referring to Fig. 13, another embodiment of the topology is shown as converter 404, which is configured to supply multiple voltage supply rails, V1, V2, to the semiconductor load 101. In some examples, the power converter can include output circuits that form multiple output rails configured to supply power at multiple respective rail voltages to one or more loads. In some examples, each of at least some of the output rails is configured to provide a rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential. For examples, the rail voltages can be 1V, 1.1V, 1.2V, and 1.3V. For example, the rail voltages can be 1V, 1.2V, 1.5V, and 2V. For example, the rail voltages can be 0.66V, 0.75V, 0.86V, and 1V. Converter 404 includes two driver circuits 490-1, 490-2 for driving two POL circuits, 431-1 and 431-2, respectively. The driver circuits 490-1, 490-2 and POL circuits 431-1, 431-2, may be of the same type as driver circuit 490 (shown in Fig. 9) and POL circuits 431 (shown in Fig. 6), respectively. For simplicity, the connections between the drivers and POL circuits and semiconductor package are shown as single connections, it being understood that each may include an AC power bus, a control bus, and communications bus as described above. Preferably, the drivers may be synchronized to the same clock as shown in Fig. 13 by the broken arrow from driver 490-1 (the master) to driver 490-2 (slave). Preferably, the POL circuits 431-1 and 431-2 may be co-packaged as a single POL circuit module 431. Each driver may, in response to commands received from the semiconductor die 101, adjust the DC output voltage of the POL circuit associated with it to comply with the requirements of the semiconductor die 101. For example, driver 490-1 may adjust the output V1 of POL circuit 431-1 and driver 490-2 may adjust the output V2 of POL circuit 431-2.

F. Driver Compensation

[0072] Separation of the driver 481 and integrated controller 425 (Fig. 6) from the POL circuits 431 may introduce parasitic capacitances and inductances, which, depending upon the layout of the customer's system board, e.g. the distance between

driver and POL circuits and the size and routing of the electrical connections between them, may adversely affect operation of the converter. For example, the parasitic inductance may lower (raise) the resonant frequency (period) of the resonant circuit (formed by the resonant capacitors 441, 442 and the transformer 440) which if uncompensated can lead to timing errors in the operation of the switches disrupting zero-voltage switching (ZVS) and zero-current switching (ZCS) operation, which in turn may lead to increased losses, power dissipation, and noise.

[0073] Preferably, the driver 481 may include compensation circuitry able to detect and adjust for the effects of parasitic capacitances and inductances introduced by the separation of driver and POL circuits and the vagaries of different system board layouts on converter operation. One method uses current detection, e.g. in one or more of the primary switches (e.g. switches 421, 422, 423, 424 in Fig. 6) of the driver circuit (using known techniques such as sensing the voltage across a switch while in the ON state) to detect errors in the switch timing. For example, if the compensation circuitry detects that the resonant current at the end of a power transfer interval has not returned to zero, the controller may incrementally increase the duration of the power transfer intervals until the current returns to zero or within a tolerance band of zero, e.g. 1% of the maximum resonant current. For the clamped version described in the CSAC patent application, the compensation circuitry may additionally sense the rate of change of the switch current at the end of the first resonant interval, extending it until the rate of change of the current returns to zero, or within a tolerance band of zero, or within a percentage of the maximum rate of change, e.g. 10%, 5%, or 1 %. In this way, the compensation circuitry may adjust the overall timing of the converter operating cycle and/or specific aspects of the converter operating cycle, e.g. the power transfer intervals (described in the SAC, POL-SAC, and Controller patents), or the first and second resonant intervals (described in the CSAC patent application), etc.

[0074] Fig. 21A shows a generalized block diagram of a power converter system 407 for supplying power to a load circuit on a semiconductor chip 101-5, including three power conversion stages, 471, 472, and 473. As shown, the first conversion stage, 471, is located outside of the semiconductor package 100-2, the second stage,

472, resides within the semiconductor package 100-2 with the semiconductor die 101-5 (which may be a CPU, GPU, ASIC, or other high load semiconductor device), and the third stage, 473, resides on the semiconductor die 101-5. The first conversion stage, 471, converts power received from a source (not shown) at a source voltage, V_{src} , at the input of the first conversion stage, 471, and delivers power to the semiconductor package 100-2 at a package voltage, V_{pkg} , preferably performing voltage transformation, e.g. reducing the voltage from a high source voltage, e.g. 48V, conducive to efficient distribution in the system, e.g. within a server rack, to a lower voltage, e.g. 25V or 6V or less, that may be appropriate for the semiconductor package or the system board on which the semiconductor package may be mounted. The second conversion stage, 472, residing within the semiconductor package 100-2, converts power received from the output of the first stage at the package voltage, V_{pkg} , for delivery to the semiconductor die, 101-5, at a chip voltage, V_{chip} , preferably further reducing the voltage by a factor of 5, 6, or more. On the semiconductor die 101-5, a third power conversion stage, 473, further processes the power for delivery to one or more loads on the die at a load voltage, V_L . The on-chip power processor, 473, may provide regulation and some voltage transformation, e.g. by a factor of two or less. In some examples, the source voltage can be greater than 40 VDC, the package voltage, V_{pkg} , can be greater than 20 VDC and the load voltage, V_L , can be less than 3 VDC. In some examples, the package voltage, V_{pkg} , can be greater than 5 VDC and the load voltage, V_L , can be less than 2 VDC.

[0075] The generalized converter system 407 may be better understood using the following specific examples. As a first example, consider a modified version of the system 403 (shown in Fig. 9) in which the semiconductor die 101 is replaced with a semiconductor die 101-5 that includes an internal power processing stage 473 as shown Fig. 21A. The integrated driver 490 of Fig. 9, corresponding to the first power conversion stage 471 of Fig. 21A, is located outside of the semiconductor package, receives power from the source at the input voltage, V_{in} , and supplies power to the semiconductor package 100 at the voltage V_F (V_{pkg} as shown in Fig. 21A). As described above in connection with Fig. 9, the integrated driver 490 includes a

regulation function in 482 and also includes a driver function 481 which generates an AC voltage for distribution to the semiconductor package 100-2. In this example, the voltage V_{pkg} is an AC voltage having a regulated peak value. The one or more POL output circuits, 431-1, 431-2, shown within the semiconductor package 100 in Fig. 9, corresponding to the second power conversion stage 472 situated within the semiconductor package 100-2 as shown in Fig. 21A, receive power at the higher voltage, V_{pkg} , and converts it down to a lower voltage for delivery to the semiconductor chip 101-5. The example of Fig. 9 may use a voltage transformation ratio of 5, 10, 20, or even 40 or greater as described above. The third power conversion stage 473 is located in the semiconductor chip as shown in Fig. 21A and may provide voltage selection and optionally additional regulation as discussed below. Although a single on-chip power processing stage 473 is shown in Fig. 21A feeding a single load, it should be appreciated that the chip may include a plurality of power processing stages feeding one or more respective loads as discussed further below.

[0076] In an alternate embodiment power system 407 (Fig. 21A) may include a fixed ratio converter for the first stage power conversion 471 providing voltage transformation and optionally galvanic isolation and regulation, converting power from a relatively high voltage bus, e.g. 48 VDC, for delivery to a lower package voltage, e.g. 5, 6, or 12VDC, or even lower. The second stage 472 may provide voltage transformation and optionally voltage regulation and galvanic isolation, converting power from the package voltage for delivery to the lower chip voltage, V_{chip} . The second stage 472 preferably provides a voltage reduction of 5 or more such that V_{pkg} is at least 5 times greater than V_{chip} . Other voltage transformation factors are possible, e.g. 4, or 6, or 12, etc. In such an example, the second stage conversion may comprise a multi-phase regulator operating in buck mode. The on-chip regulation stage 473 may comprise switching or linear regulators as appropriate for the voltage and power levels and efficiencies.

G. Multi-Rail POL with On-Chip Rail Selection

[0077] Referring to Fig. 21B, a power system 406 is shown, as another example of the power system 407 (Fig. 21A), including a driver circuit 490B connected to drive a multi-rail output POL circuit 431-4 having four outputs for supplying four unique voltages, V1, V2, V3, V4, to a large semiconductor die 101-4, which may be a CPU, GPU, ASIC or other semiconductor device. Although a single die is shown, the die 101-4 may comprise a plurality of chips. Power system 406 (Fig. 21B) is similar to the previous example of system 407 discussed in connection with Fig. 9 above, however, system 406 is modified to provide multiple voltages to the semiconductor chip 101-4. Although not shown in Fig. 21B, the semiconductor device 101-4 includes a third power processing stage similar to that shown in Fig. 21A to be described in greater detail in connection with Fig. 24 below. As shown, the POL circuit 431-4 and semiconductor die 101-4 may preferably be co-packaged together in semiconductor package 100-1, e.g. as described in connection with Figs. 1-4, 10, 11. The driver 490-B may be of the type described above in connection with Fig. 9 including an AC power bus 410, a small signal bus 415 for switch timing and control signals, and an optional communication bus 497. As shown, the multi-rail output POL circuit 431-4 includes a plurality of outputs connected to supply power to the semiconductor die 101-4. In the example shown in Fig. 21B, four outputs, V1, V2, V3, V4, and a common terminal, GND, are provided for supplying power at four separate voltages, V1, V2, V3, V4, to the semiconductor die 101-4. Embodiments of the multi-rail output circuit 431-4 are described in greater detail in connection with Figs. 22 and 23 and details of the third power conversion stage residing on the semiconductor die 101-4 are described in greater detail in connection with Figs. 24 and 25 below.

[0078] Referring to the schematic diagram of Fig. 22, a first embodiment of the multi-rail output POL circuit 431-4 is shown having a plurality of independent transformers, 440-1, 440-2, 440-3, and 440-4, each preferably having a two-turn center-tapped secondary winding connected to a respective rectifier circuit, 450-1, 450-2, 450-3, 450-4, for supplying power to its respective output terminal, 413-1, 413-2, 413-3, 413-4 at a respective output voltage V1, V2, V3, V4. Each transformer may

include a multi-turn primary winding connected to the AC input terminals 447, 448, the number of turns being chosen to establish a respective desired turns ration, $N_{Pri-1} : N_{S-1}$, $N_{Pri-2} : N_{S-2}$, $N_{Pri-3} : N_{S-3}$, $N_{Pri-4} : N_{S-4}$. For example, the primary windings of transformers 440-1, 440-2, 440-3, and 440-4 may have 36, 40, 44, and 48 turns producing turns ratios of 36:1, 40:1, 44:1, and 48:1 and voltage transformation ratios of $1/36$, $1/40$, $1/44$, and $1/48$ respectively. With a nominal driving voltage of 48 Volts, the POL circuit would produce nominal output voltages: $V1 = 1.3 \text{ V}$, $V2 = 1.2 \text{ V}$, $V3 = 1.1 \text{ V}$ and $V4 = 1.0 \text{ V}$. As the driver circuit 490B changes the AC power bus voltage, the output voltages of the POL circuit will change according to their respective voltage transformation ratios with each rail tracking the others up or downward. In this way all voltages being supplied to the semiconductor die may be adjusted upward or downward together. As explained in more detail below, power management circuitry on the semiconductor die may change the voltage of individual loads efficiently by selecting the appropriate voltage rail or change all of the voltages in unison by sending voltage commands to the driver circuit which will accordingly adjust the drive voltage.

[0079] Other turns ratios may be chosen to provide different rail voltages, e.g. turns ratios of 24:1, 32:1, 40:1, and 48:1 may be used to provide rail voltages $V1 = 2\text{V}$, $V2 = 1.5\text{V}$, $V3 = 1.2\text{V}$, and $V4 = 1\text{V}$, or turns ratios of 48:1, 56:1, 64:1, and 72:1 may be used to provide rail voltages $V1 = 1.0\text{V}$, $V2 = 0.86\text{V}$, $V3 = 0.75\text{V}$, and $V4 = 0.66\text{V}$ respectively, from a 48 Volt drive. Although the above examples of turn ratios use uniform increments of 8 turns between adjacent rails, it will be appreciated that other, either uniform or non-uniform, increments of turns may be used between some or all of the rails to set the desired rail voltage spacing according to the design requirements. Similarly, lower turns ratios may be used to accommodate lower output voltages from the first power conversion stage.

[0080] As shown in Fig. 22, each rectifier circuit, 450-1, 450-2, 450-3, 450-4, comprises two controlled rectifier switches which are controlled by switch drive circuit 460. The rectifier circuits can convert AC signals received from the respective secondary windings to unipolar voltages. As shown, the controlled rectifier switches may preferably comprise MOSFET devices connected in common source

configuration to simplify the gate drive circuitry which may be ground referenced. Because the primary windings of all four transformers, 440-1, 440-2, 440-3, and 440-4 are driven by the same AC signal, their secondary waveforms will be synchronized allowing the respective secondary switches to be driven in unison by the same signal further simplifying the drive circuitry. The switch drive circuit 460 may be integrated into the POL circuit 431-4 as shown having inputs for receiving a control signal from the driver circuit 490B and a bias voltage. Alternatively, the drive circuit may be integrated into the driver circuit 490B as described above which may provide gate drive signals directly to the switches 450-1, 450-2, 450-3, 450-4 as described above in connection with Fig. 6. Common terminal 414 provides the ground or common connection for each of the multiple outputs.

[0081] Referring to Fig. 24, a block diagram of the power system of semiconductor die 101-4 is shown including a power management control circuit 130, and a plurality of rail selection and regulation circuits 120-1, 120-2, 120-3, and 120-n, each supplying power to a respective load 129-1, 129-2, 129-3, and 129-n on the semiconductor die. The power management control circuit 130 may provide one or more signals to rail selection and regulation circuits, 120-1, 120-2, 120-3, 120-n, to set the desired voltage for its respective load, 129-1, 129-2, 129-3, and 129-n. The die 101-4 may include internal voltage supply rails, V1, V2, V3, V4, as shown in Fig. 24, connected to receive power from the respective outputs of the POL circuit 431-4 (Fig. 21B).

[0082] The circuitry of a representative one of the rail selection and regulation circuits, 120-n, is shown in greater detail in Fig. 25 having an input 131-n for receiving voltage commands from the power management circuit 130, a control circuit 128, a comparator 127, a voltage reference 125, switches 121, 122, 123, and 124, inputs connected to the voltage rails, and output 132-n connected to supply power to load 129-n. In operation, the control circuit 128 may enable one of the switches 121, 122, 123, or 124, at a time to select the appropriate input voltage (rail voltage V1, V2, V3 or V4) from which power is drawn to supply power to load 129-n. Preferably the lowest rail voltage that provides sufficient headroom for the rail selection and

regulation circuit 120-n to regulate the voltage supplied to load 129-n is chosen to minimize power dissipation in the regulator. Consistent with the description of Fig. 22 above, it is assumed that $V1 > V2 > V3 > V4$. As shown in Fig. 25, switches 122, 123, and 124 may comprise dual MOSFETs connected back-to-back in a common source configuration to block current in both directions when OFF to isolate its respective rail from the higher voltage rails. Since rail voltage $V1$ is greater than the three other rail voltages $V2$, $V3$, and $V4$, the control circuit 128 may easily drive switches 122, 123, and 124 with sufficient voltage to turn on both of the MOSFETs in the respective switches. Switch 121 may comprise a single MOSFET device since current need only be blocked in one direction when OFF. Although shown as dual MOSFETs connected back-to-back in a common source configuration, switches 122, 123, and 124 may be implemented using any appropriate technology to block current flow in either direction.

[0083] The switches 121, 122, 123 and 124 may serve dual functions of (1) rail selection (isolating the load from the unselected voltage rails when OFF and connecting to the appropriate rail when ON) and (2) regulation (acting as a series pass device in a linear regulator when ON). In such a system, the control circuit 128 may connect the gate(s) of the selected switch 121, 122, 123, 124 to the output of error amplifier 127 which may provide a feedback signal by comparing the load voltage with the reference voltage, V_{Ref} , produced by reference 125. The voltage reference 125 may adjust the reference voltage according to the commands received via an optional control bus 126 either directly from the power management circuit 130 or as shown through control circuit 128. For example, a single reference may be used with a voltage divider, which may be adjusted by commands received from the power management circuit 130 or control circuit 128, to scale the reference voltage output to the error amplifier as needed for regulation of the selected load voltage. Alternatively, individual reference voltages may be provided.

[0084] In operation, the control circuit 128 may select the appropriate rail, $V1$, $V2$, $V3$, $V4$, by enabling the respective switch, 121, 122, 123, 124, and scaling the reference voltage according to the voltage command received from the power

management circuit 130. The error amplifier provides the drive signal for the selected switch, modulating the conductivity of the switch to adjust the voltage drop across the switch, thereby regulating the load voltage according to the commands received from the power management circuit 130.

[0085] It will be appreciated that with the primary windings connected as shown in Fig. 22, the outputs of the POL circuit 431-4 each will track changes in the drive voltage allowing all outputs to be adjusted in unison. Thus the power management circuit 130 may adjust the supply voltages for the entire semiconductor die 101-4 by issuing commands to the driver 490B, e.g. over the communication bus 497. For example, in a typical ASIC or processor, a great multiplicity of independent loads may each require adjustment of the supply voltage according to their respective operational status and the overall power dissipation on the die. The linked-output POL circuit advantageously may use a single driver, with a single AC bus, to drive all of the primary windings in the POL circuit as shown. However, the outputs of the POL circuit 431-4 may be controlled independently by using separate drivers for each transformer primary winding, e.g. as described above in the multi-driver multi-rail system (Section E).

[0086] Referring to Fig. 23, an alternate embodiment is shown comprising two multi-rail POL circuits 431-5A, 431-5B. As shown, the POL circuits 431-5A, 431-5B are identical, each having a transformer 440 a switch driver 460, and secondary switches 450. The transformer is shown having a single primary winding and a multi-tap secondary winding. Each tap of the secondary is connected to a half-wave rectifier, which may preferably be a MOSFET switch. The two POL circuits are connected in parallel at the input and output except that the primary windings are connected in phase opposition. As a result, each of the two secondary windings and their respective switches 450 form half of a center-tapped full-wave rectified four rail output circuit. It will be appreciated that the POL circuit 431-5 as shown may be combined in pairs to provide a scalable architecture able to address increasing power requirements.

[0087] In an alternative embodiment of the rail selection and regulation circuits, the switches 121, 122, 123, 124 (Fig. 25) may be used exclusively to connect and disconnect the regulation circuitry to the appropriate rail voltage and a dedicated down-stream regulator, preferably a series-pass regulator, may be used to regulate the selected rail voltage.

H. Ground Referenced Driver

[0088] Referring to Fig. 26A, ground-referenced driver circuit 481-2 is shown. Instead of the full-bridge configuration as shown in Figs. 5, 6, and 9, driver circuit 481-2 uses a half-bridge configuration comprising switches S1 421 and S2 422 to drive output terminal 427 through capacitor 441, which may be a resonant capacitor, e.g. for implementing a Sine Amplitude Converter as described in the SAC Patent. As a result of the half bridge configuration, the primary windings of the POL circuits 431-6 are driven with a peak-to-peak voltage that is half of the input voltage, V_{IN} , feeding the driver circuit. Because one end of the primary windings is tied to ground potential through output terminal, 428, and the other end is capacitively coupled through capacitor 441 to the half-bridge driver, the voltage at terminal 427 will have an average of essentially zero volts and positive and negative excursions that are approximately $+ \frac{1}{2} V_{IN}$ and $- \frac{1}{2} V_{IN}$. For a $V_{IN} = 50$ VDC, the AC drive signal will have maximum excursions of 25V or less from ground maintaining a safe low voltage (SELV) drive signal. Furthermore, because the average voltage at either terminal will be zero (the transformer primary winding may be said to be driven by an average voltage of essentially zero volts), electro-migration concerns along the power bus and in the semiconductor package, e.g. the substrate, are eliminated.

[0089] In Fig. 26A, the input voltage, V_{IN} , is received at a source terminal 411 and a common terminal 412. Depending on which of the switches S1 421 and S2 422 is turned on, a node 257, referred to as the “switching node,” is connected to either the source terminal 411 or the common terminal 412, and drives the output terminal 427 through the capacitor 441.

[0090] Referring to Fig. 26B, a second ground-referenced driver circuit 481-3, which maintains the low voltage benefits of driver 481-2 (Fig 26A), is shown including a resonant capacitor 441, a DC blocking capacitor 443, and a clamp switch 444 connected in parallel with the resonant capacitor 441 for implementing the clamped resonant capacitor control described in the CSAC Patent. The clamp switch 444 may consist of two MOSFETs connected to block current in both directions as shown, preferably with the gates connected together. The DC blocking capacitor is assumed to be at least an order of magnitude larger than the resonant capacitor such that resonant period of the transformer and DC blocking capacitor combination is large compared with the duration of the longest clamp phase (switch 444 ON time) that may be implemented in the converter. Preferably, the resonant capacitor 441 may be sized relative to the primary-reflected leakage inductance of the transformer to keep the voltage across the resonant capacitor 441 to within a small percentage, e.g. to about 10 percent or less, of the driver input voltage, V_{IN} , to keep the size of the clamp switch small. For example, in a driver adapted for operation with an input voltage of 50V, the resonant capacitor may be chosen to produce a maximum excursion of 5V or less from ground. Limiting the voltage excursions across the resonant capacitor also has the beneficial effect of reducing the peak-to-peak voltage difference between terminals 427 and 428 while maximizing the peak-to-peak voltage impressed across the primary winding(s) of the POL circuits 431-6. An optional high value resistor 445, e.g. 1 mega Ohm, may be connected in parallel with the resonant capacitor 441 to keep the voltage at terminals 427 and 428 definite in the event that the clamp switch 444 is not used. Like the drive voltage at terminal 427, the voltage at terminal 428 will have an average voltage of essentially zero volts albeit with a much smaller, e.g. 10% or less, maximum excursion from ground potential. Thus the driver circuit shown in Fig. 26B has the advantage of providing ground referenced control terminals for operation of the clamp switch while also providing a safe low voltage drive signal and an average drive voltage of essentially zero volts on each of the drive terminals.

[0091] In some examples, the driver circuit 481-2 and/or 481-3 provides a driver voltage having a maximum peak-to-peak voltage, V_{peak} , that is at least a multiple, $X1$, times greater than at least one of the rail voltages (e.g., $V1$, $V2$, $V3$, $V4$ of Figs. 23

and 24) and $X1$ is at least six. In some examples, $X1$ is at least 10. In some examples, $X1$ is at least 20. In some examples, the driver circuit 481-2 and/or 481-3 provides power to the semiconductor package 100-1 at a bus voltage, V_{bus} , that is at least a multiple, $X2$, times greater than at least one of the rail voltages (e.g., $V1$, $V2$, $V3$, $V4$ of Figs. 23 and 24) and $X2$ is at least 15. For example, the terminal 428 can be referenced to a voltage potential (e.g., voltage potential across the resonant capacitor 441) within a predetermined range of the voltage at the common terminal 412, in which the predetermined range is not more than 10% of the source voltage V_{IN} .

[0092] The switch control circuitry, e.g. switch control 425 (Figs. 5, 6, 9) and optional communication 483 and regulation circuitry 482 circuitry (Fig. 9) are not shown in Figs. 26A and 26B for clarity. However, it should be understood these features may be used in combination with the ground-referenced driver circuits in Figs. 26A and 26B.

II. Semiconductor Package with Top-Mounted Integrated POL Circuits

[0093] In Figs. 1, 2 and 3, a semiconductor package 100 is shown (in isometric, top plan, and side views, respectively) including a multilayer substrate 102, a large semiconductor die 101, such as an ASIC, CPU, or GPU, and a plurality of POL modules 110-1, 110-2 mounted to the substrate adjacent to the semiconductor die 101. As shown in Fig. 8, the POL modules 110 may be packaged as a leadless module, such as described in Vinciarelli, et al., *Panel Molded Electronic Assemblies with Multi-Surface Conductive Contacts*, Application Ser. No. 14/731,287, filed June 4, 2015, and incorporated here by reference (the “Panel Mold” application”), having connections 115 (Figs. 1, 8) for surface mount soldering to respective conductive pads on the substrate 102 and preferably include shielding for improved low noise performance. (See, e.g. leadless electronic module 100 described in connection with Figs. 1-3 of the Panel Mold application.) The POL modules 110-1 and 110-2 (Figs. 1-3) may each include a respective POL circuit, e.g. 431-1, 431-2, as described in connection with Fig. 7.

[0094] Referring to Fig. 1, the POL modules 110-1, 110-2 are shown having a multiplicity of electrical contacts 115 arranged along their respective perimeters. The contacts may be formed as shown in Figs. 1-3 and 8 (and as described in the Panel Mold application) to extend along the entire vertical span of the perimeter walls and onto the top and bottom module surfaces. The common terminals may be extended as shown onto the top and bottom surfaces to form shielding. As shown for the two POL Module example in Figs. 1-3, a plurality of discrete components 105, such as capacitors, e.g. for filtering, may be provided in the free space along the semiconductor die 101. It may be preferable for very high current applications to use four POL modules, each mounted along a respective one of the four sides of the semiconductor die 101 to lower the interconnection resistance between the POL module outputs and the die. Using four POL modules to power the die may allow a reduction in the length of each POL module, e.g. by a factor of two, leaving space for discrete components such as capacitors along a respective one of the four sides of the semiconductor die 101.

[0095] In Fig. 2, several of the contacts 115 are labeled to show their preferred function. For example, at the left end of each POL module as shown in Fig. 2, two terminals are labeled consistent with Fig. 7 showing the AC power inputs 447-1 and 448-1 for module 110-1 and 447-2 and 448-2 for module 110-2; similarly, two terminals are labeled showing the respective bias and control inputs: 449-1 and 446-1 for POL module 110-1 and 449-2 and 446-2 for POL module 110-2. A multiplicity of output terminals 413-1, 413-2 and ground terminals 414-1, 414-2 are provided along the perimeter of the POL modules 110-1, 110-2 respectively to provide a low impedance distributed connection to the substrate.

[0096] In Fig. 7, the substrate 102 of semiconductor package 101 is represented with broken lines: electrical connections between the substrate 102 and a system board (to which it may be connected) are represented by interface connections 465, 466, 467-1, 468-1, 467-2, and 468-2; connections contained within the broken lines may be formed by the substrate 102; and connections outside the broken lines may be formed by the system board which typically may be a multi-layer printed circuit board

(“PCB”). The driver circuit 481 may be mounted away from the point-of-load, e.g. the semiconductor die 101 or the semiconductor package 100, on the system level board, which may provide electrical connections between the driver and the semiconductor package 100 or substrate 102 as shown symbolically in Fig. 7. Note that connections to the semiconductor die 101, which may be great in number, are not shown in Fig. 7 for clarity.

[0097] The substrate 102, in typical applications, carries a multitude of electrical connections between the semiconductor die 101 and a system-level PCB using, e.g. connector pins, ball grid array, land grid array, or other connection schemes. The breakdown voltage of the substrate 102 may be very low, e.g. on the order of 3 to 5 volts; the number of interface connections available for power connections between the substrate 102 and the system PCB may be limited due to the large number of input/output signals (“I/Os”) required by the semiconductor die 101; and consequently the ability to efficiently conduct large power supply currents may be limited. In Fig. 7, the POL circuit bias power connections 449-1 and 449-2 and control connections 446-1 and 446-2 are shown connected to bias and control interface connections 465, 466 respectively. The POL circuit power connections 447-1, 447-2, 448-1, and 448-2 are shown connected to power interface connections 467-1, 467-2, 468-1, 468-2.

[0098] As shown, the bias and control signals (Fig. 7), which are relatively low in voltage and power, may be handled by the substrate 102 and interface connections 465, 466 in the same manner as normal I/O signals to, and within, the package, e.g. the bias and control signals may be routed laterally along the substrate as shown by the small signal bus 415 within the substrate 102 (Fig. 7). However, connections to the AC power bus 410, which may need to carry voltages exceeding the voltage capabilities of the substrate 102, may not be suitable for wiring on the substrate. Accordingly, Fig. 7 shows a separate set of interface connections (467, 468) for carrying AC power from the bus 410 to each of the POL circuits (431-1, 431-2): interface connections 467-1 and 468-1 for POL circuit 431-1; and interface connections 467-2 and 468-2 for POL circuit 431-2. The AC power interface connections 467 and 468 will be described in greater detail with reference to the semiconductor package drawings of Figs. 1-4.

A. High Voltage Connections

[0099] The section taken along lines 4-4 (Fig. 2) is shown enlarged in the cross-sectional view of Fig. 4 providing physical detail of the AC power bus 410-2 (Figs. 4, 7) in the substrate 102. Because its voltage may exceed the voltage rating of the substrate, the AC power bus, e.g. 410-2, may be kept as short as possible in the substrate preferably extending vertically through the thickness of the substrate 102 from the interface connections 467-2, 468-2 on the bottom of the substrate to the conductive pads 117-2, 118-2 on top of the substrate for mating with POL module terminals 447-2, 448-2, respectively. The interface connections (e.g., 467-2, 468-2) on the bottom of the substrate 102 may be electrically coupled to connectors on the system board using, e.g. connector pins, ball grid array, land grid array, or other connection schemes. All lateral travel of the AC power bus in the substrate is preferably eliminated or, if unavoidable, then minimalized. In the example shown in Figs. 1-4, the AC power bus is divided into two sections 410-1 and 410-2 each of which consists of a pair of plated vertical through holes in the substrate 102 around which a minimum keep-out distance is maintained to account for the low interlayer breakdown voltage of the substrate. Thus as shown in Fig. 4, AC power bus 410-2 includes two vertical conductive through holes (collectively 410-2) connected to interface contacts 467-2, 468-2 on the bottom of the substrate 102 and conductive pads 117-2, 118-2 on the top of the substrate 102, all of which are preferably vertically aligned minimizing or eliminating any lateral conduction requirements for the AC power bus. The heavy broken lines 103 in Fig. 4 indicate a volume 104 of the substrate 102 defined by a projection around the high voltage connection in which no other electrical features should be formed to manage the relatively high voltage requirements of the AC power bus and the low breakdown voltage of the substrate. Ground referenced through holes may be formed in the substrate around the keep-out area to provide shielding.

B. Magnetic Field Management

[0100] Referring to Fig. 12, an example of a magnetically permeable core structure 150 in the POL modules 110-1, 110-2 is shown including a multilayer printed circuit board 151 in which the transformer windings (not shown) may be formed around a plurality of holes 155. The transformer may incorporate self-aligned windings as described in Vinciarelli, *Self-Aligned Planar Magnetic Structure and Method*, U.S. Patent Application Ser. No. 14/822,561, filed Aug. 10, 2015, (the “Self-Aligned” patent application) assigned to VLT, Inc. of Sunnyvale, CA, the entire disclosure of which is incorporated herein by reference). Core legs 154 may be placed in the holes 155 and mated with top and bottom core plates 152 and 153 when assembled to form complete magnetic loops. A small gap of 1 mil or less may be provided between one or both of the top and bottom core plates and the legs. Preferably, the effective magnetic permeability (μ) of the core legs and core plates is greater than 25 and preferably greater than 100, and more preferably greater than 200 to contain the magnetic flux during operation. Lower effective permeability core structures may result in greater flux leakage which can couple to signal conductors in the substrate, semiconductor die, or system board creating noise problems.

[0101] Referring to Figs. 14A and 14B, an alternate transformer structure 160 is shown having essentially the same PCB 151 in which the transformer windings (not shown), preferably self-aligned windings described in the Self-Aligned patent application, may be formed around a plurality of holes 155 to accommodate core legs such as the core legs 154 shown in Fig. 12, and bottom core plates 153 on the bottom surface of the PCB 151. The top core plates 162 may as shown include apertures 166 arranged to align with apertures 155 in the PCB when the core is in position. The transformer 160 may be fabricated by affixing the bottom core plates 153 to the bottom surface of the PCB 151 with a suitable adhesive such as epoxy. The top core plates 162 may be similarly affixed to the top surface of the PCB 151 with apertures 166 aligned with apertures 155. A magnetically permeable fluid, such as a powder with or without a suitable binder material or other injectable material, preferably having a permeability of 10 or greater, may be injected through apertures 166 to fill

the PCB apertures 155 and alternatively some or all of the core apertures 166. After the apertures are filled with the magnetically permeable material, the core apertures may be sealed with an epoxy or other suitable material to prevent the magnetically permeable material from escaping. One or more plugs can be disposed in one or more of the apertures 166, in which the one or more plugs form seals covering the respective one or more of the apertures 166. Optionally, the PCB may be heated before the apertures are filled and sealed to ensure that the apertures are completely filled after the PCB is cooled. For example, the core plate 153 can be a ferrite core plate, and the core plate 162 can be a ferrite core plate having one or more apertures that is/are filled with one or more plugs.

[0102] The transformer structure 160 of Figs. 14A and 14B may be particularly well suited to low voltage and high frequency applications such as the POL circuits discussed above. Using a material that may be injected as a powder or fluid into the apertures in the PCB overcomes the mechanical tolerances of conventional planar magnetic core structures where small solid core legs have to fit within small PCB apertures. By replacing small solid magnetic core legs with magnetic powder or fluid, PCB apertures can be filled with a permeable medium providing greater PCB aperture utilization and converter efficiency.

[0103] Additionally, as mentioned above, the output circuit may be covered with a conductive covering, preferably connected to a common terminal, to provide additional shielding.

C. Noise Management

[0104] The POL modules 110-1, 110-2, and preferably the driver circuits also, may use zero-current switching and/or zero-voltage switching to minimize the slew rate of voltages and currents in and around the semiconductor package 100 and system board. The power converter topologies shown in Figs. 6, 7, and 9 may preferably be based on the Sine Amplitude Converter (“SAC”) topology described in the SAC patent or on the clamped capacitor resonant topology described in the CSAC patent application. The SAC topology is preferred for the sinusoidal current and voltage

waveforms and zero-current switching (“ZCS”) and zero-voltage switching (“ZVS”) and the ability to constrain the slew rates of voltages and currents in the converter. For example, the slew rates may be limited to $dV/dT \leq V_{pk} / (T_{op} * 0.2)$ and $dI/dT \leq I_{pk} / (T_{op} * 0.2)$ in the output circuits shown in Figs. 6 and 7 using the SAC topology. In contrast, multiphase buck regulators typically exhibit characteristic current slew rates an order of magnitude greater than those in the output circuits.

[0105] In one example, the POL modules 110-1, 110-2 in the semiconductor package 100 may use a current multiplication factor K of 48 and an input voltage of 48 volts to supply 1 VDC at 100 A to the semiconductor die. Using a SAC topology operating at 1 MHz ($T_{op} = 1 \mu S$), the maximum voltage is 48V and the maximum current $I_{in} = 100/48 = 2.1$ Amps. Thus, the voltage and current slew rates for the output circuit may be limited to 240 V/ μS and 10.4 A/ μS .

D. Thermal Management

[0106] The semiconductor package 100 may include a lid 103, preferably made of thermally conductive material such as aluminum, copper, or other metallic or non-metallic thermally conductive material as shown in Fig. 10. The lid as shown may have a stepped lower surface to mate or accommodate the difference in height between the relatively short semiconductor die 101 and the relatively tall POL modules 110-1, 110-2. Lower surfaces 103A of lid 103 may mate with the top surfaces of the POL modules 110-1, 110-2 and lower surface 103B may mate with the top surface of the semiconductor die 101. Referring to the side view of the semiconductor package 100 in Fig. 11, dotted arrows 104, 105 and 106 show the direction of heat flow in the package. Heat generated by the semiconductor die 101 typically flows up through the lid as shown by arrow 106. Other proposed in-package power solutions rely on removal of the heat generated in the regulator circuits through the substrate, thus heating the substrate and the semiconductor die. However, the POL modules 110-1, 110-2, as shown provide thermally conductive conduits between the substrate and the lid, facilitating removal of heat generated by the output circuits directly through the lid and further provide a path for heat flow from the die 101 through the substrate 102 and

up through the POL modules 110-1, 110-2 as shown by arrows 105 and 104. As a result, the package 100 provides thermally enhanced operation over other solutions.

III. Semiconductor Package with Bottom-Mounted Integrated POL Module

[0107] Referring to Figs. 15A, 15B, 16A, 16B, 17, and 20, a second semiconductor package 300 is shown in bottom and top isometric exploded, top and bottom isometric assembled, and side views, respectively. Like the package 100 described above (Figs. 1-3, 10, 11), the second semiconductor package 300 includes a multilayer substrate 302 and a large semiconductor die 301, such as an ASIC, CPU, or GPU, mounted on a top surface 302A of the substrate 302 and also includes a POL module 310. However, as shown in Figs. 15A, 15B, 16A, 16B, and 17, the POL module 310 may be mounted on the bottom surface 302B of the substrate 302 beneath the die 301 in semiconductor package 300. Although Figs. 15A, 15B, 16A, 16B, and 17 show a single large POL module 310, a plurality of smaller POL modules may be used in place of the single large module shown.

[0108] Preferably, the POL module 310 (or POL modules) may occupy substantially all of the area beneath the semiconductor die 301 in the same, or a very similar, footprint allowing the remaining area, i.e. outside of the projection of the semiconductor die footprint, on the bottom surface, to be used for making connections between the substrate 302 and a system board. For example, the POL module 310 may preferably be smaller than, and fit completely within the footprint of, the die 301 as shown in the side view of Fig. 17. Note the difference 304 between the edge of the die 301 (on the top surface 302A of the substrate) and the edge of the POL module 310 on the bottom 302B of the substrate, which appears in Fig. 17 as symmetrical overhang on the left and right sides of the POL module. Although the POL module is shown slightly smaller than the die, it may be the same size as or slightly larger than the die, provided sufficient area remains for making connections between the die and the system board.

[0109] As shown in Figs. 15-17, the POL module 310 may be preferably packaged as a leadless module, such as described in the Panel Mold application,

having electrical connections for surface mount soldering to respective conductive pads on the substrate 302. For example, conductive terminations, e.g. terminations 315, may be provided along two or more sides of the POL module 310 for surface mount soldering to conductive pads 305 on the bottom 302B of the substrate 302. Additionally, conductive terminations, e.g. terminations 325A may be arranged along the top surface 310A of the POL module 310 for surface mount soldering to conductive pads 305 on the bottom 302B of the substrate 302. A pair of the conductive terminations, preferably located centrally and near an edge of the POL module 310 may be provided for making AC power connections, e.g. terminals 326A shown in Fig. 15B along a centerline and close to one edge of the POL module 310. Conductive terminations for the relatively low power bias and control signals may be provided along an edge of the module, e.g. terminals 317 are shown formed in an edge of the POL module 310 in Fig. 15B.

[0110] Preferably the conductive pads 305 on the bottom surface 302B may be electrically and thermally connected to conductive pads 307 on the top surface 302A of the substrate 302 (e.g. using conductive vias between the substrate layers (not shown)), which connect with power terminals 303 located on the bottom surface 301B of die 301. As shown in Figs. 15A, 15B, and 17, direct vertical alignment may be established between power terminals 303, conductive pads 307, conductive pads 305, and conductive terminations 315 providing the shortest electrical and thermal path between the POL module 310 and the die 301. Additionally, the power terminals 303, conductive pads 307, conductive pads 305, and conductive terminations 315 may be spatially arranged to allow signal connections to be routed between them on the inner and optionally outer layers of the substrate 302. For example, the power connections including terminals 303, pads 307, vias (not shown), pads 305, and terminations 315 and 325A are shown spaced apart and generally arranged in columns, e.g. five columns are shown in Figs. 15A, 15B. As shown in Fig. 15B and 16B, the POL module 310 may include conductive coverings 316A, 316B extending over the top 310A and bottom 310B surfaces of the POL module 310, which as discussed above may enhance the thermal and noise performance of the POL module 310 and help contain the magnetic fields in the transformer cores.

[0111] Referring to Fig. 16B, the POL module 310 may optionally include conductive terminals, 325B, 326B on the bottom surface 310B preferably aligned with the respective conductive terminations 325A, 326A on the top surface 310A of the POL module 310. Such through-module terminations, described in the Panel Mold application, may enhance the thermal conductivity, providing heat conductors through the POL module 310. As described in the Panel Mold application, a conduit (an example of which, conduit 265, is shown in Fig. 11 of the Panel Mold application) may expose conductive features to provide thermal and electrical connections between a substrate and the top and/or bottom conductive layers. The conductive terminations 325B, 326B may be used to engage with suitable terminations on the system board, directly or through a socket, to provide electrical connections and or allow heat to be conducted away through the system board as described below in connection with Fig. 20A.

Remote Gate Driver

[0112] Referring to Fig. 19, another power converter system is shown including a driver 490B, the semiconductor package 300 including the die 301 and a POL circuit 331 which may include the output transformer 440 and secondary switches 451-454 of the POL circuit 431 (Fig. 6). Preferably, the driver 490B may include transformer driver circuitry (481, Fig. 9) including switch control circuitry (e.g. 425, Fig. 9), power regulator circuitry (482, Fig. 9), and supervisory circuitry (483, Fig. 9) as described above and may be electrically connected to the semiconductor package 300 via connections formed by a system PCB (e.g. system board 11A or 11B in Figs. 20A, 20B) on which the driver 490B and semiconductor package 300 may be mounted. Preferably, the switch driver (460: Fig. 6) which is shown as part of the POL circuits described above (431: Fig. 6) may be incorporated into the driver 490B (Fig. 19) to further reduce the size of the POL module 310.

[0113] Although counterintuitive because of the high switching frequency and the parasitic inductances introduced by the system board, the package of the driver 490B, the POL module package, and the semiconductor package 300, the switch driver may

incorporate resonant gate driver techniques (*See, e.g.* Controller patent: Col. 13, ln 56 – Col. 15, ln 24; Figs. 8, 9) to use the inductances introduced by the connections between the separated driver 490B and the POL module 310 in semiconductor package 300 as some or all of the inductance required to resonantly charge and discharge the gate capacitances of the secondary switches in the rectification circuit of the POL module 310. Preferably additional inductance may be added in series with gate runs of the gate drive circuit using discrete, e.g. chip, inductors, to trim the circuit to the desired resonant frequency. For example, a power converter of the type shown in Fig. 19 configured to operate at 2 MHz with 250 nS half-cycles may require gate-voltage rise and fall times of 60-80 nS. Secondary switches configured to deliver 350 to 400 Amperes to the load may require 60 nC of gate charge to turn ON or OFF, representing approximately 12nF of gate capacitance on each terminal 446, 449. The total inductance in the driver circuit between gate drivers in the driver 490B and the secondary switches in the POL module 310 therefore should be limited to 7.6 to 13.5 nH. Allowing for 1 nH of package inductance for each of the driver and POL module and discrete inductance of approximately 3 nH, the maximum distribution inductance budget for the gate runs is between 2.6 and 8.5 nH. Design rules may be provided to keep the gate runs within the allocated inductance budget, e.g. limiting the length and width of the conductive traces, requiring a ground plane in an adjacent layer separated by an acceptable dielectric thickness.

[0114] As described above, the controller may detect and adjust for small timing errors produced by differences in the parasitic inductances in the AC power bus 410 and in the gate drive signal bus 415. Preferably, more than 50%, or 75%, or most preferably more than 90% of the energy stored in the gate capacitances of the secondary switches in the POL module 310 may be recycled using at least in part the inductances introduced by the wiring between driver and POL module.

[0115] As shown in Fig. 19, the control outputs 426, 429 of driver 490B (e.g. either bias and control or gate driver outputs) may be connected to a signal bus 415 on the system board which may in turn be connected to terminals 465, 466 on the substrate 302 of the package 300 and carried through the substrate 302 to input

terminals 449, 446, respectively. Similarly, a communication bus 497 as described above in connection with Fig. 9 may provide communication between the semiconductor package 300, e.g. with the die 301, and supervisory circuitry in the driver 490B.

[0116] The AC power connections 410 between the driver 490B (outputs 427, 428) and the POL module 310 (AC Inputs 447, 448) may be provided in part by the substrate 302 in the manner described above in connection with Fig. 4 using suitable conductor free zones around the AC connections formed in the substrate. Providing connections between the POL module 310 and the driver 490B through the substrate 302 may require routing the AC power laterally in or on the substrate in turn requiring elimination of any conductive features in a volume surrounding the AC power conductors, e.g. in a space radially around the conductors including a number of adjacent layers above and below the layer in which the AC power connections are made. Alternatively, the AC power connections 410 may be established between the system board (e.g. system board 11A, Fig. 20A) and the POL module 310 directly, e.g. using a socket 15 with suitable conductive terminations 16B to engage with terminations 326B on the bottom surface 310B of the POL module 310.

[0117] Alternatively, the AC power connections 410 between the system board and the POL module 310 may be formed using a wire harness 18 including a connector body configured to engage with AC power pins 327 preferably protruding from the bottom surface 310B of the POL module as shown in Fig. 18. The AC power pins 327 may be configured to be inserted into, engage with, and soldered to conductive features, e.g. plated holes, formed in the POL module 310. For example, a conduit, such as the feature 265 shown in Fig. 11 of the Panel Mold application, may be appropriately shaped and sized to provide a conductive receptacle for a conductive pin.

Underside Thermal Management

[0118] Referring to the side view of Fig. 20A, the package 300 may be assembled onto a surface of system board 11A using a socket 15 which may include contacts 16

for establishing electrical connections to the bottom surface 302B of the substrate. Heat may be conducted from the POL module 310 up through the substrate 302, e.g. through conductive features, 305, 307 and the connections there between, up through, and out the top of, the die 301, into a heat sink or lid (not shown) as represented by arrow 21 in Fig. 20A. Optionally, heat may be conducted away from the POL module through the system board 11A as represented by arrows 23 in Fig. 20A.

[0119] Referring to the side view of Fig. 20B, the package 300 may be assembled onto a surface of system board 11B using a low profile socket 15B having an aperture sized to accommodate the POL module 310 and contacts 16 for establishing electrical connections to the bottom surface 302B of the substrate. However, the system board 11B may include an aperture 17 sized to accommodate the POL module 310, providing a lower profile mounting alternative. In the embodiment of Fig. 20B, heat may be removed from the POL module 310 through the aperture 17 as represented by arrow 22, e.g. using forced air convection, a heat sink, or a cold plate (not shown).

[0120] A number of embodiments of the invention have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, a single resonant capacitor may be used instead of the two resonant capacitors shown in the symmetrical balanced circuit of Fig. 5 and Fig. 6. A center-tapped secondary circuit may be used in place of the full-bridge circuit shown. Although full-bridge driver circuits are shown in Figs. 5, 6, and 9, a half-bridge primary circuit may be used to drive the power transformer. Although converter topologies having one, and two POL circuits have been shown, it will be appreciated that a larger number of POL circuits may be used. The POL outputs may be connected in parallel as shown to supply higher current loads, or independently for multiple loads. The POL circuits may be deployed within the semiconductor package or at locations near, or adjacent to the semiconductor package or other loads. The magnetically permeable fluid injected into the apertures 155 of the PCB 151 can include a powder with or without a curable medium. There can be two or more semiconductor chips mounted on the substrate 102, in which the

semiconductor chips are all powered by the POL modules 110. The magnetically permeable fluid can be a liquid material.

[0121] Accordingly, other embodiments are within the scope of the following claims.

WHAT IS CLAIMED IS:

1. A method of supplying power received from an input source at an input voltage to circuitry on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising:

providing a driver circuit for deployment outside the semiconductor package having an input for receiving power from the source, circuitry adapted to drive a transformer, and a driver output;

providing one or more output circuits for deployment in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips;

providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits in the semiconductor package;

operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and

deploying the driver circuit at a first location outside of the semiconductor package;

wherein power is supplied to the semiconductor package by the driver circuit at a bus voltage that is a multiple, X, times greater than a load voltage required by the one or more semiconductor chips.

2. The method of claim 1 wherein the semiconductor package includes a package substrate on which the one or more semiconductor chips are mounted; and the providing the power bus further comprises providing a high voltage connection through the package substrate, the high voltage connection including a volume in the package substrate defined by a projection around the high voltage connection that is free of conductive features.

3. The method of claim 1 further comprising:
forming a resonant circuit including a winding of the transformer, the resonant circuit being characterized by a characteristic resonant period; and
wherein operating the driver circuit further comprises providing power during power transfer intervals, and each power transfer interval has a duration less than the characteristic resonant period.

4. The method of claim 3 further comprising:
providing at least one resonant capacitor electrically connected to the power bus.

5. The method of claim 1 further comprising:
providing a bias power output in the driver circuit for supplying power to operate circuitry in the one or more output circuits.

6. The method of claim 1 further comprising:
providing a control output in the driver circuit;
providing a control circuit in the one or more output circuits,
sending control signals from the driver circuit to the one or more output circuits;
and
synchronizing operation of the one or more output circuits using the control signals.

7. The method of claim 1 wherein:
the semiconductor package includes a package substrate;

the one or more semiconductor chips comprise a single very-large-scale-integrated (“VLSI”) semiconductor die mounted to the package substrate;

the one or more output circuits comprise a first and a second output circuit, each mounted to the package substrate adjacent to the VLSI semiconductor die; and

the power bus further comprises a first and a second high voltage connection, each including a vertical-extension through the package substrate located adjacent to a respective one of the first and the second output circuits and terminating on a top surface of the package substrate in a pad positioned to align with a respective input terminal of the respective output circuit and terminating on a bottom surface of the package substrate in an interface connection, and including a respective conductor-free volume in the package substrate defined by a projection around the respective the vertical extension that is free of conductive features.

8. The method of claim 7 further comprising providing the package substrate with an inter-layer breakdown voltage that is lower than the bus voltage.

9. The method of claim 7 further comprising:

providing the power transformer with a step down ratio, N , equal to K ; and
connecting the outputs of the one or more output circuits in parallel.

10. The method of claim 7 further comprising:

providing the power transformer with a step down ratio, N , that is less than K ; and
connecting the inputs of the one or more output circuits in series.

11. The method of claim 1 further comprising:

providing a regulation stage in the driver adapted to control the output voltage of the driver output.

12. The method of claim 11 further comprising:

receiving a control signal from the one or more semiconductor chips for adjustment of the DC output voltage;

adjusting the output voltage of the driver output in response to the received control signal.

13. The method of claim 1 wherein X is greater than or equal to 10.

14. The method of claim 1 wherein X is greater than or equal to 20.

15. The method of claim 1 wherein X is greater than or equal to 40.

16. The method of claim 1 wherein the rectification circuit comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

17. The method of claim 16 wherein the rectification circuit further comprises a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

18. The method of claim 1 wherein the one or more semiconductor chips further comprises a processor core.

19. The method of claim 1 wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and

the method comprises turning one or more of the primary switches OFF essentially at times when the resonant current returns to zero.

20. The method of claim 19 wherein a magnetizing current flows through the primary winding during the power transfer phases, and each converter operating cycle comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF, and

the method comprises using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

21. The method of claim 4 further comprising:

providing a clamp switch configured to clamp the resonant capacitor; and

wherein each power transfer interval comprises first and second resonant intervals, each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

wherein the converter operating cycles further comprise a clamp interval occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

22. The method of claim 21 further comprising adjusting the clamp duration as a function of power delivered to the one or more semiconductor chips.

23. The method of claim 22 further comprising reducing the clamp duration in response to a decrease in the power delivered to the one or more semiconductor chips.

24. The method of claim 1 further comprising:

providing a second driver circuit for deployment outside the semiconductor package having an input for receiving power from the source, circuitry adapted to drive a transformer, and a second driver output;

providing one or more additional output circuits for deployment in the semiconductor package, each additional output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the second driver output, a rectification

circuit connected to the second winding for rectifying power received from the transformer, and a second output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips at a second load voltage;

providing a second power bus for carrying AC power from the second driver output to the input of each of the one or more additional output circuits in the semiconductor package;

operating the second driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and

deploying the second driver circuit as a self-contained unit at a second location outside of the semiconductor package;

wherein power is supplied to the semiconductor package by the second driver circuit at a bus voltage that is a multiple, X , times greater than the second load voltage required by the one or more semiconductor chips.

25. The method of claim 1 further comprising establishing X as an integer.

26. The method of claim 1 further comprising establishing X as a non-integer rational number.

27. The method of claim 1 further comprising deploying the driver circuit as a self-contained unit at the first location.

28. The method of claim 1, further comprising detecting a resonant current at the end of a selected one of the power transfer phases and adjusting the timing of the converter operating cycles.

29. The method of claim 21 further comprising detecting a resonant current at the end of a selected one or more of the first or second resonant intervals and adjusting the timing of the converter operating cycles.

30. An apparatus comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more output circuits housed in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; and

a power bus connected to selected interface connections for carrying AC power at a bus voltage from the system board to the input of each of the one or more output circuits in the semiconductor package;

wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips.

31. The apparatus of claim 30 further comprising:

a driver circuit located outside the semiconductor package having an input for receiving power from a source, circuitry adapted to drive the transformer, and a driver output for supplying AC power to the transformer via the power bus; and

the driver circuit including a controller configured to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the one or more output circuits via the respective transformer of the one or more output circuits.

32. The apparatus of claim 30 wherein the power bus further comprises a high voltage connection extending vertically through the package substrate and having a volume in the package substrate defined by a projection around the high voltage connection that is free of conductive features.

33. The apparatus of claim 31 further comprising a resonant circuit including a winding of the transformer, the resonant circuit being characterized by a characteristic resonant period; and wherein each power transfer phase includes a duration that is less than the characteristic resonant period.

34. The apparatus of claim 32 further comprising at least one resonant capacitor electrically connected in series with the first winding.

35. The apparatus of claim 31 wherein the driver circuit further comprises a bias power output for supplying power to operate circuitry in the one or more output circuits.

36. The apparatus of claim 31 wherein:
the driver circuit further comprises a control output for sending a control signal to the one or more output circuits;
the one or more output circuits further comprise a control circuit adapted to synchronize operation of the one or more output circuits using the control signals.

37. The apparatus of claim 30 wherein:
the one or more semiconductor chips comprise a single very-large-scale-integrated (“VLSI”) semiconductor die;
the one or more output circuits comprise a first and a second output circuit, each mounted to the package substrate adjacent to the VLSI semiconductor die; and
the power bus further comprises a first and a second high voltage connection, each including a vertical-extension through the package substrate located adjacent to a respective one of the first and the second output circuits and terminating on a top surface

of the package substrate in a pad positioned to align with a respective input terminal of the respective output circuit and terminating on a bottom surface of the package substrate in an interface connection, and including a respective conductor-free volume in the package substrate defined by a projection around the respective vertical extension that is free of conductive features.

38. The apparatus of claim 30 wherein the package substrate has an inter-layer breakdown voltage that is lower than the bus voltage.

39. The apparatus of claim 30 wherein the power transformer comprises a step down ration, N , equal to X ; and
the outputs of the one or more output circuits are connected in parallel.

40. The apparatus of claim 30 wherein the power transformer comprises a step down ration, N , that is less than X ; and
the inputs of the one or more output circuits are connected in series.

41. The apparatus of claim 31 wherein the driver circuit further comprises a regulation stage adapted to control the output voltage of the driver output.

42. The apparatus of claim 41 wherein the driver circuit is connected to receive a control signal from the one or more semiconductor chips and is adapted to adjust the output voltage of the driver output in response to the control signal.

43. The apparatus of claim 30 wherein X is greater than or equal to 10.

44. The apparatus of claim 30 wherein X is greater than or equal to 20.

45. The apparatus of claim 30 wherein X is greater than or equal to 40.

46. The apparatus of claim 30 wherein the rectification circuit comprises a full-bridge configuration.

47. The apparatus of claim 30 wherein the second winding comprises a center-tap and the rectification circuit comprises a center-tap configuration.

48. The apparatus of claim 30 wherein the one or more semiconductor chips comprise a processor core.

49. The apparatus of claim 30 wherein the rectification circuit further comprises one or more secondary switches and a control circuit adapted to turn the one or more secondary switches OFF essentially at times when a current in the second winding returns to zero.

50. The apparatus of claim 31, wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, a resonant current flows in the first winding during the power transfer phases, and the driver circuit comprises a control circuit configured to turn one or more of the primary switches OFF essentially at times when the resonant current returns to zero.

51. The apparatus of claim 30 in which X is an integer.

52. The apparatus of claim 30 in which X is a non-integer rational number.

53. The apparatus of claim 31 in which the driver circuit comprises compensation circuitry to detect and adjust for effects of at least one of parasitic capacitances or inductances introduced by separation of the driver circuit and the one or more output circuits.

54. The apparatus of claim 53 further comprising circuitry configured to detect a resonant current at the end of a selected one or more of the power transfer phases and adjust timing of the converter operating cycles.

55. The apparatus of claim 30, comprising the system board.

56. The apparatus of claim 30 in which the interface connections comprise at least one of connector pins, ball grid array, or land grid array.

57. A method of packaging a semiconductor die comprising:
providing a substrate for mounting to a bottom surface of the semiconductor die;
providing a lid to extract heat from a top surface of the semiconductor die;
providing one or more output circuits having a bottom surface mounted to the substrate for supplying power to the semiconductor die; and
conducting heat from the bottom of the semiconductor die, into the substrate, from the substrate into the bottom of the one or more output circuits, out of the top of the one or more output circuits, into the lid.

58. The method of claim 57 further comprising providing a package for each of the one or more output circuits, the package including a thermal conductive path between the top and the bottom of the output circuits, wherein power dissipation in the output circuits causes a temperature rise between the bottom and the top of the output circuits, and wherein power dissipation in the semiconductor die causes a temperature rise between the bottom and the top of the semiconductor die that is greater than the temperature rise in the output circuits.

59. The method of claim 57 in which each output circuit includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and

an output connected to the rectification circuit for supplying DC power to the semiconductor die.

60. An apparatus comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, the inductive component having a magnetically permeable core having an effective permeability of at least 25; and

wherein the ratio of input voltage to output voltage is fixed, subject to a series resistance, and is at least 5.

61. The apparatus of claim 60 wherein the effective permeability is at least 100.

62. The apparatus of claim 60 wherein the effective permeability is at least 200.

63. The apparatus of claim 60 wherein the inductive component further comprises a conductive shield.

64. The apparatus of claim 60, comprising a driver circuit located outside the semiconductor package having an input for receiving power from a source, circuitry adapted to drive the one or more power conversion circuits, and a driver output for supplying AC power to the one or more power conversion circuits via a power bus.

65. An apparatus comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more

semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, wherein the switch controller is configured to turn the at least one switch ON or OFF at essentially zero voltage; and

wherein the power conversion circuit ratio of input voltage to output voltage is fixed, subject to a series resistance, and is at least 5.

66. The apparatus of claim 65 in which the one or more power conversion circuits and the one or more semiconductor chips are mounted on a top surface of the substrate, the semiconductor package comprises a high voltage connection extending vertically through the substrate from the top surface to a bottom surface of the substrate, and the interface connections are connected to the bottom surface of the substrate.

67. The apparatus of claim 65 in which the inductive component comprises a power transformer including a first winding and a second winding.

68. The apparatus of claim 65, comprising the system board.

69. The apparatus of claim 65 in which the interface connections comprise at least one of connector pins, ball grid array, or land grid array.

70. The apparatus of claim 65, comprising a driver circuit located outside the semiconductor package having an input for receiving power from a source, circuitry adapted to drive the one or more power conversion circuits, and a driver output for supplying AC power to the one or more power conversion circuits via a power bus.

71. An apparatus comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, wherein the switch controller is configured to turn the at least one switch ON or OFF at essentially zero current; and

wherein the power conversion circuit ratio of input voltage to output voltage is fixed, subject to a series resistance, and is at least 5.

72. The apparatus of claim 71 in which the one or more power conversion circuits and the one or more semiconductor chips are mounted on a top surface of the substrate, the semiconductor package comprises a high voltage connection extending vertically through the substrate from the top surface to a bottom surface of the substrate, and the interface connections are connected to the bottom surface of the substrate.

73. The apparatus of claim 71 in which the inductive component comprises a power transformer including a first winding and a second winding.

74. The apparatus of claim 71, comprising the system board.

75. The apparatus of claim 71 in which the interface connections comprise at least one of connector pins, ball grid array, or land grid array.

76. The apparatus of claim 71, comprising a driver circuit located outside the semiconductor package having an input for receiving power from a source, circuitry adapted to drive the one or more power conversion circuits, and a driver output for supplying AC power to the one or more power conversion circuits via a power bus.

77. An apparatus comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips;

wherein the switch controller is configured to operate the at least one switch to limit slew rates of voltage in the converter to $5 (V_{peak}/Top)$.

78. The apparatus of claim 77 in which the one or more power conversion circuits and the one or more semiconductor chips are mounted on a top surface of the substrate, the semiconductor package comprises a high voltage connection extending vertically through the substrate from the top surface to a bottom surface of the substrate, and the interface connections are connected to the bottom surface of the substrate.

79. The apparatus of claim 77 in which the inductive component comprises a power transformer including a first winding and a second winding.

80. The apparatus of claim 77, comprising the system board.

81. The apparatus of claim 77 in which the interface connections comprise at least one of connector pins, ball grid array, or land grid array.

82. The apparatus of claim 77, comprising a driver circuit located outside the semiconductor package having an input for receiving power from a source, circuitry adapted to drive the one or more power conversion circuits, and a driver output for supplying AC power to the one or more power conversion circuits via a power bus.

83. An apparatus comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips;

wherein the switch controller is configured to operate the at least one switch to limit slew rates of current in the converter to less than or equal to $5 * (I_{peak}/Top)$.

84. The apparatus of claim 83 in which the one or more power conversion circuits and the one or more semiconductor chips are mounted on a top surface of the substrate, the semiconductor package comprises a high voltage connection extending vertically through the substrate from the top surface to a bottom surface of the substrate, and the interface connections are connected to the bottom surface of the substrate.

85. The apparatus of claim 83 in which the inductive component comprises a power transformer including a first winding and a second winding.

86. The apparatus of claim 83, comprising the system board.

87. The apparatus of claim 83 in which the interface connections comprise at least one of connector pins, ball grid array, or land grid array.

88. The apparatus of claim 83, comprising a driver circuit located outside the semiconductor package having an input for receiving power from a source, circuitry adapted to drive the one or more power conversion circuits, and a driver output for supplying AC power to the one or more power conversion circuits via a power bus.

89. A method of supplying power received from an input source at an input voltage to circuitry on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising:

providing one or more output circuits for deployment at one or more locations near or adjacent to the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips;

providing a driver circuit for deployment at a location spaced apart from the one or more output circuits, the driver circuit having an input for receiving power from the source, circuitry configured to drive a transformer, a driver output for supplying AC power to the one or more output circuits, and circuitry configured to control the power supplied to the one or more output circuits;

providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits;

operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and

deploying the driver circuit at a first location outside of an area immediately near or adjacent to the one or more output circuits;

wherein power is supplied to the one or more output circuits by the driver circuit at a bus voltage that is a multiple, X, times greater than the load voltage required by the one or more semiconductor chips.

90. The method of claim 89 in which the driver circuit is deployed as a self-contained unit at the first location.

91. The method of claim 89 wherein the semiconductor package includes a package substrate on which the one or more semiconductor chips are mounted; and the providing the power bus further comprises providing a high voltage connection through the package substrate, the high voltage connection including a volume in the package substrate defined by a projection around the high voltage connection that is free of conductive features.

92. The method of claim 89 further comprising:
providing a bias power output in the driver circuit for supplying power to operate circuitry in the one or more output circuits.

93. The method of claim 89 further comprising:
providing a control output in the driver circuit;
providing a control circuit in the one or more output circuits,
sending control signals from the driver circuit to the one or more output circuits;
and
synchronizing operation of the one or more output circuits using the control signals.

94. The method of claim 89 wherein:
the semiconductor package includes a package substrate;
the one or more semiconductor chips comprise a single very-large-scale-integrated (“VLSI”) semiconductor die mounted to the package substrate;
the one or more output circuits comprise a first and a second output circuit, each mounted to the package substrate adjacent to the VLSI semiconductor die; and
the power bus further comprises a first and a second high voltage connection, each including a vertical-extension through the package substrate located adjacent to a respective one of the first and the second output circuits and terminating on a top surface of the package substrate in a pad positioned to align with a respective input terminal of the respective output circuit and terminating on a bottom surface of the package substrate in an interface connection, and including a respective conductor-free volume in the

package substrate defined by a projection around the respective the vertical extension that is free of conductive features.

95. The method of claim 94 further comprising providing the package substrate with an inter-layer breakdown voltage that is lower than the bus voltage.

96. The method of claim 94 further comprising:
providing the power transformer with a step down ratio, N , equal to X ; and
connecting the outputs of the one or more output circuits in parallel.

97. The method of claim 94 further comprising:
providing the power transformer with a step down ratio, N , that is less than X ; and
connecting the inputs of the one or more output circuits in series.

98. The method of claim 89 further comprising:
providing a regulation stage in the driver circuit adapted to control the output voltage of the driver output.

99. The method of claim 98 further comprising:
receiving a control signal from the one or more semiconductor chips for
adjustment of the DC output voltage; and
adjusting the output voltage of the driver output in response to the received
control signal.

100. The method of claim 89 wherein X is greater than or equal to 10.

101. The method of claim 89 wherein X is greater than or equal to 20.

102. The method of claim 89 wherein X is greater than or equal to 40.

103. A method of supplying power received from an input source at an input voltage to a load at a DC output voltage, comprising:

providing one or more current multiplier modules at one or more locations in close proximity to the load, the one or more current multiplier modules having a transformer and circuitry for supply DC power to the load; and

providing a driver module at a location spaced apart from the one or more current multiplier modules, the driver module having circuitry including an input for receiving power from the input source, control circuitry for generating a controlled driver voltage, and driver circuitry for generating AC power for driving the transformer in each of the one or more current multiplier modules.

104. The method of claim 103 in which the load comprises one or more semiconductor chips, and the method comprises mounting the one or more current multiplier modules and the one or more semiconductor chips on a substrate.

105. The method of claim 104, comprising mounting the substrate on a system board, and mounting the driver module on the system board.

106. The method of claim 104, comprising providing a power bus for carrying AC power from an output of the driver module to an input of each of the one or more current multiplier modules, the power bus comprising a high voltage connection through the substrate.

107. The method of claim 106 in which the high voltage connection includes a volume in the substrate defined by a projection around the high voltage connection that is free of conductive features.

108. A method of making a planar magnetic component comprising:
providing a multilayer printed circuit board (PCB) comprising conductive features arranged on conductive layers of the PCB to form one or more windings around one or more predetermined axes;

forming a hole in the PCB at each of the one or more predetermined axes to accommodate one or more core legs, wherein for each hole, an inner edge of one of the windings overlaps an edge of the hole in a lateral direction after the hole is formed;

assembling a first magnetically permeable plate to a first surface of the PCB covering a selected one or more of the holes at the one or more predetermined axes;

assembling a second magnetically permeable plate to a second surface of the PCB covering the selected one or more of the holes at the one or more predetermined axes;

filling the selected one or more of the holes with at least one of a magnetically permeable fluid or powder; and

sealing the selected one or more holes to prevent the at least one of magnetically permeable fluid or powder from escaping.

109. The method of claim 108 further comprising removing a portion of the conductive features after forming the hole to provide a clearance between the edge of the hole and the inner edge of the winding.

110. The method of claim 109 wherein the hole has a circular cross-section, and wherein removing the portion of the conductive features to provide the clearance comprises removing a same amount of the conductive features around the outside of the hole such that the clearance is uniform around a circumference of the hole and the inner edge of the winding has a complementary cross-section to the hole.

111. The method of claim 109 further comprising:

providing at least one aperture in the second magnetically permeable plate aligned with each of the selected one or more of the holes at the one or more predetermined axes;

wherein the filling is performed after the second magnetically permeable plate is assembled to the second surface of the PCB; and

wherein the sealing comprises sealing the at least one aperture after the filling.

112. An apparatus comprising:

a planar magnetic structure including a multilayer printed circuit board (PCB) having a first surface, a second surface, and conductive features arranged on conductive layers of the PCB to form one or more windings around one or more predetermined axes;

a hole in the PCB at each of the one or more predetermined axes, each hole having an inner edge aligned with an inner circumference of one of the windings;

a first magnetically permeable section affixed to the first surface of the PCB covering a selected one or more of the holes;

a second magnetically permeable section affixed to the second surface of the PCB covering the selected one or more of the holes;

at least one of a magnetically permeable fluid or powder disposed in the selected one or more of the holes; and

wherein the at least one of magnetically permeable fluid or powder is contained within the selected one or more holes.

113. The apparatus of claim 112 further comprising a clearance between the inner edge of the hole and the inner circumference of the one of the windings.

114. The apparatus of claim 113 wherein the hole has a circular cross-section, and the clearance comprises a uniform distance around a circumference of the hole and the inner edge of the winding has a complementary cross-section to the hole.

115. The apparatus of claim 112 further comprising at least one aperture formed in the second magnetically permeable section aligned with each of the selected one or more of the holes, and having a plug disposed in the at least one aperture wherein the plug forms a seal covering the respective selected one or more of the holes.

116. The apparatus of claim 112 wherein the first magnetically permeable section comprises a ferrite core plate.

117. The apparatus of claim 116 wherein the second magnetically permeable section comprises a ferrite core plate having an aperture formed therein and filled with a plug.

118. The apparatus of claim 117 wherein the fluid is a liquid.

119. The apparatus of claim 117, wherein a powder and a curable medium are disposed in the selected one or more of the holes.

120. The apparatus of claim 112 further comprising:

a semiconductor package including a substrate having a plurality of interface connections connected to at least one surface of the package substrate and one or more semiconductor chips mounted to the substrate, the interface connections adapted for electrical connection to a system board;

one or more power conversion circuits housed in the semiconductor package, each power conversion circuit including at least one switch, a switch controller, and an inductive component for converting power from an input at an input voltage to an output at an output voltage for delivery to the one or more semiconductor chips, wherein the switch controller is configured to turn the at least one switch ON or OFF at essentially zero current; and

wherein the inductive component comprises the planar magnetic structure.

121. The method of claim 1 further comprising:

providing a switch driver output in the driver circuit;

providing a control input connected to one or more secondary switches in the rectification circuit,

using the switch driver output to resonantly charge and discharge capacitances associated with the control input to operate the one or more switches in the rectification circuit.

122. The method of claim 1 wherein:
the semiconductor package includes a package substrate;
the one or more semiconductor chips comprise a single very-large-scale-integrated (“VLSI”) semiconductor die mounted to the package substrate; and
the one or more output circuits comprise a point-of-load (POL) module mounted to the package substrate beneath the VLSI semiconductor die.

123. The method of claim 122 wherein:
the power bus further comprises a first high voltage connection and a second high voltage connection, each including an extension laterally along the package substrate terminating on a bottom surface of the package substrate in a pad positioned to align with a respective input terminal of the POL module and terminating in an interface connection, and including a respective conductor-free volume in the package substrate defined by a projection around the respective the extension that is free of conductive features.

124. The method of claim 122 further comprising:
providing a conductive terminal extending from a surface of the POL module electrically connected to the input of the output circuit;
providing a connector adapted to mate with the conductive terminal;
wherein the power bus further comprises the connector and the conductive terminal.

125. The method of claim 123 further comprising:
providing a switch driver output in the driver circuit;
providing a control input connected to one or more secondary switches in the rectification circuit,
using the switch driver output to resonantly charge and discharge capacitances associated with the control input to operate the one or more switches in the rectification circuit.

126. The method of claim 121 further comprising:

providing a discrete inductor in series between the switch driver output and the control input.

127. The method of claim 122 wherein the POL module further comprises a conductive shield on one or more outer surfaces of the POL module.

128. A method of making an inductive component comprising:
forming one or more electrical windings in one or more layers of a printed circuit board ("PCB");
forming one or more holes extending through the PCB, each hole being surrounded by the one or more windings;
affixing a first magnetically permeable material to cover the one or more holes on a first side of the PCB;
filling the one or more holes with a second magnetically permeable fluid;
affixing a third magnetically permeable material to a second side of the PCB; and
sealing the one or more holes from the second side of the PCB.

129. The method of claim 128 further comprising:
providing one or more holes in the third magnetically permeable material;
wherein the third magnetically permeable material is affixed before the one or more holes are filled.

130. The method of claim 108 further comprising heating the PCB prior to the filling and sealing steps.

131. The method of claim 1 wherein:
the semiconductor package includes a package substrate;
the one or more semiconductor chips comprise a single very-large-scale-integrated ("VLSI") semiconductor die mounted to the package substrate;

the one or more output circuits comprise a first, a second, a third and a fourth output circuit, each mounted to the package substrate adjacent to the VLSI semiconductor die; and

the power bus further comprises a first and a second high voltage connection, each including a vertical-extension through the package substrate located adjacent to a respective one of the output circuits and terminating on a top surface of the package substrate in a pad positioned to align with a respective input terminal of the respective output circuit and terminating on a bottom surface of the package substrate in an interface connection, and including a respective conductor-free volume in the package substrate defined by a projection around the respective the vertical extension that is free of conductive features.

132. A method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising:

providing one or more driver circuits for deployment outside the semiconductor package, each having an input for receiving power from the source and a driver output;

providing one or more output circuits for deployment in or near the semiconductor package, being connected to receive power from the driver circuit at a driver voltage;

arranging the output circuits to form a plurality of output rails configured to supply power at a plurality of respective unique rail voltages to the one or more semiconductor chips;

configuring each output rail to provide a rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential;

configuring the output circuits to provide the rail voltages as a function of the driver voltage;

providing one or more rail selection circuits on the one or more semiconductor chips, each having inputs connected to the output rails and an output connected to deliver power to a respective load at a respective load voltage, each rail selection circuit having a

plurality of rail switches connected to provide a selectable path for current flow from a selected one of the rails to the output;

providing a power bus for carrying power from the driver output to the input of each of the one or more output circuits; and

deploying the driver circuit at a first location outside of the semiconductor package;

wherein each rail selection circuit is operative to enable a respective rail switch to conduct current from selected output rail to its respective output.

133. The method of claim 132, further comprising:

providing power management circuitry on the one or more semiconductor chips, the power management circuitry having one or more outputs for sending commands to the one or more rail selection circuits, the commands including a voltage command for setting a desired output voltage of respective rail selection circuits;

wherein each rail selection circuit is further adapted to receive the voltage command and enable the respective rail switch according to predetermined criteria, the predetermined criteria including selection of the rail voltage that is closest to the desired output voltage.

134. The method of claim 133 further comprising:

providing regulation circuitry adapted to regulate power conducted from the selected rail to the respective output; and

wherein the predetermined criteria further include selection of the rail voltage that is greater than the desired output voltage by a minimum voltage drop required for the regulation.

135. The method of claim 134 wherein the regulation circuitry is further adapted to adjust a drive signal to the selected rail selection switch as a function of an error signal derived from a difference between the respective output voltage and a reference signal to vary a voltage drop across the selected rail selection switch.

136. The method of claim 135 further comprising providing circuitry adapted to adjust the reference signal as a function of the respective voltage command.

137. The method of claim 132 further comprising providing one or more regulation stages for regulating the driver voltage delivered to the one or more driver circuits.

138. The method of claim 137 wherein the one or more driver circuits are further adapted to adjust the driver voltage as a function of commands received from the one or more semiconductor chips.

139. The method of claim 132 wherein the one or more output circuits further comprise:

one or more transformers, each having a primary winding connected to receive the driver voltage and one or more secondary windings;

rectification circuitry connected to convert AC signals received from the one or more secondary windings to unipolar voltages.

140. The method of claim 139 wherein the one or more transformers comprise at least one transformer per output rail.

141. The method of claim 139 wherein the one or more output circuits comprise at least one pair of output circuits, each having at least one transformer.

142. The method of claim 139 wherein the one or more output circuits comprise at least one pair of output circuits, each having at least one transformer, and at least two of the output rails are connected in parallel.

143. The method of claim 132 wherein the one or more output circuits are provided in the semiconductor package and each output circuit includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output and an output connected to the second winding for supplying power to respective ones of the plurality of output rails.

144. The method of claim 143 further comprising:

providing a controller adapted to operate the one or more driver circuits and the one or more output circuits in a series of converter operating cycles, each converter operating cycle being characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the one or more driver circuits to the plurality of output rails via the transformers.

145. The method of claim 144 wherein the one or more driver circuits and the one or more output circuits are adapted for power to be supplied to the semiconductor package by the one or more driver circuits at a bus voltage, V_{bus} , that is a multiple, X , times greater than at least one of the rail voltages.

146. The method of claim 132 wherein the one or more output circuits each includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to conduct AC power received from the driver output to the first winding for driving the power transformer and an output connected to conduct power from the second winding to respective ones of the plurality of output rails and further comprising:

providing a controller adapted to operate the one or more driver circuits and output circuits in a series of converter operating cycles, each converter operating cycle being characterized by two power transfer phases of essentially equal duration during which a magnetizing current flows in a winding of a respective power transformer and

power is transferred from the one or more driver circuits to at least one of the plurality of output rails via the respective power transformer.

147. The method of claim 146 further comprising:

forming a resonant circuit including the respective power transformer and a resonant capacitor connected in series with the first winding, the resonant circuit being characterized by a characteristic resonant period and frequency;

wherein each power transfer phase is further characterized in having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency.

148. The method of claim 132 further comprising:

providing a controller adapted to operate the one or more driver circuits and output circuits in a series of converter operating cycles;

providing a clamp switch connected across the resonant capacitor;

wherein the one or more output circuits each includes a printed circuit board, a power transformer including a first winding and a second winding, an input connected to conduct AC power received from the driver output to the first winding for driving the power transformer and an output connected to conduct power from the second winding to respective ones of the plurality of output rails;

wherein the driver output includes a first terminal driven by a switching node and the driver input includes a source terminal and a common terminal;

and each converter operating cycle is characterized by

(a) two power transfer phases of essentially equal duration during which the switching node is connected to either the source or the common terminal, a magnetizing current flows in a winding of a respective transformer, and power is transferred from the one or more driver circuits to at least one of the plurality of output rails via the respective transformer, each power transfer phase including

(i) first and second resonant intervals, each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

(ii) a clamp interval occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

149. The method of claim 132:

wherein the driver output includes a first terminal driven by a switching node and a second terminal, the driver input includes a source terminal and a second terminal referenced to a voltage potential within a predetermined range of a common terminal; and further comprising

providing a controller to operate the driver circuit in a series of converter operating cycles;

providing a first capacitor connected to block direct current between the switching node and the primary winding, wherein the driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts;

wherein the driver voltage comprises a maximum peak-to-peak voltage, V_{peak} , that is at least a multiple, X , times greater than at least one of the rail voltages and X is at least six; and

each converter operating cycle is characterized by two power transfer intervals of essentially equal duration during which the switching node remains connected to either the source or the common terminal, a magnetizing current flows in a winding of the transformer, and power is transferred between the driver input and at least one of the multiplicity of output rails via the transformer.

150. The method of claim 145 wherein X is greater than or equal to 15.

151. A method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising:

- providing one or more driver circuits for deployment outside the semiconductor package, each driver circuit having an input for receiving power from the source, and a driver output having a first terminal driven by a switching node and a second terminal referenced to a voltage potential within a predetermined range of a common terminal;

- deploying the driver circuit at a first location outside of the semiconductor package;

- providing one or more output circuits for deployment in or near the semiconductor package for supplying power to the one or more semiconductor chips, at least one of the output circuits including a transformer having a primary winding adapted for connection to the first and second terminals of the driver output and a secondary winding;

- providing a power bus for carrying power from the first and second terminals of the driver output to the primary winding; and

- providing a first capacitor connected to block direct current between the switching node and the primary winding;

- wherein driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts;

- wherein the one or more output circuits are adapted to receive power from the driver circuit at a peak-to-peak bus voltage, V_{peak} , and supply power to the one or more semiconductor chips at a load voltage, V_{load} , wherein V_{peak} is a multiple, X , of V_{load} and X is at least six.

152. The method of claim 151 wherein the one or more output circuits are deployed in the semiconductor package.

153. The method of claim 151 wherein the one or more driver circuits comprise a first semiconductor switch connected between a positive terminal of the driver input and the switching node and a second semiconductor switch connected between the common terminal and the switching node, the first and second semiconductor switches being adapted to alternatively connect the switching node to the positive terminal or the common terminal in a succession of switching cycles.

154. The method of claim 151 wherein the first capacitor is connected between the switching node and the driver output.

155. The method of claim 154 further comprising:

forming a resonant circuit including the transformer and a resonant capacitor connected in series with the primary winding, the resonant circuit being characterized by a characteristic resonant period and frequency.

156. The method of claim 155 wherein the first capacitor comprises the resonant capacitor.

157. The method of claim 155 wherein the resonant capacitor is connected between the transformer and the second terminal of the driver output and further comprising providing a clamp switch connected across the resonant capacitor and the first capacitor is at least an order of magnitude greater than the resonant capacitor.

158. The method of claim 157 further comprising providing a resistance in parallel with the resonant capacitor.

159. The method of claim 157 further comprising:

providing a controller to operate the driver circuit in a series of converter operating cycles, each converter operating cycle characterized by:

(a) two power transfer phases of essentially equal duration during which the switching node is connected to either the source or the common terminal, a magnetizing current flows in a winding of the transformer, and power is transferred between the driver input and the output circuit via the transformer, each power transfer interval including

(i) first and second resonant intervals, each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

(ii) a clamp interval occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

160. The method of claim 151 wherein the predetermined range is less than or equal to 10% of the source voltage.

161. The method of claim 155 further comprising:

providing a controller to operate the driver circuit in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which the switching node remains connected to either the source or the common terminal, a magnetizing current flows in a winding of the transformer, and power is transferred between the driver input and the POL circuit output via the transformer.

162. The method of claim 161 wherein each power transfer phase is further characterized in having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency.

163. The method of claim 151 wherein X is greater than or equal to 20.

164. The method of claim 151 wherein X is greater than or equal to 10.

165. A method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage, comprising:

- performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage;

- conducting power from the first output through a portion of the semiconductor package to a point of load (POL) circuit located within the semiconductor package;

- performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering power to the semiconductor chip at a chip voltage; and

- performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip;

- wherein the package voltage is at least, 5, times greater than the load voltage.

166. The method of claim 165 wherein the third power conversion comprises voltage regulation.

167. The method of claim 166 wherein the third power conversion comprises linear voltage regulation.

168. The method of claim 166 wherein the third power conversion comprises switching voltage regulation.

169. The method of claim 166 wherein the second power conversion comprises voltage regulation.

170. The method of claim 169 wherein the POL circuit comprises a multi-phase switching regulator.

171. The method of claim 169 wherein the first power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current, in which V_{in} is the source voltage and V_o is the package voltage.

172. The method of claim 169 wherein the package voltage is greater than 20 VDC and the chip voltage is less than 3 VDC.

173. The method of claim 169 wherein the package voltage is greater than 5 VDC and the chip voltage is less than 2 VDC.

174. The method of claim 165 wherein the second power conversion comprises voltage regulation.

175. The method of claim 174 wherein the POL circuit comprises a multi-phase switching regulator.

176. The method of claim 165 wherein the second power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current, in which V_{in} is the package voltage and V_o is the chip voltage.

177. A method of supplying power received from an input source at a source voltage to circuitry on a semiconductor chip mounted in a semiconductor package at a DC load voltage, comprising:

- performing a first power conversion outside of the semiconductor package, the first conversion including receiving power from the source at the source voltage, making a voltage adjustment, and delivering power from a first output to the semiconductor package at a package voltage;

- conducting power from the first output through a portion of the semiconductor package to a point-of-load (POL) circuit located within the semiconductor package and outside of the semiconductor chip;

- performing a second power conversion in the POL circuit, the second power conversion including receiving the conducted power, making a second voltage adjustment, and delivering power to the semiconductor chip at a chip voltage; and

- performing a third power conversion in circuitry located on the semiconductor chip, the third conversion including receiving power from the POL circuit, making a third voltage adjustment, and delivering power at the load voltage to circuitry on the semiconductor chip;

- wherein the package voltage is at least, 5, times greater than the load voltage.

178. The method of claim 177 wherein the third power conversion comprises voltage regulation.

179. The method of claim 178 wherein the third power conversion comprises linear voltage regulation.

180. The method of claim 178 wherein the third power conversion comprises switching voltage regulation.

181. The method of claim 178 wherein the second power conversion comprises voltage regulation.

182. The method of claim 181 wherein the POL circuit comprises a multi-phase switching regulator.

183. The method of claim 181 wherein the first power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current.

184. The method of claim 181 wherein the package voltage is greater than 20 VDC and the chip voltage is less than 3 VDC.

185. The method of claim 181 wherein the package voltage is greater than 5 VDC and the chip voltage is less than 2 VDC.

186. The method of claim 177 wherein the second power conversion comprises voltage regulation.

187. The method of claim 186 wherein the second stage comprises a multi-phase switching regulator.

188. The method of claim 177 wherein the second power conversion comprises an essentially fixed ratio voltage conversion, $K = V_o/V_{in}$, at a load current, in which V_{in} is the source voltage and V_o is the package voltage.

189. The method of claim 188 wherein the source voltage is greater than 40 VDC.

190. The method of claim 189 wherein the package voltage is greater than 20 VDC and the chip voltage is less than 3 VDC.

191. The method of claim 188 wherein the package voltage is greater than 5 VDC and the chip voltage is less than 2 VDC.

192. The method of claim 177 wherein:

the semiconductor package comprises a substrate and the semiconductor chip is mounted to the substrate; and

the POL circuit further comprises one or more semiconductor switches and at least one inductive component, the POL circuit being substantially contained within an electronic package mounted to the first substrate.

193. An apparatus comprising:

a semiconductor package comprising one or more semiconductor chips, each semiconductor chip comprising one or more circuit loads;

one or more driver circuits disposed outside of the semiconductor package, each driver circuit having a driver input and a driver output, in which the driver input is configured to receive power from an input source at an input voltage;

one or more output circuits disposed inside or in a vicinity of the semiconductor package and configured to receive power from the driver circuit at a driver voltage, in which the output circuits are configured to form a plurality of output rails to supply power at a plurality of respective unique rail voltages to the one or more semiconductor chips, each output rail is configured to provide a DC rail voltage that is greater than or less than an adjacent rail voltage by at least a predetermined minimum differential, and the output circuits provide the rail voltages as a function of the driver voltage;

one or more rail selection circuits disposed on the one or more semiconductor chips, each having inputs electrically coupled to the output rails and an output electrically coupled to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches electrically coupled to provide a selectable path for current to flow from a selected one of the output rails to the respective output; and

a power bus configured to carry power from the driver output to the input of each of the one or more output circuits.

194. An apparatus comprising:

a semiconductor package comprising one or more semiconductor chips, each semiconductor chip comprising one or more circuit loads;

a first power conversion stage disposed outside of the semiconductor package and configured to perform a first power conversion, in which the first power conversion stage is configured to receive power from an input source at a source voltage, make a voltage adjustment, and deliver power from a first output to the semiconductor package at a package voltage;

a power bus configured to conduct power from the first output through a portion of the semiconductor package to a point-of-load (POL) circuit located within the semiconductor package;

the POL circuit configured to perform a second power conversion, in which the POL circuit is configured to receive the conducted power, make a second voltage adjustment, and deliver power to at least one of the semiconductor chips mounted in the semiconductor package at a chip voltage; and

a third power conversion stage located on the semiconductor chip and configured to perform a third power conversion, in which the third power conversion stage is configured to receive power from the POL circuit, make a third voltage adjustment, and deliver power at a load voltage to at least one circuit load on the semiconductor chip, and the package voltage is at least 5 times greater than the load voltage.

ABSTRACT

A semiconductor package includes a VLSI semiconductor die and one or more output circuits connected to supply power to the die mounted to a package substrate. The output circuit(s), which include a transformer and rectification circuitry, provide current multiplication at an essentially fixed conversion ratio, K , in the semiconductor package, receiving AC power at a relatively high voltage and delivering DC power at a relatively low voltage to the die. The output circuits may be connected in series or parallel as needed. A driver circuit may be provided outside the semiconductor package for receiving power from a source and driving the transformer in the output circuit(s), preferably with sinusoidal currents. The driver circuit may drive a plurality of output circuits. The semiconductor package may require far fewer interface connections for supplying power to the die. Multi-output POL circuits may be used in conjunction with on-chip rail-selection and regulation circuitry to further improve efficiency. A three-stage power conversion system includes off-package, on-package and on-chip conversion stages.

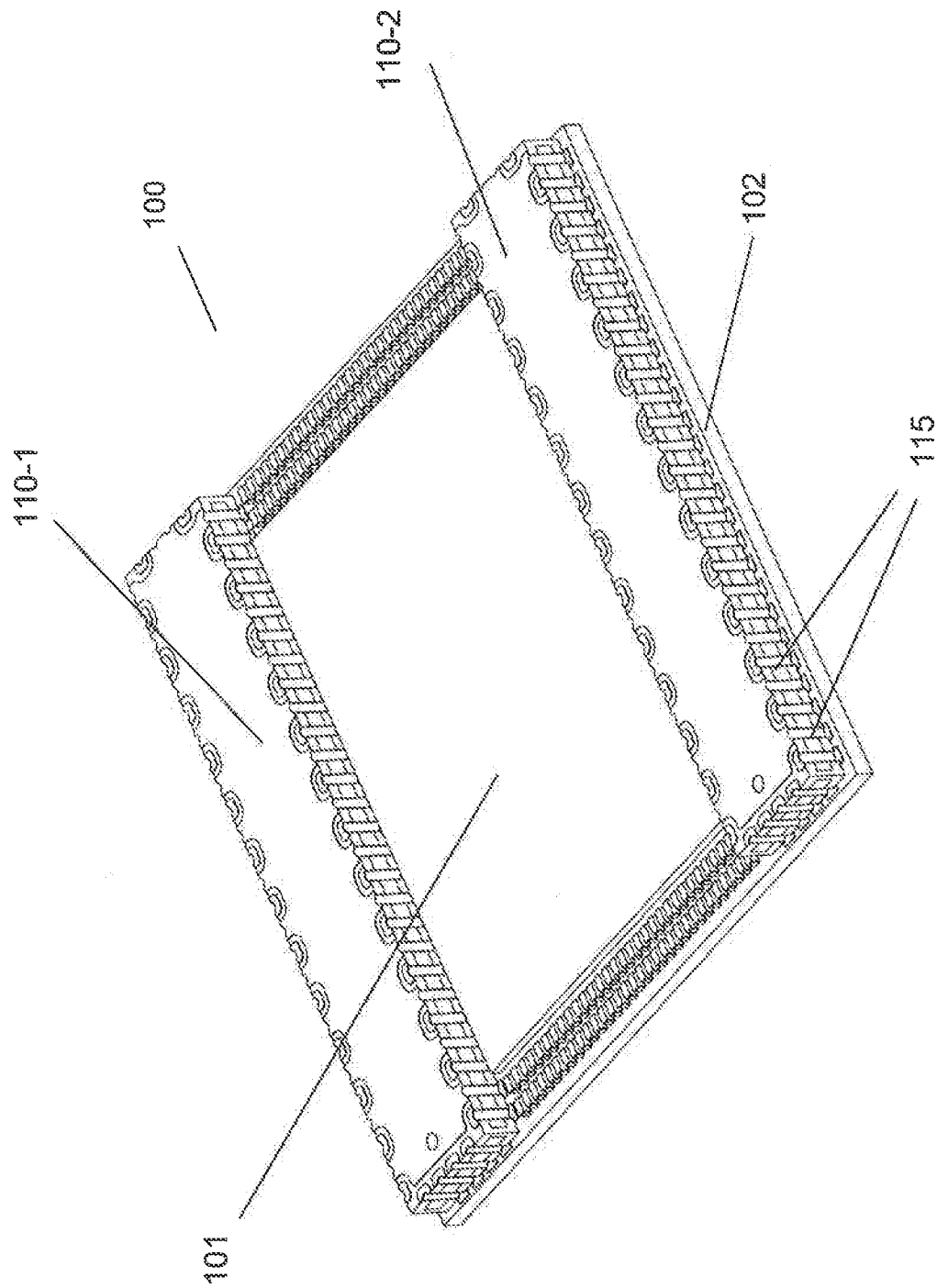


FIG. 1

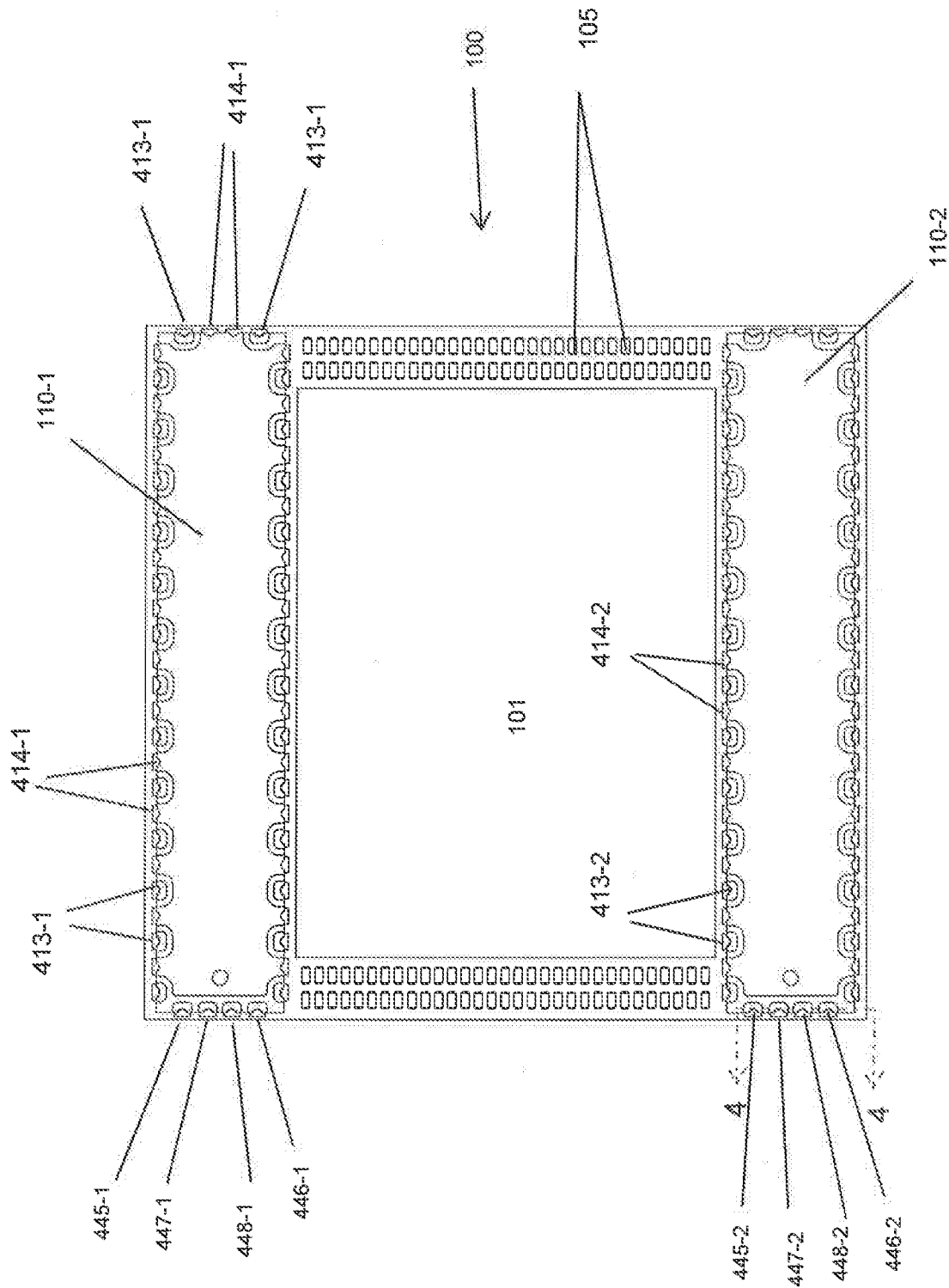


FIG. 2

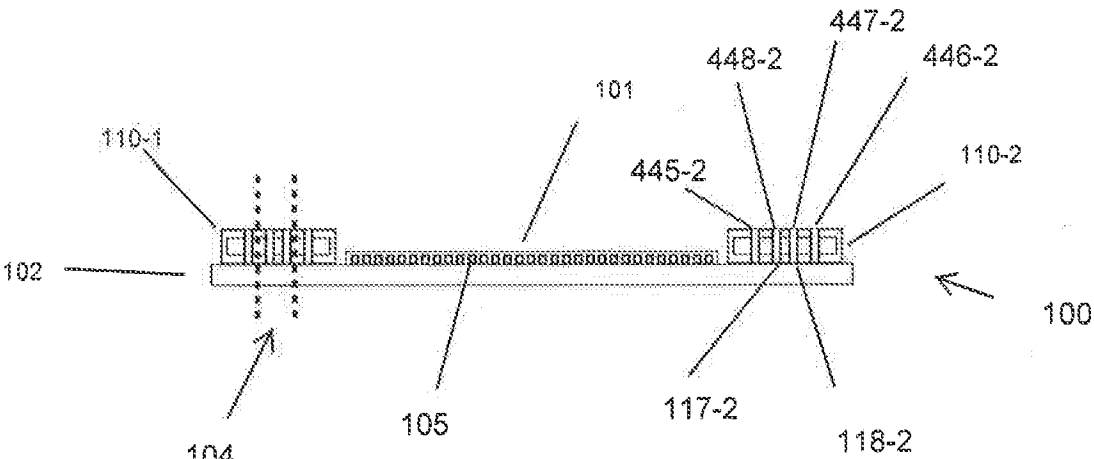


FIG. 3

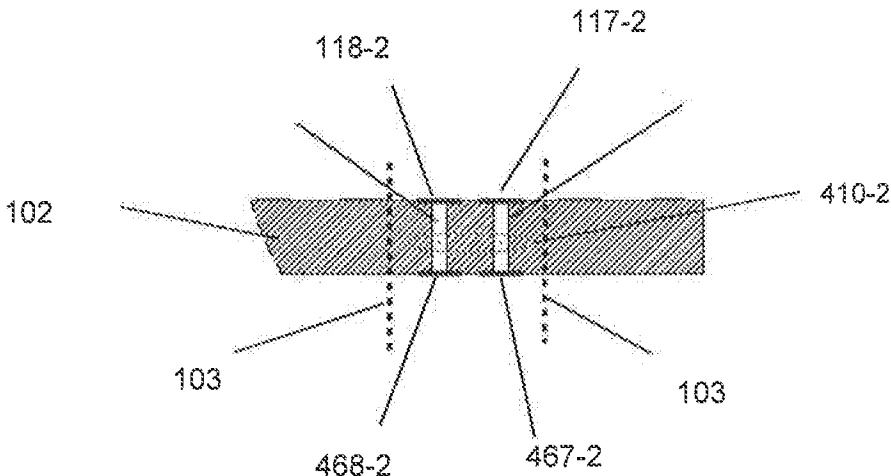


FIG. 4

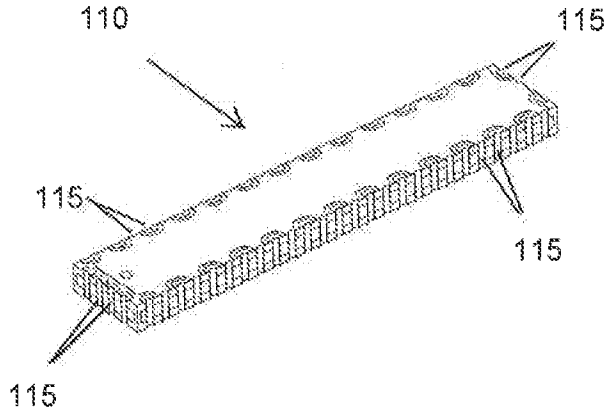


FIG. 8

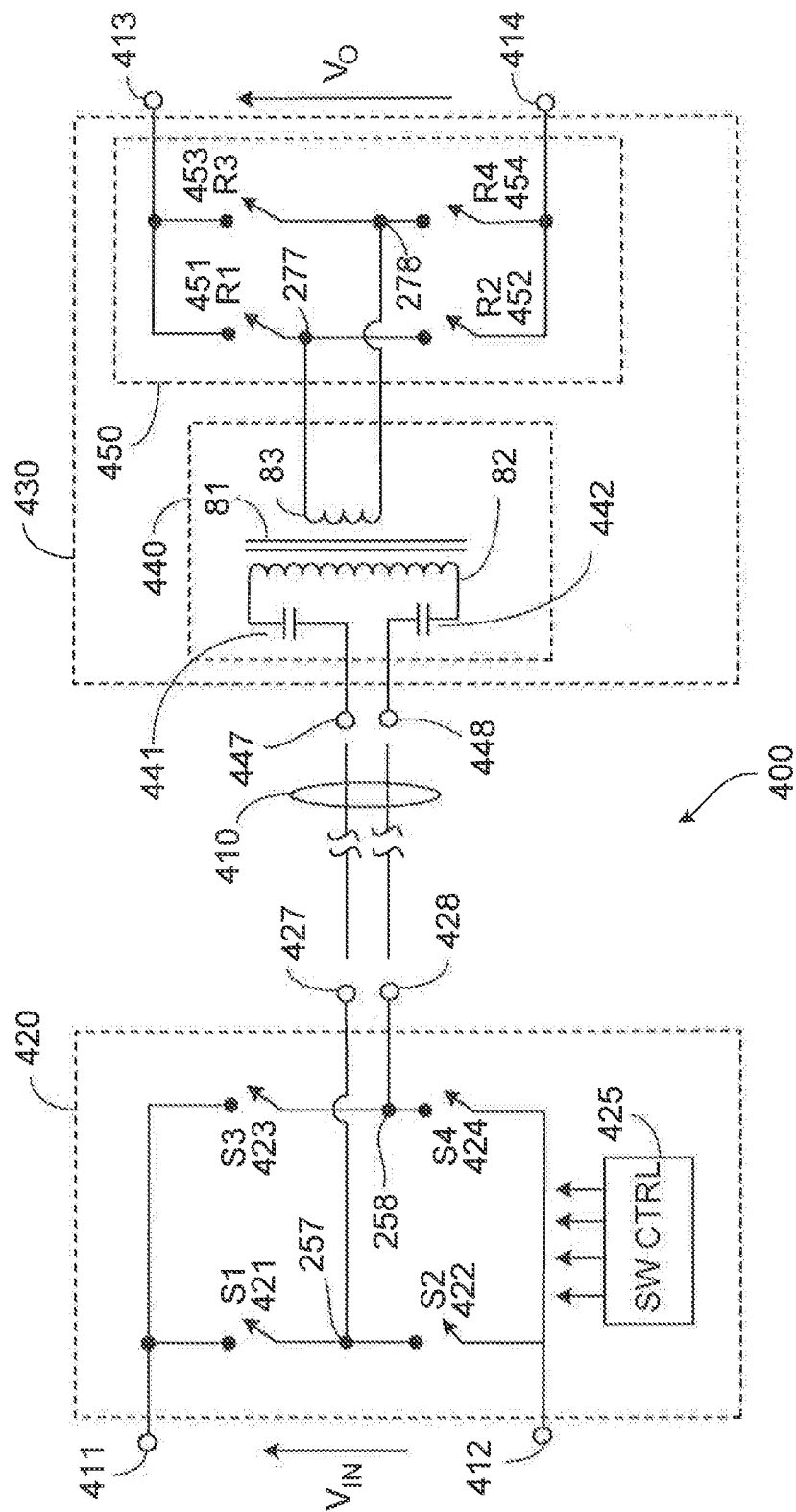


FIG. 5

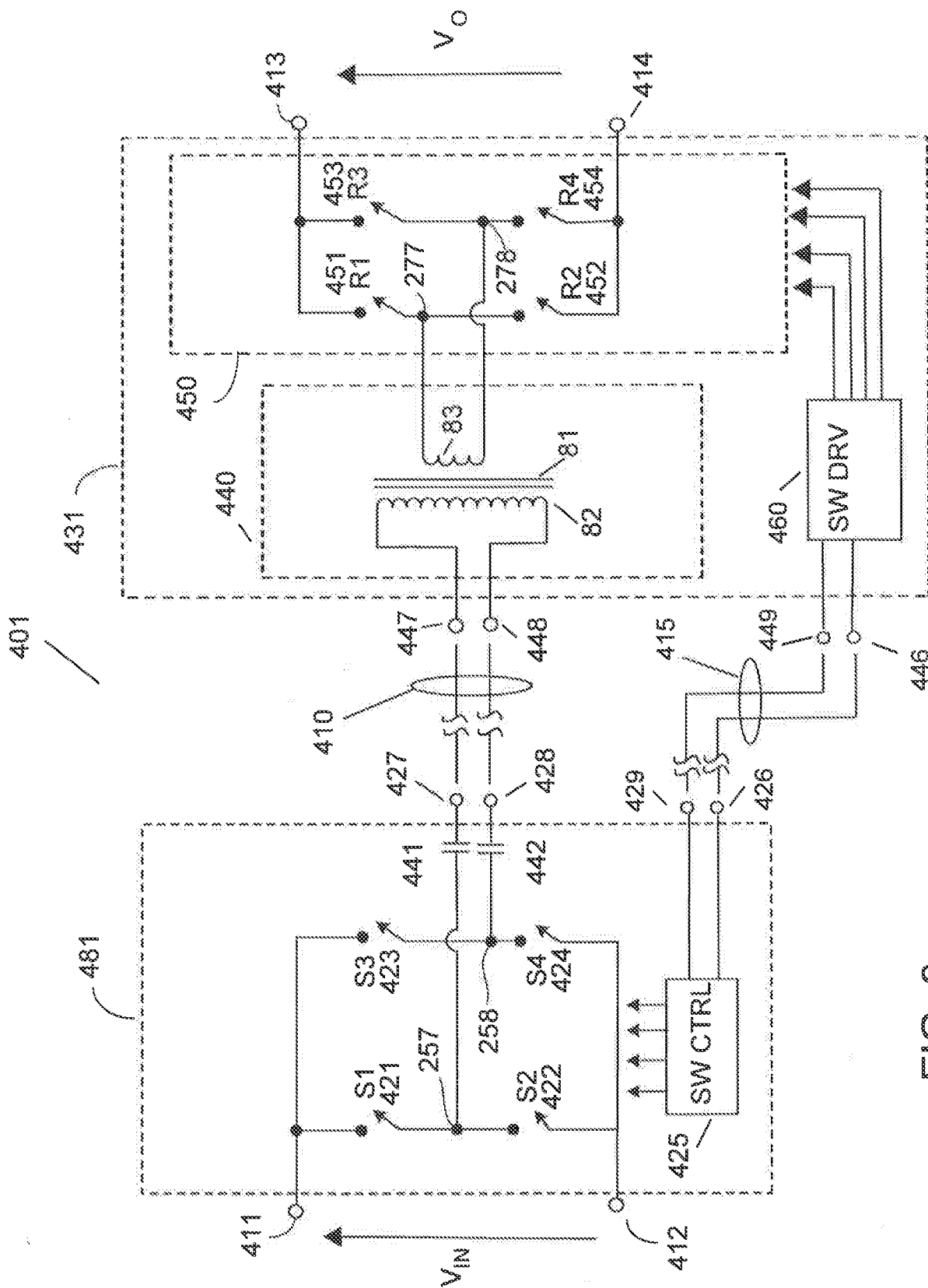


FIG. 6

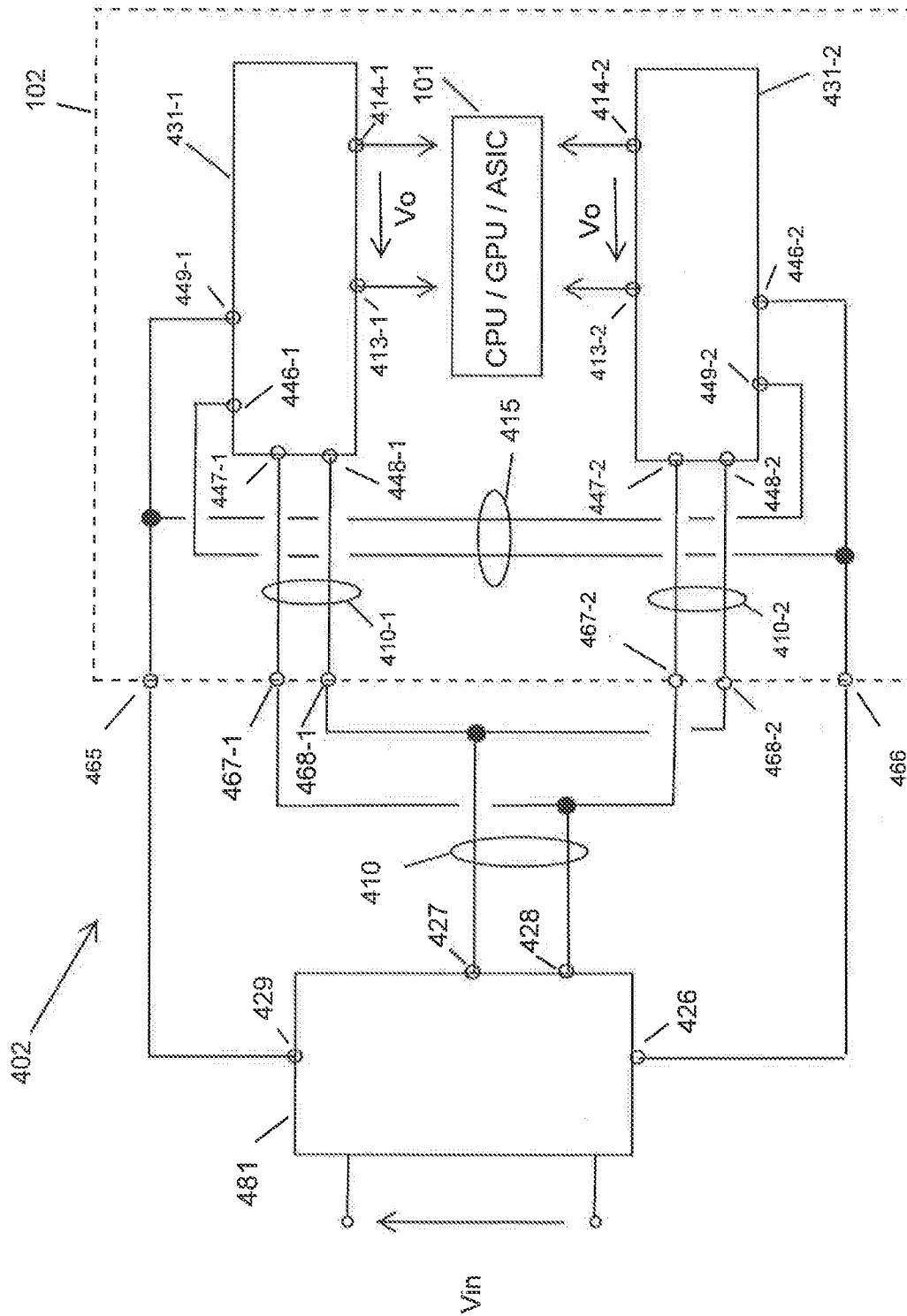


FIG. 7

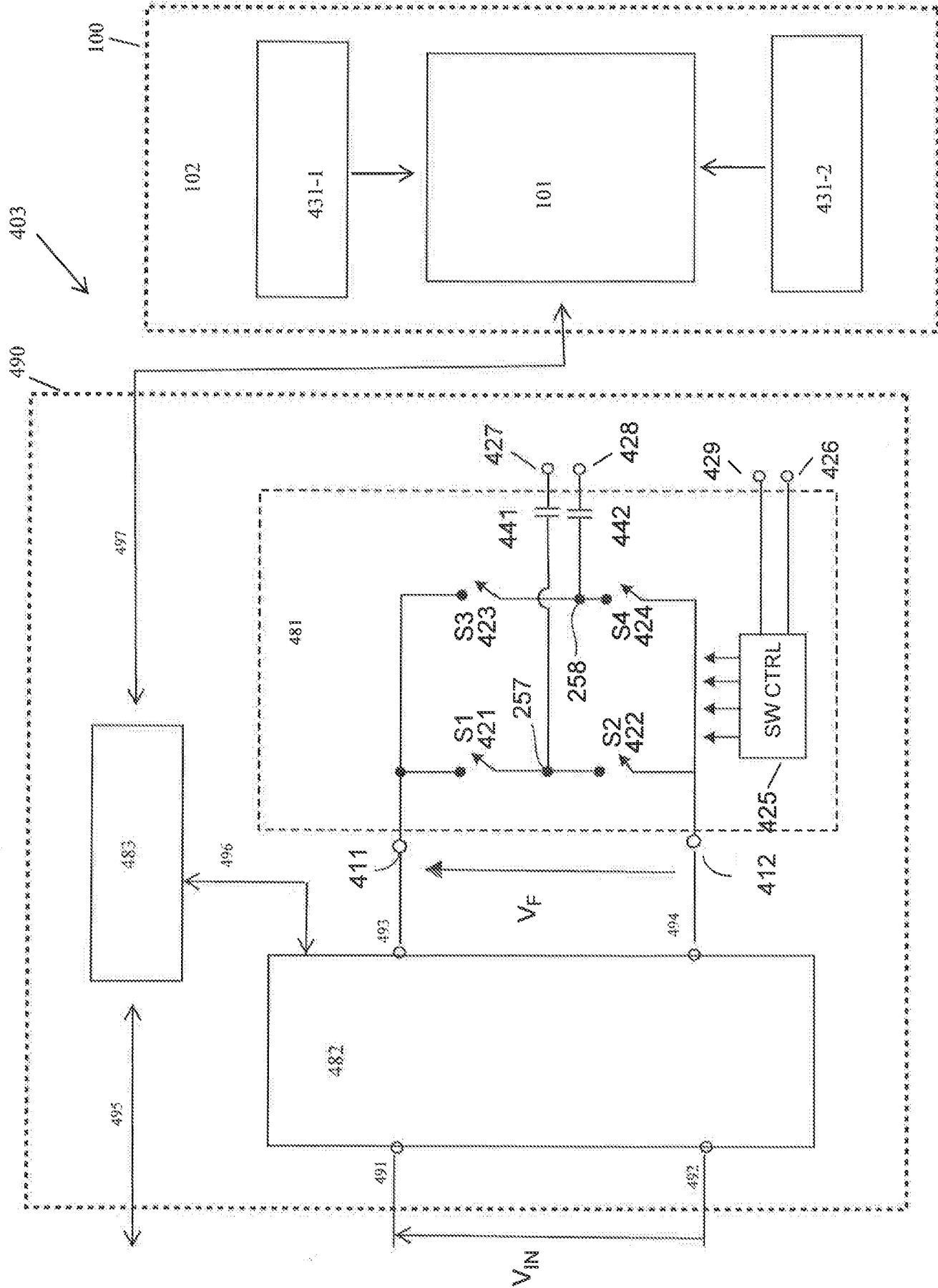


FIG. 9

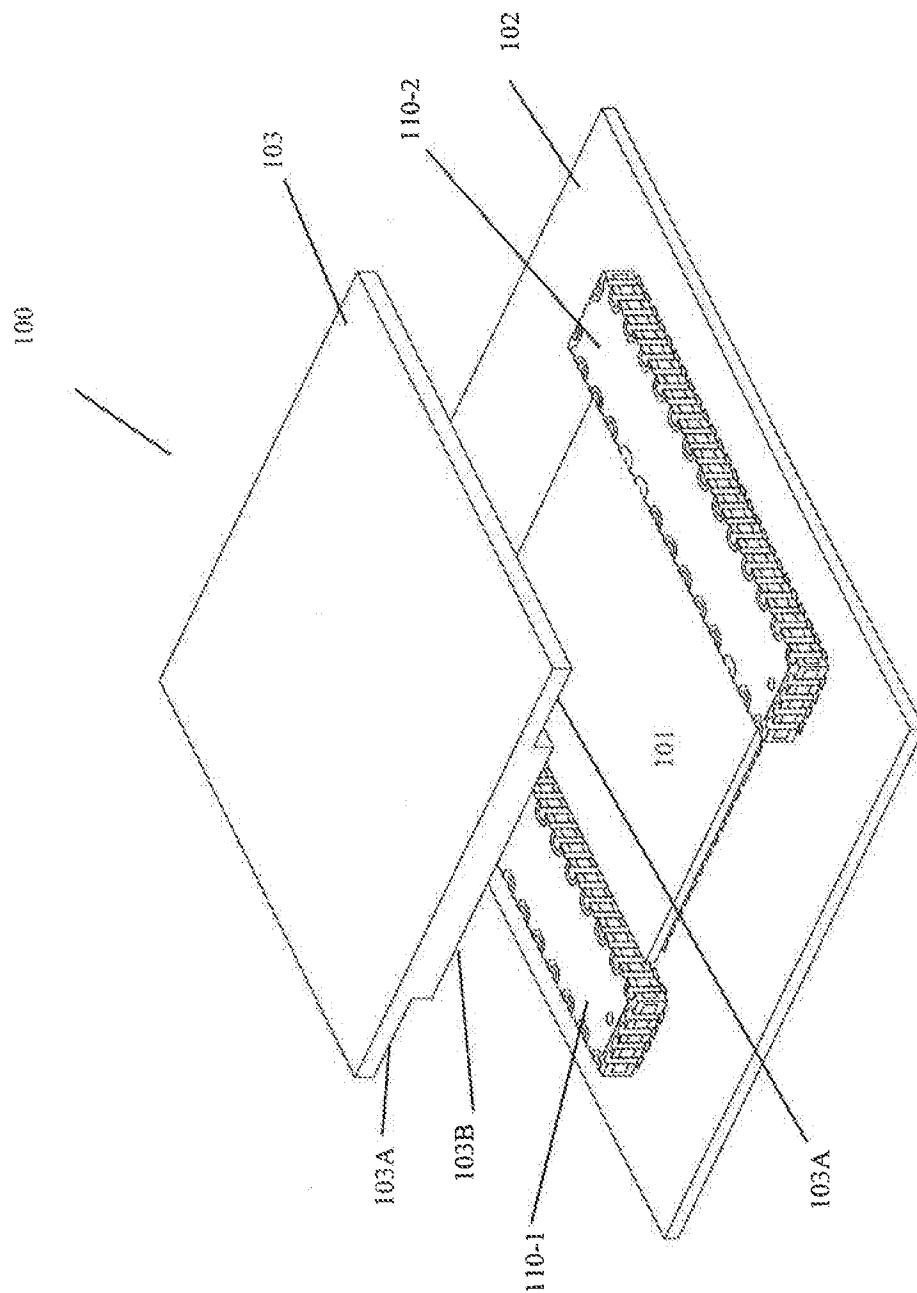


FIG. 10

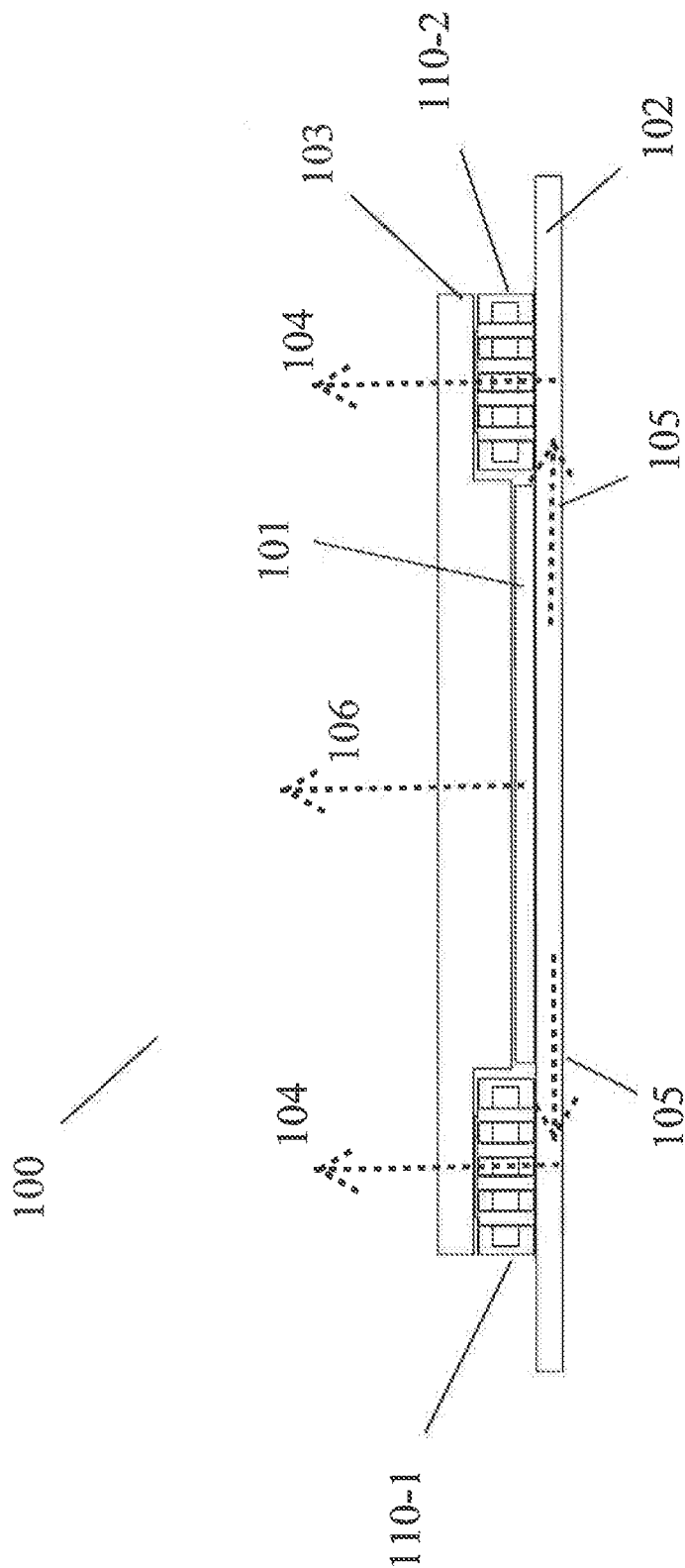


FIG. 11

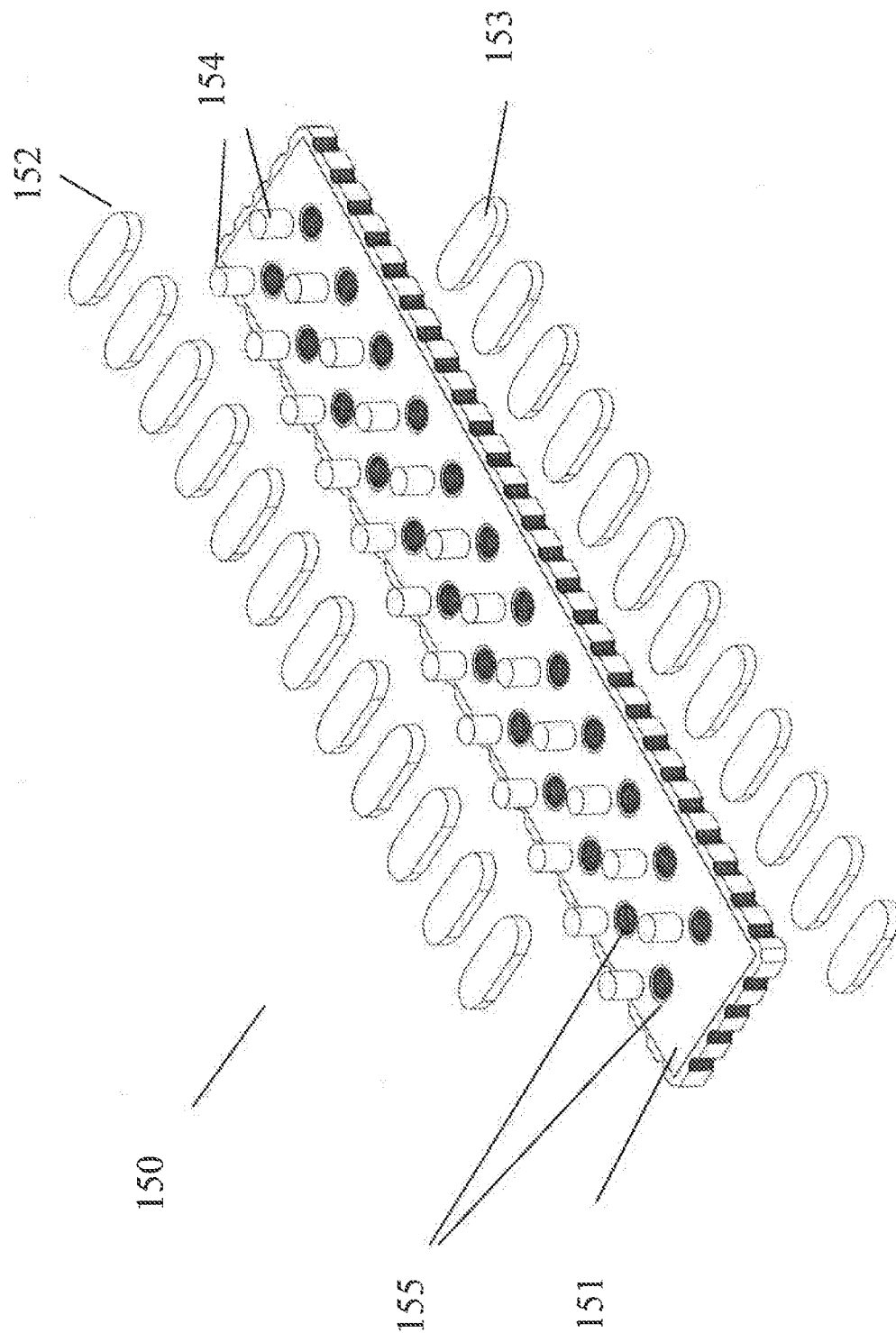


FIG. 12

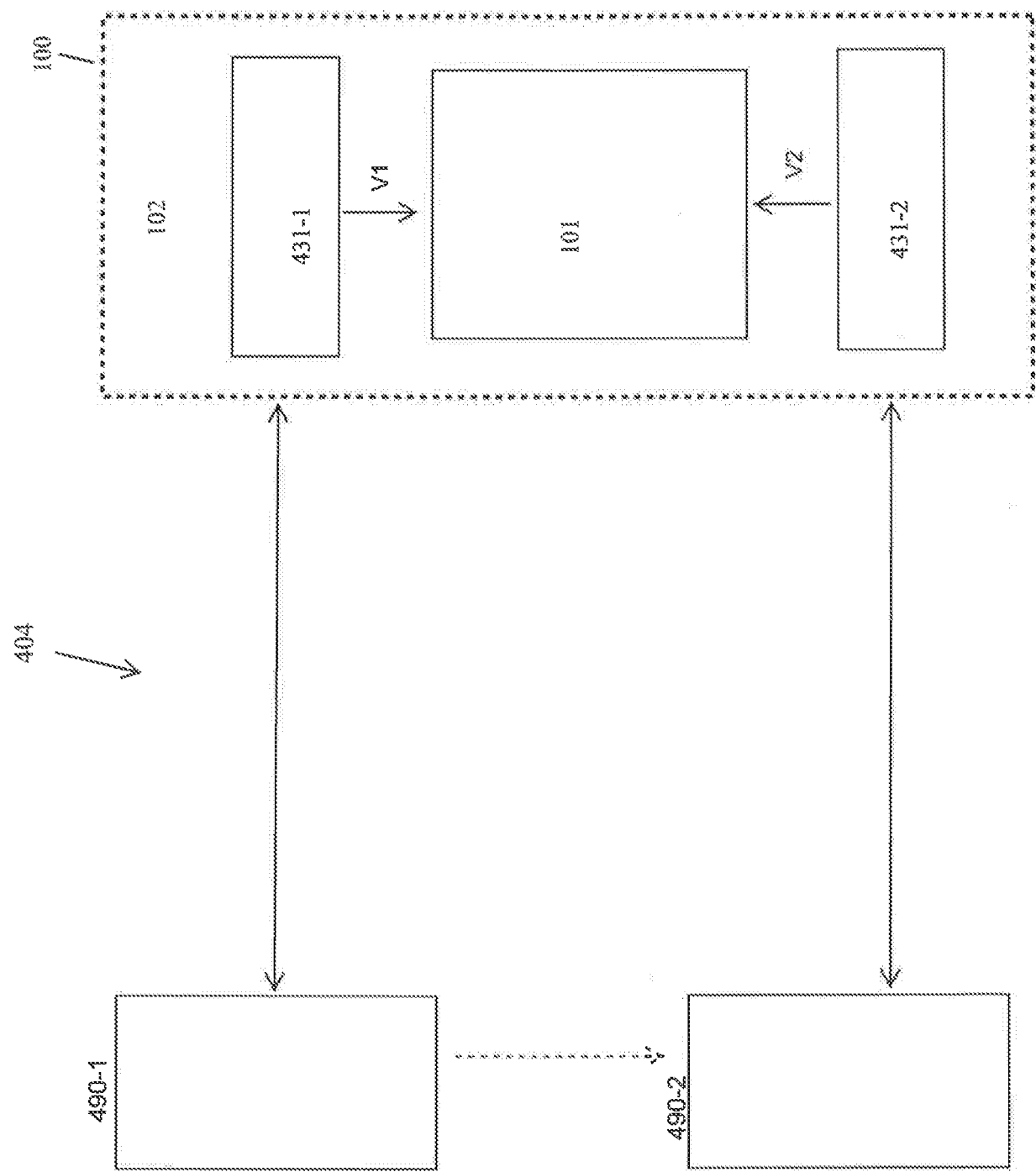


FIG. 13

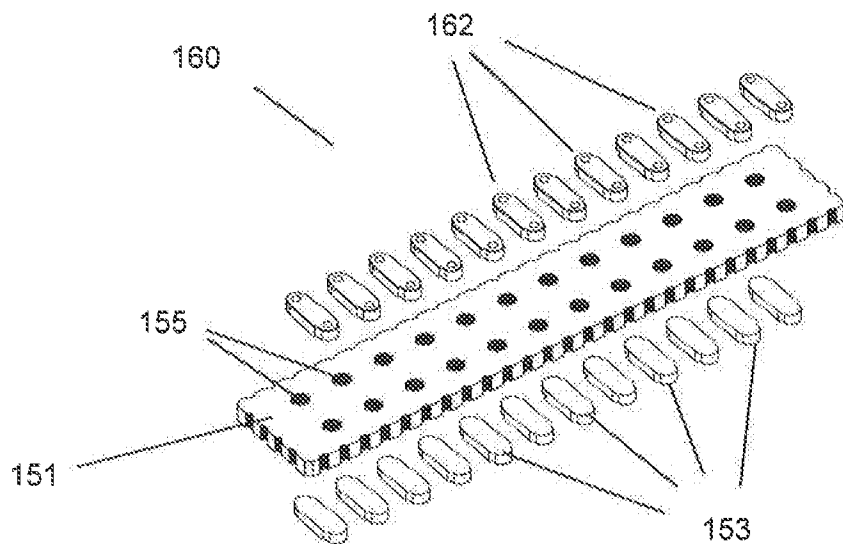


FIG. 14A

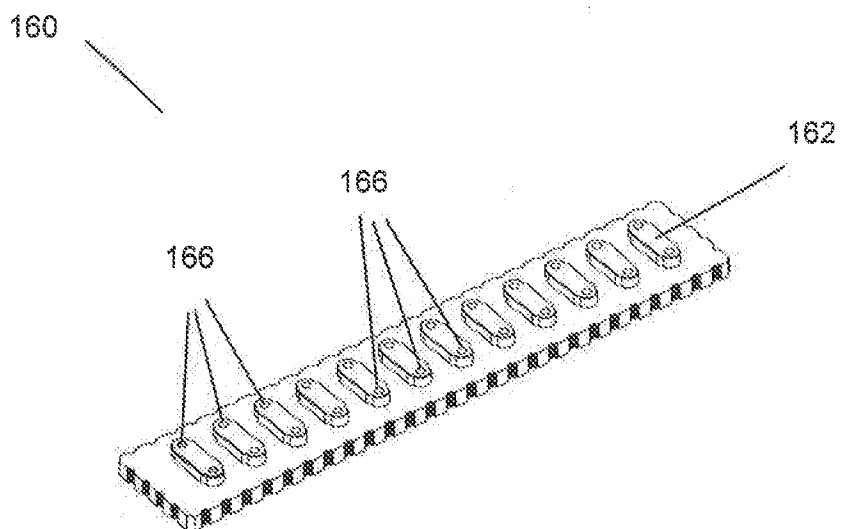
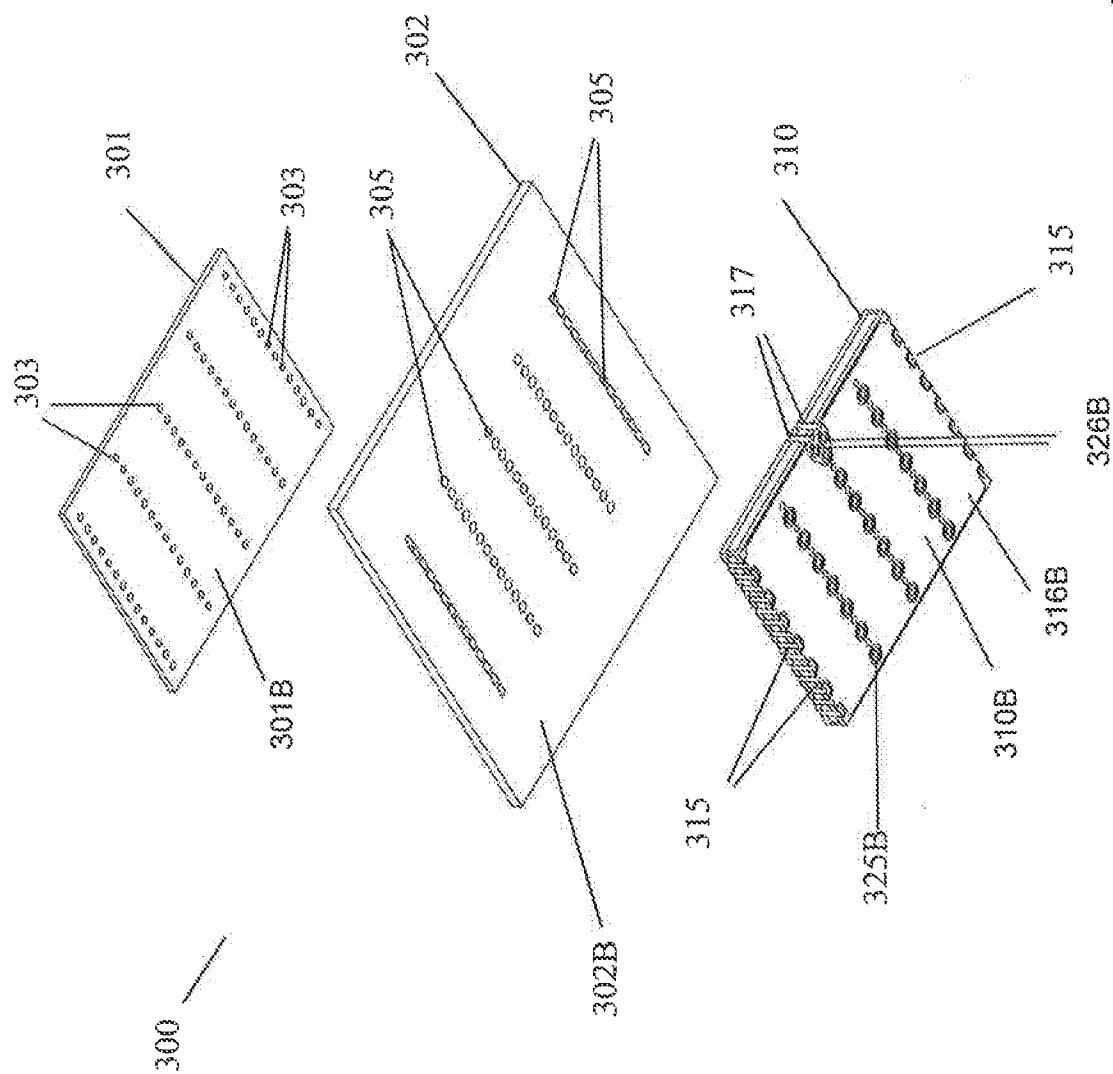
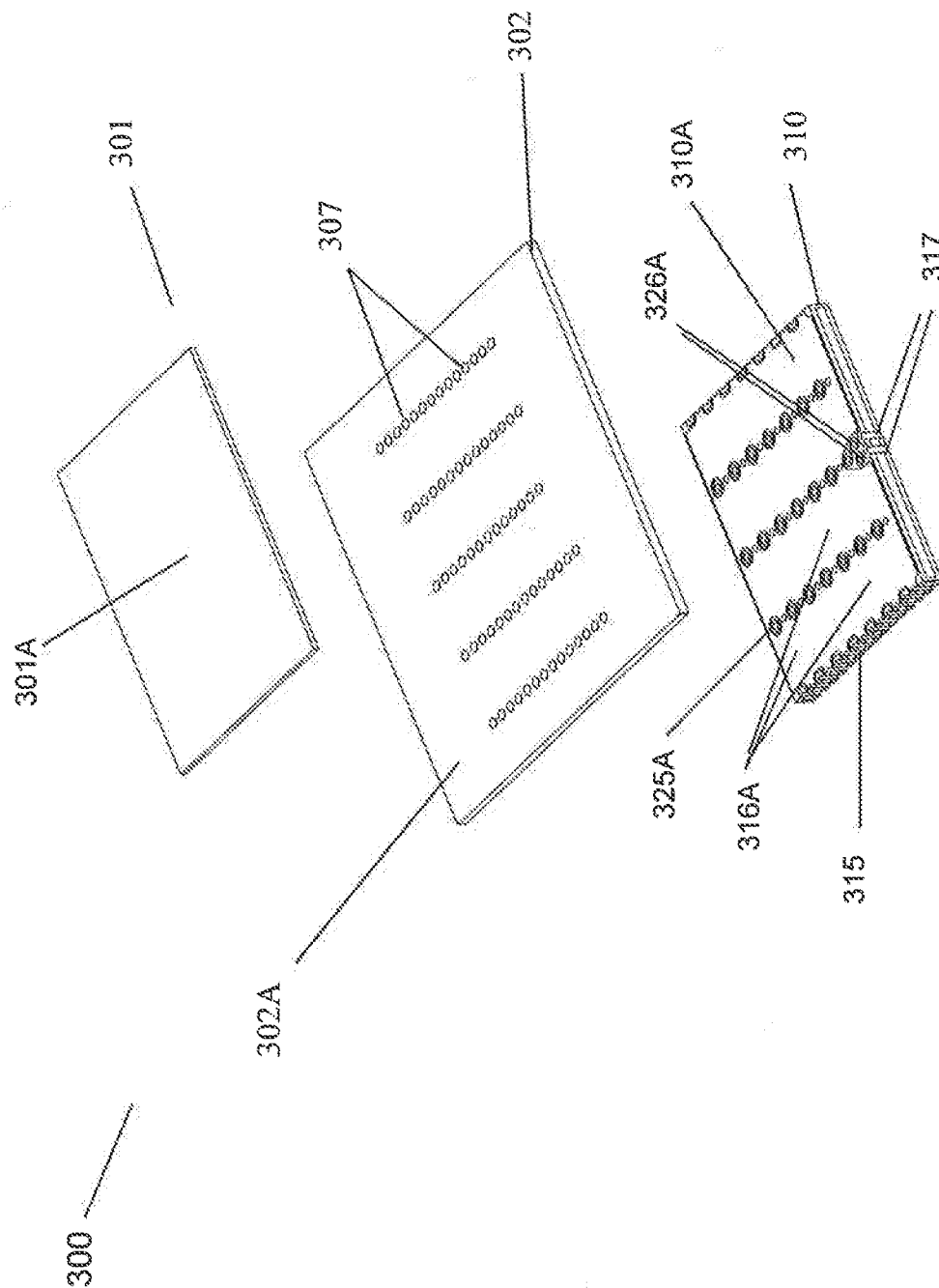


FIG. 14B



ASL
100



1500

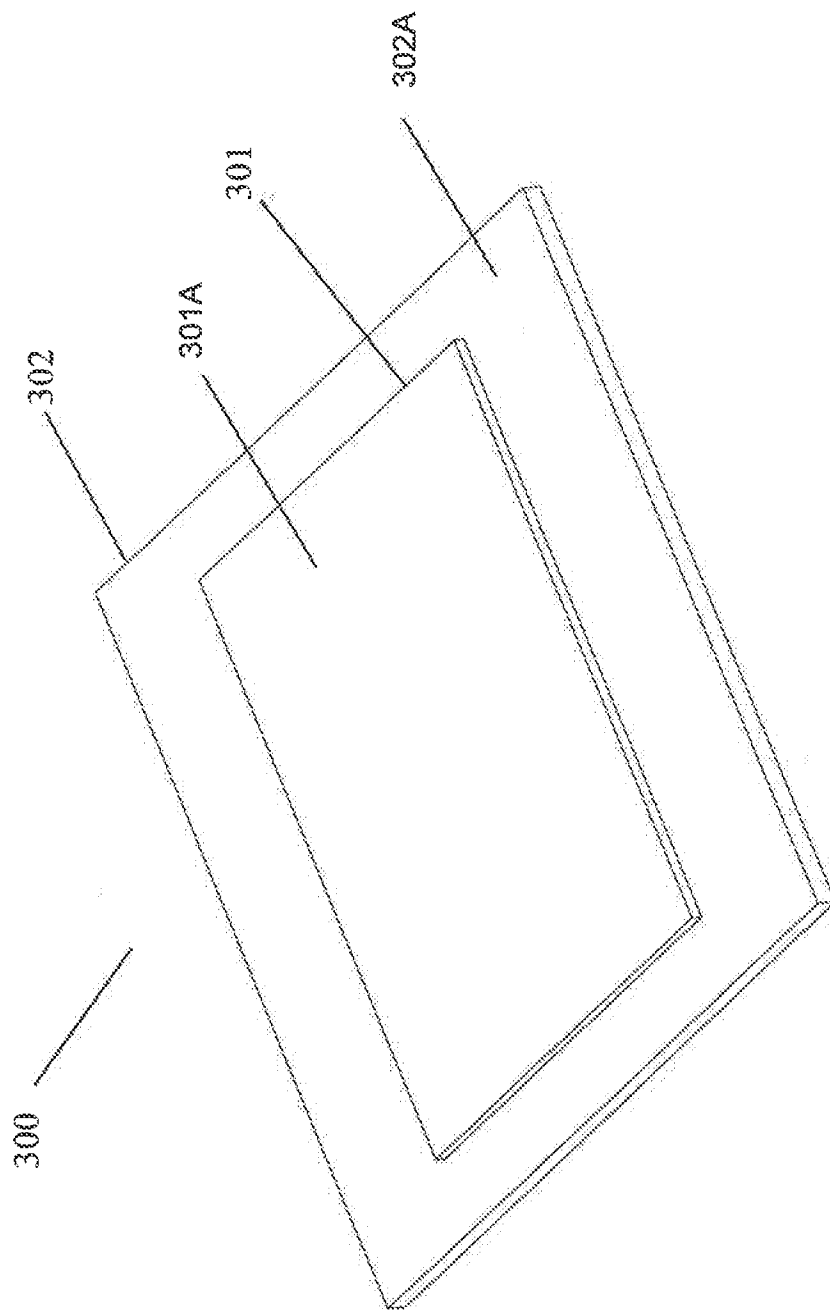


Fig. 16A

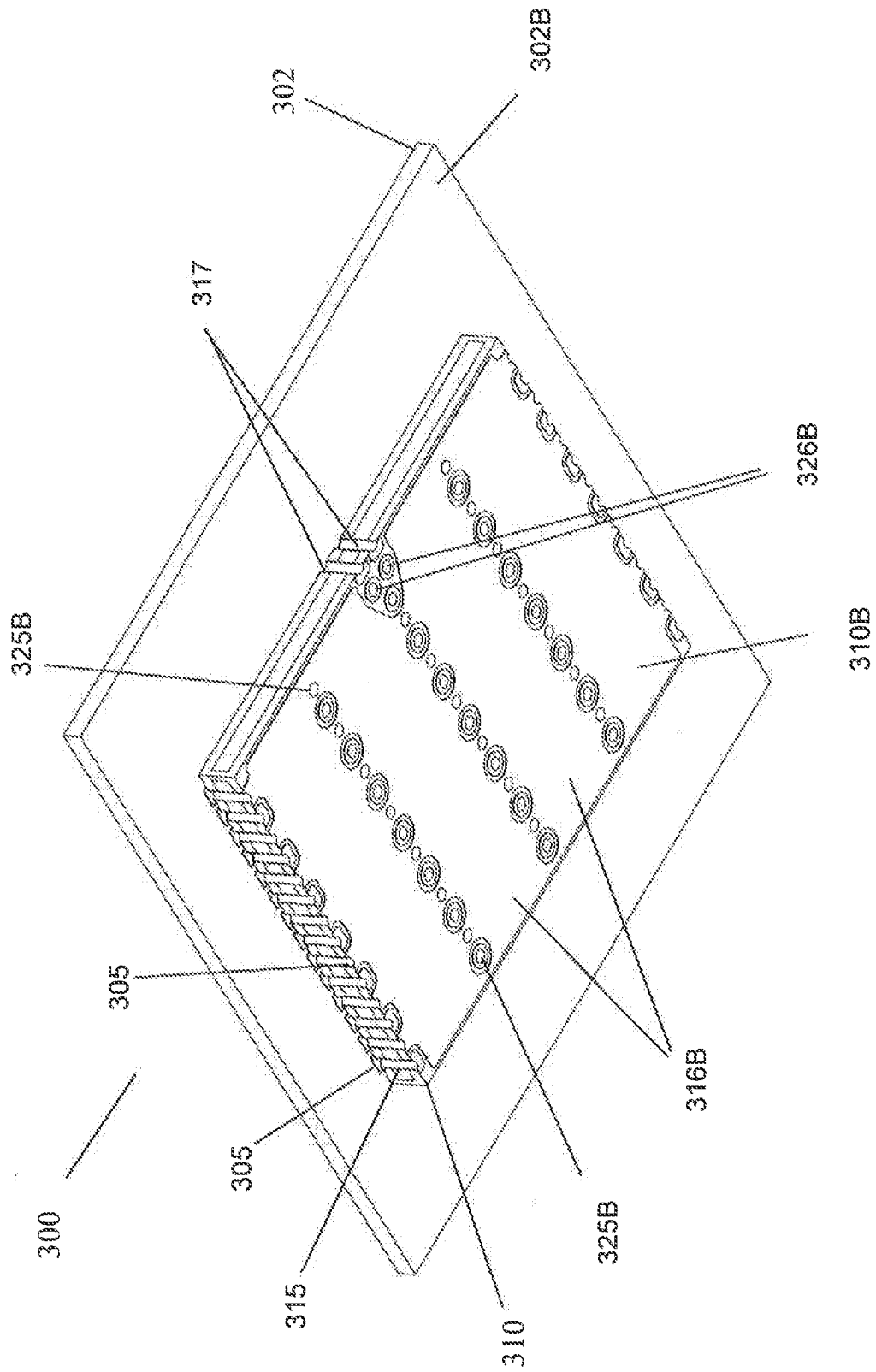


Fig. 16B

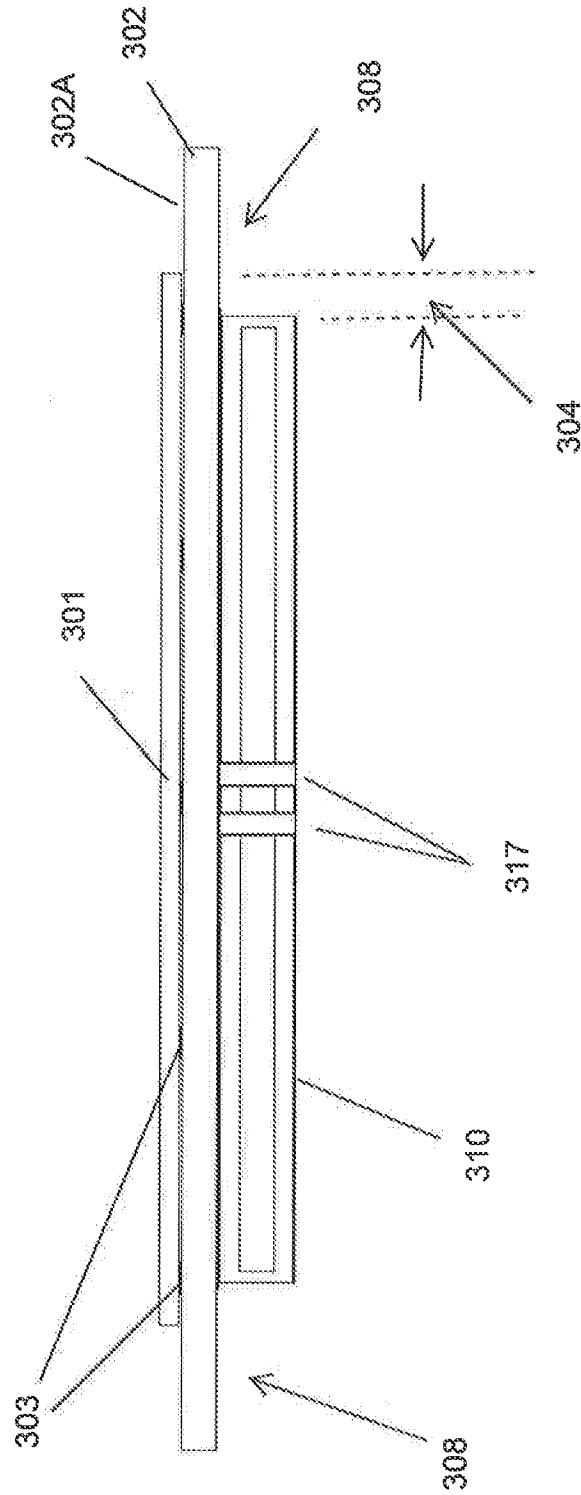


Fig. 17

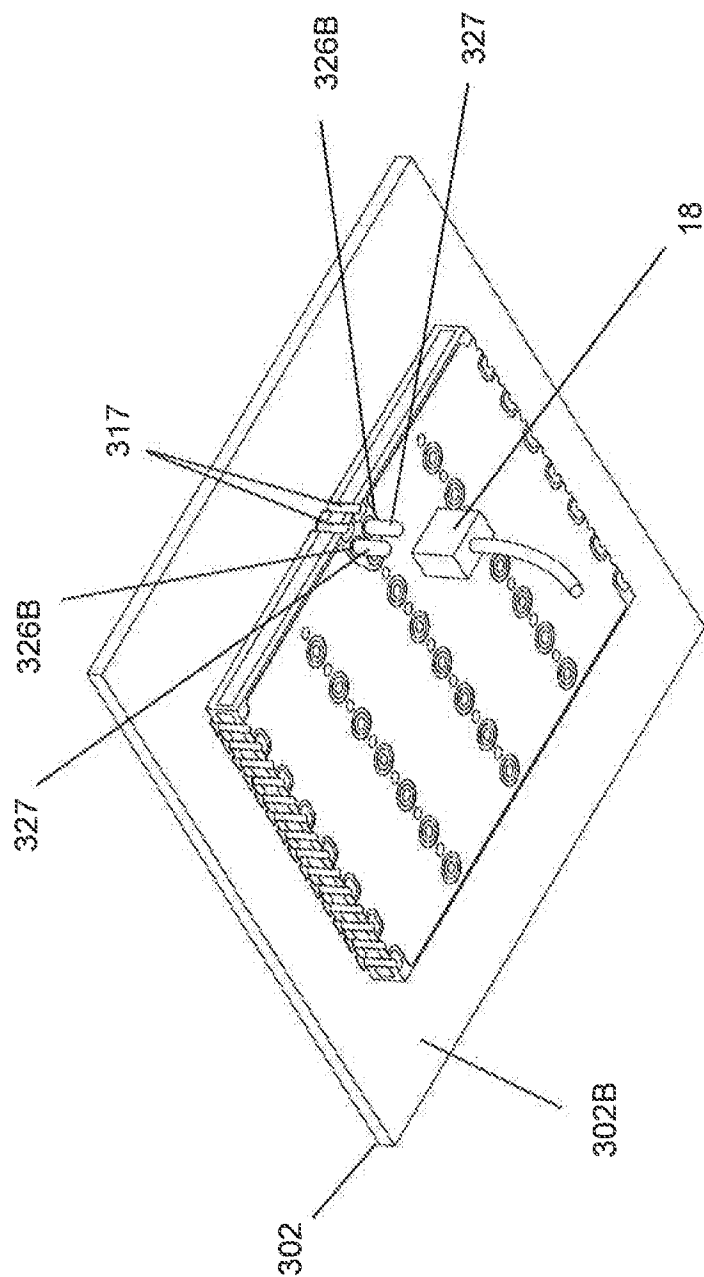


Fig. 18

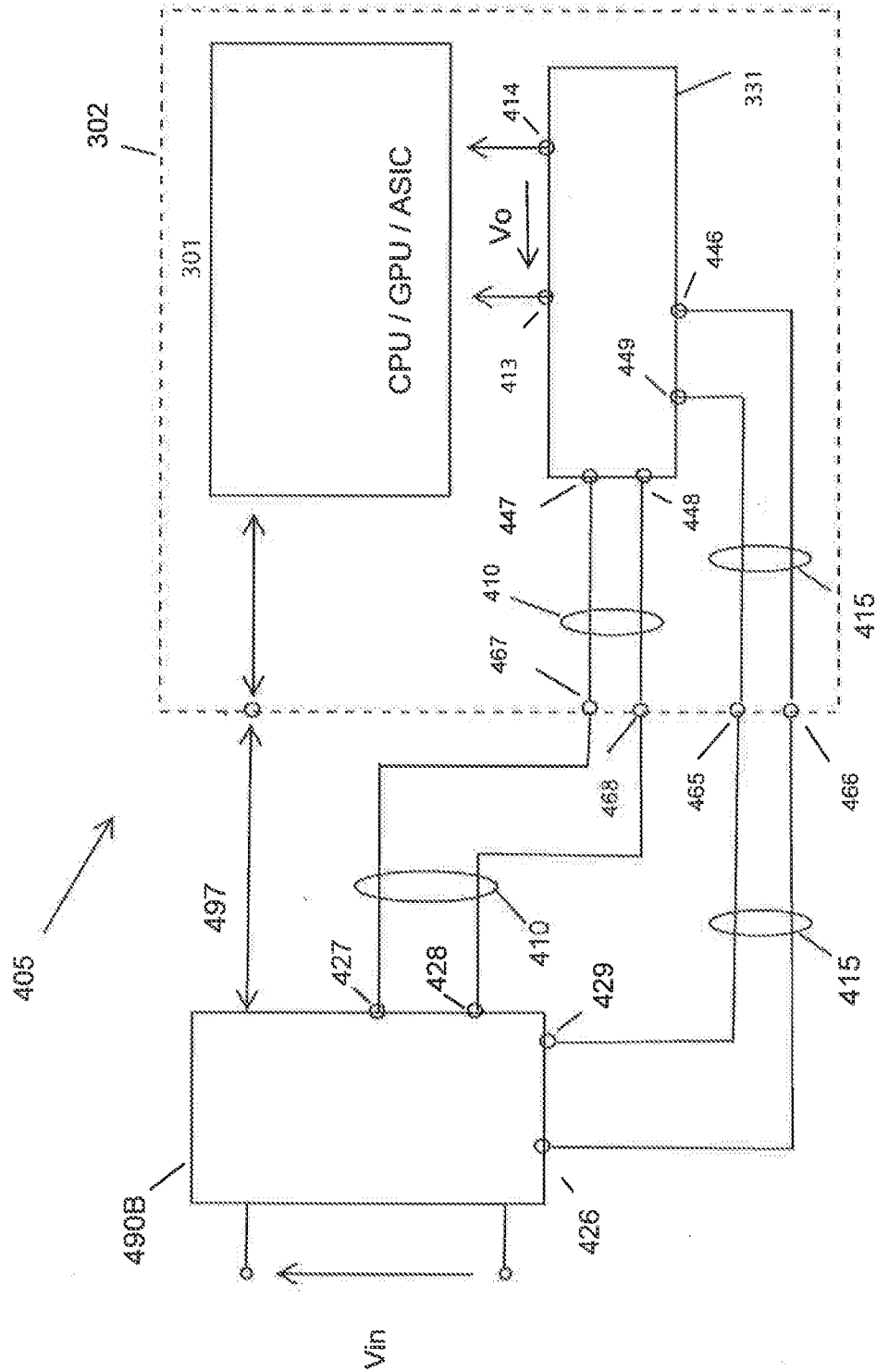


FIG. 19

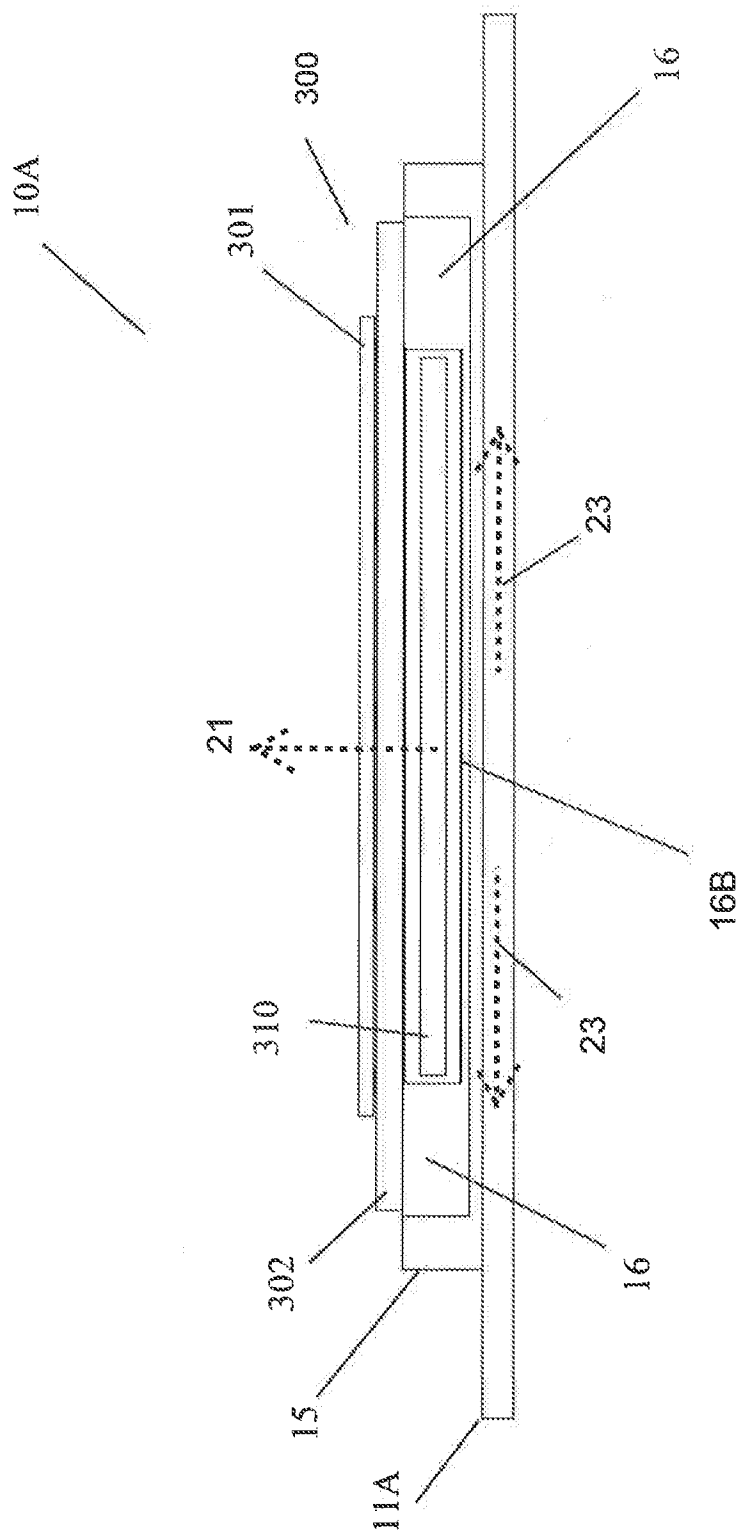


Fig. 20A

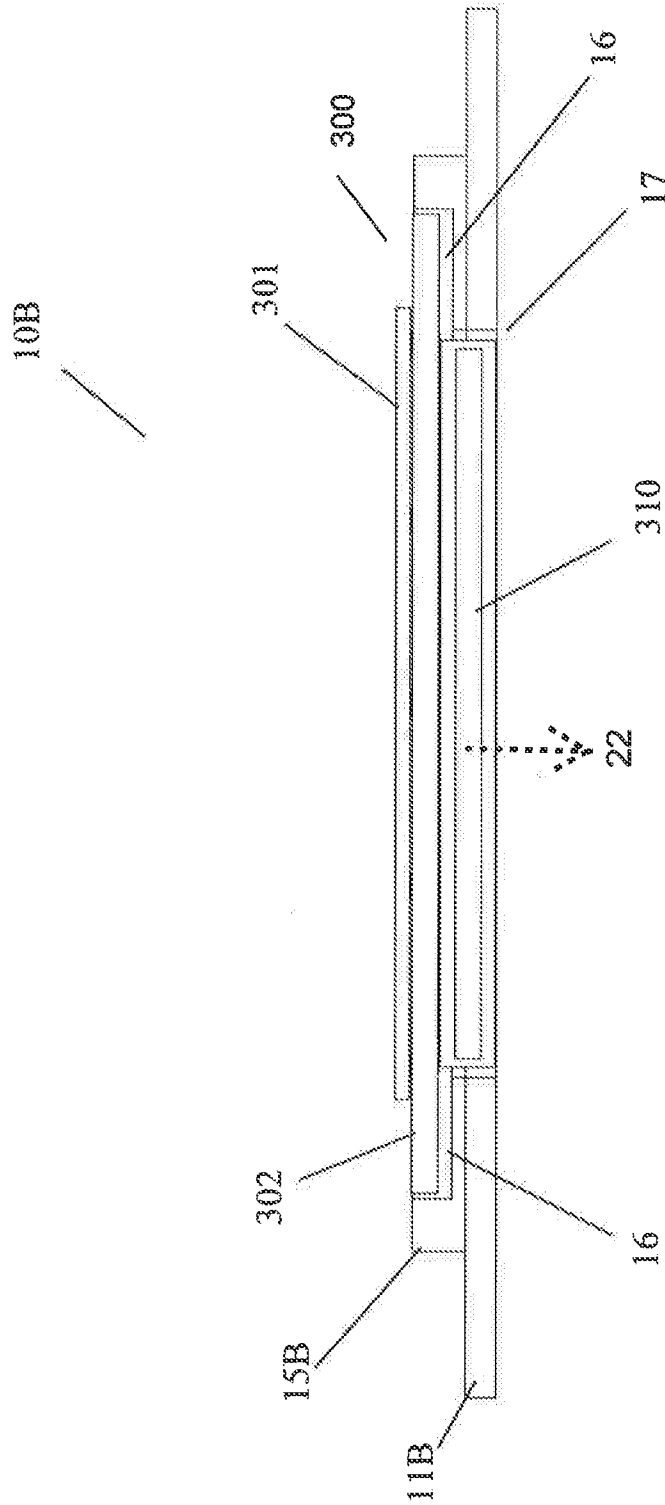


Fig. 20B

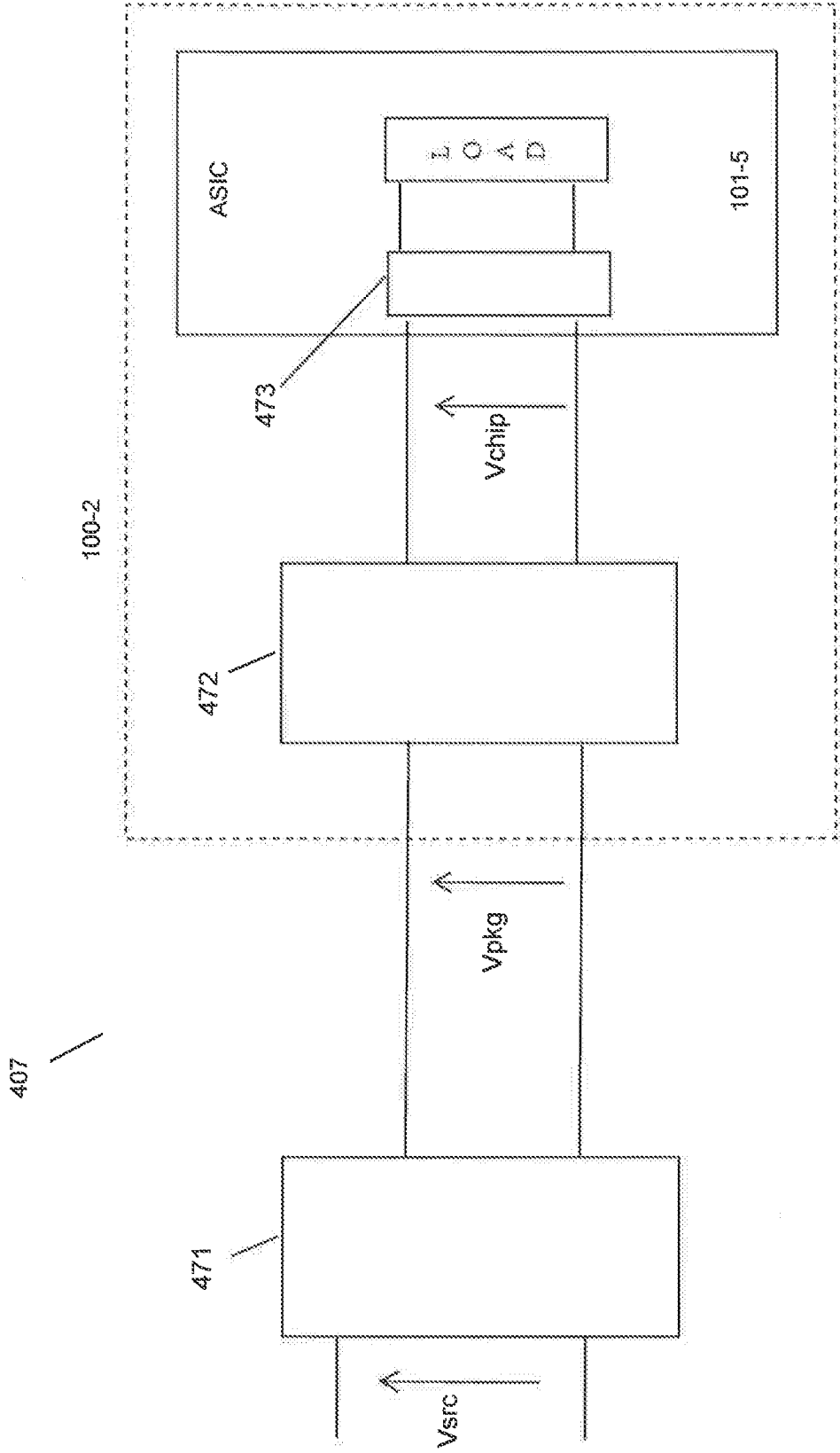


FIG. 21A

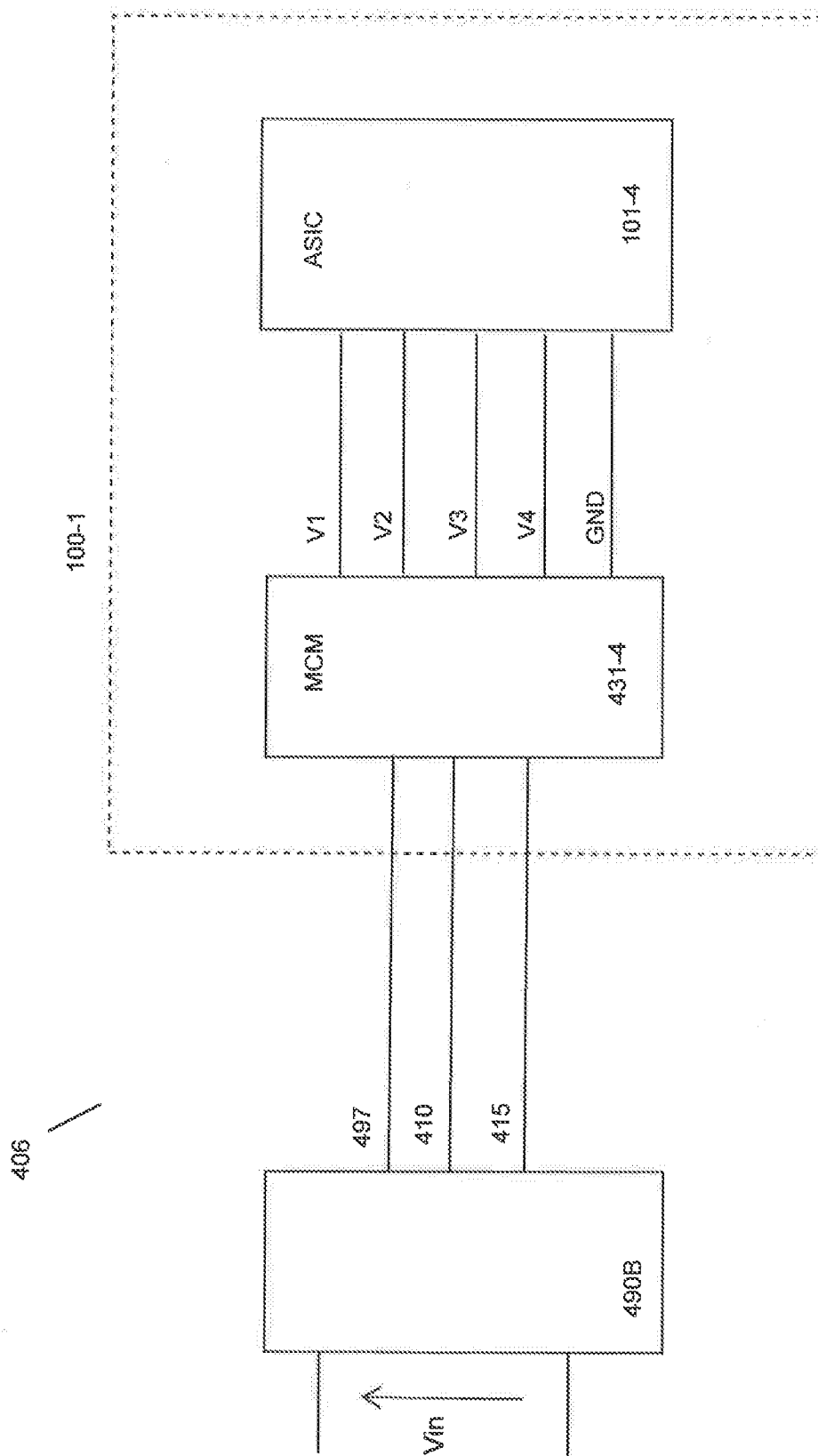


FIG. 21B

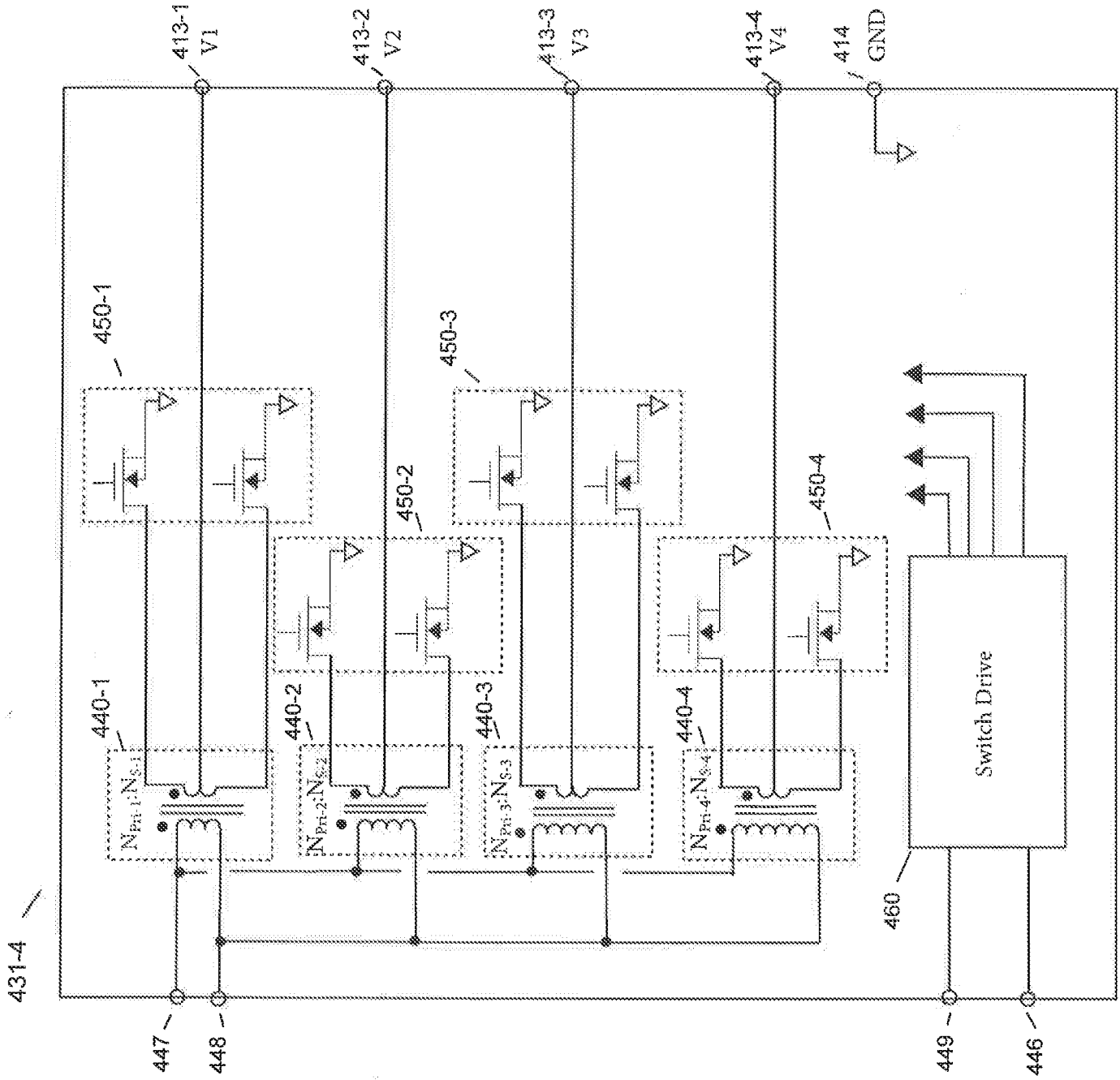


FIG. 22

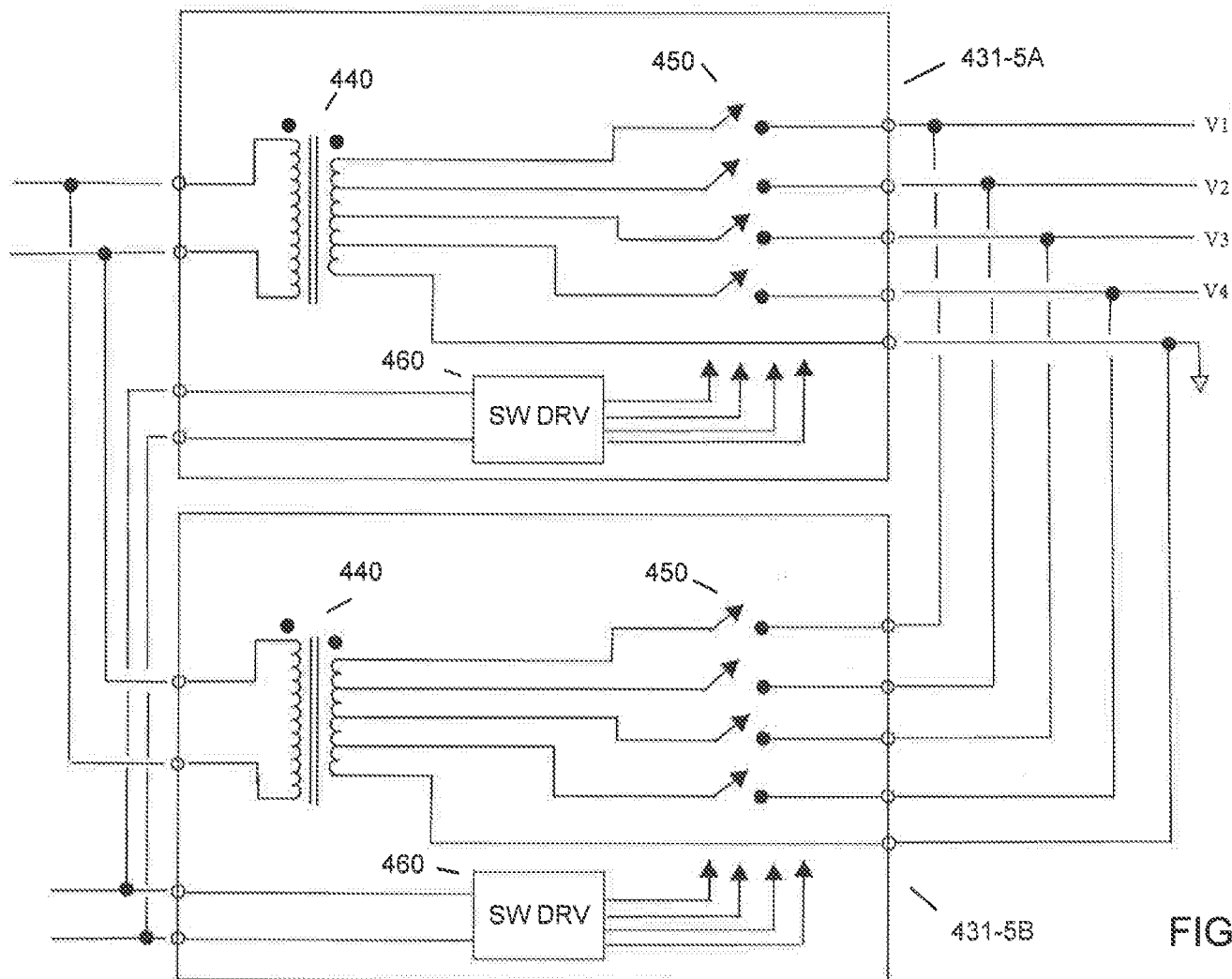


FIG. 23

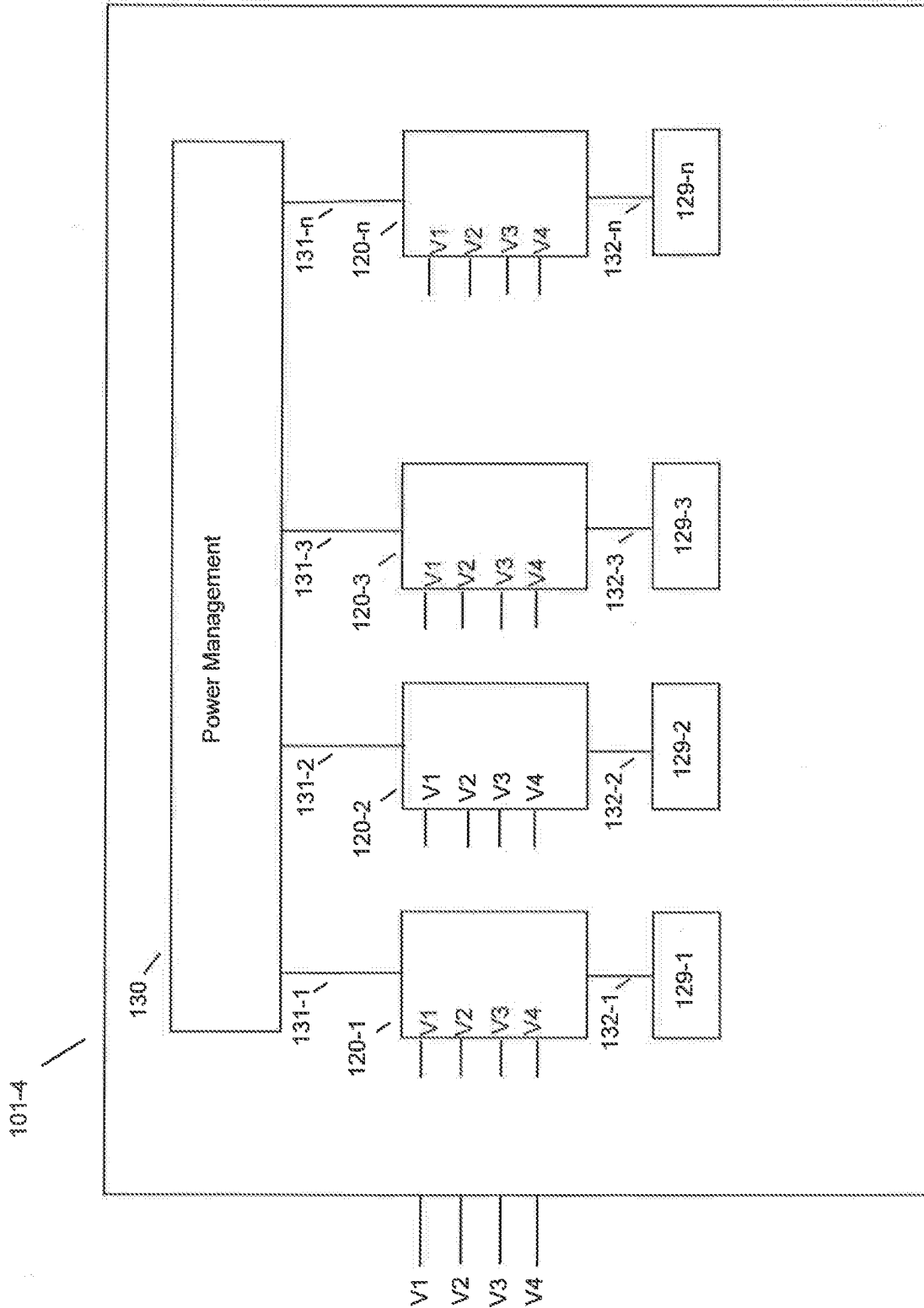


FIG. 24

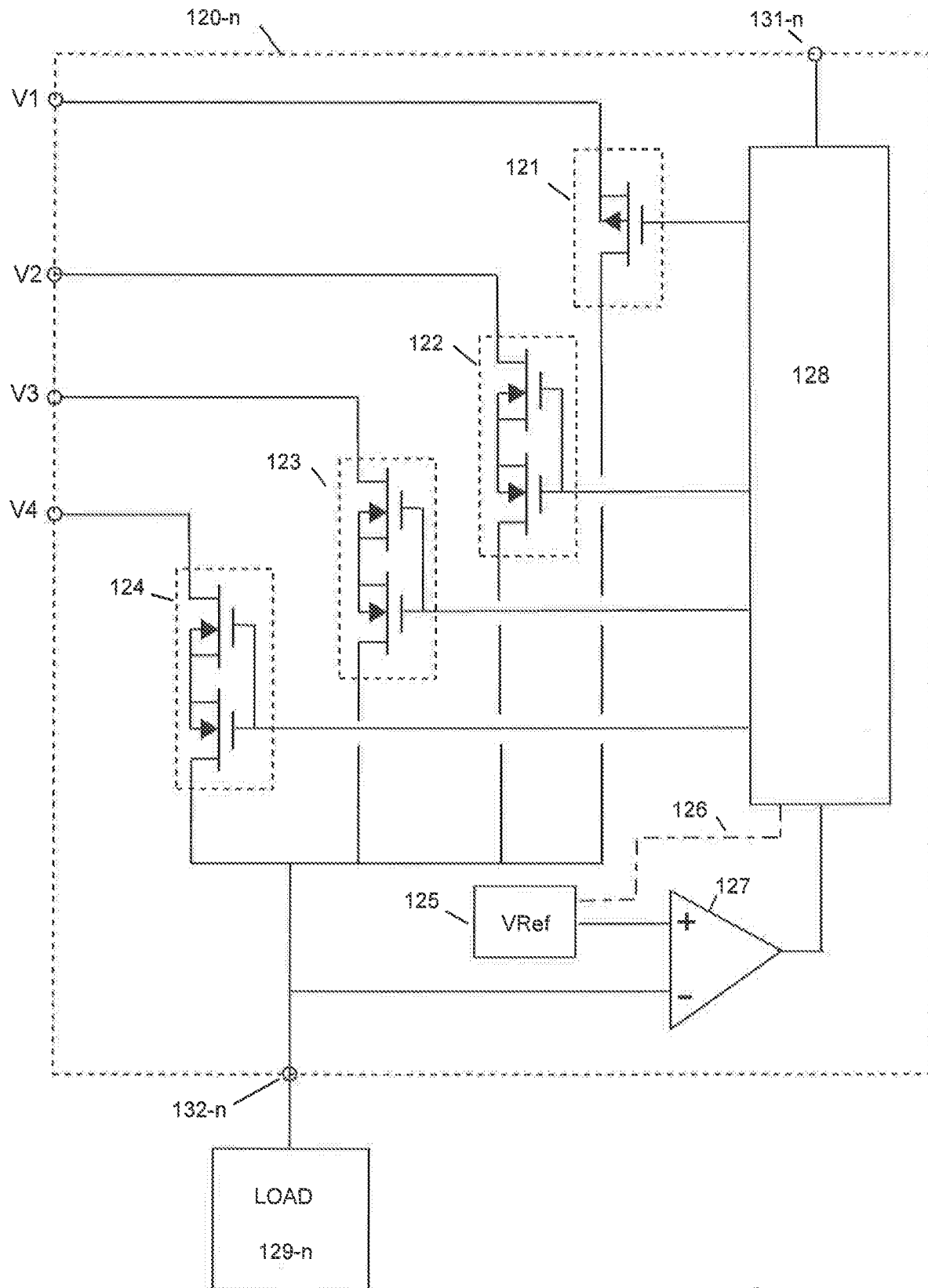


FIG. 25

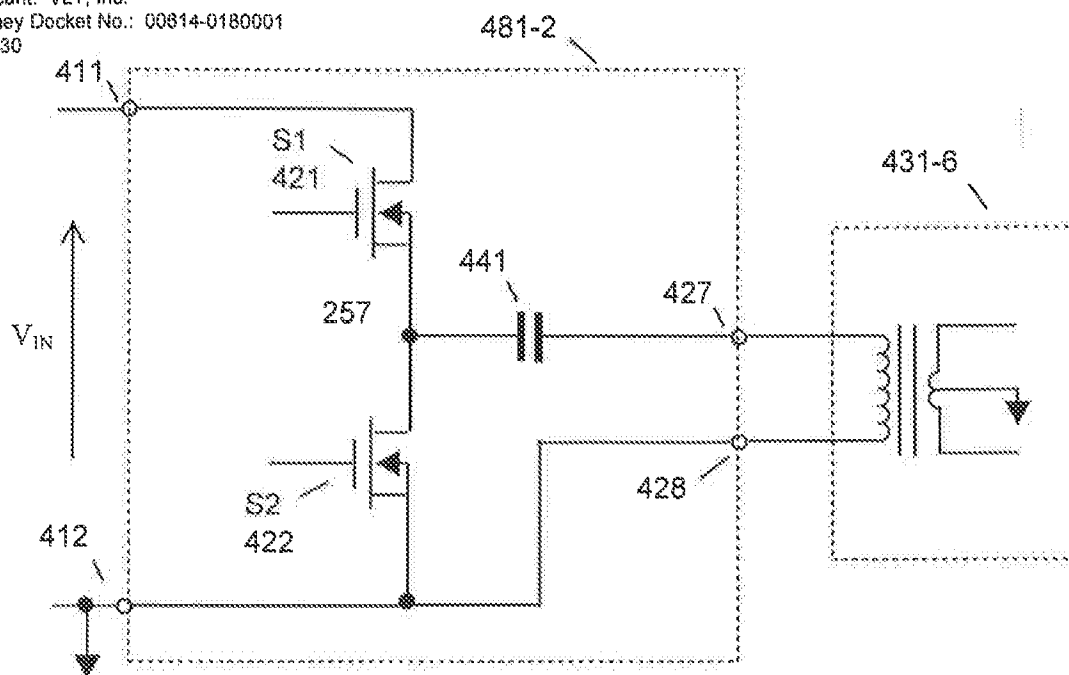


FIG. 26A

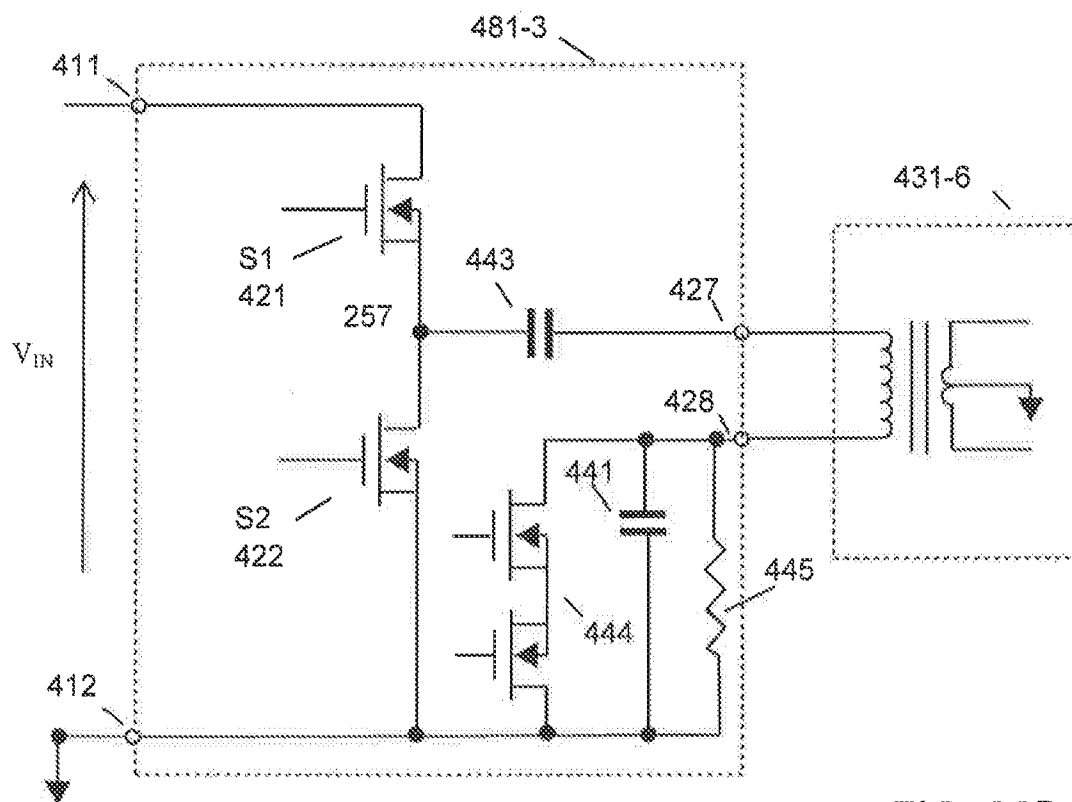


FIG. 26B

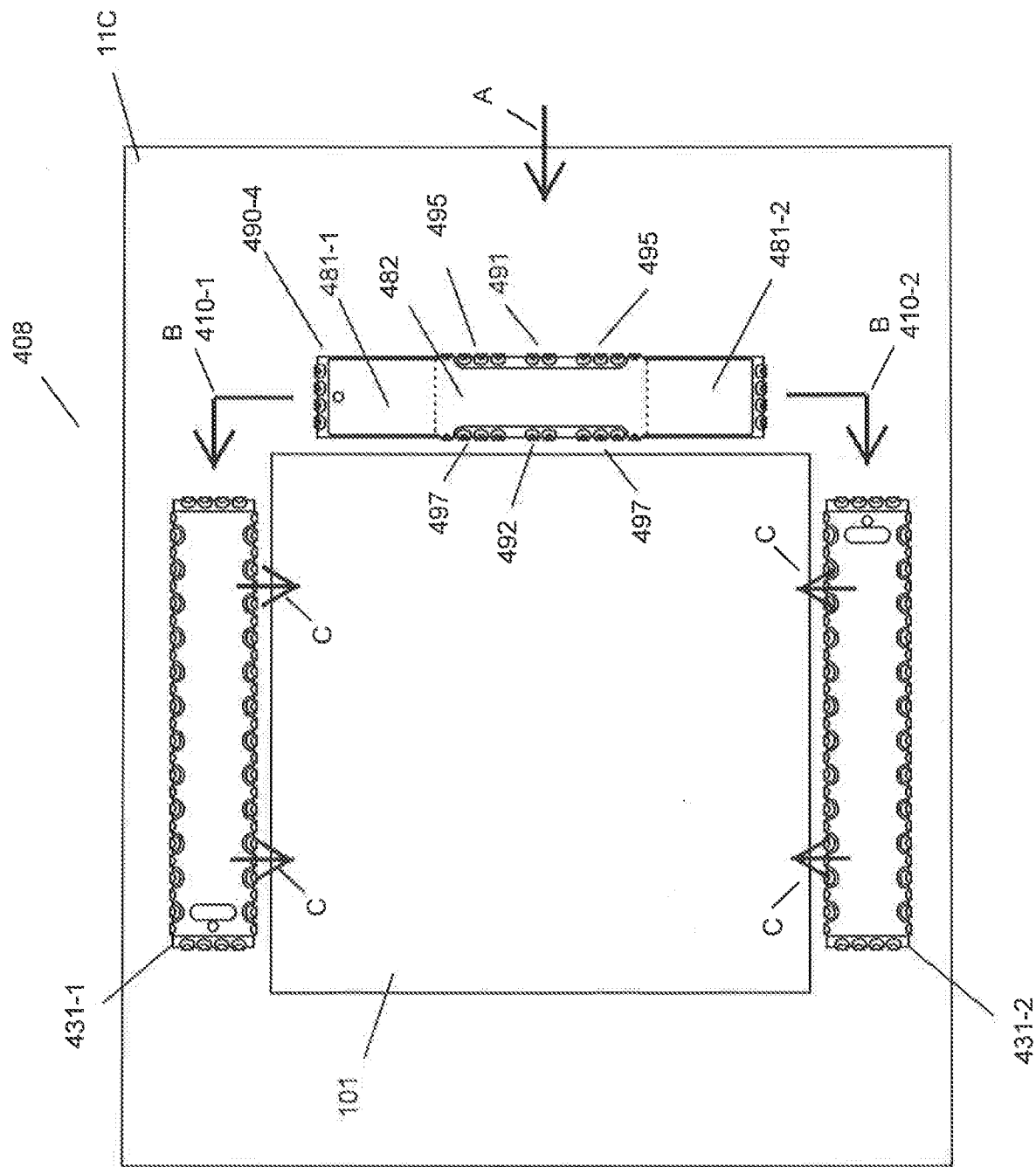


FIG. 27

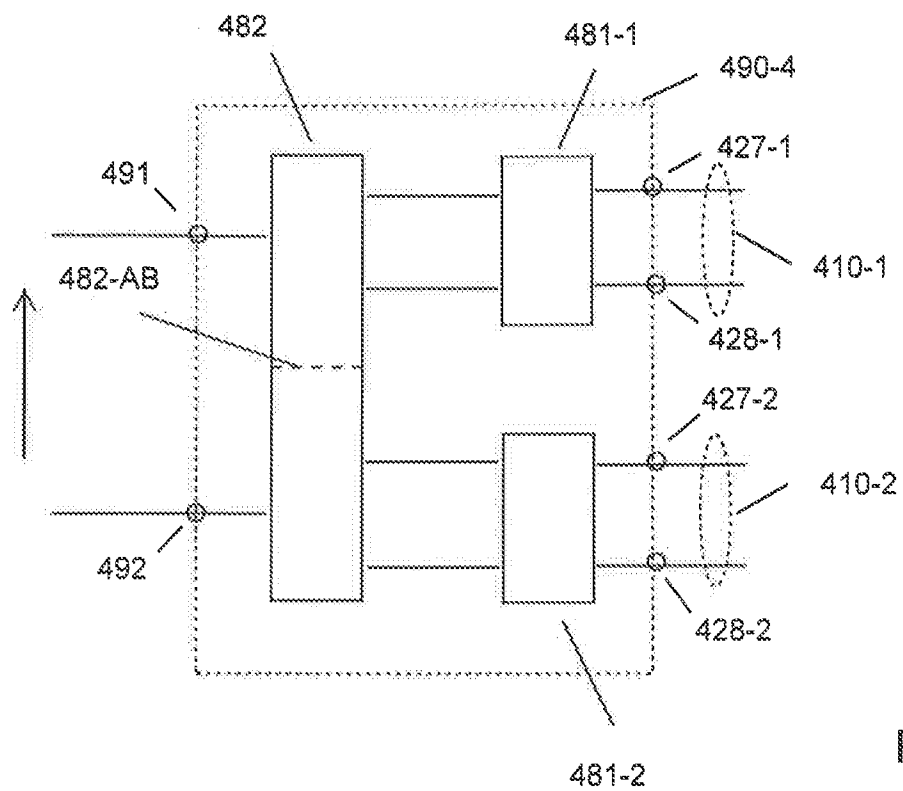


FIG. 28



Espacenet

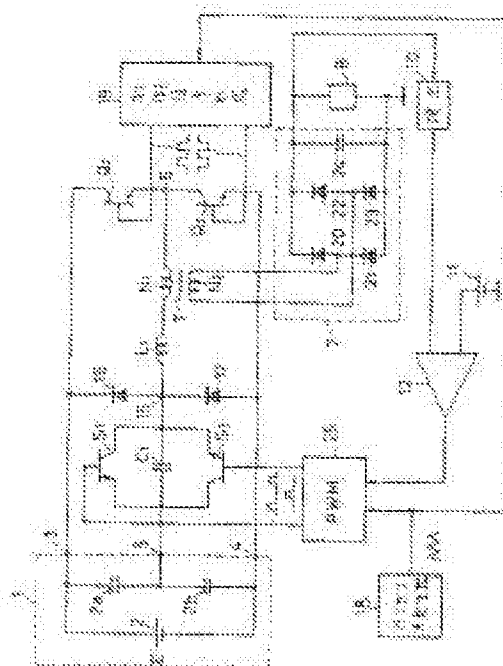
Bibliographic data: JPH02280666 (A) — 1990-11-16

RESONANT SWITCHING POWER SOURCE

Inventor(s): TANIYAMA KOREYASU ± (TANIYAMA KOREYASU)
Applicant(s): SANKEN ELECTRIC CO LTD ± (SANKEN ELECTRIC CO LTD)
Classification: - **international:** *H02M3/28; H02M3/335*; (IPC1-7): H02M3/28;
H02M3/335
- **cooperative:**
Application number: JP19890100847 19890420
Priority number(s): JP19890100847 19890420
Also published as: JPH0746903 (B2)

Abstract of JPH02280666 (A)

PURPOSE: To reduce power loss by connecting a switching element in parallel with a resonant capacitor. **CONSTITUTION:** A resonant DC-DC converter comprising a DC power source 1, first and second transistor switching elements Q1, Q2, a series resonant inductance element L1, a transformer T, and the like, feeds power through an output smoothing/rectifying circuit 7 to a load 8. First and second switching elements S1, S2 for controlling charge/discharge operation are connected in parallel with the resonant capacitor C1, and a PWM pulse forming circuit 25 is provided. By such arrangement, power to be fed to the resonant circuit can be controlled through control of ON time width of the switching elements S1, S2.



⑤Int.Cl.⁵H 02 M 3/335
3/28

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Q 7829-5H
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庁内整理番号

⑬公開 平成2年(1990)11月16日

審査請求 未請求 請求項の数 7 (全11頁)

⑭発明の名称 共振型スイッチング電源回路

⑰特 願 平1-100847

⑱出 願 平1(1989)4月20日

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㉑代 理 人 弁理士 高野 則次

明 細 書

1. 発明の名称

共振型スイッチング電源回路

2. 特許請求の範囲

〔1〕 第1の端子と第2の端子と前記第1及び第2の端子の midpoint 電位を与える第3の端子とを有する直流電源回路と、

前記第1の端子と前記第2の端子との間に互いに直列になるように接続された第1及び第2のスイッチング素子と、

前記第1及び第2のスイッチング素子を交互にオン・オフ制御するための制御回路と、

一端が前記第3の端子に接続されている共振用コンデンサと、

前記共振用コンデンサの他端と前記第1及び第2のスイッチング素子の接続 midpoint との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか又は前記インダクタンス素子に電磁結合されている負荷回路と

から成る共振型スイッチング電源回路において、

前記コンデンサに並列に前記コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路。

〔2〕 更に、前記コンデンサの他端と前記電源回路の第1及び第2の端子との間に前記コンデンサの電圧をクランプするためのクランプ用素子をそれぞれ接続したことを特徴とする請求項1記載の共振型スイッチング電源回路。

〔3〕 第1及び第2の端子を有する直流電源回路と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のスイッチング素子と、

前記第1及び第2の端子の間に互いに直列になるように接続された第3及び第4のスイッチング素子と、

前記第1及び第4のスイッチング素子と前記第2及び第3のスイッチング素子とを交互にオン・オフ制御するための制御回路と、

一端が前記第3及び第4のスイッチング素子の接続中点に接続された共振用コンデンサと、

前記共振用コンデンサの他端と前記第1及び第2のスイッチング素子の接続中点との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか、又は前記インダクタンス素子に電磁結合されている負荷回路と

から成る共振型スイッチング電源回路において、

前記コンデンサに並列に前記コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路。

〔4〕 更に、前記コンデンサの他端と前記電源回路の第1及び第2の端子との間に前記コンデンサの電圧をクランプするためのクランプ用素子をそれぞれ接続したことを特徴とする請求項3記載の共振型スイッチング電源回路。

〔5〕 第1及び第2の端子を有する直流電源回路と、

前記第1及び第2の端子の間に互いに直列にな

路と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のスイッチング素子と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のダイオードと、

前記第1のスイッチング素子と前記第2のスイッチング素子とを交互にオン・オフ制御するための制御回路と、

前記第1及び第2のダイオードに並列に接続されている第1及び第2のコンデンサと、

前記第1及び第2のスイッチング素子の接続中点と前記第1及び第2のダイオードの接続中点との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか又は前記インダクタンス素子に電磁結合されている負荷回路と、

前記第1及び第2のコンデンサに並列に接続された前記コンデンサの充電及び／又は放電を制御するための第1及び第2のコンデンサ制御素子と

るように接続された第1及び第2のスイッチング素子と、

前記第1及び第2のスイッチング素子を交互にオン・オフするための制御回路と、

前記第1及び第2の端子の間に互いに直列になるように接続された第1及び第2のダイオードと、

前記第2のダイオードに並列に接続された共振用コンデンサと、

前記第1及び第2のスイッチング素子の接続中点と前記第1及び第2のダイオードの接続中点との間に接続された共振用インダクタンス素子と、

前記インダクタンス素子に直列に接続されているか、又は前記インダクタンス素子に電磁結合されている負荷回路と

から成る共振型スイッチング電源回路において、

前記共振用コンデンサに並列に前記共振用コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路。

〔6〕 第1及び第2の端子を有する直流電源回

を具備した共振型スイッチング電源回路。

〔7〕 第1の端子と第2の端子と前記第1及び第2の端子の中点電位を与える第3の端子とを有する直流電源回路と、

センタタップを有する1次巻線を備えているトランスと、

前記電源回路の前記第3の端子と前記1次巻線のセンタタップとの間に接続されたインダクタンス素子とコンデンサとから成る直列共振回路と、

前記第1の端子と前記1次巻線の一端との間に接続された第1のスイッチング素子と、

前記第2の端子と前記1次巻線の他端との間に接続された第2のスイッチング素子と、

前記コンデンサに並列に接続された前記コンデンサの充電及び／又は放電を制御するための素子と

から成る共振型スイッチング電源回路。

3. 発明の詳細な説明

〔産業上の利用分野〕

本発明は、直流安定化電源等に使用される共振

振型スイッチング電源回路に関するものである。

〔従来の技術〕

現在、電子計算機、通信機器等に使用される電子通信用電源は、小型、軽量、高効率の利点を持つPWM方式のスイッチングレギュレータが多く使用されている。しかし、スイッチング周波数が20kHz以下の場合には、大きな問題が生じないが、100kHz以上になると、電力伝送用トランスの洩れインダクタンスや、スイッチング素子の浮遊容量の影響により、スイッチング素子のオン・オフ転換時におけるスイッチング素子の両端の電圧波形と電流波形とが重なり合う割合が大きくなり、この電圧、電流波形が互いに重なり合う部分がスイッチング損失となり、発振周波数が高くなるに従って損失が増加する。スイッチング損失が増加すると、電力変換効率が低下すると共に、発熱する。このため大きな放熱器を必要とし、電源装置の小型化の妨げとなる。

上述の如き問題点を解決するための方式として、第11図に示す共振型スイッチング電源回路があ

る。この共振型スイッチング電源回路の直流電源1は第1及び第2の電源2a、2bの直列回路から成り、第1、第2及び第3の端子3、4、5を有するセンタタップ型に形成されている。第1の端子3と第2の端子4との間にはトランジスタから成る第1及び第2のスイッチング素子 Q_1 、 Q_2 の直列回路が接続されている。交互にオン・オフ制御される第1及び第2のスイッチング素子 Q_1 、 Q_2 にはバイパス用ダイオード D_1 、 D_2 がそれぞれ逆並列に接続されている。

電源1の第3の端子5と第1及び第2のスイッチング素子 Q_1 、 Q_2 の接続 midpoint 6との間には共振用コンデンサ C_1 と直列共振用インダクタンス素子(リアクトル) L_1 と負荷回路を構成するためのトランスTの1次巻線 N_1 との直列回路が接続されている。即ち、コンデンサ C_1 の一端が第3の端子5に接続され、コンデンサ C_1 の他端がインダクタンス素子 L_1 と1次巻線 N_1 とを介してスイッチング素子 Q_1 、 Q_2 の接続 midpoint 6に接続されている。

トランスTは2次巻線 N_2 、 N_3 を有し、このセン

タタップ構成の2次巻線 N_2 、 N_3 には、ダイオード D_3 、 D_4 と平滑用コンデンサ C_2 とから成る出力整流平滑回路7を介して負荷8が接続されている。

第1及び第2のスイッチング素子 Q_1 、 Q_2 をオン・オフ制御するための制御回路9は、出力平滑整流回路7の出力電圧を検出する電圧検出回路10と、この電圧検出回路10から得られる検出電圧と基準電圧源11の基準電圧との差に対応する出力を得るための誤差増幅器12と、この誤差増幅器12の出力電圧に対応した周波数信号を出力するVCO(電圧制御発振器)13と、このVCO13の出力周波数でスイッチング素子 Q_1 、 Q_2 を交互にオン・オフ制御するためのスイッチ制御信号形成回路14とから成る。スイッチ制御信号形成回路14はVCOの出力に基づいて第1のスイッチング素子 Q_1 の制御信号を形成し、又この反転信号で第2のスイッチング素子 Q_2 の制御を形成し、各スイッチング素子 Q_1 、 Q_2 即ちトランジスタのベースに供給するものである。

スイッチング素子 Q_1 、 Q_2 は、直列共振回路の共

振周波数よりも低い周波数で第12図(A)(B)に示すように交互にオン・オフ制御される。第1のスイッチング素子 Q_1 がオン、第2のスイッチング素子 Q_2 がオフの期間(第7図の $t_1 \sim t_2$)では、まず、直列共振に基づく電流Iが、コンデンサ C_1 、電源2a、第1のスイッチング素子 Q_1 、1次巻線 N_1 、インダクタンス素子 L_1 から成る回路で流れる。コンデンサ C_1 の電圧は、第12図(C)に示すように t_1 時点でほぼ-Eであるが、零に向つて減少し、しかる後ほぼ+Eになる。コンデンサ C_1 及びインダクタンス素子 L_1 を通つて流れる電流Iは、第12図(D)に示すようにコンデンサ電圧 V_C が零になる時点で正の最大値になり、その後減少する。 $t_2 \sim t_3$ で示すコンデンサ C_1 の放電期間においては、コンデンサ C_1 、インダクタンス素子 L_1 、1次巻線 N_1 、ダイオード D_1 、電源2aから成る回路で逆方向電流が流れる。第2のスイッチング素子 Q_2 がオンになる $t_3 \sim t_4$ 期間においても $t_1 \sim t_2$ 期間と同様な動作が生じる。

〔発明が解決しようとする課題〕

ところで、共振型スイッチング電源回路の出力電圧又は電力の制御は、周波数制御方式で行われる。即ち、第1及び第2のスイッチング素子 Q_1 、 Q_2 のオン・オフ制御の周波数を変えることによつて負荷8の電圧を変える。従つて、最低周波数で回路定数を決めなければならず、損失を大幅に低減することが困難になつた。また、ダイオード D_1 が $t_2 \sim t_3$ でオンしている状態において、第2のスイッチング素子 Q_2 がオンになると、ダイオード D_1 の蓄積キャリアに基づくリカバリー電流（短絡電流）がスイッチング素子 Q_2 に流れ込む。このリカバリー電流（短絡電流）が第2のスイッチング素子 Q_2 のコレクタ・エミッタ間電圧がEボルトから零ボルトに変化する期間に流れるので、損失となり、効率が低下する。ダイオード D_1 、 D_2 を省くとリカバリー電流（短絡電流）による損失が生じなくなるが、コンデンサ C_1 がEよりも高くなる。即ち、電源投入時、負荷急変時等における第1及び第2のスイッチング素子 Q_1 、 Q_2 に流れる共振電流のアンバランスによつてコンデンサ C_1 の電圧がE

続中点との間に接続された共振用インダクタンス素子と、前記インダクタンス素子に直列に接続されているか又は前記インダクタンス素子に電磁結合されている負荷回路とから成る共振型スイッチング電源回路において、前記コンデンサに並列に前記コンデンサの充電及び／又は放電を制御するための素子を接続したことを特徴とする共振型スイッチング電源回路に係わるものである。

請求項2に示すようにクランプ用ダイオードを接続することが望ましい。

請求項3に示すように第1～第4のスイッチング素子をブリッジ型に接続する場合にも請求項1の技術思想を適用することができる。

請求項4に示すように、請求項3の回路にクランプ用ダイオードを付加することが望ましい。

請求項5に示すように1つのダイオードに並列にコンデンサを接続する変形ハーフブリッジ型の回路にも請求項1の技術思想を適用することができる。

請求項6に示すように2つのダイオードにコン

よりも高くなり、スイッチング素子 Q_1 、 Q_2 及びダイオード D_1 、 D_2 に耐圧の高いものを使用することが必要になる。高耐圧のスイッチング素子 Q_1 、 Q_2 及びダイオード D_1 、 D_2 を使用すると、オン時における抵抗（電圧降下）が大きくなり、損失も多くなるので、共振型に基づく損失低減効果が期待出来なくなる。

そこで、本発明の目的は効率の良い共振型スイッチング電源回路を提供することにある。

〔課題を解決するための手段〕

上記目的を達成するための本発明は、第1の端子と第2の端子と前記第1及び第2の端子の中点電位を与える第3の端子とを有する直流電源回路と、前記第1の端子と前記第2の端子との間に互いに直列になるように接続された第1及び第2のスイッチング素子と、前記第1及び第2のスイッチング素子を交互にオン・オフ制御するための制御回路と、一端が前記第3の端子に接続されている共振用コンデンサと、前記共振用コンデンサの他端と前記第1及び第2のスイッチング素子の接

デンサをそれぞれ並列接続する回路にも請求項1の技術思想を適用することができる。

請求項7に示すように並列型のスイッチング回路にも請求項1の技術思想を適用することができる。

〔作用〕

各請求項に従う発明において、コンデンサを短絡させるためのスイッチング素子のオン時間幅を制御することによつて1周期中に電源から共振回路に供給する電力量が制御される。これにより、出力電力、又は電圧又は電流の制御をスイッチング周波数固定状態で行うことが可能になる。

請求項2、4、5、6に従う発明ではコンデンサの電圧の上昇をダイオードで抑制することができる。

〔第1の実施例〕

次に、第1図及び第2図を参照して本発明の第1の実施例に係わる共振型スイッチング電源回路即ち共振型DC-DCコンバータを説明する。但し、第1図において第11図と実質的に同一の部分には同一の符号を付してその説明を省略する。

第1図の本実施例の共振型スイッチング電源回路は、コンデンサ C_1 に並列接続された第1及び第2のコンデンサ制御用スイッチング素子 S_1 、 S_2 を有し、更に、コンデンサ C_1 の右端15と電源1の第1の端子3との間及びコンデンサ C_1 の右端15と電源1の第2の端子4との間にそれぞれ接続された第1及び第2のクランプ用ダイオード16、17を有する。第1図の第1及び第2のスイッチング素子 Q_1 、 Q_2 は出力電圧の変化に無関係に一定の周波数で交互にオン・オフ制御される。

18はのこぎり波(三角波)発生器であつて、この実施例では第1及び第2のスイッチング素子 Q_1 、 Q_2 のオン・オフ繰返し周波数の2倍の周波数を有するのこぎり波を発生する。19はスイッチ制御信号形成回路(スイッチ制御回路)であつて、のこぎり波発生回路18から供給されたのこぎり波に同期した第2図(A)(B)に示す固定の周波数の方形波の制御信号を形成し、第1及び第2のスイッチング素子 Q_1 、 Q_2 に供給する。なお、第2図(A)(B)の制御信号の周波数は C_1L_1 共振回路の共振周波数

は第1及び第2のコンデンサ制御スイッチング素子 S_1 、 S_2 に供給される。なお、のこぎり波発生回路18の周波数が第1及び第2のスイッチング素子 Q_1 、 Q_2 のスイッチング周波数と同一になるように構成することも可能である。

第1図の回路の動作を第2図を参照して説明すると、第2図(A)に示すように t_0 時点で第1のスイッチング素子 Q_1 がオンになつた時には従来の回路と同様に共振による電流 I が第2図(B)に示すように流れ始める。これにより、コンデンサ C_1 の電圧が第2図(C)に示すように $-E/2$ から0に向つて変化する。なお、コンデンサ C_1 の充電電圧の変化範囲は、クランプ用ダイオード16、17の働きで $-E/2$ から $+E/2$ までとなる。 t_0 時点で第1のコンデンサ制御用スイッチング素子 S_1 が第2図(C)のパルスでオン制御されているが、コンデンサ C_1 の左端が正になるように充電されているために、第1のコンデンサ制御用スイッチング素子 S_1 が逆バイアス状態にあり、 $t_0 \sim t_1$ 期間はオフに保たれる。電源回路1の第1の端子3、第1のスイッ

よりも低い。本実施例では出力電圧の制御を第1及び第2のスイッチング素子 Q_1 、 Q_2 のオン・オフ繰返し周波数の制御によつて行わずに、共振コンデンサ C_1 の充放電制御によつて行う。

出力整流平滑回路7は4つのダイオード20、21、22、23と、平滑用コンデンサ24とから成り、トランスTの2次巻線 N_2 と負荷8との間に接続されている。出力整流平滑回路7の出力電圧は検出回路10で検出され、誤差増幅器12の入力となり、基準電圧源11の基準電圧と比較される。PWM(パルス幅変調)パルス形成回路25は、のこぎり波発生回路18と誤差増幅器12とに接続され、のこぎり波と誤差出力とを比較してPWMパルスを出力する電圧比較器を含んでおり、第2図(C)(D)に示すPWMパルスを発生する。なお、のこぎり波は第2図(A)～(D)に示すパルスの繰返し周波数の2倍の周波数を有するので、比較器の出力を交互に分配することによつて第2図(C)(D)に示す180度位相差の2つのPWMパルス列を得る。PWMパルス形成回路25の出力ライン

ング素子 Q_1 、トランスT、インダクタンス素子 L_1 、コンデンサ C_1 の回路でコンデンサ C_1 の逆充電が進み、 t_2 時点で充電電圧が零になると、第1のコンデンサ制御用スイッチング素子 S_1 の逆バイアスが解除され、これがオン状態になる。この結果、スイッチング素子 Q_1 、 Q_2 、 S_1 、 S_2 の電圧降下を無視して考えると、コンデンサ C_1 の充電電圧は第2図(C)のパルスが終了する t_2 時点まで零に保たれる。 $t_1 \sim t_2$ 期間にはインダクタンス素子 L_1 に制限された電流が第2図(B)に示すように流れ続ける。 t_2 時点で第1のコンデンサ制御用スイッチング素子 S_1 がオフに転換すると、 C_1L_1 共振回路に基づく電流が流れ始め、コンデンサ C_1 の電圧は第2図(D)に示すように正方向に上昇し、電流 I は徐々に減少する。 t_3 時点でコンデンサ C_1 の電圧が $E/2$ よりも僅かに高くなるとクランプ用ダイオード16がオンになり、コンデンサ C_1 の電圧はほぼ $E/2$ にクランプされる。 $t_3 \sim t_4$ 期間ではインダクタンス素子 L_1 のエネルギーの放出で電流が流れるが、第2図(D)に示すように徐々に減少し、 t_4 で零になる。第

1 図では第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 にダイオードが逆並列接続されていないが、第 1 図に示すようにダイオード D_1 、 D_2 が接続されている場合又は FET のためにダイオードが内蔵されている場合であっても、第 1 図ではコンデンサ C_1 の電圧が $E/2$ にクランプされているために、 $t_3 \sim t_5$ の期間に第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 に並列のダイオードがオンにならない。従つて、第 2 図の t_5 時点での第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 の切換時にダイオードの蓄積キャリアに基づく過大電流が流れない。

第 2 図の $t_5 \sim t_6$ 期間には第 2 のスイッチング素子 Q_2 がオン制御され、 $t_6 \sim t_5$ 期間と同様な動作が生じる。勿論この期間の電流 I 、電圧 V_C の向きは $t_0 \sim t_5$ 期間と逆になる。

出力電圧が変動すると、誤差増幅器 12 の出力が変化し、第 2 図(C)(D)のパルスの後縁が変化し、第 2 図の $t_1 \sim t_2$ 期間も変化する。この結果、第 1 のスイッチング素子 Q_1 の固定されたオン期間 $t_0 \sim t_5$ 内における電源回路 1 からのエネルギー供給時

質的に同一部分には同一の符号を付してその説明を省略する。この実施例では第 1 ～第 4 のスイッチング素子 $Q_1 \sim Q_4$ でスイッチング回路がフルブリッジ型に構成されている。第 3 及び第 4 のスイッチング素子 Q_3 、 Q_4 は第 1 図のコンデンサ 2 a、2 b の位置に接続されている。第 1 ～第 4 のスイッチング素子 $Q_1 \sim Q_4$ は第 4 図(A)～(D)に示すように制御され、第 1 及び第 2 のコンデンサ制御用スイッチング素子 S_1 、 S_2 は第 4 図(E)(F)に示すように制御される。

第 4 図の $t_0 \sim t_1$ 期間中における共振回路の電流 I は、 Q_1 、 N_1 、 L_1 、 C_1 、 Q_4 から成る回路で流れ、 $t_1 \sim t_2$ 期間中における共振回路の電流 I は、 Q_3 、 C_1 、 L_1 、 N_1 、 Q_2 から成る回路で流れる。共振による電流 I 及び電圧 V_C の変化の原理は第 1 図と同一である。また、出力電圧調整方式の原理も第 1 図と同一であり、第 4 図(G)(H)に示す第 1 及び第 2 のコンデンサ制御用スイッチング素子 S_1 、 S_2 のオン時間幅を変えることによつてコンデンサ C_1 の電圧が零になる時間幅が変化し、出力電圧が変化

間及び量に変化し、出力整流平滑回路 7 に与えられる電力量も変化し、出力電圧が変化する。

上述のように、本方式は周波数一定制御方式であるので、トランス T 等を最適設計することが可能になり、効率の高い共振型スイッチング電源回路を提供することができる。

また、クランプ用ダイオード 16、17 の働きにより、コンデンサ C_1 の電圧が $-E/2 \sim +E/2$ に制限されるため、第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 の耐圧を下げる事が可能になり、これに基づく電力損失低減効果も生じる。また、第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 に逆並列にダイオードを接続しなくても、コンデンサ C_1 の電圧上昇を抑えることが可能になり、且つダイオードの蓄積キャリアによる過大電流の問題も前述したように解決される。

〔第 2 の実施例〕

次に、第 3 図及び第 4 図を参照して本発明の第 2 の実施例に係わる共振型スイッチング電源回路を説明する。但し、第 3 図において、第 1 図と実

する。従つて、本実施例によつても、第 1 の実施例と同一の作用効果を得ることができる。

〔第 3 の実施例〕

次に、第 5 図及び第 6 図を参照して本発明の第 3 の実施例の共振型スイッチング電源回路を説明する。但し、第 5 図において第 1 図及び第 3 図と実質的に同一の部分には同一符号を付してその説明を省略する。

この実施例では第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 の接続中点 6 と第 1 及び第 2 のダイオード 16、17 の接続中点 15 との間にトランス T とインダクタンス素子 L_1 とが接続され、第 2 のダイオード 17 に並列にコンデンサ C_1 とこの制御用スイッチング素子 S_1 とが接続されている。

第 1 及び第 2 のスイッチング素子 Q_1 、 Q_2 は第 6 図(A)(B)に示すように第 1 図の回路と同様に一定の周波数で交互にオン・オフ制御される。コンデンサ制御用スイッチング素子 S_1 は第 6 図(C)に示す PWM パルスで制御される。第 6 図の t_0 時点で $I = 0$ 、 $V_C = 0$ であるとすれば、 $t_0 \sim t_1$ 期間では Q_1 、

Q_3 がオン、 Q_2 がオフであるので、電流 I は零から直線上に上昇する。なお、この電流 I は Q_1 、 N_1 、 L_1 、 S_1 の回路で流れる。

$t_1 \sim t_2$ 期間では、 Q_1 がオン、 Q_2 、 Q_3 がオフであり、 Q_1 、 N_1 、 L_1 、 C_1 の回路で電流 I が流れる。

$t_2 \sim t_3$ 期間では、コンデンサ C_1 の充電電圧 V_C が電源電圧 E よりも僅かに高くなっているために、第1のダイオード16がオンになり、コンデンサ C_1 の電圧 V_C がほぼ電源電圧 E にクランプされる。インダクタンス素子 L_1 の蓄積エネルギーの放出は t_3 時点よりも前に終了し、電流 I は t_3 で零である。従つて、第2のスイッチング素子 Q_2 が電流零の状態でもオンに転換する。

$t_3 \sim t_4$ 期間では、 Q_2 がオン、 Q_1 、 Q_3 がオフであるので、 C_1 、 L_1 、 N_1 、 Q_2 の閉回路が形成され、コンデンサ C_1 の放電が起きる。 t_4 時点になつてコンデンサ C_1 の電圧が零になり、その後逆方向に充電されるとダイオード17がオンになり、コンデンサ C_1 の電圧 V_C はほぼ零になる。コンデンサ制御用スイッチング素子 S_1 を駆動するための第6図

のコンデンサ制御用スイッチング素子 S_1 、 S_2 は第8図(C)(D)のPWMパルスで制御されるため、出力電圧が第1、第2及び第3の実施例と同様に制御され、同様な作用効果が得られる。

〔変形例〕

本発明は上述の実施例に限定されるものでなく、例えば次の変形が可能なるものである。

(1) 第9図に示すようにトランス T の1次巻線にセンタタップを設け、このセンタタップと第3の端子5との間にインダクタンス素子 L_1 とコンデンサ C_1 との共振回路を接続し、第1の端子3とセンタタップとの間に第1のスイッチング素子 Q_1 を介して1次巻線の半分 N_{1a} を接続し、第2の端子4とセンタタップとの間に第2のスイッチング素子 Q_2 を介して1次巻線の残りの半分 N_{1b} を接続し、第1及び第2のスイッチング素子 Q_1 、 Q_2 と第1及び第2のコンデンサ制御用スイッチング素子 S_1 、 S_2 とを第1図と同様に制御してもよい。

(2) コンデンサ C_1 を制御するためのスイッチング素子 S_1 、 S_2 を第10図に示すように2つのFE

(C) のパルスの後縁を制御すると、1サイクル ($t_0 \sim t_5$) 中に電源回路1から供給される電力量が変化し、出力電圧も変化する。従つて、第1及び第2の実施例と同一の作用効果を得ることができる。
〔第4の実施例〕

次に、第7図及び第8図を参照して第4の実施例の共振型スイッチング電源回路を説明する。但し、第7図において、第1図、第3図、第5図と実質的に同一の部分には同一の符号を付してその説明を省略する。第7図の回路は第5図のダイオード16に対して並列にコンデンサ C_2 とコンデンサ制御用スイッチング素子 S_2 とを接続した回路と同一である。

第8図の $t_0 \sim t_3$ 期間の動作は第6図の $t_0 \sim t_3$ 期間と同一である。 $t_3 \sim t_6$ 期間は、 $t_0 \sim t_3$ 期間の反対極性の動作であり、 $t_3 \sim t_4$ で Q_1 、 S_1 がオフ、 Q_2 及び S_2 がオンになり、 $t_4 \sim t_6$ 期間で Q_1 、 S_1 、 S_2 がオフ、 Q_2 がオンになり、電流 I は第8図(D)に示すように対称的に変化し、コンデンサ C_1 の電圧は第8図(E)に示すように変化する。第1及び第2

T31、32と2つのダイオード33、34とで構成してもよい。

(3) 第1～第4のスイッチング素子 $Q_1 \sim Q_4$ を FETとしてもよい。

(4) 第1図、及び第3図の第1及び第2のスイッチング素子 Q_1 、 Q_2 に逆並列にダイオードを接続してもよい。

(5) 第1図の電源回路1を2個の $E/2$ の電源によつて構成することができる。

(6) S_1 、 S_2 を制御するためのPWMパルスを例えば第2図の $t_1 \sim t_2$ 期間に対応するパルス幅を有するものにしてもよい。

(7) トランス T が大きなインダクタンスを有する場合にはこれを共振用のインダクタンスに兼用し、インダクタンス素子 L_1 を省いてもよい。

(8) トランス T の位置に負荷8を直接に接続し、負荷8に交流を供給する場合にも適用可能である。
〔発明の効果〕

上述のようにいずれの請求項の発明によつてもスイッチング周波数を固定した状態で電力又は電

圧又は電流を調整することが可能になる。また、請求項2、4、5、6によれば、コンデンサの電圧の上昇を抑えることができると共に、スイッチング素子のオン・オフ切換時の電力損失の低減を図ることができる。

4. 図面の簡単な説明

第1図は本発明の第1の実施例の共振型スイッチング電源回路を示す回路図、

第2図は第1図の各部の状態を原理的に示す波形図、

第3図は第2の実施例の共振型スイッチング電源回路を示す回路図、

第4図は第3図の各部の状態を原理的に示す波形図、

第5図は第3の実施例の共振型スイッチング電源回路を示す回路図、

第6図は第5図の各部の状態を原理的に示す波形図、

第7図は第4の実施例の共振型スイッチング電源回路を示す波形図、

第8図は第7図の各部の状態を原理的に示す波形図、

第9図は変形例の共振型スイッチング電源回路を示す回路図、

第10図は変形例のコンデンサ制御用スイッチ回路を示す回路図、

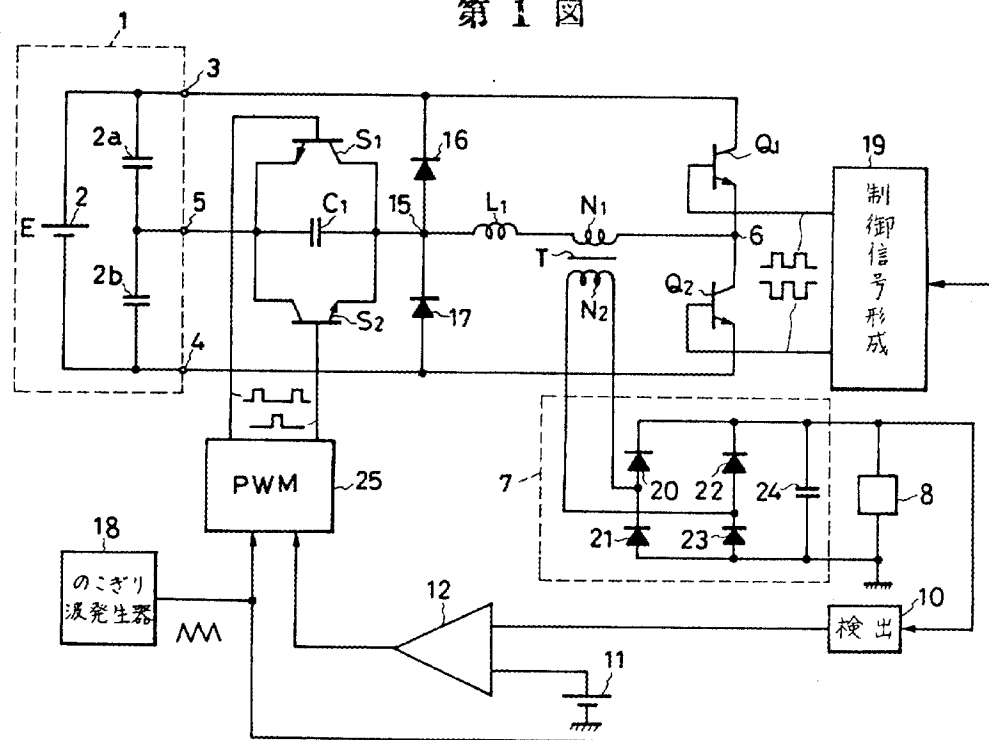
第11図は従来の共振型スイッチング電源回路を示す回路図、

第12図は第11図の各部の状態を原理的に示す波形図である。

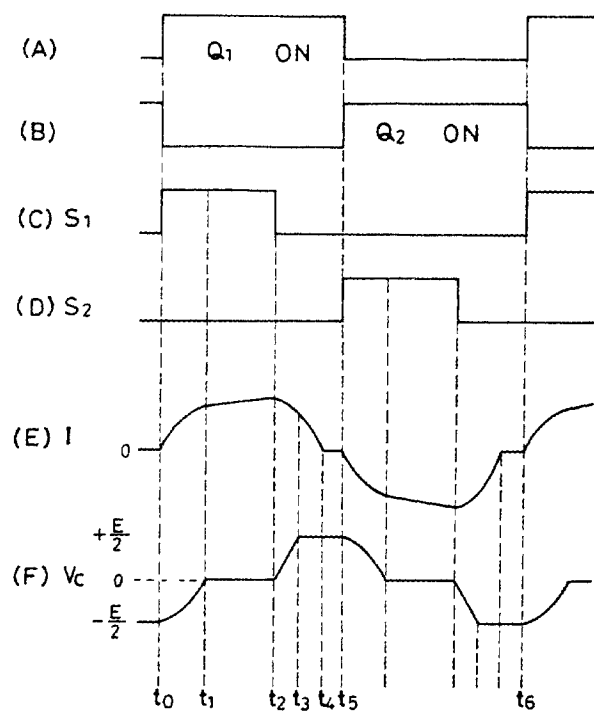
1…直流電源回路、3…第1の端子、4…第2の端子、 Q_1 …第1のスイッチング素子、 Q_2 …第2のスイッチング素子、 C_1 …共振用コンデンサ、 L_1 …直列共振用インダクタンス素子、 S_1 、 S_2 …コンデンサ制御用スイッチング素子。

代理人 高野 則 次

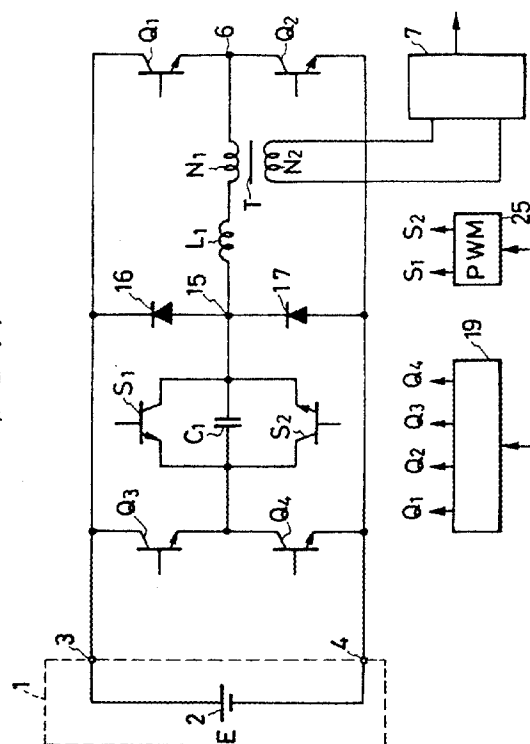
第1図



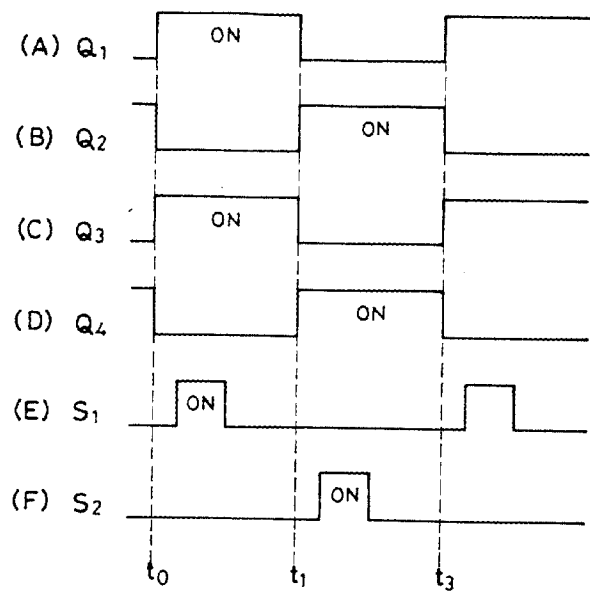
第 2 図



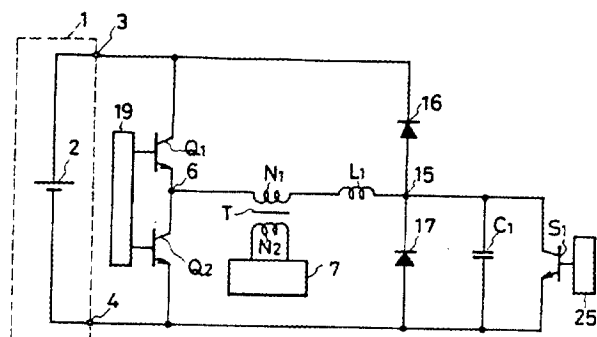
第 3 図



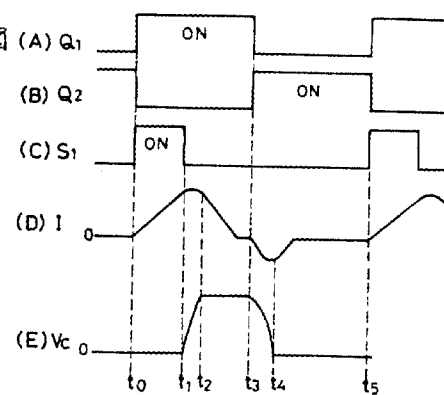
第 4 図

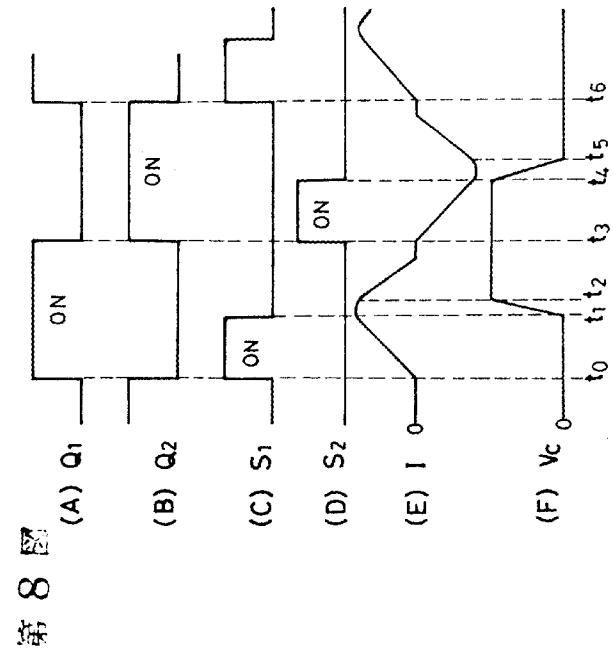
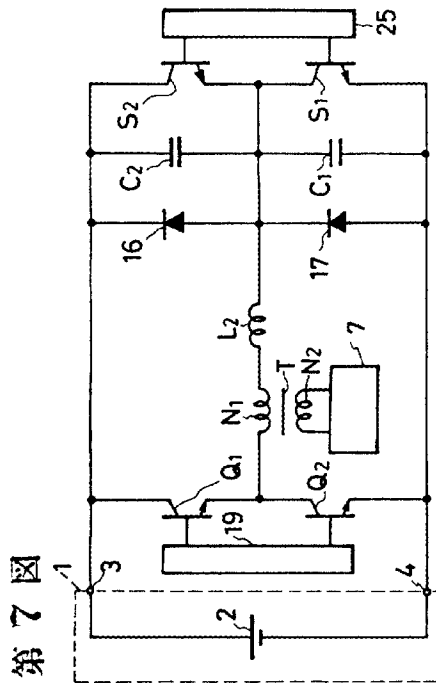


第 5 図

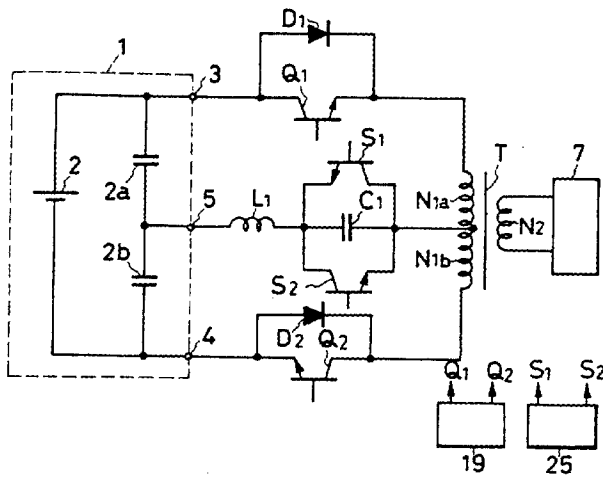


第 6 図

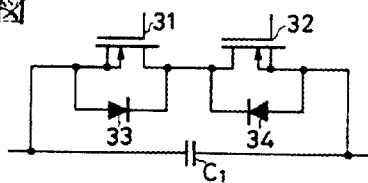




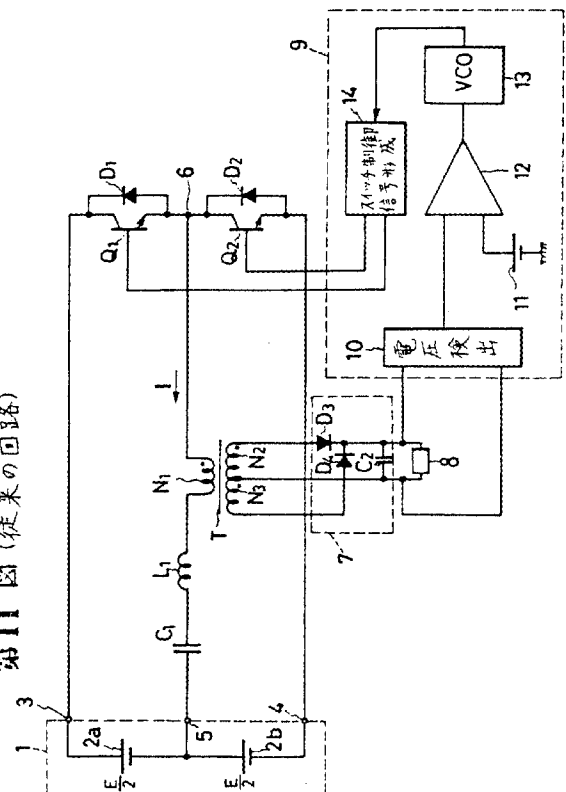
第9図



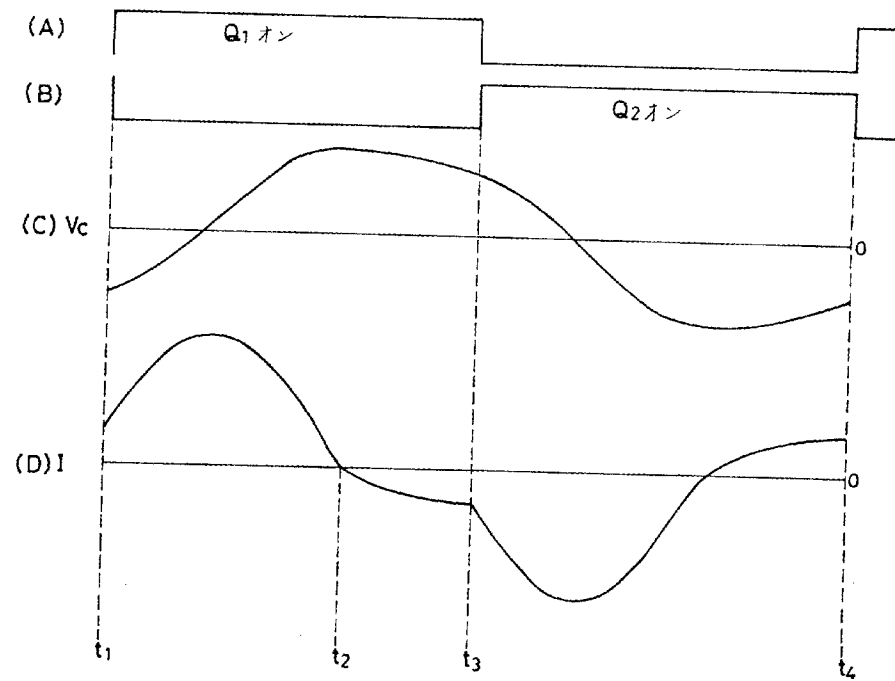
第10図



第11図 (従来の回路)



第 12 図





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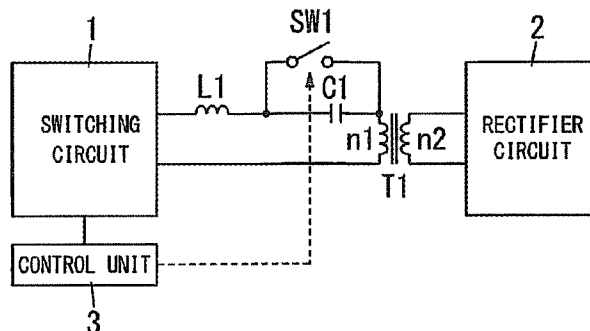
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(54) **ELECTRIC POWER CONVERSION DEVICE**

(57) A switching circuit is configured to switch on/off a current that flows through a resonance circuit constituted by a primary winding of a transformer, a capacitance element, and an inductor that are connected in series. Electric power that is induced in a secondary winding of the transformer is rectified by a rectifier circuit. A switch is connected in parallel with the capacitance element. A control unit is configured to control the switching

circuit and the switch. The control unit is configured to select a first operation in which an operation frequency of the switching circuit is controlled by switching off the switch, and select a second operation in which a period during which a current flows from the switching circuit to the primary winding of the transformer is controlled by switching on the switch.

FIG. 1



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Description

TECHNICAL FIELD

[0001] The present invention relates to an electric power conversion device that performs electric power conversion using a switching element and transmits electric power between an input and an output via a transformer.

BACKGROUND ART

[0002] Conventionally, electric power conversion devices are known that perform electric power conversion using a switching element and transmit electric power between an input and an output via a transformer. For example, a bidirectional DC/DC converter that transmits DC power bi-directionally is disclosed in JP 2011-234541 A (hereinafter referred to as "Document 1") as this type of electric power conversion device. The electric power conversion device described in Document 1 is provided with a primary circuit and a secondary circuit that sandwich a transformer.

[0003] The primary circuit and the secondary circuit each include a bridge circuit configured by a reverse conducting type semiconductor switch (switching element). The electric power conversion device can transmit desired DC power bi-directionally by controlling duty cycles (on duties) of the switching element of the primary circuit and the switching element of the secondary circuit.

[0004] Incidentally, although the electric power conversion device described in Document 1 can transmit desired DC power between the primary circuit and the secondary circuit by controlling the duty cycles of the switching elements, a voltage ratio between an input voltage and an output voltage depends on the winding ratio of the transformer. That is, the electric power conversion device described in Document 1 cannot output voltage by boosting the input voltage and only performs current adjustment, since bridge circuits are used in the primary circuit and the secondary circuit, and only the duty cycles are controlled.

[0005] On the other hand, an electric power conversion device that is used for utilizing electric power of a storage battery that is mounted in an automobile having an electric motor serving as a power source, a storage battery used by being attached to a building, or the like is required to have a function of adjusting a relationship between an input voltage and an output voltage over a wide range.

DISCLOSURE OF INVENTION

[0006] An object of the present invention is to provide an electric power conversion device configured to perform electric power conversion by switching and transmit electric power between an input and an output via a transformer, and is capable of adjusting a relationship between an input voltage and an output voltage over a wide range according to the use.

[0007] An electric power conversion device according to one aspect of the present invention includes a transformer, a resonance capacitance element, a switching circuit, a rectifier circuit, a switch, and a control unit. The transformer is configured to transmit electric power between an input and an output. The capacitance element is connected in series with a primary winding of the transformer. The switching circuit is configured to switch on/off current that flows through a series circuit constituted by the primary winding of the transformer and the capacitance element. The rectifier circuit is configured to rectify and output electric power induced in a secondary winding of the transformer. The switch is connected in parallel with the capacitance element. The control unit is configured to control a period during which current flows through the series circuit constituted by the primary winding of the transformer and the capacitance element and on/off of the switch. The control unit is configured to switch off the switch so as to configure, as an input-side circuit of the rectifier circuit, a first conversion circuit configured to change a relationship between an input voltage and an output voltage based on an operation frequency of the switching circuit. The control unit is configured to switch on the switch so as to configure, as the input-side circuit of the rectifier circuit, a second conversion circuit configured to change a relationship between an input voltage and an output voltage based on a period during which current flows from the switching circuit to the primary winding of the transformer. The control unit is configured to select an operation of the switching circuit from a first operation for adjusting the relationship in the first conversion circuit by controlling the operation frequency and a second operation for adjusting the relationship in the second conversion circuit by controlling the period during which the current flows.

[0008] In the electric power conversion device, the control unit is preferably configured to match a timing at which the switch is switched on with a zero-crossing point of the current that flows from the switching circuit to the series circuit constituted by the primary winding of the transformer and the capacitance element, when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.

[0009] Preferably, the electric power conversion device further includes a series circuit constituted by a second switch and a resistor. The series circuit is preferably connected in parallel with the capacitance element. The control unit is preferably configured to switch on the switch after switching on the second switch when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.

[0010] In the electric power conversion device, the control unit is preferably configured to provide a transition period during which an operation of the switching circuit is an operation different from the first operation of the first conversion circuit and the second operation of the

second conversion circuit, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.

[0011] In the electric power conversion device, the control unit is preferably configured to store the operation frequency of the switching circuit in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store the period during which current flows from the switching circuit to the primary winding of the transformer in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit. The control unit is preferably configured to control the switching circuit in accordance with the stored operation frequency and the stored period during which current flows, such that the relationship between the input voltage and the output voltage after switching coincides with the relationship between the input voltage and the output voltage before switching, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.

[0012] In the electric power conversion device, the control unit is preferably configured to store a conversion efficiency in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store a conversion efficiency in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit. The control unit is preferably configured to select whichever of the first operation and the second operation has a higher stored conversion efficiency, when both the first operation of the first conversion circuit and the second operation of the second conversion circuit are operable, in an overlapping region between the relationship in the first conversion circuit and the relationship in the second conversion circuit.

[0013] Preferably, the electric power conversion device further includes a resonance inductor. The inductor is preferably connected between the primary winding of the transformer and the switching circuit.

[0014] Preferably, the electric power conversion device further includes a plurality of resonance inductors and a short-circuit switch. The plurality of inductors are preferably connected between the primary winding of the transformer and the switching circuit. The short-circuit switch is preferably configured to short-circuit at least one of the plurality of inductors.

[0015] In the electric power conversion device, the capacitance element preferably includes a plurality of resonance capacitors. The plurality of capacitors are preferably connected between the primary winding of the transformer and the switching circuit. The switch is preferably configured to short-circuit at least one of the plurality of capacitors.

[0016] In the electric power conversion device, the capacitance element preferably includes a resonance capacitor. The capacitor is preferably connected between the primary winding of the transformer and the switching

circuit. The switch is preferably configured to short-circuit the capacitor.

[0017] In the electric power conversion device, both the relationship in the first conversion circuit and the relationship in the second conversion circuit are preferably a voltage ratio between the input voltage and the output voltage.

[0018] In the electric power conversion device, the first conversion circuit is preferably a resonance type conversion circuit. The resonance type conversion circuit is preferably configured to change the relationship in the first conversion circuit using a resonance circuit constituted by the capacitance element and the transformer by adjusting the operation frequency of the switching circuit. The second conversion circuit is preferably a phase shift type conversion circuit. The phase shift type conversion circuit is preferably configured to change the relationship in the second conversion circuit by adjusting the period during which current flows from the switching circuit to the primary winding of the transformer through changing a phase of switching on a switching element of the switching circuit.

[0019] According to the one aspect of the present invention, the electric power conversion is performed by switching, and the electric power is transmitted between the input and the output via the transformer. It is possible to adjust the relationship between the input voltage and the output voltage over a wide range according to the use by selecting one of the operation of the first conversion circuit and the operation of the second conversion circuit.

BRIEF DESCRIPTION OF DRAWINGS

[0020] Preferable embodiments according to the present invention will be described in more detail. Other features and advantages of the present invention will be better understood with reference to the following detailed description and the attached drawings:

FIG. 1 is a circuit diagram illustrating an electric power conversion device according to Embodiment 1; FIG. 2 is a specific circuit diagram illustrating an example of the electric power conversion device according to Embodiment 1; FIG. 3 is an explanatory diagram illustrating an operation of the electric power conversion device according to Embodiment 1 as an LLC resonance type conversion circuit; "A" of FIG. 4 is an explanatory diagram illustrating an operation of a switching element Q11 in a phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; "B" of FIG. 4 is an explanatory diagram illustrating an operation of a switching element Q12 in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; "C" of FIG. 4 is an explanatory diagram illustrating an operation

of a switching element Q13 in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; "D" of FIG. 4 is an explanatory diagram illustrating an operation of a switching element Q14 in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1; and "E" of FIG. 4 is a diagram illustrating a conduction state of a primary winding of a transformer in the phase shift type conversion circuit of the electric power conversion device according to Embodiment 1;

FIG. 5 is an explanatory diagram illustrating an operation of the electric power conversion device according to Embodiment 1 as the phase shift type conversion circuit;

"A" of FIG. 6 is a diagram illustrating a waveform of current flowing through a resonance circuit of the electric power conversion device according to Embodiment 1; and "B" of FIG. 6 is an explanatory operation diagram illustrating a switching timing of a switching element used in the electric power conversion device according to Embodiment 1;

"A" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q11 of a switching circuit in the electric power conversion device according to Embodiment 1; "B" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q12 of the switching circuit in the electric power conversion device according to Embodiment 1; "C" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q13 of the switching circuit in the electric power conversion device according to Embodiment 1; and "D" of FIG. 7 is an explanatory diagram illustrating an operation of a switching element Q14 of the switching circuit in the electric power conversion device according to Embodiment 1;

FIG. 8 is a circuit diagram illustrating an electric power conversion device according to Embodiment 2;

FIG. 9 is a circuit diagram illustrating an electric power conversion device according to Embodiment 3;

FIG. 10 is a circuit diagram illustrating another configuration example of the electric power conversion device according to Embodiment 3;

FIG. 11 is a circuit diagram illustrating an electric power conversion device according to Embodiment 4; and

FIG. 12 is a circuit diagram illustrating another configuration example of the electric power conversion device according to Embodiment 4.

BEST MODE FOR CARRYING OUT INVENTION

Embodiment 1

[0021] Although, in an embodiment described below, a case where an electric power conversion device performs DC/DC conversion uni-directionally will be de-

scribed as an example, the technical idea described below can also be applicable to an electric power conversion device that performs DC/AC conversion or an electric power conversion device that performs electric power conversion bi-directionally. Also, although the electric power conversion device described below is assumed to be used for utilizing electric power of a storage battery mounted in an automobile having an electric motor as a power source, a storage battery used by being attached to a building, or the like, the application is not limited thereto. For example, the technique of the electric power conversion device described below may be adopted in a distributed power source such as a solar power generation device or a fuel cell.

[0022] The electric power conversion device according to the present embodiment includes a switching circuit 1 and a rectifier circuit 2 with a transformer T1 interposed therebetween, as shown in FIG. 1. The switching circuit 1 includes four switching elements Q11 to Q14 (refer to FIG. 2) described later, and a control unit 3 is attached to the switching circuit 1 so as to control the switching elements Q11 to Q14. A primary winding n1 of the transformer T1 is connected to the switching circuit 1 via a series circuit constituted by a capacitor C1 serving as a capacitance element and an inductor L1. The rectifier circuit 2 is connected to a secondary winding n2 of the transformer T1. Furthermore, a switch SW1 is connected in parallel with the capacitor C1. The control unit 3 is also configured to control on/off of the switch SW1.

[0023] The switching circuit 1 and the rectifier circuit 2 includes a configuration shown in FIG. 2, for example. The configuration shown in FIG. 2 is an example of the switching circuit 1 and the rectifier circuit 2, and the configuration of the switching circuit 1 and the rectifier circuit 2 is not limited thereto. Also, a DC power supply serving as an input power supply is connected to the switching circuit 1. The DC power supply may be a DC power supply obtained by rectifying a commercial AC power supply, other than being a storage battery, a fuel cell, a solar cell, or the like. Also, the DC power outputted from the rectifier circuit 2 may be used for charging a storage battery, in addition to being supplied to a load.

[0024] The switching circuit 1 shown in the diagram includes a bridge circuit constituted by the four switching elements Q11 to Q14. A series circuit of the switching elements Q11 and Q12 and a series circuit of the switching elements Q13 and Q14 are connected in parallel with a smoothing capacitor C11.

[0025] Also, the rectifier circuit 2 shown in the diagram uses a circuit having the same configuration as the switching circuit 1. That is, the rectifier circuit 2 includes a bridge circuit constituted by four switching elements Q21 to Q24. A series circuit of the switching elements Q21 and Q22 and a series circuit of the switching elements Q23 and Q24 are connected in parallel with a smoothing capacitor C21. By controlling on/off of the switching elements Q21 to Q24 in the rectifier circuit 2, electric power can be bi-directionally transmitted via the

transformer T1. A description of operations to perform an electric power conversion bi-directionally will, however, be omitted in the present embodiment.

[0026] Each of the switching elements Q11 to Q14 and Q21 to Q24 is assumed to be a MOSFET. Note that a configuration in which a diode is connected anti-parallelly between an emitter and a collector of a bipolar transistor, an IGBT (Insulated Gate Bipolar Transistor), or the like may be used as each of the switching elements Q11 to Q14 and Q21 to Q24. These switching elements Q11 to Q14 and Q21 to Q24 can pass current in one direction or bi-directionally in an on-state, and can pass current in the other direction in an off-state. For example, in the case where the switching elements Q11 to Q14 and Q21 to Q24 are MOSFETs, the switching elements Q11 to Q14 and Q21 to Q24 can pass current via a path passing through a body diode (parasitic diode) in the off-state.

[0027] The rectifier circuit 2 shown in the diagram performs full-wave rectification using a function in the off-state of the switching elements Q21 to Q24. Note that synchronous rectification can also be performed if on/off of the switching elements Q21 to Q24 is controlled.

[0028] The control unit 3 includes a processor (selected from a microcontroller, a DSP (Digital Signal Processor), an FPGA (Field-Programmable Gate Array), or the like) that operates according to a program. The control unit 3 has a function of monitoring an output current of the rectifier circuit 2 using an output from a current detector (not shown) provided at an appropriate location, and includes an AD converter that converts analog information outputted from the current detector to digital information so as to be handled by the processor. On/off of each of the switching elements Q11 to Q14 is controlled by a pulse signal given by the control unit 3, and the frequency and the phase of the pulse signal are adjustable. That is, the control unit 3 includes a pulse generator that generates pulse signals, and performs control on the switching circuit 1, selecting from PFM (Pulse Frequency Modulation) control and PPM (Pulse Phase shift Modulation) control, by giving instructions to the pulse generator from the processor. Furthermore, the control unit 3 includes a memory that stores a program for causing the processor to operate, data for determining an operation condition of the processor, data that is generated during an operation of the processor, and the like.

[0029] A frequency for switching on/off the switching elements Q11 to Q14 is selected appropriately from a range of several tens of Hz to several hundreds of kHz. Here, since the switch SW1 is controlled based on the on/off timings of the switching elements Q11 to Q14, the switch SW1 is required to respond in a time period of approximately several μ s to several tens of ms from the time when an instruction is given. Accordingly, it is desirable that a semiconductor switch having a short response time is used for the switch SW1, instead of using an electromagnetic relay including a mechanical contact.

[0030] Incidentally, when the switch SW1 is off, the capacitor C1 is electrically connected between the inductor

L1 and the primary winding n1 of the transformer T1, and a resonance circuit is formed by the capacitor C1, the inductor L1, and the transformer T1. Accordingly, if the frequency of the switching circuit 1 is adjusted appropriately with respect to the resonance frequency of the resonance circuit, this resonance circuit and the switching circuit 1 operate as a first conversion circuit that changes the voltage that is outputted from the secondary winding n2 of the transformer T1. That is, a circuit that is on the input side of the rectifier circuit 2 operates as the first conversion circuit. The first conversion circuit is configured to change the relationship (voltage ratio) between an input voltage and an output voltage based on the operation frequency of the switching circuit 1. In this way, a configuration, in which the resonance circuit constituted by the inductor L1 and the transformer T1 with one capacitor C1 is used to change the output voltage of the rectifier circuit 2 by adjusting the operation frequency of the switching circuit 1, will be hereinafter referred to as "LLC resonance type conversion circuit". Note that the inductor L1 is not an essential constituent element in the first conversion circuit.

[0031] Hereinafter, in the configuration shown in FIG. 2, the two switching elements Q11 and Q14 and the two switching elements Q12 and Q13 that each have a positional relationship of being connected in series with the primary winding n1 of the transformer T1 interposed therebetween are each referred to as "opposing switching elements". When operating as the LLC resonance type conversion circuit, on/off of the opposing switching elements Q11 and Q14 are switched at the same time, and on/off of the opposing switching elements Q12 and Q13 are switched at the same time. Also, the two switching elements Q11 and Q12 that are connected in series are not made to be in on-states at the same time, and the two switching elements Q13 and Q14 that are connected in series are not made to be in on-states at the same time.

[0032] FIG. 3 shows a relationship between the operation frequency of the switching circuit 1 in the LLC resonance type conversion circuit and a voltage gain of the output voltage of the rectifier circuit 2. Also, the curves denoted by reference signs A1 to A6 in FIG. 3 show results that are taken on different load sizes. The size of the load increases in the order from the curve A1 to the curve A6. That is, the load corresponding to the curve A1 is the smallest and the load corresponding to the curve A6 is the largest, among the curves A1 to A6.

[0033] In the illustrated example, when the load is light, the voltage gain reaches a peak when an operation frequency ratio is approximately 0.4, and the voltage gain decreases as the operation frequency ratio moves away from 0.4. The operation frequency ratio is a ratio of the operation frequency of the switching circuit 1 with respect to the resonance frequency determined by the capacitor C1 and the inductor L1. The lighter the load is, the stronger the resonance is that occurs in the LLC resonance type conversion circuit, and thus, in a case of a light load, the output voltage changes largely by the operation fre-

quency of the switching circuit 1 merely changing slightly in the vicinity of the resonance point, possibly resulting in the output voltage becoming unstable. Furthermore, in the case of a light load, many noise components are included in the output of the rectifier circuit 2.

[0034] On the other hand, in an example shown in FIG. 3, when the load increases, the change of the output voltage with respect to the change of the operation frequency of the switching circuit 1 decreases, and an adjustable range of the voltage gain also decreases. That is, in the LLC resonance type conversion circuit, the resonance weakens in a region in which the load is large, and a change of the output voltage with respect to the change in the operation frequency of the switching circuit 1 decreases. At this time, the noise component included in the output of the rectifier circuit 2 decreases compared with the case of a light load.

[0035] As described above, in the LLC resonance type conversion circuit, the voltage ratio (voltage gain) of the output voltage to the input voltage can, depending on the size of the load, be changed by changing the operation frequency ratio. Furthermore, a voltage boost is possible, and adjustment of the voltage boost ratio becomes possible. However, since the operation becomes unstable when the load is light, the adjustable range of the voltage gain is relatively narrow in the load range where a stable operation can be achieved.

[0036] On the other hand, when the switch SW1 is in an on-state, the capacitor C1 is short-circuited, and thus only the inductor L1 and the transformer T1 are electrically interposed between the switching circuit 1 and the rectifier circuit 2. In this case, due to a change in the current flowing through the inductor L1 and the primary winding n1 of the transformer T1, the output voltage of the rectifier circuit 2 changes. In this operation, electromagnetic energy accumulated in the transformer T1 due to the current flowing through the primary winding n1 of the transformer T1 is transferred to the secondary winding n2. Accordingly, by adjusting the electromagnetic energy accumulated in the transformer T1 per unit time period, the circuit constituted by the inductor L1, the transformer T1, and the switching circuit 1 functions as a second conversion circuit that adjusts the output voltage of the rectifier circuit 2. That is, the input side circuit of the rectifier circuit 2 functions as the second conversion circuit. The second conversion circuit changes a relationship (voltage ratio) between the input voltage and the output voltage based on the period during which current flows from the switching circuit 1 to the primary winding n1 of the transformer T1. Note that the inductor L1 is not an essential constituent element in the second conversion circuit.

[0037] In order to adjust the period during which current flows through the primary winding n1 of the transformer T1, the phases of switching on the switching elements Q11 to Q14, as shown in "A" to "D" of FIG. 4, are adjusted, for example. In this way, a configuration in which the phases of switching on the switching elements Q11 to

Q14 of the switching circuit 1 are changed in order to adjust the current that flows through the primary winding n1 of the transformer T1 per unit time period (that is, second conversion circuit) will hereinafter be referred to as a "phase shift type conversion circuit".

[0038] In the phase shift type conversion circuit, without changing the duty cycles of the four switching elements Q11 to Q14 that constitute the switching circuit 1, the timings at which the switching elements are turned on in each pair of switching elements that need to be on to allow current to flow through the transformer T1, out of the switching elements Q11 to Q14, are changed such that the period in which the opposing switching elements are on at the same time is changed. That is, the output electric power is adjusted by adjusting the timings at which the opposing switching elements Q11 and Q14 are turned on such that the period of being on at the same time is adjusted, and by adjusting the timings at which the opposing switching elements Q12 and Q13 are turned on such that the period of being on at the same time is adjusted.

[0039] In the circuit shown in FIG. 2, current flows through the primary winding n1 of the transformer T1 in the period during which the opposing switching elements Q11 and Q14 are on at the same time, and the period during which the opposing switching elements Q12 and Q13 are on at the same time (refer to "E" of FIG. 4). Accordingly, by adjusting the timings at which the opposing switching elements Q11 and Q14 are turned on, or adjusting the timings at which the opposing switching elements Q12 and Q13 are turned on, the period during which current flows through the primary winding n1 is adjusted, and as a result the output voltage of the rectifier circuit 2 is adjusted. Note that the switching elements Q11 and Q12 that are connected in series or the switching elements Q13 and Q14 that are connected in series are prohibited from being on at the same time, and the switching on/off of the switching elements Q11 and Q12 or the switching elements Q13 and Q14 are controlled such that the phases thereof being on/off are inverted. That is, the on/off of the switching element Q11 is inverted from the on/off of the switching element Q12, and the on/off of the switching element Q13 is inverted from the on/off of the switching element Q14.

[0040] Referring to operations shown in "A" to "E" of FIG. 4, duty cycles of the switching elements Q11 to Q14 are set to 50% and the on period of the switching element Q11 differs from the on period of the switching element Q13 by a phase difference T_s , as shown in "A" to "D" of FIG. 4. Accordingly, the on periods of the opposing switching elements Q11 and Q14 overlap during a period corresponding to the phase difference T_s , and the on periods of the opposing switching elements Q12 and Q13 overlap during a period corresponding to the phase difference T_s . The phase difference T_s is assumed to be 0 degrees when the on periods of the switching element Q11 and the switching element Q13 do not overlap, and to be 180 degrees when the on periods of the switching

element Q11 and the switching element Q13 coincide. That is, the phase difference T_s corresponds to a value represented by an angle the period in which the opposing switching elements Q11 and Q14 are on at the same time and the period in which the opposing switching elements Q12 and Q13 are on at the same time, with the angle being 180 degrees when the on periods of the switching element Q11 and the switching element Q13 coincide.

[0041] When the switching circuit 1 performs the operations described above, a current flows through the primary winding n1 of the transformer T1 only in the period corresponding to the phase difference T_s among the on periods of the switching elements Q11 to Q14, as shown in "E" of FIG. 4. The direction of the current flowing through the primary winding n1 of the transformer T1 in the period during which the opposing switching elements Q11 and Q14 are on at the same time is inverted from the direction of the current in the period during which the opposing switching elements Q12 and Q13 are on at the same time. According to these operations, an alternating voltage is induced in the secondary winding n2 of the transformer T1.

[0042] FIG. 5 shows the relationship between the phase difference T_s and the voltage gain of the output voltage of the rectifier circuit 2 in the phase shift type conversion circuit. As shown in FIG. 5, the voltage gain has an approximately linear relationship with the phase difference T_s in the phase shift type conversion circuit. That is, in the phase shift type conversion circuit, the output voltage from the rectifier circuit 2 can be changed by adjusting the phase difference T_s relating to the period during which the opposing switching elements Q11 and Q14 are on at the same time and the period during which the opposing switching elements Q12 and Q13 are on at the same time.

[0043] Also, the phase shift type conversion circuit is characterized in that the adjustable range of the voltage ratio (voltage gain) between the input voltage and the output voltage is relatively wide. When the configuration shown in FIG. 2 is operated as the phase shift type conversion circuit, noise is reduced compared with forward type converters or flyback type converters, since a period during which current does not flow is unlikely to occur. Note that the voltage adjustment can only be performed in a voltage range not exceeding an upper limit value, with the upper limit value being a voltage determined by the winding ratio (number of winding of secondary winding n2/number of winding of primary winding n1) of the transformer T1.

[0044] As described above, the control unit 3 is configured to select the operation of the switching circuit 1 from a first operation and a second operation. The first operation is an operation in which the relationship (voltage ratio) between the input voltage and the output voltage is adjusted by controlling the operation frequency of the switching circuit 1. The second operation is an operation in which the relationship (voltage ratio) between the input voltage and the output voltage is adjusted by con-

trolling the period during which a current flows from the switching circuit 1 to the primary winding n1 of the transformer T1. Operations of both the LLC resonance type conversion circuit and the phase shift type conversion circuit are possible by only changing the operation of the switching circuit 1 by switching on/off the switch SW1. Accordingly, both features of the LLC resonance type conversion circuit and the phase shift type conversion circuit can be utilized so as to complement each other according to the state of the load.

[0045] Also, the configuration according to the present embodiment is realized by adding the switch SW1 to the LLC resonance type conversion circuit and adding, to the control unit 3, a program for enabling an operation as the phase shift type conversion circuit. Accordingly, in a case where the LLC resonance type conversion circuit has already been designed, an electric power conversion device having the functions of both the LLC resonance type conversion circuit and the phase shift type conversion circuit can be provided without a large-scale design change.

[0046] For example, the case is assumed where the input voltage is 400V and the output voltage is changed in a range from 100 to 500V. In this case, in a range of the output voltage from 100 to 400V, it is possible to adjust the output voltage in a wide range by causing the conversion circuit to operate as the phase shift type conversion circuit by switching on the switch SW1. On the other hand, in a range of the output voltage from 400 to 500V, it is possible to obtain an output voltage in a voltage range higher than the input voltage by causing the conversion circuit to operate as the LLC resonance type conversion circuit by switching off the switch SW1.

[0047] Also, in a case where the current detector is provided as described above, the on/off of the switch SW1 may be determined based on the output current of the rectifier circuit 2 (electric power conversion device). The input side circuit of the rectifier circuit 2 may be caused to operate as the phase shift type conversion circuit if the output current is a prescribed value (1A, for example) or less in a case where the load is light, and the input side circuit of the rectifier circuit 2 may be caused to operate as the LLC resonance type conversion circuit if the output current exceeds the prescribed value. In this operation, a stable operation with a small noise component is possible by causing the input side circuit of the rectifier circuit 2 to operate as the phase shift type conversion circuit in a light load region in which the operation of the LLC resonance type conversion circuit becomes unstable and a noise component increases in the output of the rectifier circuit 2.

[0048] There are cases where an electric power conversion efficiency is considered as a condition to switch on/off the switch SW1. For example, in a case where both the LLC resonance type conversion circuit and the phase shift type conversion circuit can be used such that a ratio of the input voltage and the output voltage becomes a desired voltage ratio, it is possible to determine

the on/off of the switch SW1 based on the conversion efficiency. For example, assuming that the conversion efficiency of the LLC resonance type conversion circuit is E1, the conversion efficiency of the phase shift type conversion circuit is E2, and E1 and E2 satisfy a condition of $E1 > E2$ when the input voltage and the output voltage are desired values, it is desirable to use the LLC resonance type conversion circuit, since the conversion efficiency is higher compared with a case where the phase shift type conversion circuit is used.

[0049] That is to say, the control unit 3 is configured to select whichever of the first operation and the second operation has a higher stored conversion efficiency, when both the first operation of the first conversion circuit and the second operation of the second conversion circuit are operable, in an overlapping region between the relationship (voltage ratio) in the first conversion circuit and the relationship (voltage ratio) in the second conversion circuit.

[0050] Note that, in a case where the on/off of the switch SW1 is determined while taking the conversion efficiency into consideration as described above, the relationship between the efficiency and the input voltage and the output voltage needs to be stored in the control unit 3. Also, a function of notifying the input voltage and the output voltage of the control unit 3 is required.

[0051] As described above, an operation as the LLC resonance type conversion circuit and an operation as the phase shift type conversion circuit can be switched by selecting the on/off of the switch SW1 so as to change the operation of the switching circuit 1. Accordingly, usability is improved by switching on/off the switch SW1 while focusing on the output voltage range, the size of the load, the conversion efficiency, or the like.

[0052] Incidentally, when the switch SW1 is switched from off to on in order to switch from operating as the LLC resonance type conversion circuit to operating as the phase shift type conversion circuit, there are cases where charges are accumulated in the capacitor C1. When the switch SW1 is switched on in a state in which charges are accumulated in the capacitor C1, it is possible that a rush current flows through the switch SW1. The occurrence of such a rush current puts much stress on the switch SW1. Also, since the voltage gain differs in the operation of the LLC resonance type conversion circuit and in the operation of the phase shift type conversion circuit, the output voltage changes when the on/off of the switch SW1 is switched if the voltage gain is not considered when the on/off of the switch SW1 is switched.

[0053] In order to prevent the rush current, the control unit 3 should switch on the switch SW1 at the time of a zero-cross point of the current flowing through the resonance circuit constituted by the inductor L1, the capacitor C1, and the transformer T1 in a period during which the conversion circuit is operated as the LLC resonance type conversion circuit. The current Iq flowing through the resonance circuit has a waveform as shown in "A" of FIG.

6. The waveform of the current Iq is aligned with the on/off timings of the switching elements Q11 and Q14 shown in "B" of FIG. 6, and the current Iq becomes zero at the timings of switching on/off the switching elements Q11 and Q14, and the polarity of the current Iq inverts.

[0054] Accordingly, switching on the switch SW1 at the timing of switching on/off the switching element Q11 and Q14 prevents the rush current caused by the charges in the capacitor C1 from flowing through the switch SW1. In the illustrated example, the timing at which the switch SW1 is switched on is designated by an arrow, and this timing is matched with the timing at which the switching elements Q11 and Q14 are switched on. Although the timing shown in the diagram (timing at which switch SW1 is switched on) is matched with the timing at which the switching element Q11 and Q14 are switched on, the timing may be matched with the timing at which the switching element Q11 and Q14 are switched off.

[0055] Also, since the above operations are aligned with the on/off timing of the switching elements Q11 and Q14, the control unit 3 can easily determine the timing at which the switch SW1 is switched on. Note that it is possible, by detecting the voltage between two terminals of the capacitor C1 or by detecting the resonance current that flows through the resonance circuit, to determine the timing at which the switch SW1 is switched on according to the detected state.

[0056] Due to the above operations, the switch SW1 is switched on in a state in which charges are not accumulated in the capacitor C1 when transitioning from the LLC resonance type conversion circuit to the phase shift type conversion circuit. Accordingly, a rush current is prevented from flowing from the capacitor C1 to the switch SW1 when the switch SW1 is switched on. That is, the transition from the LLC resonance type conversion circuit to the phase shift type conversion circuit can be performed seamlessly.

[0057] Incidentally, the on/off timings of the switching elements Q11 to Q14 differs between a period during which the conversion circuit operates as the LLC resonance type conversion circuit and a period during which the conversion circuit operates as the phase shift type conversion circuit, as described above. Accordingly, the control unit 3 needs to also change the on/off timings of the switching elements Q11 to Q14 when switching on/off the switch SW1. However, it is difficult to adjust the phases of on/off of the switching elements Q11 to Q14 instantly when shifting between the LLC resonance type conversion circuit and the phase shift type conversion circuit. Therefore, it is desirable that a transition period during which the on/off timings of the switching elements Q11 to Q14 are changed is provided before and after the timing at which the switch SW1 is switched on/off. In the transition period, the on/off of the switching elements Q11 to Q14 is temporarily suspended or the on/off timings of the switching elements Q11 to Q14 are forcedly changed.

[0058] An example of on/off timings of the switching elements Q11 to Q14 is shown in "A" to "D" of FIG. 7 in

the case of transitioning from operating as the LLC resonance type conversion circuit to operating as the phase shift type conversion circuit. When operating as the LLC resonance type conversion circuit, the on/off timings are the same for the opposing switching elements Q11 and Q14, and the on/off timings are the same for the opposing switching elements Q12 and Q13. Also, when operating as the phase shift type conversion circuit, an appropriate phase difference is given between on timings of the opposing switching elements Q11 and Q14, and an appropriate phase difference is given between on timings of the opposing switching elements Q12 and Q13.

[0059] When transitioning from the operation as the LLC resonance type conversion circuit to the operation as the phase shift type conversion circuit, a transition period P_s during which all the switching elements Q11 to Q14 are switched off is provided. The control unit 3 is configured to switch on the switch SW1 in this transition period P_s .

[0060] Incidentally, when transitioning between the LLC resonance type conversion circuit and the phase shift type conversion circuit, the output voltage of the rectifier circuit 2 preferably does not vary. Accordingly, when the operation is transitioned, the control unit 3 needs to adjust the on/off timings of the switching elements Q11 to Q14 such that the output voltage of the rectifier circuit 2 remains the same. In order to realize this operation, the relationship between the operation frequency and the voltage gain in the LLC resonance type conversion circuit and the relationship between the phase difference T_s and the voltage gain in the phase shift type conversion circuit are stored in a storage unit provided in the control unit 3. Also, the control unit 3 is configured to monitor the input voltage, the output voltage, and the output current, and determine the on/off timings of the switching elements Q11 to Q14 according to the voltage gain immediately before transitioning between the LLC resonance type conversion circuit and the phase shift type conversion circuit by referring to the storage unit. Variation in the output voltage of the rectifier circuit 2 is suppressed by this operation when switching between operations of the LLC resonance type conversion circuit and the phase shift type conversion circuit.

[0061] The relationship between the voltage gain and the conversion efficiency of the LLC resonance type conversion circuit and the relationship between the voltage gain and the conversion efficiency of the phase shift type conversion circuit may be stored in the storage unit provided in the control unit 3. That is, if the relationships between the conversion efficiency and the voltage ratio (voltage gain), which is a ratio of the input voltage and the output voltage, are known for both the LLC resonance type conversion circuit and the phase shift type conversion circuit, an operation having higher conversion efficiency can be selected out of the operation of the LLC resonance type conversion circuit and the operation of the phase shift type conversion circuit according to the relationship between the input voltage and the output

voltage. That is, when both the LLC resonance type conversion circuit and the phase shift type conversion circuit are operable in order to obtain a desired voltage ratio between the input voltage and the output voltage, the control unit 3 determines the on/off of the switch SW1 based on the conversion efficiency, as described above.

[0062] In the embodiment described above, although the capacitor C1 is inserted between the inductor L1 and the transformer T1, a similar operation can be obtained in a configuration in which an inductor L1 is inserted between the capacitor C1 and the transformer T1.

[0063] Note that although, in the present embodiment, only a configuration in which electric power is supplied from the switching circuit 1 to the transformer T1 is described, it is possible to supply electric power from the rectifier circuit 2 to the transformer T1 by controlling the on/off of the switching elements Q21 to Q24 provided in the rectifier circuit 2. That is, electric power can be transmitted bi-directionally. Also, the switching circuit 1 is not limited to the configuration shown in FIG. 2. The switching circuit 1 may include a configuration in which a period during which current flows through the series circuit constituted by the primary winding n_1 of the transformer T1 and the capacitor C1 is controlled by switching.

Embodiment 2

[0064] As shown in FIG. 8, an electric power conversion device according to the present embodiment includes a configuration in which a series circuit constituted by a switch SW2 and a resistor R2 is connected in parallel with the switch SW1 in the configuration of Embodiment 1.

[0065] When transitioning from an LLC resonance type conversion circuit to a phase shift type conversion circuit in the configuration according to the present embodiment, a control unit 3 is configured to switch on the switch SW2 before switching on the switch SW1. That is, the control unit 3 is configured to switch on the switch SW1 after switching on the switch SW2 when transitioning from a first operation of the LLC resonance type conversion circuit to a second operation of the phase shift type conversion circuit. A timing at which the switch SW2 is switched on need not be matched with the on/off timings of switching elements Q11 to Q14. When the switch SW2 is switched on, charges accumulated in the capacitor C1 are discharged via the resistor R2, and the current flowing through the switch SW2 is limited by the resistance value of the resistor R2.

[0066] Furthermore, after the switch SW2 is switched on and the charge amount in the capacitor C1 decreases, the control unit 3 is configured to switch on the switch SW1. The charge amount over time in the capacitor C1 after the switch SW2 is switched on is determined by a voltage between the two terminals and the capacitance of the capacitor C1 and a resistance value of the resistor R2. Accordingly, the time period from switching on the switch SW2 to switching on the switch SW1 is dependent

on the voltage between the two terminals of the capacitor C1. When simplified control is desired, the time period may be set to a fixed time period using the capacitance of the capacitor C1 and the resistance value of the resistor R2.

[0067] In this configuration, since the timing at which the switch SW1 is switched on does not depend on the waveform of the current flowing through the resonance circuit, the response time of the switch SW1 need not be taken into consideration, and it is possible to use an electromagnetic relay having a longer response time than the semiconductor switch as the switch SW1. The other configurations and operations of the present embodiment are similar to those of Embodiment 1.

Embodiment 3

[0068] An electric power conversion device according to the present embodiment is a modification of Embodiment 1, and uses a plurality of (three each in diagram) inductors L31 to L33 and switches (short-circuit switches) SW31 to SW33, as shown in FIG. 9, instead of using the inductor L1 shown in FIG. 1. The switches SW31 to SW33 are respectively connected in parallel with the inductors L31 to L33. That is, the switch SW31 is connected in parallel with the inductor L31, the switch SW32 is connected in parallel with the inductor L32, and the switch SW33 is connected in parallel with the inductor L33. Accordingly, at least one of the inductors L31 to L33 is used in a resonance circuit, depending on the combination of on/off of the switches SW31 to SW33. In other words, the inductance used in the resonance circuit is adjusted by combining the on/off of the switches SW31 to SW33, and as a result the resonance frequency can be selected from a plurality of frequencies.

[0069] For example, in the configuration shown in the diagram, if the switch SW31 is switched off and the switches SW32 and SW33 are switched on, only the inductor L31 effectively functions in the resonance circuit. Also, if all the switches SW31 to SW33 are switched off, a series circuit of the three inductors L31 to L33 effectively functions in the resonance circuit. Accordingly, it is possible to change the resonance frequency of the resonance circuit in a wide range.

[0070] Instead of the configuration shown in FIG. 9, the electric power conversion device may use a plurality of (three each in diagram) inductors L41 to L43 and switches SW41 to SW43 so as to have the connection relationship shown in FIG. 10, instead of using the inductor L1 shown in FIG. 1. In this configuration, the three inductors L41 to L43 are connected in series, and one terminal (first terminal) of each of the three switches SW41 to SW43 is connected to one terminal of the series circuit of the inductors L41 to L43. Also, the other terminal (second terminal) of each of the three switches SW41 to SW43 is respectively connected to one terminal of the three inductors L41 to L43. That is, the switch SW41 is connected in parallel with the inductor L41, the switch

SW42 is connected in parallel with a series circuit of the inductors L41 and L42, and the switch SW43 is connected in parallel with the series circuit of the inductors L41 to L43.

[0071] In the configuration shown in FIG. 9, the number of types of selectable inductances is determined by the number of combinations of on/off of the switches SW31 to SW33. In the configuration shown in FIG. 10, the number of types of selectable inductances is one more than the number of switches SW41 to SW43. That is, in the configuration shown in FIG. 10, only four states are selectable, namely a state in which the switch SW43 is on, a state in which the switch SW43 is off and the switch SW42 is on, a state in which the switches SW42 and SW43 are off and the switch SW41 is on, and a state in which the switches SW41 to SW43 are off. However, whereas the number of switches SW41 to SW43 included in the resonance circuit is one or less in the configuration shown in FIG. 10, the number of switches SW31 to SW33 included in the resonance circuit changes in a range from zero to three in the configuration shown in FIG. 9. Accordingly, an average loss of the resonance circuit caused by the switch in the configuration shown in FIG. 10 is smaller than that in the configuration shown in FIG. 9.

[0072] Although the electric power conversion device according to the present embodiment includes three each of the inductors and the switches, the numbers of the inductors and the switches are selected arbitrarily according to the number of frequencies to which the resonance frequency is changed. The other configurations and operations of the present embodiment are similar to those of Embodiment 1, and also, the configuration of the present embodiment can be combined with a configuration in which the switch SW2 and the resistor R2 are provided similarly to Embodiment 2.

Embodiment 4

[0073] The present embodiment is a modification of Embodiment 1, and uses a plurality of (three each in diagram) capacitors C51 to C53 and switches SW51 to SW53 as shown in FIG. 11 as a capacitance element, instead of using the capacitor C1 shown in FIG. 1. The switches SW51 to SW53 are respectively connected in parallel with the capacitors C51 to C53. That is, the switch SW51 is connected in parallel with the capacitor C51, the switch SW52 is connected in parallel with the capacitor C52, and the switch SW53 is connected in parallel with the capacitor C53. Accordingly, at least one of the capacitors C51 to C53 is used in the resonance circuit depending on the combination of on/off of the switches SW51 to SW53. In other words, the capacitance used in the resonance circuit is adjusted by combining the on/off of the switches SW51 to SW53, and as a result the resonance frequency can be selected from a plurality of frequencies.

[0074] Also, in the diagram shown, an inductor L3 is

used instead of the inductor L1, and a switch SW3 is connected in parallel with the inductor L3. Accordingly, whether or not the inductor L3 is included in the resonance circuit can be selected by the on/off of the switch SW3.

[0075] For example, in the configuration shown in the diagram, if the switch SW51 is switched off and the switches SW52 and SW53 are switched on, only the capacitor C51 effectively functions in the resonance circuit. Also, if all the switches SW51 to SW53 are switched off, a series circuit of the three capacitors C51 to C53 effectively functions in the resonance circuit. Accordingly, it is possible to change the resonance frequency of the resonance circuit in a wide range.

[0076] Instead of the configuration shown in FIG. 11, the electric power conversion device may use a plurality of (three each in diagram) capacitors C61 to C63 and the switches SW61 to SW63 so as to have the connection relationship shown in FIG. 12, instead of using the capacitor C1 shown in FIG. 1. In this configuration, the three capacitors C61 to C63 are connected in series, and one terminal (first terminal) of each of the three switches SW61 to SW63 is connected to one terminal of the series circuit of the capacitors C61 to C63. Also, the other terminal (second terminal) of each of the three switches SW61 to SW63 is respectively connected to one terminal of the three capacitors C61 to C63. That is, the switch SW61 is connected in parallel with the capacitor C61, the switch SW62 is connected in parallel with the series circuit of capacitors C61 and C62, and the switch SW63 is connected in parallel with the series circuit of the capacitors C61 to C63.

[0077] In the configuration shown in FIG. 11, the number of types of selectable capacitances is determined by the number of combinations of on/off of the switches SW51 to SW53. In the configuration shown in FIG. 12, the number of types of selectable capacitances is one more than the number of switches SW61 to SW63. That is, in the configuration shown in FIG. 12, only four states are selectable, namely a state in which the switch SW63 is on, a state in which the switch SW63 is off and the switch SW62 is on, a state in which the switches SW62 and SW63 are off and the switch SW61 is on, and a state in which the switches SW61 to SW63 are off. However, whereas the number of switches SW61 to SW63 included in the resonance circuit is one or less in the configuration shown in FIG. 12, the number of switches SW51 to SW53 included in the resonance circuit changes in a range from zero to three in the configuration shown in FIG. 11. Accordingly, an average loss of the resonance circuit due to the switch in the configuration shown in FIG. 12 is smaller than that in the configuration shown in FIG. 11.

[0078] Also, as described in Embodiment 1, the timings at which the switches SW51 to SW53 or SW61 to SW63 are switched on are controlled so as to be matched with the on/off timings of the switching elements Q11 to Q14 provided in the switching circuit 1. Therefore, a rush cur-

rent caused by the charges in the capacitors C51 to C53 or C61 to C63 is prevented from flowing through the switches SW51 to SW53 or SW61 to SW63.

[0079] Although the electric power conversion device according to the present embodiment includes three each of the capacitors and the switches, the numbers of the capacitors and the switches are selected arbitrarily according to the number of frequencies to which the resonance frequency is changed. The other configurations and operations of the present embodiment are similar to those of Embodiment 1, and also, the configuration of the present embodiment may be combined with a configuration in which the switch SW2 and the resistor R2 are provided similarly to Embodiment 2. Furthermore, the configuration of the present embodiment and the configuration shown in Embodiment 3 may be combined.

[0080] Although the present invention has been described in a preferred embodiment, various modifications and variations are possible by those skilled in the art without departing from the spirit or scope of this invention, that is, the claims.

Claims

1. An electric power conversion device comprising:

a transformer configured to transmit electric power between an input and an output;
a resonance capacitance element connected in series with a primary winding of the transformer;
a switching circuit configured to switch on/off current that flows through a series circuit constituted by the primary winding of the transformer and the capacitance element;
a rectifier circuit configured to rectify and output electric power induced in a secondary winding of the transformer;
a switch connected in parallel with the capacitance element; and
a control unit configured to control a period during which current flows through the series circuit constituted by the primary winding of the transformer and the capacitance element and on/off of the switch,
wherein the control unit is configured to switch off the switch so as to configure, as an input-side circuit of the rectifier circuit, a first conversion circuit configured to change a relationship between an input voltage and an output voltage based on an operation frequency of the switching circuit,
the control unit being configured to switch on the switch so as to configure, as the input-side circuit of the rectifier circuit, a second conversion circuit configured to change a relationship between an input voltage and an output voltage based on a period during which current flows from the

switching circuit to the primary winding of the transformer,

the control unit being configured to select an operation of the switching circuit from a first operation for adjusting the relationship in the first conversion circuit by controlling the operation frequency and a second operation for adjusting the relationship in the second conversion circuit by controlling the period during which the current flows.

2. The electric power conversion device according to claim 1, wherein the control unit is configured to match a timing at which the switch is switched on with a zero-crossing point of current that flows from the switching circuit to the series circuit constituted by the primary winding of the transformer and the capacitance element, when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.
3. The electric power conversion device according to claim 1, further comprising a series circuit constituted by a second switch and a resistor and connected in parallel with the capacitance element, wherein the control unit is configured to switch on the switch after switching on the second switch when transitioning from the first operation of the first conversion circuit to the second operation of the second conversion circuit.
4. The electric power conversion device according to any one of claims 1 to 3, wherein the control unit is configured to provide a transition period during which an operation of the switching circuit is an operation different from the first operation of the first conversion circuit and the second operation of the second conversion circuit, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.
5. The electric power conversion device according to any one of claims 1 to 4, wherein the control unit is configured to store the operation frequency of the switching circuit in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store the period during which current flows from the switching circuit to the primary winding of the transformer in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit, the control unit being configured to control the switching circuit in accordance with the stored operation frequency and the stored period during which current flows, such that the relationship between the input voltage and the output voltage after switching coincides with the relationship between the input

voltage and the output voltage before switching, when switching between the first operation of the first conversion circuit and the second operation of the second conversion circuit.

6. The electric power conversion device according to any one of claims 1 to 5, wherein the control unit is configured to store a conversion efficiency in the first operation of the first conversion circuit in association with the relationship in the first conversion circuit, and store a conversion efficiency in the second operation of the second conversion circuit in association with the relationship in the second conversion circuit, the control unit being configured to select whichever of the first operation and the second operation has a higher stored conversion efficiency, when both the first operation of the first conversion circuit and the second operation of the second conversion circuit are operable, in an overlapping region between the relationship in the first conversion circuit and the relationship in the second conversion circuit.
7. The electric power conversion device according to any one of claims 1 to 6, further comprising a resonance inductor that is connected between the primary winding of the transformer and the switching circuit.
8. The electric power conversion device according to any one of claims 1 to 6, further comprising:
 - a plurality of resonance inductors that are connected between the primary winding of the transformer and the switching circuit; and
 - a short-circuit switch configured to short-circuit at least one of the plurality of inductors.
9. The electric power conversion device according to any one of claims 1 to 8, wherein the capacitance element comprises a plurality of resonance capacitors that are connected between the primary winding of the transformer and the switching circuit, and wherein the switch is configured to short-circuit at least one of the plurality of capacitors.
10. The electric power conversion device according to any one of claims 1 to 8, wherein the capacitance element comprises a resonance capacitor that is connected between the primary winding of the transformer and the switching circuit, and wherein the switch is configured to short-circuit the capacitor.
11. The electric power conversion device according to any one of claims 1 to 10,

wherein both the relationship in the first conversion circuit and the relationship in the second conversion circuit are a voltage ratio between the input voltage and the output voltage.

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12. The electric power conversion device according to any one of claims 1 to 11,

wherein the first conversion circuit is a resonance type conversion circuit configured to change the relationship in the first conversion circuit using a resonance circuit constituted by the capacitance element and the transformer by adjusting the operation frequency of the switching circuit, and

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wherein the second conversion circuit is a phase shift type conversion circuit configured to change the relationship in the second conversion circuit by adjusting the period during which current flows from the switching circuit to the primary winding of the transformer through changing a phase of switching on a switching element of the switching circuit.

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FIG. 1

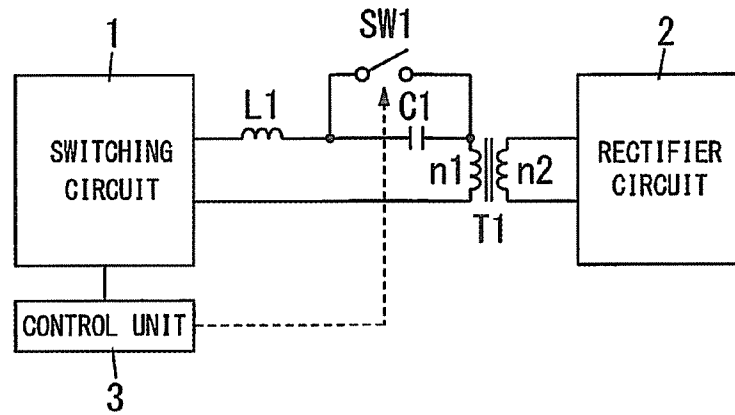


FIG. 2

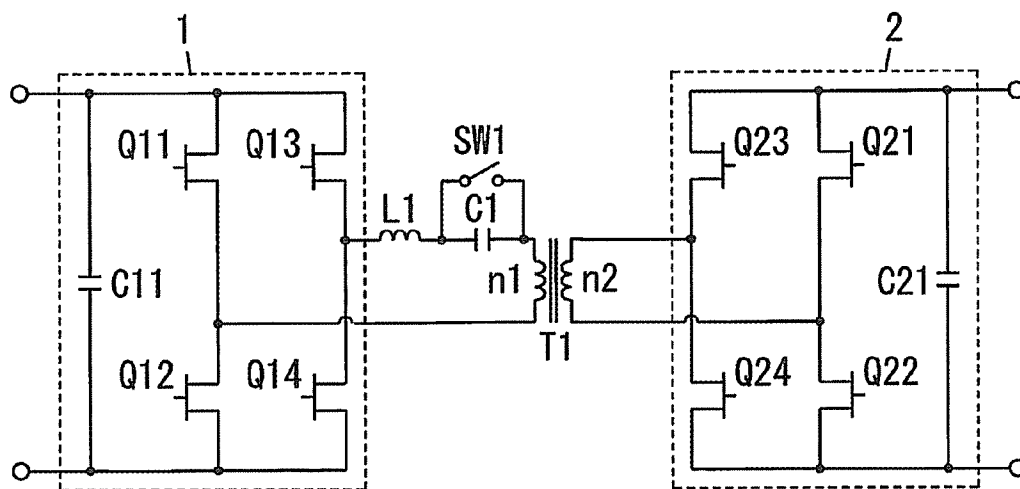


FIG. 3

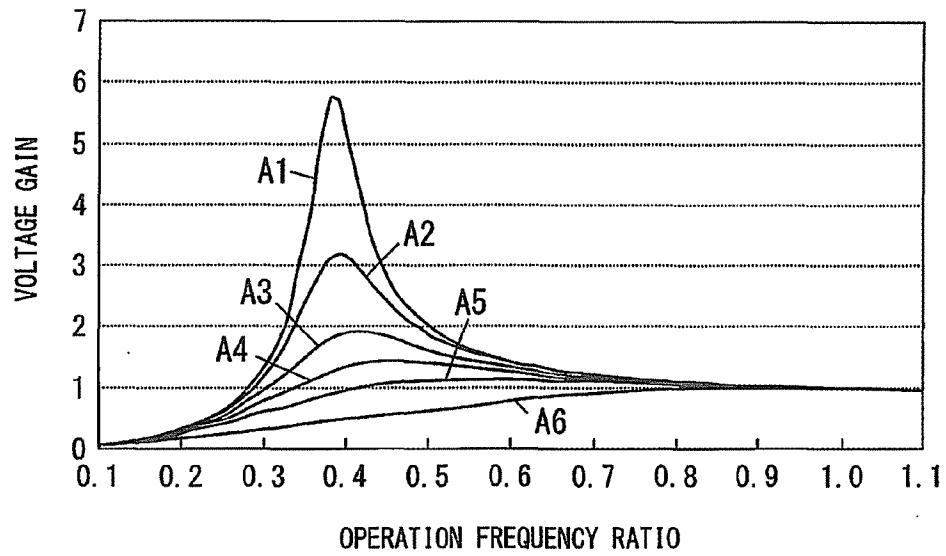


FIG. 4

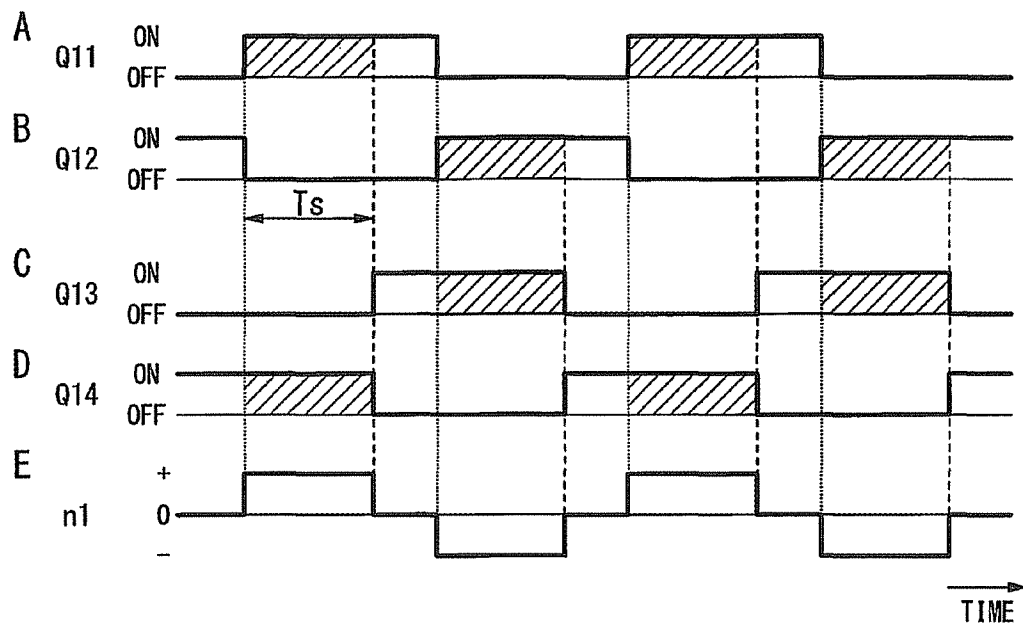


FIG. 5

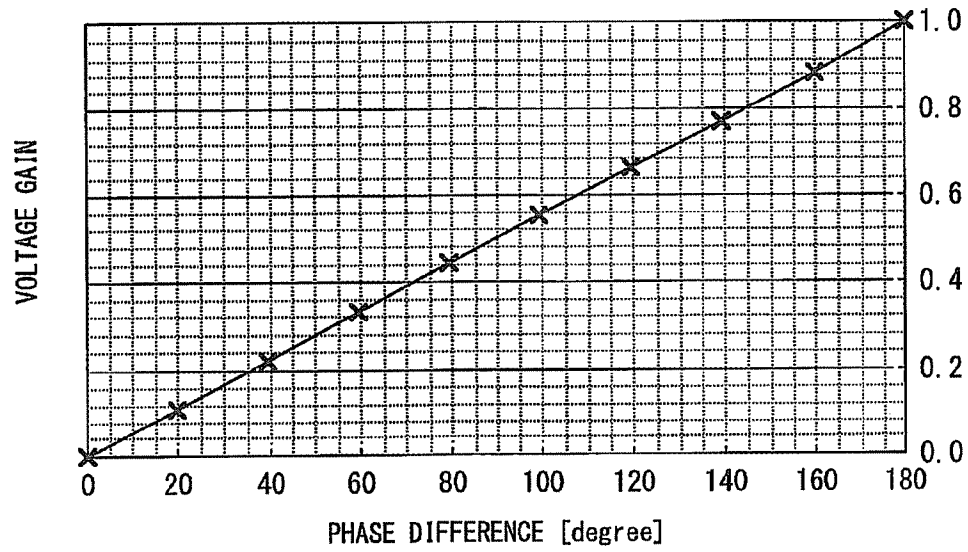


FIG. 6

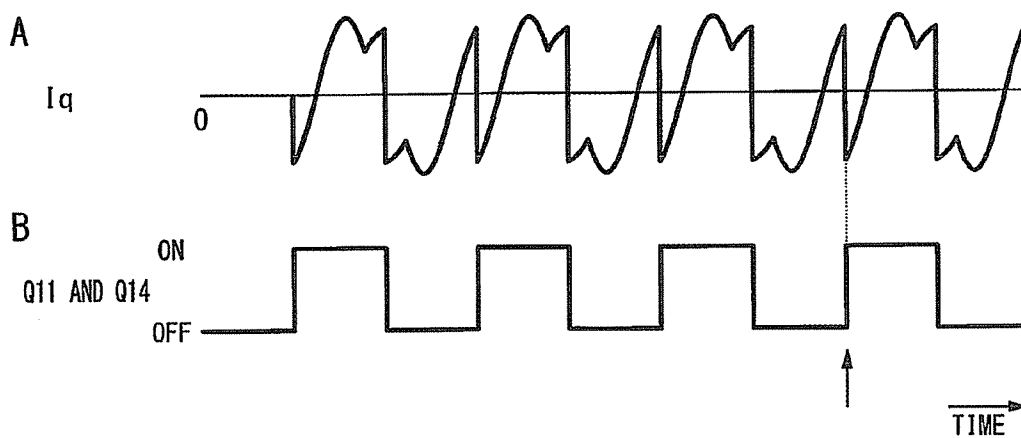


FIG. 7

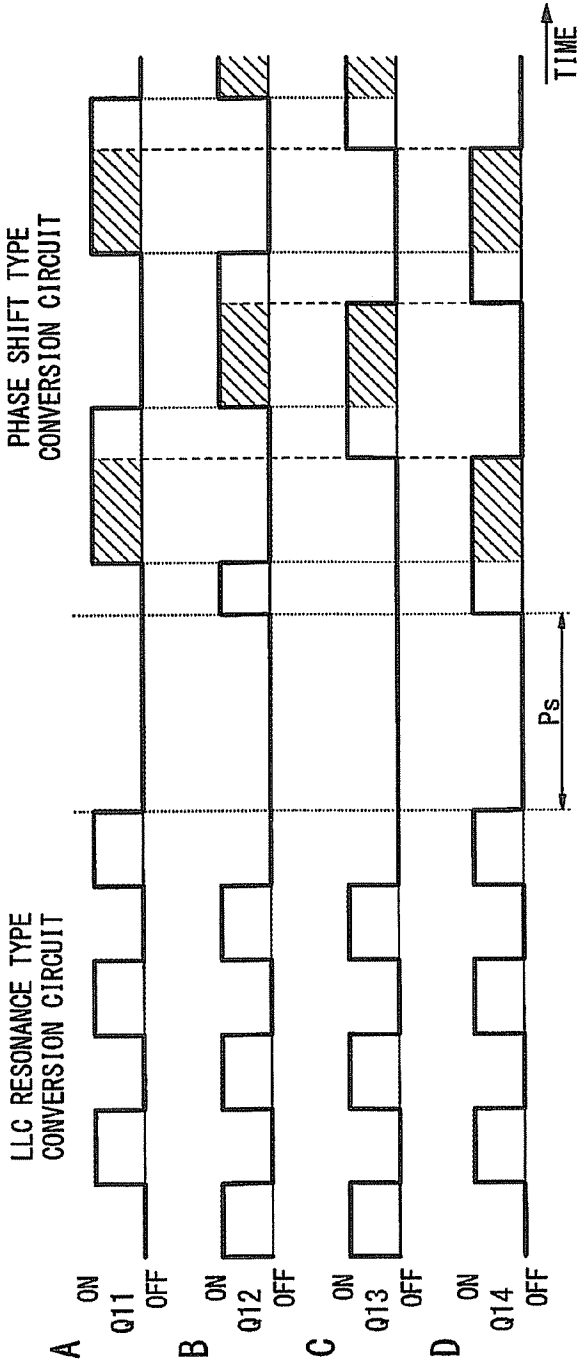


FIG. 8

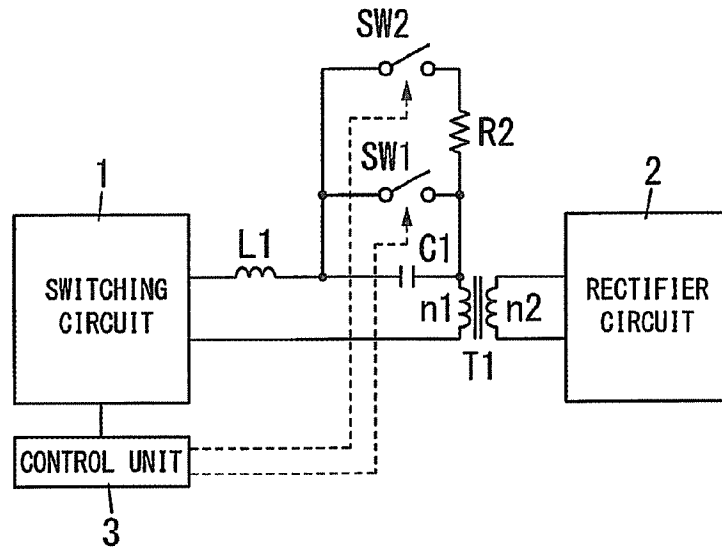


FIG. 9

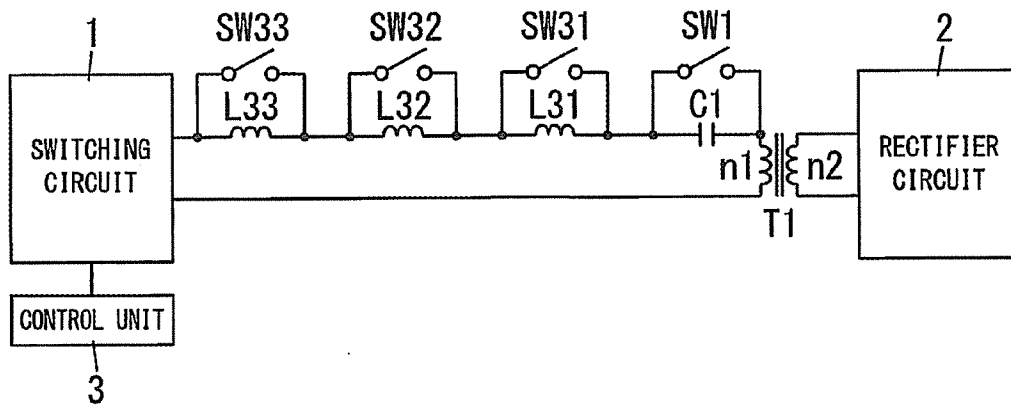


FIG. 10

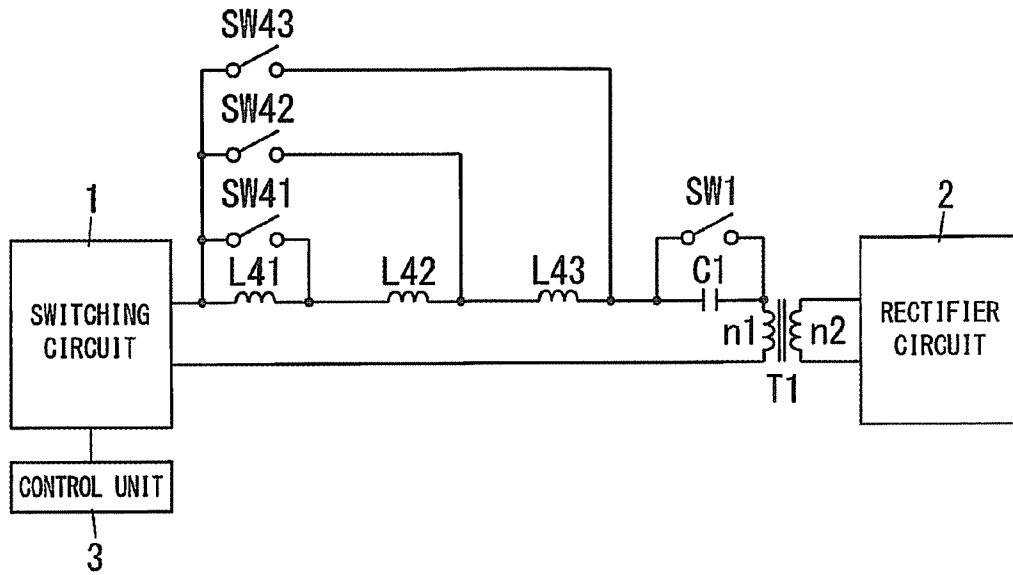


FIG. 11

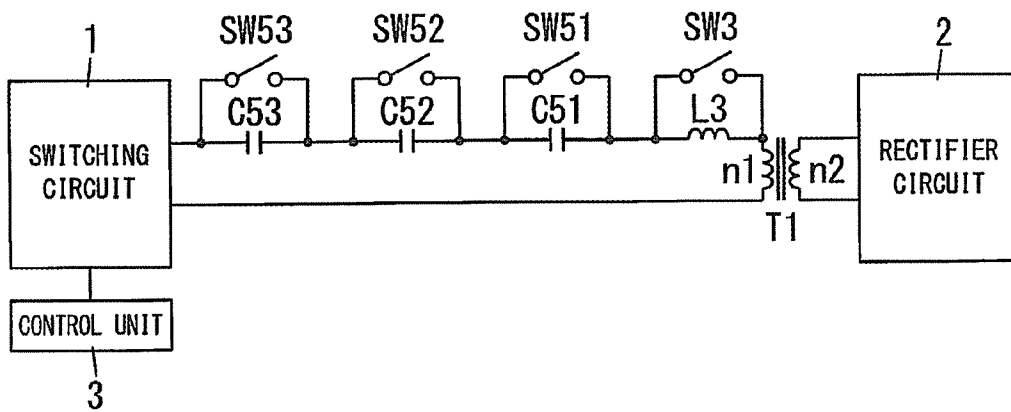
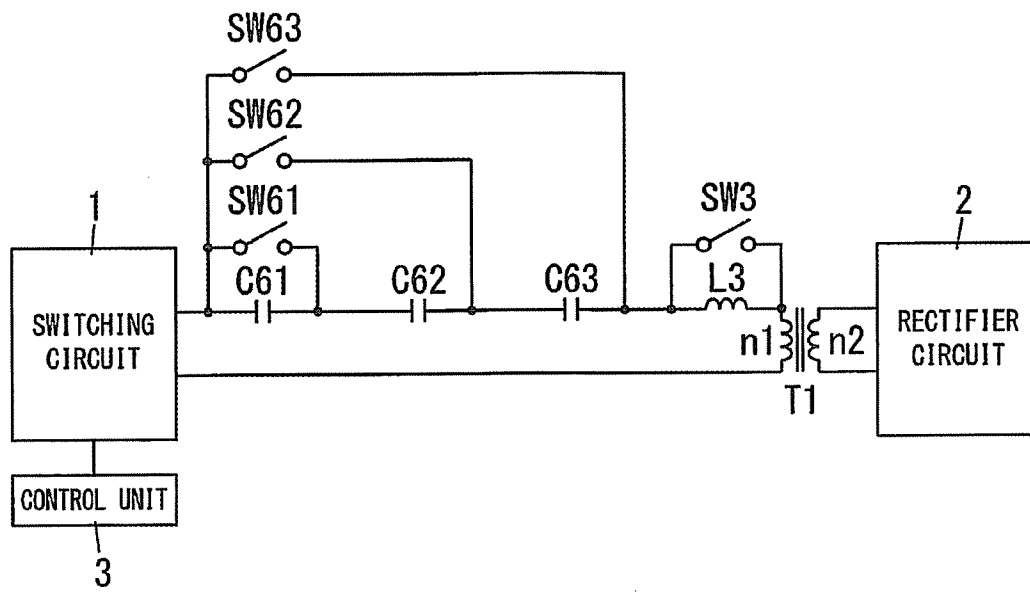


FIG. 12



INTERNATIONAL SEARCH REPORT

International application No.

PCT/JP2013/003211

5	A. CLASSIFICATION OF SUBJECT MATTER H02M3/335(2006.01) i, H02M3/28(2006.01) i	
	According to International Patent Classification (IPC) or to both national classification and IPC	
10	B. FIELDS SEARCHED	
	Minimum documentation searched (classification system followed by classification symbols) H02M3/00-3/44	
15	Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched Jitsuyo Shinan Koho 1922-1996 Jitsuyo Shinan Toroku Koho 1996-2013 Kokai Jitsuyo Shinan Koho 1971-2013 Toroku Jitsuyo Shinan Koho 1994-2013	
	Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)	
20	C. DOCUMENTS CONSIDERED TO BE RELEVANT	
	Category*	Relevant to claim No.
25	A	JP 2006-158137 A (TDK Corp.), 15 June 2006 (15.06.2006), abstract; fig. 1 (Family: none)
30	A	JP 4-125062 A (Toyoda Automatic Loom Works, Ltd.), 24 April 1992 (24.04.1992), claims; drawings (Family: none)
35	A	US 2009/0046482 A1 (SMITH, Stephen William), 19 February 2009 (19.02.2009), fig. 1 (Family: none)
40	☒ Further documents are listed in the continuation of Box C. ☐ See patent family annex.	
45	* Special categories of cited documents: "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family	
50	Date of the actual completion of the international search 22 July, 2013 (22.07.13)	Date of mailing of the international search report 30 July, 2013 (30.07.13)
	Name and mailing address of the ISA/ Japanese Patent Office	Authorized officer
55	Facsimile No.	Telephone No.

INTERNATIONAL SEARCH REPORT

International application No.
PCT/JP2013/003211

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C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

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Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	JP 2010-207014 A (Fujitsu Telecom Networks Ltd.), 16 September 2010 (16.09.2010), abstract; paragraphs [0017], [0019] to [0024]; fig. 1 (Family: none)	1-12
A	JP 2004-260928 A (Omron Corp.), 16 September 2004 (16.09.2004), abstract; drawings (Family: none)	1-12
A	JP 3-128672 A (General Electric Co.), 31 May 1991 (31.05.1991), entire text & US 4951185 A & EP 0408253 A2 & CA 2012811 A1	1-12
A	JP 3-251079 A (Origin Electric Co., Ltd.), 08 November 1991 (08.11.1991), claims; drawings (Family: none)	1-12
A	JP 2011-234541 A (Panasonic Electric Works Co., Ltd.), 17 November 2011 (17.11.2011), abstract & WO 2011/135415 A1 & US 2013/0044519 A1 & EP 2566027 A1 & CN 102986125 A & KR 10-2013-0036226 A	1-12

REFERENCES CITED IN THE DESCRIPTION

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Patent documents cited in the description

- JP 2011234541 A [0002]

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Date

10.04.17

Reference JL83874P.EPP	Application No./Patent No. 16173089.0 - 1809 / 3151411
Applicant/Proprietor VLT, Inc.	

Communication pursuant to Rules 70(2) and 70a(2) EPC and reference to Rule 39(1) EPC

The date on which the European Patent Bulletin mentions the publication of the European search report for the above-identified European patent application (publication number: 3151411) is 05.04.17.

Since the request for examination has been filed (R. 70(1), Art. 94(1) EPC) prior to the transmission of the European search report, you are hereby invited to indicate whether you desire to proceed further with the application.

If you do not reply to this invitation within **six months** of the above-mentioned publication date, the application **will be deemed to be withdrawn** (R. 70(3) EPC).

You are invited, within the above-mentioned six-month period, to comment on the objections raised in the opinion accompanying the European search report and/or to file any amendments to the description, claims and drawings correcting any deficiencies noted in the opinion (R. 70a(2), R. 137(2) EPC; Guidelines for Examination in the EPO, B-XI, 8).

If filing amendments, you must identify them and indicate the basis for them in the application as filed. Failure to meet either requirement may lead to a communication from the Examining Division requesting that you correct this deficiency (R. 137(4) EPC).

Should the reply to the invitation pursuant to Rule 70a(2) EPC be filed in an admissible non-EPO language, a translation is to be submitted within one month of its filing (R. 6(2) EPC).

Should you not comply with this invitation within the time limit, the application **will be deemed to be withdrawn** in accordance with Rule 70a(3) EPC.

Note

If not yet paid, the designation fee is also payable within **six months** of the above-mentioned publication date (R. 39(1) EPC).

This period for payment is applicable for extension and validation fees as well.

The present amount of the designation fee is EUR 585,00.

Note to users of the automatic debiting procedure

Unless the EPO receives prior instructions to the contrary, the designation fee will be debited on the last day of the period for payment. For further details see the Arrangements for the automatic debiting procedure, supplementary publication 3 - OJ EPO 2015.

Important information concerning fee amounts

Following any amendment to the Rules relating to Fees, the amount(s) mentioned in this communication may be different from the amount(s) **actually due on the date of payment**. The latest version of the Schedule of fees and expenses, published as a Supplement to the Official Journal of the EPO, is also available on the EPO website (www.epo.org) and can be found under www.epo.org/schedule-of-fees, which allows the viewing, downloading and searching for individual fee amounts, both current and previous.

Please note that procedural fees are usually adjusted every two years, on even years, with effect from 1 April.

Payments by cheque delivered or sent direct to the EPO are no longer accepted as from 1 April 2008 (see OJ EPO 2007, 626).

Note concerning renewal fees

Under Article 86 EPC renewal fees shall be paid to the EPO for the European patent application. These fees shall be due in respect of the third and each subsequent year, calculated from the date of filing. Due account should be taken that the EPO will not send invoices or reminders to pay any fees in due time.

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Date	14.02.17
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Reference JL83874P.EPP	Application No./Patent No. 16173089.0 - 1809
Applicant/Proprietor VLT, Inc.	

Communication

The extended European search report is enclosed.

The extended European search report includes, pursuant to Rule 62 EPC, the European search report (R. 61 EPC) or the partial European search report/ declaration of no search (R. 63 EPC) and the European search opinion.

Copies of documents cited in the European search report are attached.

☒ 0 additional set(s) of copies of such documents is (are) enclosed as well.

The following have been approved:

☒ Abstract ☒ Title

☐ The Abstract was modified and the definitive text is attached to this communication.

The following figure(s) will be published together with the abstract: 1

Refund of search fee

If applicable under Article 9 Rules relating to fees, a separate communication from the Receiving Section on the refund of the search fee will be sent later.

Should you wish to further prosecute this application in the examination phase, your attention is drawn to the provisions of Rule 70a EPC. An invitation to respond to the extended European search report will be issued once the date of publication of the European search report has been mentioned in the European Patent Bulletin (R. 69(1), R. 70(2) EPC).



EUROPEAN SEARCH REPORT

Application Number
EP 16 17 3089

DOCUMENTS CONSIDERED TO BE RELEVANT			
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X	JP H02 280666 A (SANKEN ELECTRIC CO LTD) 16 November 1990 (1990-11-16)	1,8, 10-12	INV. H02M3/337
A	* abstract * * figures 1,2 *	2-7,9, 13-15	ADD. H02M1/00
A	----- EP 2 863 531 A1 (PANASONIC IP MAN CO LTD [JP]) 22 April 2015 (2015-04-22) * the whole document *	1-15	
A	----- Shizumasa Okudaira ET AL: "A New Quasi-Resonant Inverter With Two-way Short-circuit Switch Across A Resonant Capacitor", Proceedings of the Power Conversion Conference, 2002. PCC-Osaka 2002., 5 April 2002 (2002-04-05), XP055340750, DOI: 10.1109/PCC.2002.998195 Retrieved from the Internet: URL:http://ieeexplore.ieee.org/ielx5/7830/ 21539/00998195.pdf?tp=&arnumber=998195&isn umber=21539 [retrieved on 2017-01-31] * the whole document *	1-15	
			TECHNICAL FIELDS SEARCHED (IPC)
			H02M H01R
The present search report has been drawn up for all claims			
Place of search		Date of completion of the search	Examiner
The Hague		7 February 2017	Lochhead, Steven
CATEGORY OF CITED DOCUMENTS X : particularly relevant if taken alone Y : particularly relevant if combined with another document of the same category A : technological background O : non-written disclosure P : intermediate document T : theory or principle underlying the invention E : earlier patent document, but published on, or after the filing date D : document cited in the application L : document cited for other reasons & : member of the same patent family, corresponding document			

2

**ANNEX TO THE EUROPEAN SEARCH REPORT
ON EUROPEAN PATENT APPLICATION NO.**

EP 16 17 3089

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report.
The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

07-02-2017

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
JP H02280666 A	16-11-1990	JP H0746903 B2	17-05-1995
		JP H02280666 A	16-11-1990

EP 2863531 A1	22-04-2015	EP 2863531 A1	22-04-2015
		JP 5906418 B2	20-04-2016
		JP 2014003764 A	09-01-2014
		US 2015124490 A1	07-05-2015
		WO 2013186991 A1	19-12-2013

The examination is being carried out on the **following application documents**

Description, Pages

1-8 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/3-3/3 as originally filed

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 JP H02 280666 A (SANKEN ELECTRIC CO LTD) 16 November 1990 (1990-11-16)

D2 EP 2 863 531 A1 (PANASONIC IP MAN CO LTD [JP]) 22 April 2015 (2015-04-22)

Conciseness

2 The application does not meet the requirements of Article 84 EPC, because claims 1, 8, 10, 11 and 12 are not concise.

2.1 Claims 1, 8, 10, 11 and 12 have been drafted as separate independent claims.

Under Article 84 in combination with Rule 43(2) EPC, an application may contain more than one independent claim in a particular category only if the subject-matter claimed falls within one or more of the exceptional situations set out in paragraph (a), (b) or (c) of Rule 43(2) EPC. This is not the case in the present application, however, the claims all appearing to relate to the same method of converting power.

- 2.2 In the further prosecution of the application, failure to file an amended set of claims which complies with Rule 43(2) EPC, or to submit convincing arguments as to why the current set of claims does in fact comply with these provisions, may lead to refusal of the application under Article 97(2) EPC.

Novelty

- 3 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1, 2, 8, 10, 11 and 12 is not new within the meaning of Article 54(1) and (2) EPC.

- 3.1 Document D1 discloses:

A power conversion apparatus for converting power between an input source and a load (abstract), where the load may vary over a normal operating range (abstract, last sentence), comprising:

a transformer (figure 1:T);

a resonant circuit including the transformer and a resonant capacitor and having a characteristic resonant frequency and period (figure 1:L₁,C₁ and abstract);

output circuitry (figure 1:7) connected to the transformer for delivering a rectified output voltage to the load;

two or more primary switches (figure 1:Q1,Q2) to drive the resonant circuit;

a clamp switch (figure 1:S1,S2) to clamp the resonant capacitor; and

a switch controller (figure 1:25) to operate the primary switches and the clamp switch in a series of converter operating cycles, each converter operating cycle comprising:

(a) two power transfer intervals (figure 2:periods t₀-t₆) of essentially equal duration during which one or more of the primary switches are ON, a magnetizing current flows in a winding of the transformer, and power is transferred between the input and the output via the transformer, each power transfer interval comprising:

(i) first and second resonant intervals (figure 2:t2-t4 and corresponding period in second half-wave), each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

(ii) a clamp interval (figure 2:S2 high period) occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

3.2 The definition of the converter operating cycles is open in claim 2 and may therefore correspond to two switching periods, each corresponding to a power transfer interval. The subject-matter of claim 1 is therefore not new.

3.3 The subject-matter of independent claim 1 corresponds in terms of method features to that of claim 2. The objections raised with respect to the latter claim, therefore, also apply, mutatis mutandis, to independent claim 1, which is therefore not allowable under Article 52(1) EPC.

3.4 The subject-matter of claims 8, 10, 11 and 12 corresponds to that of claim 1. The objections raised with respect to the latter claim, therefore, also apply, mutatis mutandis, to independent claims 8, 10, 11 and 12, which are therefore not allowable under Article 52(1) EPC.

Further Remarks

4 The invention as currently claimed includes in its scope embodiments in which the transformer and resonant capacitor are connected in parallel. Such embodiments do not, however, appear to be supported by the description (Article 84 EPC), which does not supply the skilled reader with sufficient information to allow him to adjust the durations of the output current pulses for supplying the varying load with such a configuration.

Amendments

5 The applicant is invited to file new claims which take account of the above comments.

5.1 To meet the requirements of Rule 42(1)(b) EPC, document D2 should be identified in the description and its relevant contents should be indicated. The applicant should ensure that it is clear from the description which features of the subject-matter of independent claims are known from the document D2.

5.2 Independent claims should be in the two-part form in accordance with Rule 43(1) EPC, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

5.3 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see Guidelines F-IV, 4.19).

5.4 When filing amended claims the applicant should at the same time bring the description into conformity with the amended claims. Care should be taken during revision, especially of the introductory portion and of any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

5.5 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H III, 2.1).

If the applicant considers it appropriate, these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed.

Information on Search Strategy - Pilot phase (see OJ 2015, A86)

The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number

EP 16 17 3089

TITLE: CLAMPED CAPACITOR RESONANT POWER CONVERTER

APPLICANT: VLT, Inc.

IPC CLASSIFICATION: H02M3/337, H02M1/00

EXAMINER: Lochhead, Steven

CONSULTED DATABASES: NPL, WPI

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC: H02M3/337, H02M3/3376, H02M2001/0058, H02M3/28/LOW, H02M2007/4818, 5H730/BB27, 5H730/BB66, 5H730/FG21, H02M3/33569, H02M2007/4815, H02M2005/2937, 5H750/DD06, H01R4/646

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:

full-bridge, resonant, quasi-resonant, shunt, clamp, parallel, switch

A New Quasi-Resonant Inverter

With Two-way Short-circuit Switch Across A Resonant Capacitor

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Abstract

Two frequencies are often used in induction heating, that is, a lower frequency f_L is for preheating and a higher frequency f_H is for main heating in order to decrease thermal stress. In this case, each of f_H and f_L is generated by the different inverter, and f_H approximately equals several times to ten times of f_L . Then we propose a new quasi-resonant inverter that has a two-way short-circuit switch across the resonant capacitor. The proposed inverter has features that its operating frequency is adjustable and the frequency range is so wide that the higher frequency can be ten times as much as the lower frequency. It is considered that the output current has only a fundamental wave by properly manipulating the shorting time of the resonant capacitor even though the output current is not a sinusoidal wave.

Key words : quasi-resonant inverter, adjustable frequency, capacitor shorting, dual frequency, induction heating

1 Introduction

The purpose of this study is to propose a novel quasi-resonant inverter whose operating frequency can be adjustable over a wide range for induction heating and to present a frequency control method such that higher harmonics of the output current may be suppressed.

Dual frequencies which are a higher frequency f_H and a lower frequency f_L , are often used for induction heating [1]-[5]. According to [1], it is to be desired that the ratio of f_H to f_L should be 3.3 in case of gear hardening. It is impossible for a conventional resonant inverter to generate two frequencies because its operating frequency is kept constant i.e. the resonant frequency. Then we reported a quasi-resonant inverter that the operating frequency was adjustable [6]-[9], however, the higher operating frequency f_H was about three times as much as the lower operating frequency f_L and the operating frequency range was not so wide. Practically it is also proposed that the ratio f_H/f_L should be about ten according to [2]. Further, in contour induction hardening [3] which consists of preheating and final heating, the lower frequency f_L is used in the preheating and the higher frequency f_H is used in the final heating for a result of high residual compressive stresses and a precise case of heated depth. In both cases, the ratio f_H/f_L needs to be about ten or more and the reported inverter [6]-[9] could not be used. It is desired that the operating frequency of the inverter can be widely changed for various purposes.

Thus we propose a new adjustable frequency quasi-resonant inverter that has a two-way short-circuit switch across the resonant capacitor. The proposed inverter has a feature that its operating frequency range is wide and the higher frequency can be ten times as much as the

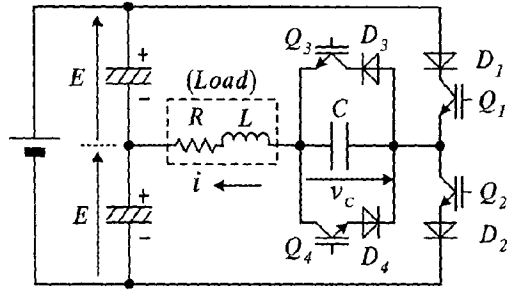


Fig.1 Circuit configuration

lower frequency. However the output current i is not a sinusoidal wave and includes many harmonics. The harmonic current causes the surface of a work to be overheated, and it is needed that the harmonics should be suppressed. It is considered that the output current has only a fundamental wave by properly manipulating the shorting time of the resonant capacitor even though the output current is not a sinusoidal wave.

2 Circuit Configuration and Operation Principle

Fig.1 shows a proposed inverter circuit using IGBTs as switching devices. In Fig.1, R means an equivalent resistance and L means an equivalent inductance of an induction heater with an iron-core matching transformer. Q_3 and Q_4 play the role of shorting the resonant capacitor C , whereas Q_1 and Q_2 work as switches for changing the polarity of the power supply voltage E .

Fig.2 shows typical operation waveforms of the output current i and the capacitor voltage v_c and mode classifications. It is supposed that the resonant capacitor has been charged to $-V_m$ before the mode 1 begins. The operation in each mode is disclosed as follows on condition that the IGBTs $Q_1 - Q_4$ and the diodes $D_1 - D_4$ are ideal switching devices.

1) Mode 1 : On-signal is supplied to Q_1 and Q_2 . But Q_3 cannot turn on because the voltage across D_3 is negative. Then the discharge current i flows under LC resonant state.

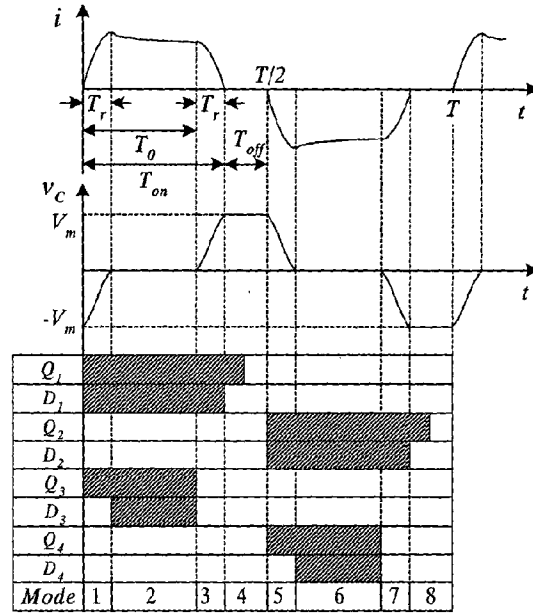


Fig.2 Typical operation waveforms and mode classifications

2) Mode 2 : When the capacitor voltage v_c returns to zero, Q_3 and D_3 turn on, and the non-oscillatory current i is supplied to the load.

3) Mode 3 : After desired time T_o passes, Q_3 is made off. Then the charge current i decreases under LC resonant state and the capacitor voltage v_c rises. When the current i returns to zero, the diode D_1 turns off.

4) Mode 4 : The circuit becomes in a standby state until the beginning of the next mode. After desired period $T/2$ passes, the antisymmetric operation starts.

In Mode 5 - 8, the circuit operates in the same manner as the operation in Mode 1 - 4. Thus the typical output current i and the capacitor voltage v_c are obtained as in Fig.2. In Fig.2, the operating frequency f is adjustable by manipulating T and the absolute value of the current i is adjustable by manipulating the capacitor shorting time T_o . At the same time, harmonics of the current i are variable by manipulating the capacitor shorting time T_o .

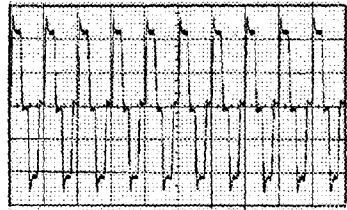
It is basically possible that the operating frequency f

Table 1 Parameters

Equivalent inductance L	82 μH
Resonant capacitance C	0.5 μF
Equivalent resistance R	about 4 Ω



(a) $f = 1 \text{ kHz}$



(b) $f = 10 \text{ kHz}$

(Abscissa : 100 $\mu\text{s}/\text{div}$, Ordinate axis : 5A/div)

Fig.3 Examples of waveforms output current i

becomes infinitely low by increasing the cycle time T then the operating frequency range can be made very wide. Besides, the harmonics can be suppressed by properly manipulating the capacitor shorting time T_o .

3 Experimental Waveforms

A prototype inverter system is manufactured and the parameters are shown in Table 1. The value of R is rough and depends on the frequency f of the supplied current. When the power supplied into the work is controlled, the value of R must be known by some means *i.e.* detecting during the operation [9].

Examples of the waveforms of the output current i are illustrated in Fig.3 when d.c. power voltage $E = 50\text{V}$. As shown in Fig.3, it is clear that the prototype inverter

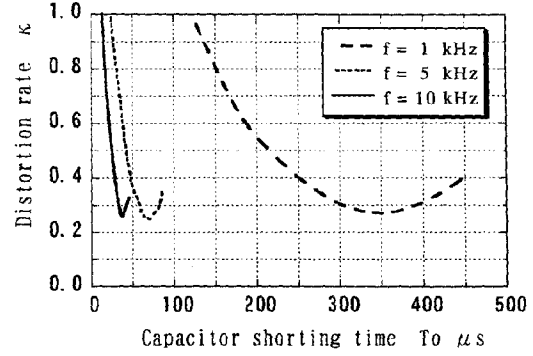


Fig.4 Relations between the capacitor shorting time T_o and the distortion rate κ of the output current (Analytical results)

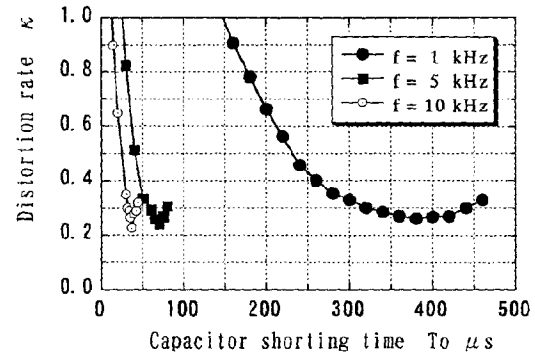


Fig.5 Relations between the capacitor shorting time T_o and the distortion rate κ of the output current (Experimental results)

operates in accordance with the theory in Fig.2 and the higher operating frequency can be ten times as much as the lower operating frequency. But the waveforms are not sinusoidal and it is assumed that many harmonics are included in the current.

4 Harmonics Characteristics and Harmonics Control

4.1 Harmonics Characteristic

Fig.4 and Fig.5 show relations between the capacitor shorting time T_o and the distortion rate κ of the output

current, where Fig.4 indicates analytical results and Fig.5 indicates experimental results. The distortion rate κ is defined as (1).

$$\kappa = \frac{\sqrt{I^2 - I_1^2}}{I_1} = \sqrt{\left(\frac{I}{I_1}\right)^2 - 1} \quad (1)$$

where I_1 is the effective value of the fundamental wave of the output current i and I is the effective value of the output current i .

The analytical characteristics are very similar to the experimental characteristics. It is clear that the distortion rate κ depends on T_0 and has a minimum value. Therefore higher harmonics can be suppressed by manipulating T_0 properly and then the surface of a work is prevented from overheating.

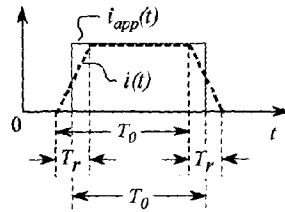
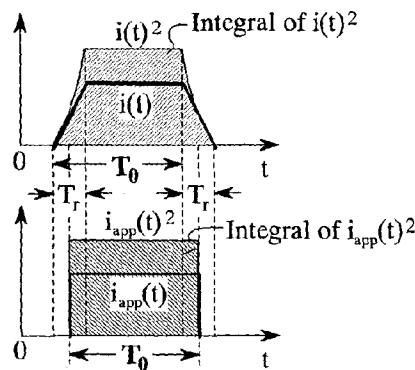
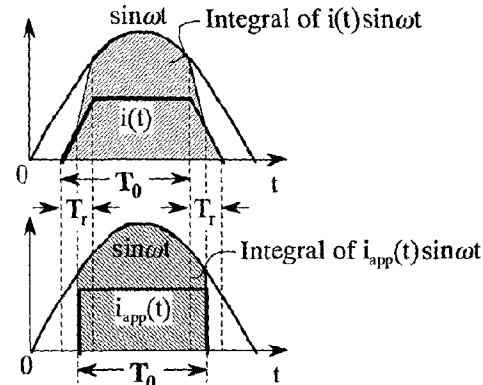


Fig.6 Approximation of i by square wave



(a) Calculation of the effective value of the current



(b) Calculation of the fundamental current

Fig.7 Validity of the approximation of i for calculating (2) and (3)

4.2 Harmonics Control

The fundamental wave content λ_1 of the output current should be the maximum in order to minimize the distortion rate κ . Thus the capacitor shorting time T_0 is such manipulated that λ_1 may be the maximum. The fundamental wave content λ_1 is defined as (2).

$$\lambda_1 = \frac{I_1}{I} \quad (2)$$

Each of I_1 and I is calculated by (3) and (4) as well known.

$$I_1 = \frac{4}{\sqrt{2}T} \int_0^{T/2} i(t) \cdot \sin \omega t dt \quad (3),$$

$$I = \sqrt{\frac{2}{T} \int_0^{T/2} i(t)^2 dt} \quad (4).$$

Strictly speaking, the equation of i should be obtained by solving the circuit equation in each mode 1 to 4 in Fig.2 but it is too complicated to solve the circuit equations during the operation. Then the waveform of the output current i is approximated by a square wave. Though the shape of the output current i is like a trapezoid wave shown in Fig.2 or Fig.3, for convenience, let

the waveform of the output current i be a square wave i_{app} which has dead time as shown in Fig.6. Let us assume that the approximated wave i_{app} has the maximum value that is equal to the trapezoid wave i and has on-time T_0 that is equal to the capacitor shorting time.

Fig.7 shows the validity of the approximation of i in order to simply calculate (3) and (4). In Fig.7, the upper sketches in (a) and (b) illustrate the calculation images of the effective value and the fundamental wave of the trapezoid wave i , and the lower sketches illustrate the images of the approximated wave i_{app} . In Fig.7(a), each shadowed portion expresses the integral of i^2 and the integral of i_{app}^2 . It is considered that both are almost equal and we can use i_{app} instead of i for calculating the effective value in (4). Likewise, In Fig.7(b), each shadowed portion expresses the integral of $i \sin \alpha$ and the integral of $i_{app} \sin \alpha$. It is deemed that both are almost equal and we can use i_{app} instead of i for calculating the fundamental wave in (3).

Fig.8 illustrates an equivalent square wave of the output current i in order to calculate the fundamental wave content λ_1 in (2) and λ_1 is given by (5).

$$\lambda_1 = \frac{2\sqrt{T}}{\pi} \cdot \frac{\sin \frac{\omega T_0}{2}}{\sqrt{T_0}} \quad (5)$$

where $\omega = 2\pi/T = 2\pi f$ and f means the operating frequency i.e. the fundamental output frequency.

The equation $d\lambda_1/dT_0 = 0$ is solved in order to obtain the capacitor shorting time T_{OM} which gives the maximum value of the fundamental wave content λ_1 . T_{OM} is obtained as (6).

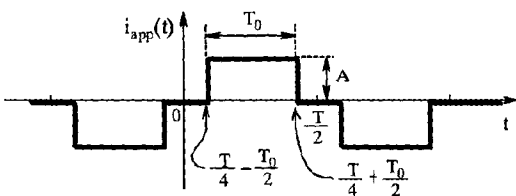


Fig.8 Equivalent square wave of the output current i

$$T_{OM} = 0.371/f \quad (6)$$

We can minimize the higher harmonics of the output current by the manipulated variable T_{OM} in (6).

4.3 Experimental Results of Harmonics Control

Fig.9 shows the relation between the operating frequency f and the capacitor shorting time T_{OM} for minimizing the distortion rate. The calculated values by (6) agree well with the experimental results. Thus (6) is appropriate for controlling the operating frequency so that the harmonics including the output current may be suppressed.

Fig.10 discloses an example of experimental results of harmonics control by using (6) in case of $f = 1$ kHz and $f = 10$ kHz. It is apparent that higher harmonics of the current are suppressed and the fundamental content becomes about 1.0 even though the operating frequency f is adjusted high or low. There is not a serious problem although a little higher harmonics of the output current are included, because the power is proportion to the square of the current and the higher harmonics of the power are very little.

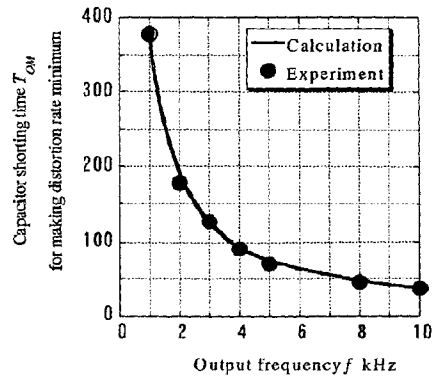


Fig.9 Relation between the operating frequency f and the capacitor shorting time T_{OM} for minimizing the distortion rate

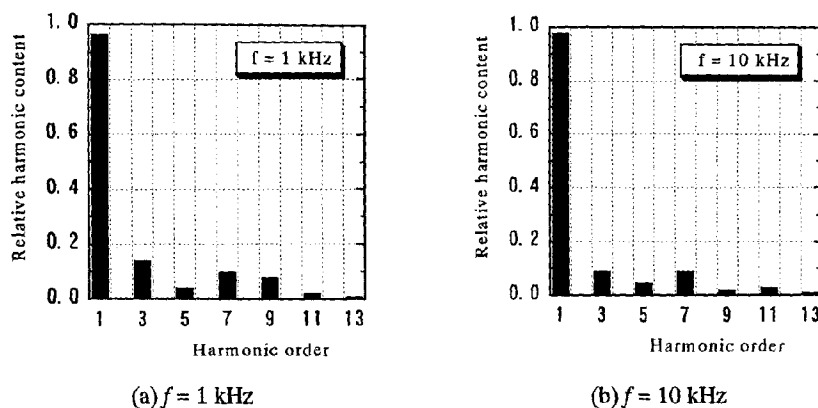


Fig.10 Example of experimental results of harmonics control

5 Conclusions

A new quasi-resonant inverter that has a two-way short-circuit switch across the resonant capacitor. The proposed inverter has a feature that its operating frequency range is wide and the higher frequency can be ten times as much as the lower frequency. Although the output current is not a sinusoidal wave and includes many harmonics, the harmonics can be suppressed by manipulating the capacitor shorting time properly and then the surface of a work is prevented from overheating. According to the present inverter, dual frequency induction heating which consists of the preheating and the final heating can be possible with one inverter system.

In the future, we will discuss how to control the output power supplied into the work, where the equivalent output resistance and the equivalent loss resistance are unknown and depend on the operating frequency [9].

References

- [1] Yoichi Matsubara, Makoto Kumakawa and Yasuo Watanabe, "Induction Hardening of Gear by the Dual Frequency Induction Heating", Transactions of Heat Treatment, Vol.29, No.2,1989 (in Japanese).
- [2] Industrial heating Inc., http://www.industrialheating.com/CDA/ArticleInformation/features/BNP_Features_Item/0,2832,27824,00.html
- [3] Neturen.Co.,Ltd., <http://ns.k-neturen.co.jp/gijutsu/gijutsu.html>
- [4] Dai-ichi High Frequency Co.,Ltd., <http://www.dhf.co.jp/>
- [5] J.Ying and K.Heumann, "Series Resonant Inverter with Cycle Control for Induction Heating", Proceeding IPEMC, Vol.1, pp.367-373, 1994.
- [6] S.Okudaira, K.Nomura and K.Matsuse: "New Quasi-Resonant Inverter for Induction Heating", in Conference Record of PCC-Yokohama, pp.117-122, 1993.
- [7] S.Okudaira and K.Matsuse: "Adjustable Frequency Quasi-Resonant Inverter and Basic Characteristics", Transactions of IEEE, Vol. D-114, 1994 (in Japanese).
- [8] S.Okudaira and K.Matsuse: "Frequency and Power Control of Induction Heater with Adjustable Frequency Quasi-Resonant Inverter", Transactions of IEEE, Vol. D-118, 1997 (in Japanese).
- [9] S.Okudaira & K.Matsuse: "Power Control of an Adjustable Frequency Quasi-Resonant Inverter for Dual Frequency Induction Heating", Proceeding of IPEMC-2000, S-34-2, pp.968-973, 2000



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Title page for ETD etd-71398-22552

Type of Document	Dissertation												
Author	Watson, Robert III												
Author's Email Address	watsonr@tce.com												
URN	etd-71398-22552												
Title	New Techniques in the Design of Distributed Power Systems												
Degree	Doctor of Philosophy												
Department	Electrical and Computer Engineering												
Advisory Committee	<table> <tr> <th>Advisor Name</th><th>Title</th></tr> <tr> <td>Fred C. Lee</td><td>Committee Chair</td></tr> <tr> <td>Dan M. Sable</td><td>Committee Member</td></tr> <tr> <td>Dan Y. Chen</td><td>Committee Member</td></tr> <tr> <td>Douglas J. Nelson</td><td>Committee Member</td></tr> <tr> <td>Jason Lai</td><td>Committee Member</td></tr> </table>	Advisor Name	Title	Fred C. Lee	Committee Chair	Dan M. Sable	Committee Member	Dan Y. Chen	Committee Member	Douglas J. Nelson	Committee Member	Jason Lai	Committee Member
Advisor Name	Title												
Fred C. Lee	Committee Chair												
Dan M. Sable	Committee Member												
Dan Y. Chen	Committee Member												
Douglas J. Nelson	Committee Member												
Jason Lai	Committee Member												

Keywords	• DC Distributed Power System • AC Distributed Power System • Soft-switching • Electromagnetic Interference
Date of Defense	1998-08-07
Availability	unrestricted

Abstract

Power conversion system design issues are expanding their role in technology equipment design philosophies. These issues include not only power conversion efficiency, but also increased concerns regarding the complexity of the power conversion design techniques utilized to satisfy a system's total performance requirements. In particular, in computer (desktop computers, workstations, and servers) designs, the power "supplies" are becoming a limiting factor in meeting overall design objectives.

This dissertation addresses the issue of simplifying the architecture of power systems incorporated into computing equipment. In the first half, the subject of the design of the distributed power system's front end is investigated from the perspective of simplifying the conversion process while simultaneously improving efficiency. This is initially accomplished by replacing the second-stage DC/DC converter in the standard two-stage front-end (consisting of a first-stage AC/DC converter followed by DC/DC conversion) through the incorporation of second-order active power factor control. Unique modifications are then made to two basic topologies (buck and boost converter topologies) that enable the two-stage front-end to be reduced to an isolated PFC conversion stage, resulting in a front-end architecture with features of reduced complexity and higher efficiency.

In the dissertation's second half, the overall DC distributed power system concept is simplified through the elimination of power processing components, the result being the creation of a high-frequency (HF) AC distributed power system. Design techniques for generating, distributing, and processing HF AC power are developed and experimentally verified. Also, an experimental comparison between both DC and AC distributed power systems is

illustrating in a succinct fashion the merits and limitations of both a

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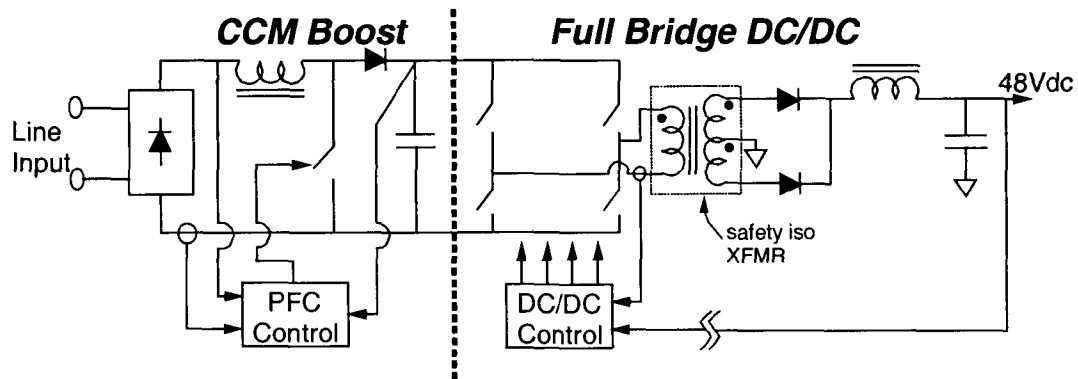
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4. ACTIVE-CLAMP BOOST AS AN ISOLATED PFC FRONT-END CONVERTER

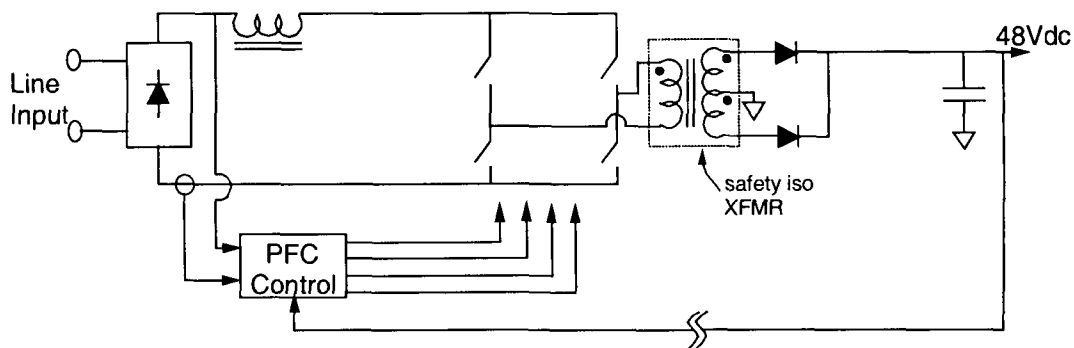
4.1 Introduction

This chapter continues the theme set by Chapter 3 - simplifying the standard two-stage front-end implementation to one that consists of just an isolated PFC converter. However, this chapter addresses this issue for higher power level system requirements.

As a starting point, shown in Fig. 4.1(a) is a typical, higher power output, two-stage front-end design topology. In Chapter 2, this design was simplified by incorporating secondary-side switching in the DC/DC converter. However, just as in Chapter 3, the "bulk" capacitor can be moved to the secondary, as shown in Fig. 4.1(b). Because kilowatt applications are being targeted, the full-bridge structure is maintained, and essentially replaces the PFC boost converter switch. Therefore, whatever the preferred PFC control method that is used in the two-stage approach is preserved in the implementation of Fig. 4.1(b). As with the front-end PFC flyback converter described in Chapter 3, the input line current's second harmonic is processed by the isolation transformer in the full-bridge boost converter shown in Fig. 4.1(b) and appears as voltage ripple on the output DC bus. However, the overall power stage design is much simpler, and, as will be experimentally demonstrated later in this chapter, this PFC implementation is more efficient than the two-stage circuit shown in Fig. 4.1(a).



(a) Topological concept of a traditional two-stage front-end design.



(b) Front-end consisting of an isolated boost PFC converter only.

Fig. 4.1 Two-stage and isolated boost PFC front-end conceptual designs.

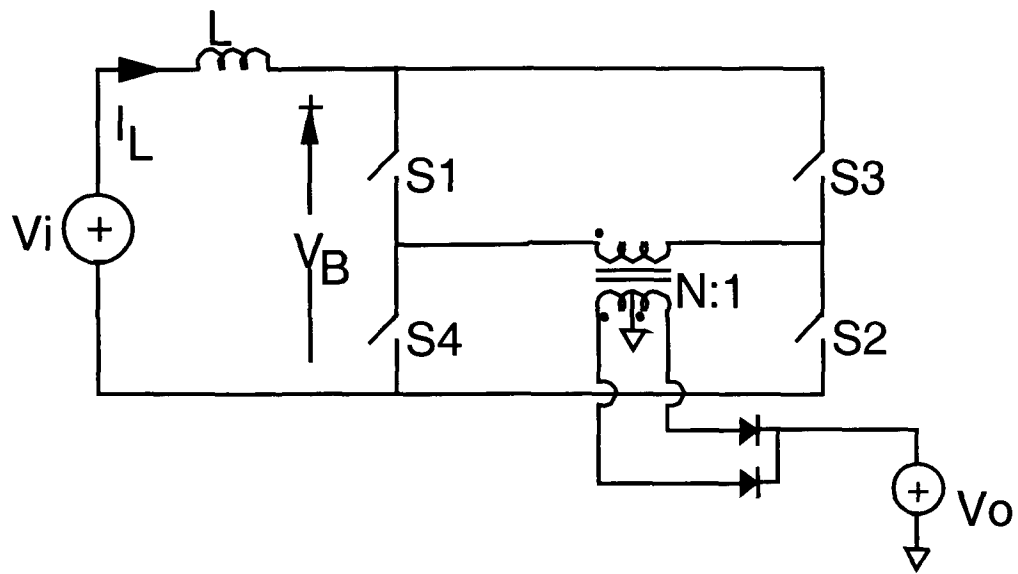
The full-bridge structure of the PFC boost converter's "switch" introduces the possibility of phase-shift control, in an analogous fashion to the phase-shift control used in voltage fed full-bridge converters. As will be discussed in the next section, phase-shifted control for the design shown in Fig. 4.1(b) can realize zero-current-switching (ZCS) for two of the four bridge switches, and a solution to the "voltage-spike" problem - so apparent in transformer isolated current fed designs, leads to ZVS for all four of the bridge switches [51].

4.2 Theory of Operation

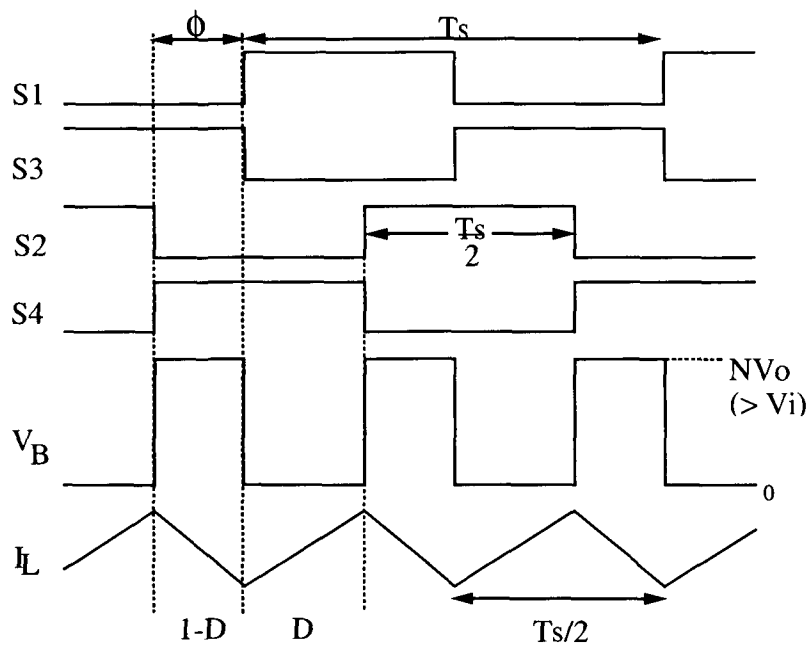
Figure 4.2 shows a simplified schematic of the active-clamp full-bridge boost topology. CCM operation for the boost inductor is assumed. Phase-shifting of the bridge switching sequence is utilized to control the boost function. The full-bridge topology, coupled with phase-shift control, offers a couple of advantages: 1) the boost inductor ripple frequency is twice the switching frequency, reducing its size, and, 2) with a slight modification to the phase-shift control ZCS becomes possible for the two of the four bridge switches [52]. Figures 4.2 and 4.3 illustrate these two concepts. For ZCS it is necessary to utilize the leakage inductance in the transformer (or possibly an external resonant inductor). Referring to Fig. 4.3, turning on S4 before S2 is turned off forces the current in L_{lk} to decrease linearly at a rate determined by:

$$\frac{di_{L_{LK}}}{dt} = \frac{NV_O}{L_{LK}}. \quad (4.1)$$

Current is increasing in S4 at the same rate (it's assumed the boost inductor can be

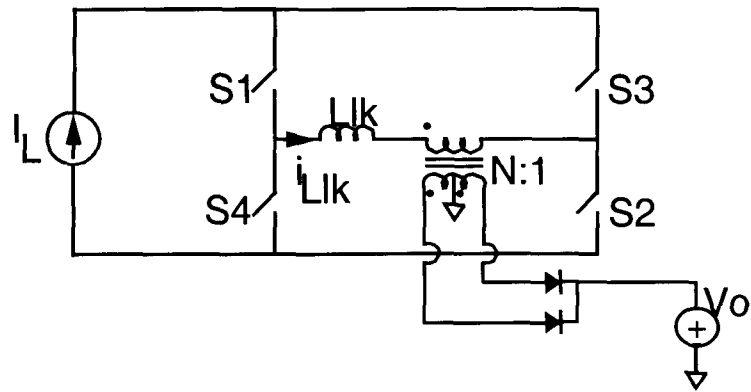


(a) Simplified circuit.

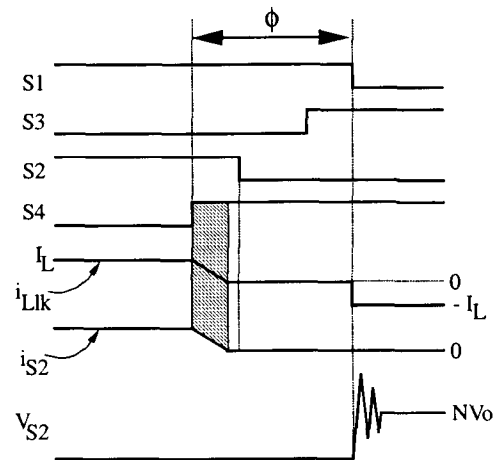


(b) Idealized waveforms.

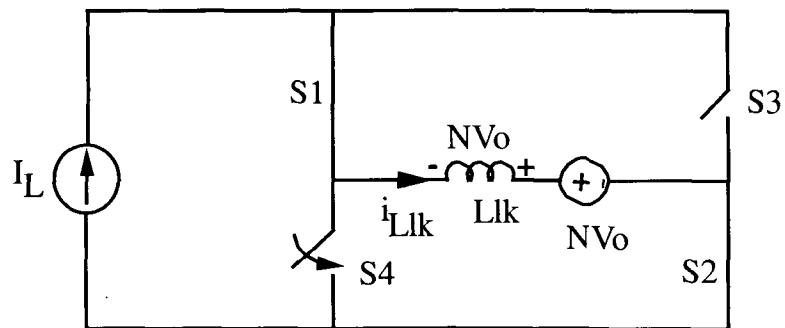
Fig. 4.2 Simplified ϕ -shifted active-clamp full-bridge boost converter operation.



(a) Simplified circuit with leakage inductance.



(b) Circuit waveforms.



(c) Converter topology for ZCS.

Fig. 4.3 Basic concept of boost full-bridge phase-shift operation [52].

idealized as a current source). When i_{s2} reaches zero the switch can be turned off with zero current.

As a practical matter, the isolated boost converter suffers from a problem at switch turn-off due to the dissimilarity of the values of the current source feeding the bridge (assuming the boost inductor can be idealized as a current source) and the current in the isolation transformer's leakage inductance. Unless an alternative path for the boost inductor current is temporarily provided, a large voltage spike, capable of destroying devices, will appear across the opened bridge switches. Some form of snubbing or clamping is necessary. For high-power applications RC snubbers are not very efficient. Therefore, an active-clamp circuit [51] (as shown in Fig. 4.4) is proposed for this application. In addition to the clamping function, the active-clamp circuit enables the isolation transformer's leakage energy to be used for zero-voltage switching (ZVS) of the bridge switches. Zero-voltage-switching is particularly important for S1 and S3, since snubber capacitors are used to reduce their turn-off losses. The ZVS mechanism recovers this turn-off energy before it is dissipated by switch turn-on. However, it should be pointed out that the circuit proposed here differs from that discussed in [51] in three fundamental ways: 1) ϕ -shift operation of the bridge switches enables zero-current switching (ZCS), 2) individual switches in the bridge operate with a fixed 50 % duty cycle, and 3) the operation of the boost inductor is in continuous conduction mode - necessary for single-phase high power operation.

Figure 4.4 shows the basic waveforms of operation with the active-clamp circuit. These waveforms are for a switching cycle and assume sufficient energy is stored in the leakage inductance to discharge C_{snub} completely. Also, it's assumed the clamp capacitor is infinite in value and the boost inductor can be replaced by a current source. The sequence of topological states is shown in Fig. 4.5, and is described below:

T_0-T_1 : At T_0 , S3 is turned off and the boost inductor current charges the switch capacitance. In order to control the switch turn-off losses, an external capacitor may be

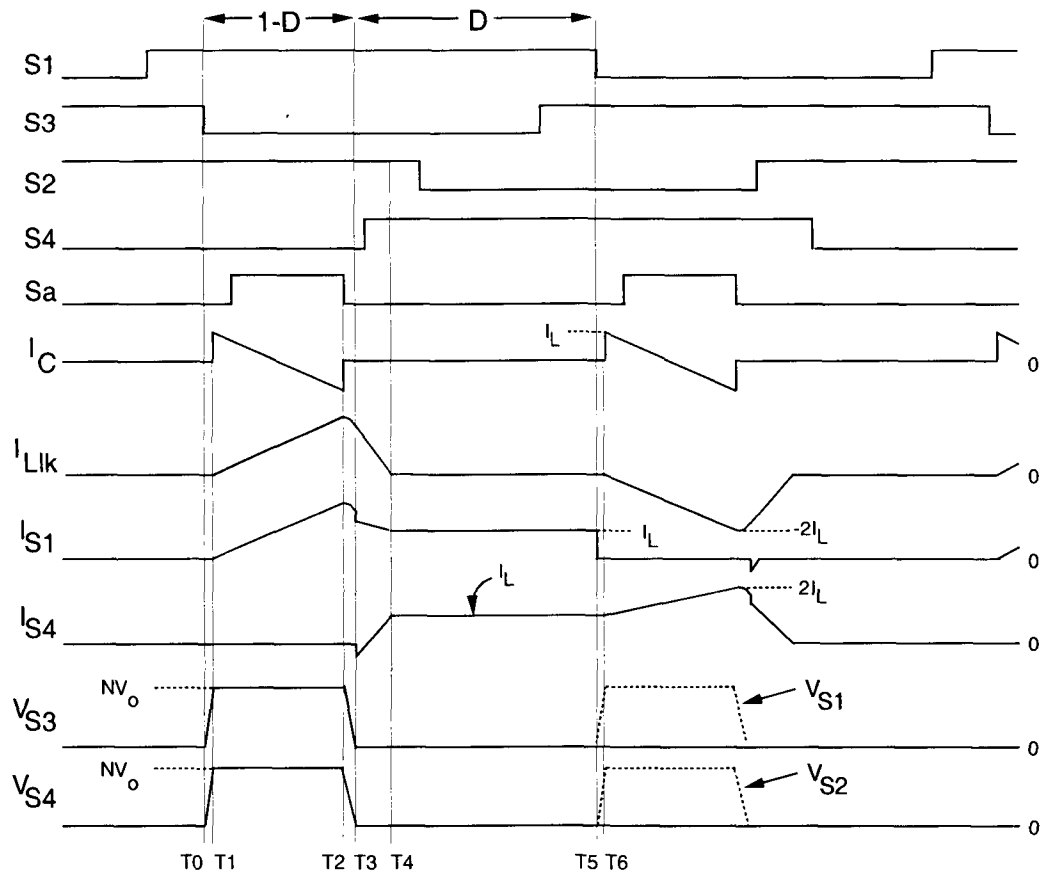
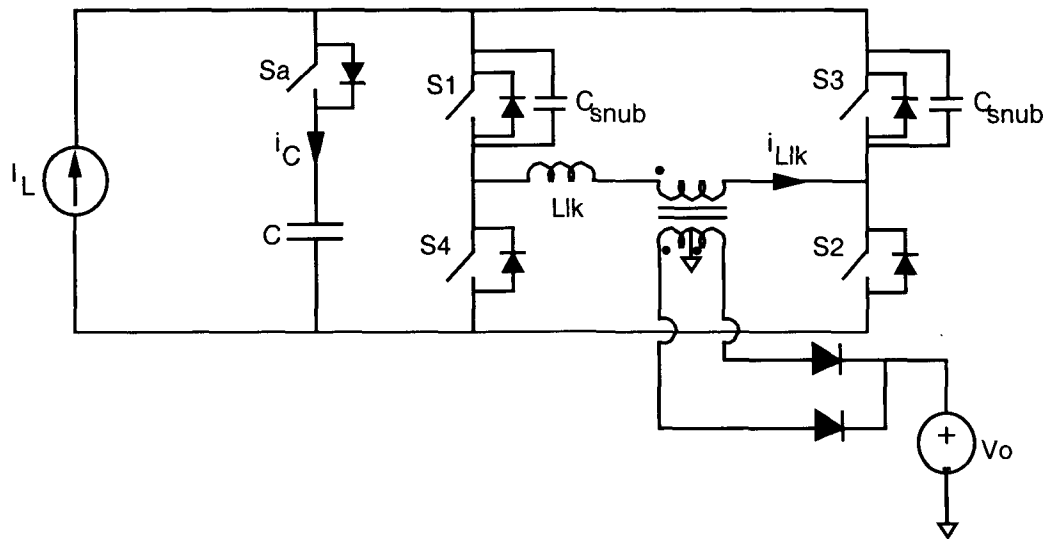


Fig. 4.4 Active-clamp FB boost converter ideal waveforms.

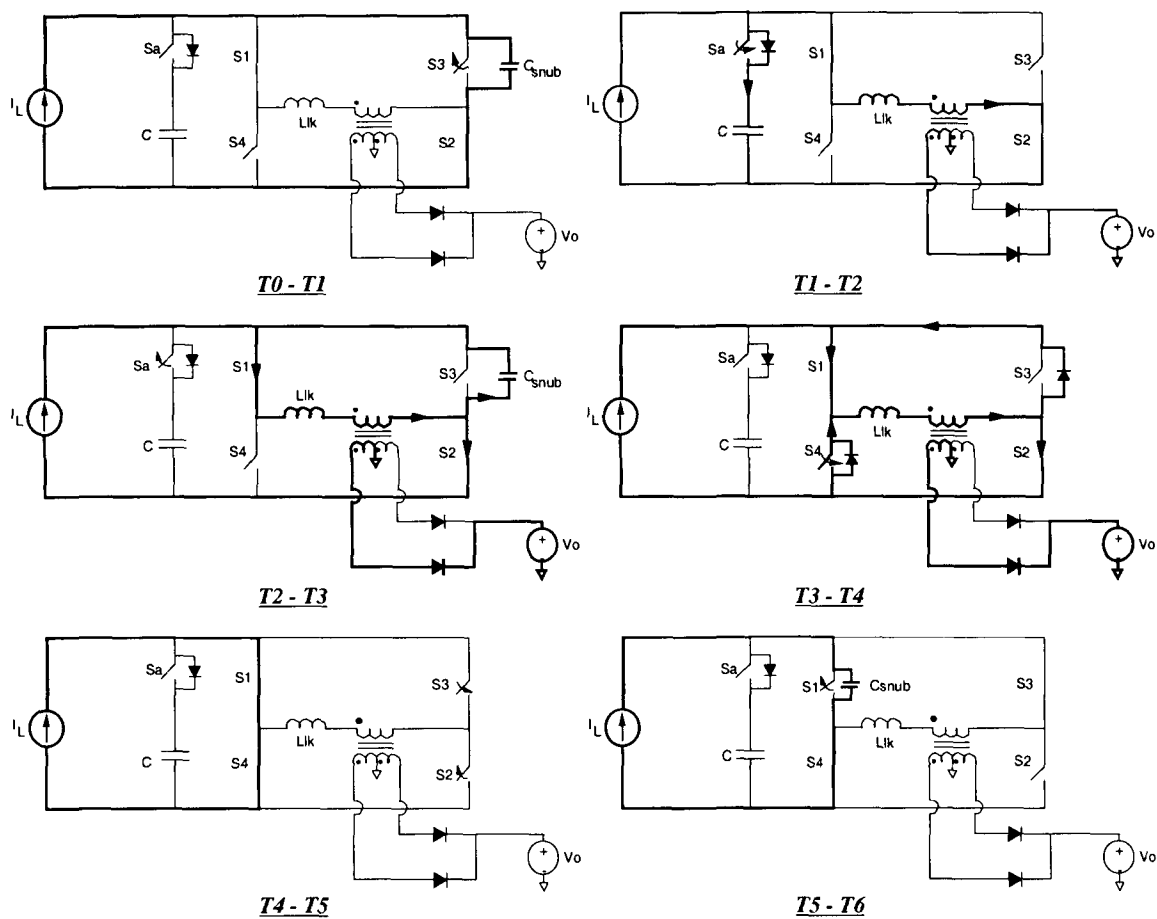


Fig. 4.5 Active-clamp full-bridge boost converter topological states.

added across the switch to form a turn-off snubber. This would be done, however, at the expense of decreasing the ZVS range.

T_1 - T_2 : At T_1 , V_{S3} reaches the clamp capacitor voltage ($\approx NV_o$), and the clamp diode conducts the boost inductor current. The current through the leakage inductance increases as L_{LK} and C start to resonate. Current starts flowing to the output. To facilitate ZVS for S_a , the switch is turned on before the clamp current reverses.

T_2 - T_3 : At T_2 the active-clamp switch is turned off. Under the assumption of an infinite value of clamp capacitance, the turn-off current is equal to the boost inductor current (which implies $i_{Llk} = 2I_L$ at T_2). The difference between i_{Llk} and I_L discharges C_{snub} (C_{snub} and L_{LK} resonate). The voltage across S_3 and S_4 starts decreasing.

T_3 - T_4 : Assuming sufficient energy has been stored in L_{LK} to discharge C_{snub} , at T_3 the body diodes of S_3 and S_4 begin to conduct. S_4 can then be turned on with zero voltage. I_{Llk} (and the current through S_2) now decrease at a linear rate determined by the reflected output voltage and the value of L_{LK} . When i_{Llk} decreases to the value of the boost inductor current, S_4 begins conducting the difference between the two currents.

T_4 - T_5 : When i_{Llk} decreases to zero, S_2 can be turned off with zero current. Sometime during this interval S_3 can be turned on with zero voltage, enabling C_{snub} to perform a lossless snubber action for turn-off of S_3 .

T_5 - T_6 : At T_5 , S_1 is turned off, ending the inductor charging interval. The sequence of topologies is then repeated for the other half of the bridge.

4.3 Design Considerations and Calculations

4.3.1 Switch timing

Other than the timing required to modulate the duty cycle through phase-shift control, it is necessary to derive the timing constraints to realize ZVS and ZCS. For ZVS, the timing between the turn off of S_a and the turn-on of S_2 or S_4 must allow the snubber capacitor voltage to reach its minimum. This is simply given by one-quarter of the L_{LK} - C_{snub} resonant period (during the T_2 - T_3 interval):

$$t_{Sa-S2} = t_{Sa-S4} = \frac{\pi}{2} \sqrt{C_{snub} L_{LK}} . \quad (4.2)$$

For ZCS, sufficient time must be given between the turn-on of S_2 or S_4 and the turn-off of the switch on the opposite side of bridge (S_4 or S_2 , respectively). The required time is determined by the value of transformer leakage inductance (referred to either the primary or the secondary), the output voltage (or reflected output voltage if leakage is considered from the primary), and the initial leakage inductance current. The worst case maximum time occurs with maximum leakage current (during the T_3 - T_4 interval):

$$t_{S2-S4} = t_{S4-S2} = \frac{2I_L^{peak} L_{LK}}{\frac{N_p}{N_s} V_o} . \quad (4.3)$$

I_L^{peak} is the peak boost inductor current.

4.3.2 Circulating energy

Due to the action of the active-clamp circuit, circulating energy is increased in the power stage components. To some degree this is beneficial, since it is desirable to operate under zero-voltage switching conditions. However, for ZVS it is not necessary for the clamp switch to conduct for almost the entire $(1-D)T_s$ time period. As long as the leakage inductance current is larger than I_L at the instant the clamp switch is turned off, ZVS may be possible. Therefore, the timing for S_a can be modified, as shown in Fig. 4.6, to reduce conduction losses. The drawback to this modified timing scheme is that ZVS is lost for S_a and some ringing will be introduced across the bridge when neither S_a nor its diode is conducting.

4.3.3 Design calculations

The design calculations developed in this section primarily revolve around determining various component losses. It is assumed insulated gate bipolar transistors (IGBTs) are used for the bridge switches. In fact, the experimental results reported in Section 4.4 are based on the use of IGBT bridge switches.

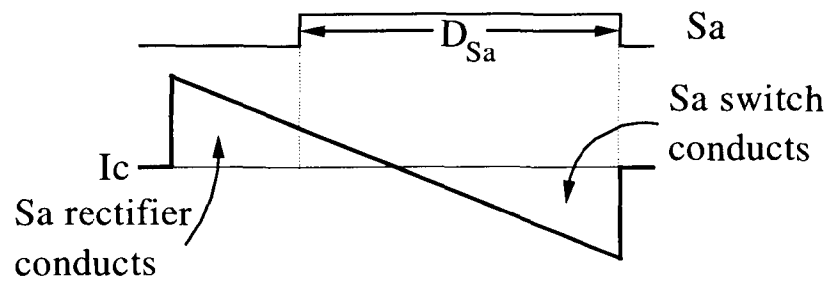
4.3.3.1 Full-bridge switch calculations

4.3.3.1.1 Conduction loss

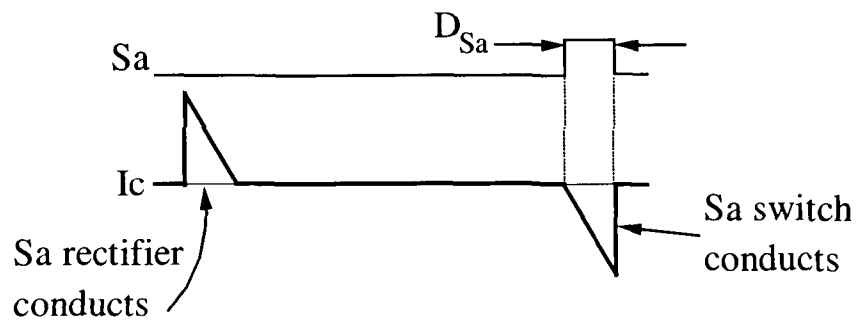
As a function of the 120 Hz line cycle, the average (switching cycle) current seen by individual switches in the bridge is given by:

$$\langle I_S \rangle(\phi) = \frac{\sqrt{2}P_o}{\eta 2V_{RMS}} \sin \phi, \quad (4.4)$$

where ϕ is the instantaneous operating point on the input line (normalized over the interval $[0, \pi]$) and η is the conversion efficiency. Due to the phase-shifted control of the



(a) Original timing.



b) Modified timing.

Fig. 4.6 Active-clamp switch timing to reduce conduction loss

full bridge, the duty cycle of each switch is approximately 50 % regardless of the operating point. Assuming $V_{CE(sat)}$ for IGBTs is constant as a function of collector current, the conduction loss averaged over a line cycle is:

$$P_{cond,bridge} = \frac{4}{\pi} \int_0^{\pi} \langle I_S \rangle(\phi) V_{CE(sat)} d\phi \quad (4.5a)$$

$$= \frac{4\sqrt{2}P_o}{\eta\pi V_{RMS}} V_{CE(sat)}. \quad (4.5b)$$

4.3.3.1.2 Switching loss

Calculating the switching loss is more involved than calculating the conduction loss. Assuming unity power factor, ZVS is a function of the input voltage as it changes over the line cycle. This dependence greatly complicates matters, so in order to simplify the arithmetic, it will be assumed that C_{snub} is so small as to have no effect on reducing the turn-off loss for S1 and S3. As a consequence, however, ZVS will be available over the entire line cycle. The result of these assumptions is probably an overestimation of the S1, S3 turn-off loss and an underestimation of the S2, S4 turn-on loss. For ZVS operation, all four switches are turned on losslessly, and the bottom two switches are turned off with zero current. An important aspect for the zero-current turn-off of S2 and S4 is that their collector voltages remain at approximately $V_{CE(sat)}$ for the duration of the boost inductor charge interval (see Fig. 4.10(a)). The top two switches suffer from hard turn-off, which is the reason for installing the snubber capacitors. The use of empirical methods is probably best to determine an optimum value for these capacitors.

For S1, S3 turn-off:

$$P_{off}(\phi)|_{S1 \text{ or } S3} = F_S E_{off}(\phi) \frac{V_{CE,off}}{V_{CE,baseline}}, \quad (4.6)$$

where $V_{CE,baseline}$ is the test voltage used for device turn-off measurements and

$$E_{off}(\phi) = f(I_{CE,off}(\phi)). \quad (4.7)$$

$I_{CE,off}(\phi)$ is the boost inductor current at device turn-off. Its value is twice that given by Eq. (4.4). The turn-off voltage is constant and equal to the reflected output voltage. A curve fit based on device turn-off loss data can then be used to develop a relationship between turn-off energy and collector current.

4.3.3.2 Output rectifier calculations

For the output rectifiers, assume that individual rectifier forward drops (V_{fw}) are independent of forward current, and that the forward drops in each rectifier are equal. In an average sense, each rectifier sees the one-half of the output current. Therefore:

$$P_{output \ rectifier} = \frac{P_o}{2V_o} V_{fw}. \quad (4.8)$$

For rectifier switching losses, the stored charge can be used to determine the reverse recovery energy:

$$E_{sw}|_{output \ rectifier} = V_R Q_{rr}. \quad (4.9)$$

Q_{rr} is a function of the peak rectifier forward current (just before turn-off) and the turn-off di_F/dt . The rate of turn-off current is fixed by the reflected value of the leakage (or resonant) inductance and the output voltage:

$$\frac{di_F}{dt} = \frac{V_O}{L_{LK} / \left(\frac{N_P}{N_S} \right)^2}. \quad (4.10)$$

The peak rectifier forward current is given by:

$$I_F^{peak}(\phi) = 2 \frac{N_P}{N_S} \frac{\sqrt{2} P_O}{\eta V_{RMS}} \frac{N_P}{N_S} \sin(\phi). \quad (4.11)$$

The factor of 2 appearing in Eq. (4.11) is due to the action of active-clamp and is worst-case conservative. The reverse voltage (V_R) is equal to the output voltage. Assuming all the reverse recovery energy is dissipated, the switching losses are given by:

$$P_{SW}|_{output\ rectifier} = \frac{1}{\pi} \int_0^\pi F_s \frac{V_o}{2} Q_{rr} \left(I_F^{peak}(\phi), V_o \right) d\phi. \quad (4.12)$$

Because of the nonlinear dependence of Q_{rr} on I_F and V_R , the integration called for in Eq. (4.12) is a math phobic's worst nightmare. Things can be simplified by using the average value (over a line cycle) of the peak rectifier current (Eq. (4.11)) as a "steady-

state" operating point. This (along with V_o) can be used to select a value for Q_{rr} . Therefore:

$$P_{SW}|_{output\ rectifier} \approx \frac{1}{2} F_S V_o Q_{rr} \langle I_F^{peak}(\phi) \rangle V_o. \quad (4.13)$$

4.3.3.3 Active-clamp circuit calculations

Active-clamp circuit losses consist of conduction losses for both the auxiliary diode and MOSFET and switching (turn-off) losses of the MOSFET. For the rectifier, or MOSFET body diode, conduction losses are given by:

$$P_{Sa, Rect} = \langle I_{Sa} \rangle_{line\ cycle} V_F = \frac{V_F}{\pi} \int_0^\pi \langle I_{Sa} \rangle_{switching\ cycle} d\phi, \quad (4.14)$$

where:

$$\langle I_{Sa} \rangle_{switching\ cycle} \cong I_L(\phi) D_{Sa}, \quad (4.15)$$

and D_{sa} is the switch's duty cycle. $I_L(\phi)$ is the boost inductor (and line) current. Carrying out the integration called for in Eq. (4.14) yields:

$$P_{Sa, Rect} = \frac{D_{Sa} \sqrt{2} P_o}{\eta \pi V_{RMS}} V_F. \quad (4.16)$$

Calculating conduction losses for the MOSFETs requires knowledge of the RMS current through the devices:

$$P_{Sa, MOSFET}(\phi) = (I_{Sa, RMS}(\phi))^2 R_{DS, on}. \quad (4.17)$$

Therefore:

$$\langle P_{Sa, MOSFET} \rangle_{cycle}^{line} = \frac{1}{\pi} \int_0^\pi P_{Sa, MOSFET}(\phi) d\phi. \quad (4.18)$$

Turn-off losses are given by:

$$P_{off, MOSFET} = \frac{1}{\pi} \int_0^\pi P_{off}(\phi) d\phi \quad (4.19a)$$

$$= \frac{1}{\pi} \int_0^\pi F_S \frac{N_P}{N_S} \frac{V_o}{3} \frac{t_f}{2} I_L(\phi) d\phi. \quad (4.19b)$$

The turn-off fall time, t_f , is assumed to be 75 ns. Turn-on losses are calculated in a similar fashion ($t_r = 50$ ns):

$$P_{on} = \frac{1}{\pi} \int_0^\pi P_{on}(\phi) d\phi = \frac{1}{\pi} \int_0^\pi F_S \frac{N_P}{N_S} \frac{V_o}{3} \frac{t_r}{2} I_L(\phi) d\phi. \quad (4.20)$$

4.4 Experimental Results

To provide experimental verification of the operating concepts incorporated into the phase-shift controlled, active-clamp FB boost converter, a breadboard was constructed to provide a 48 Vdc, 1 kW output DC bus voltage from a universal AC line input. The efficiency of this design is then compared to the efficiency obtained from two-stage front-end design consisting of a zero-voltage transition (ZVT) PFC boost converter cascaded with a ZVS FB second stage DC/DC converter.

4.4.1 Power stage design

The power stage design for the active-clamp boost converter is shown in Fig. 4.7. The bridge switches operate at 75 kHz, and are each 600 V ultra-fast IGBTs. The turn-off snubber capacitors are each 1500 pF, a value that was arrived at empirically. The clamp network consists of 600 V, 2 μ F, polypropylene capacitor. The clamp capacitor should be of high quality in order to process the HF ripple currents introduced by the action of the active clamp network with as little loss as possible. The output diodes are 100 V Schottky rectifiers. Here the single-stage boost converter has a distinct advantage over the second stage DC/DC converter in the two-stage approach, where the use of Schottky rectifiers would be prohibited due to too high of a reverse voltage. Three 4700 μ F Aluminum electrolytic capacitors form the output bulk capacitor. Peak 120 Hz voltage ripple was measured at about 2.5 V peak (± 5.2 %). In practice, the amount of output capacitance is determined by ripple current capability.

Also shown in Fig. 4.7 is the design of the boost inductor and the isolation transformer. Because of the high output current required and the additional circulating current introduced by the active clamp circuit, the design of T1's secondary required the use of Cu foil, as shown.

A block diagram of the average current mode control scheme is shown in Fig. 4.8. In order to obtain good power factor at the input, the full-bridge phase-shift is modulated.

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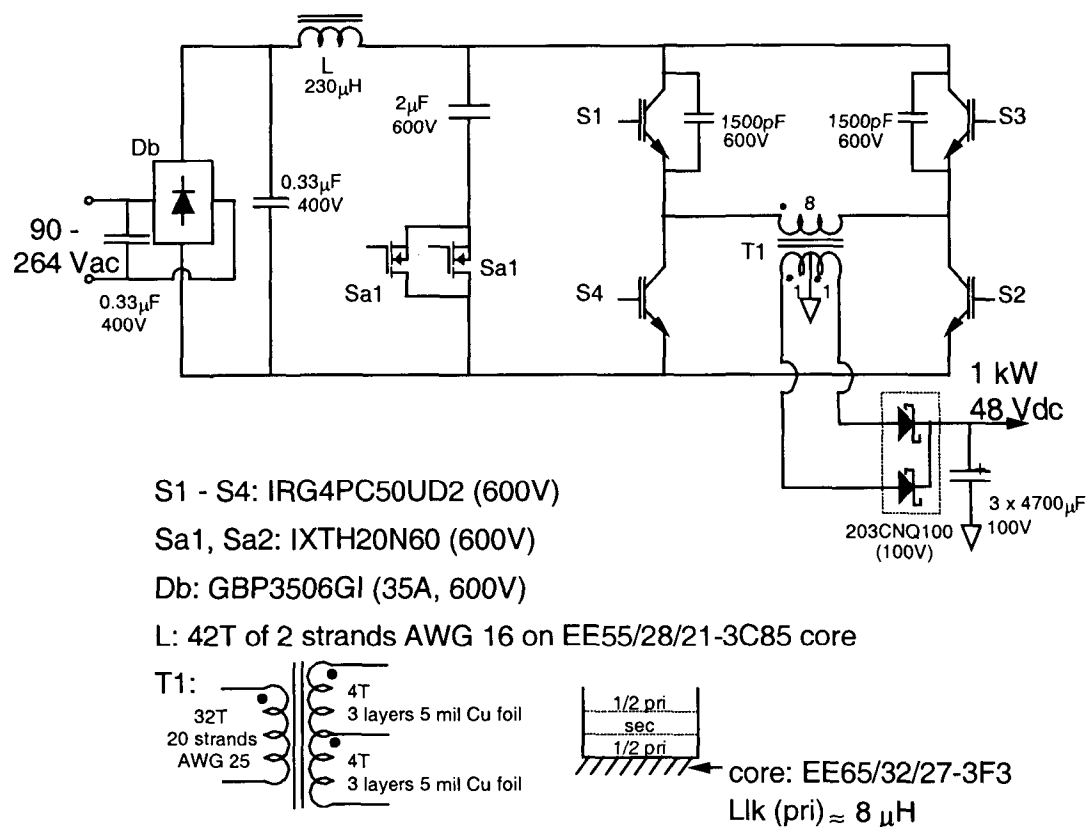


Fig. 4.7 Power stage component selection.

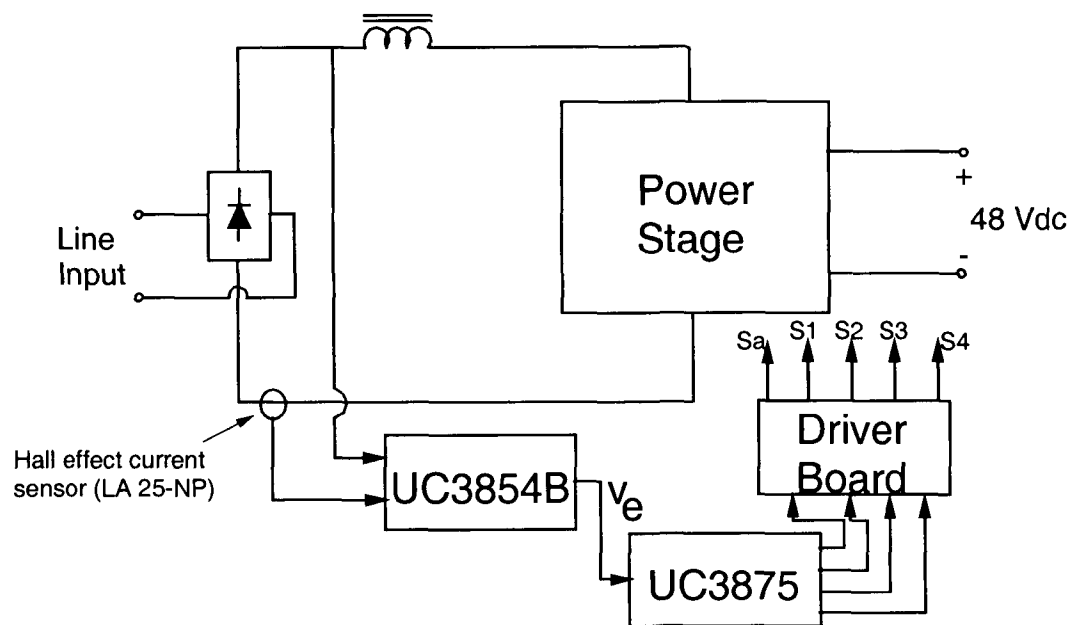


Fig. 4.8 Block diagram of FB control for PFC.

However, since there are no commercially available full-bridge ϕ -shift control integrated circuits for PFC, the method shown in Fig. 4.8 was adopted. A single-ended PFC controller IC, the UC3854B, was used to generate a control voltage, V_e , as part of the standard application of this IC [48]. This control voltage was then used as the error amplifier input for a ϕ -shift control IC, the UC3875. The UC3875 then generates the four ϕ -shifted drive signals for the bridge switches. The driver board is then responsible for developing the active-clamp switch drive based on the ϕ -shifting of the bridge switches.

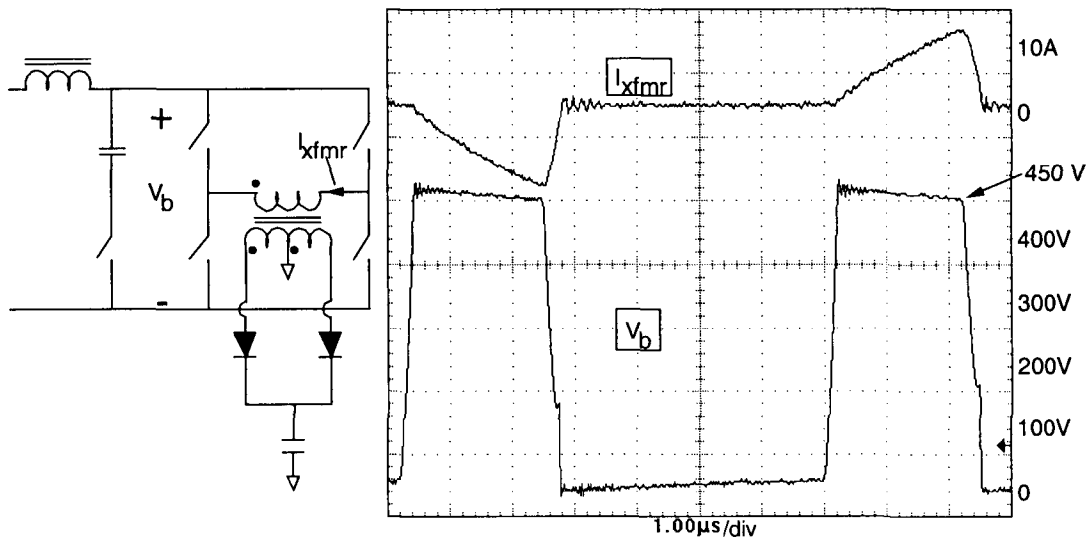
4.4.2 Experimental waveforms and efficiency measurements

Figures 4.9 through 4.11 summarize the fundamental experimental waveforms of the converter. Figures 4.9 and 4.10 were obtained with the converter operating from a DC input voltage in order to more clearly show these waveforms over a switching cycle. Figure 4.9 summarizes the operating waveforms associated with the active-clamp network. The fundamental characteristics of these waveforms is identical to those of the active-clamp flyback converter presented in Chapter 3. The waveforms illustrating the soft-switching characteristics are shown in Fig. 4.10. The line input waveforms are shown in Fig. 4.11 for $V_{in} = 120 \text{ Vac}$ and $P_o = 1 \text{ kW}$. As can be seen, the power factor is very high and has minimal harmonic distortion, both typical of average current mode control PFC and CCM operation of the boost inductor.

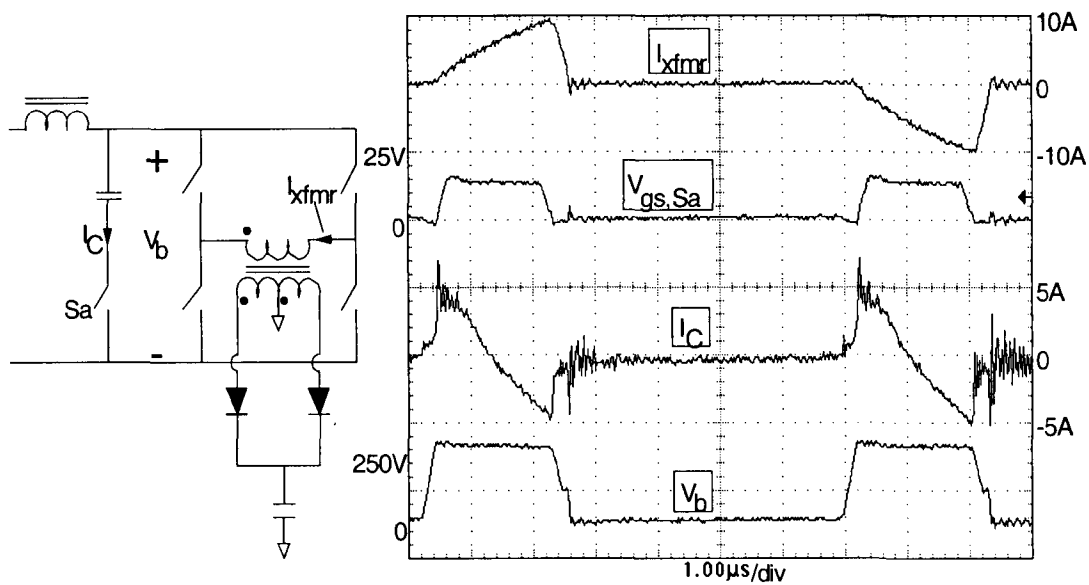
The experimental efficiencies, as a function of line and load are presented in Fig. 4.12. High-line/maximum load efficiencies are above 90 %.

4.4.3 Experimental comparison with the two-stage approach

In order to provide a relative efficiency comparison with the two-stage front-end approach, data was taken from the two-stage power train design shown in Fig. 4.13. This design cascades a ZVT PFC boost converter, switching at 140 kHz, with a standard ZVS ϕ -shifted full-bridge DC/DC converter switching at 100 kHz, using the components

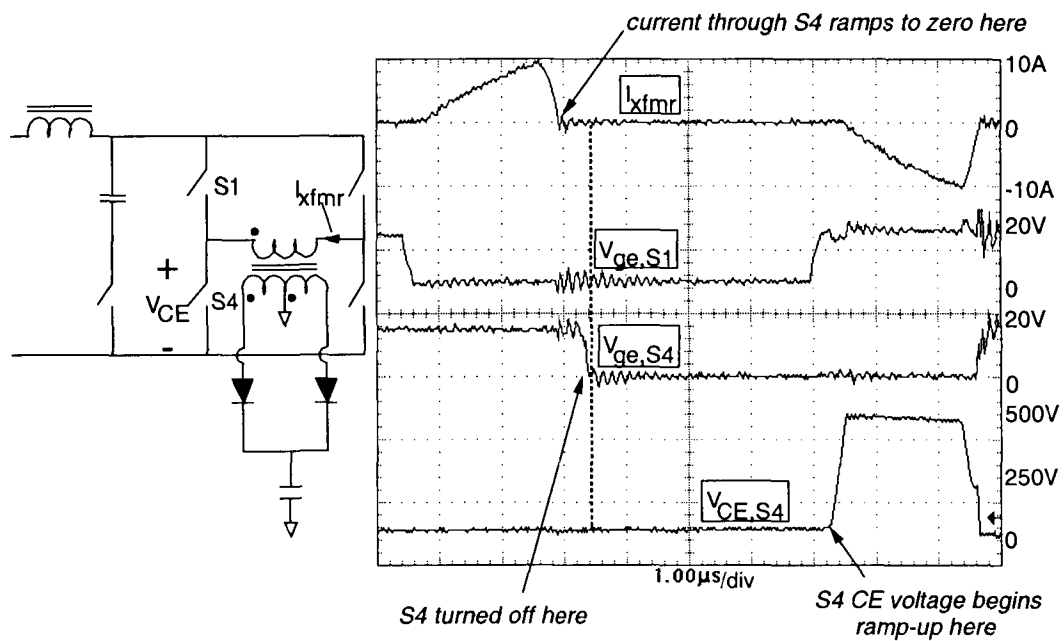


(a) Bridge-voltage clamping, $V_{in} = 150$ Vdc, $P_o = 960$ W.

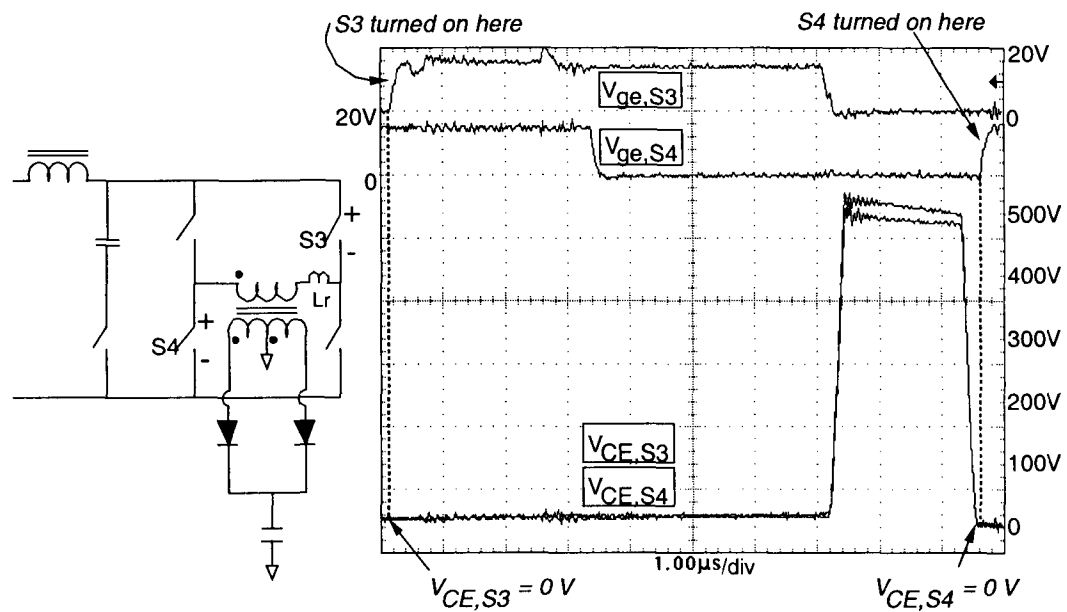


(b) Clamp circuit waveforms, $V_{in} = 140$ Vdc, $P_o = 700$ W.

Fig. 4.9 Experimental waveforms - clamp operation.



(a) ZCS for $S4$, $V_{\text{in}} = 140 \text{ Vdc}$, $P_o = 700 \text{ W}$.



(b) ZVS for $S3/S4$ ($L_r = 5 \mu\text{H}$), $V_{\text{in}} = 140 \text{ Vdc}$, $P_o = 958 \text{ W}$.

Fig. 4.10 Experimental waveforms - soft-switching.

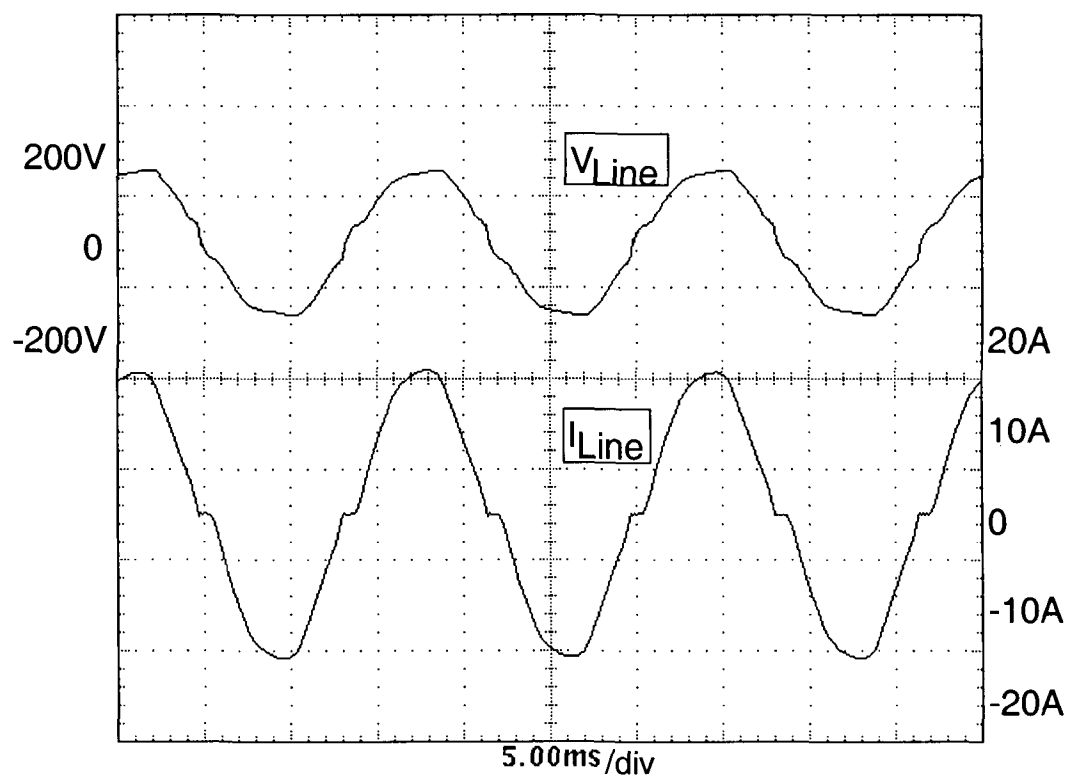


Fig. 4.11 Experimental waveforms - line voltage and current, $V_{in} = 120 \text{ Vac}$,
 $P_o = 1 \text{ kW}$.

Efficiency (%)

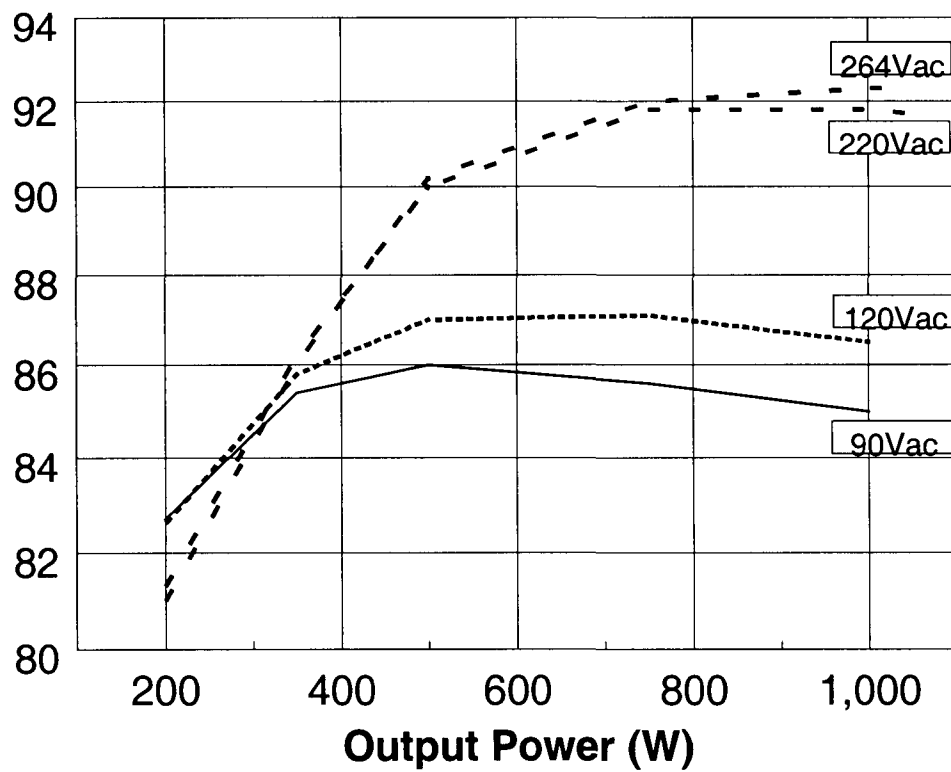


Fig. 4.12 Experimental efficiency as a function of line and load (includes control circuit losses).

indicated. The large value of bulk capacitance (close to 900 μF) used in this design is to provide for a significant hold-up function.

The efficiency data, as a function of line voltage with a constant output power of 1 kW, is given in Fig. 4.14. At low line voltages the efficiencies are very comparable, but as the line voltage is increased past about 150 Vac the two curves diverge. At $V_{in} = 220$ Vac the active-clamp boost is nearly 3 % more efficient. This makes sense as at high line the circulating energy losses introduced by the active-clamp network are reduced (boost inductor current is at a minimum) and maximum advantage can be taken of the fact that the power is processed in one conversion stage. From the efficiency data it would seem the active-clamp boost converter is best suited for applications requiring PFC and equipment operation from a single-phase 220 Vac input.

4.5 Summary

In this chapter, the phase-shift controlled, active-clamp full-bridge PFC boost converter has been shown to be an attractive solution for higher power front-end converters utilized in DC distributed power systems. The introduction of the active-clamp circuit introduces minimum complexity in the implementation of the overall converter and provides a mechanism to recover the transformer leakage energy and achieve zero-voltage switching. This approach reduces device voltage and switching stress at the expense of increasing circulating energy. It was also demonstrated how phase-shift operation of the bridge switches results in zero-current switching for two of the four bridge switches [52]. The control concept is straightforward, with constant frequency, PWM phase-shift control being utilized. Experimental results indicate a 3 % efficiency improvement (at high input line and with a 1 kW, 48 Vdc output) over a boost PFC/ZVS FB DC/DC two-stage design.

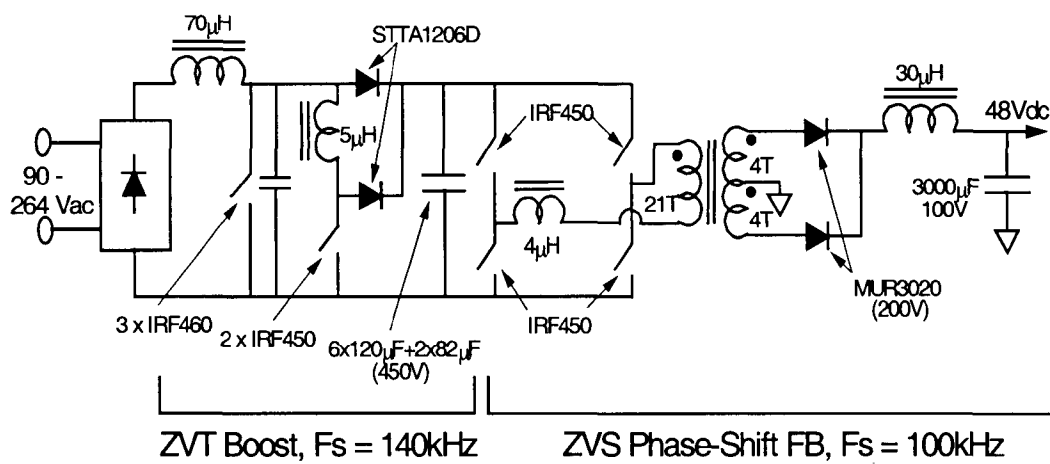


Fig. 4.13 Experimental two-stage power train design.

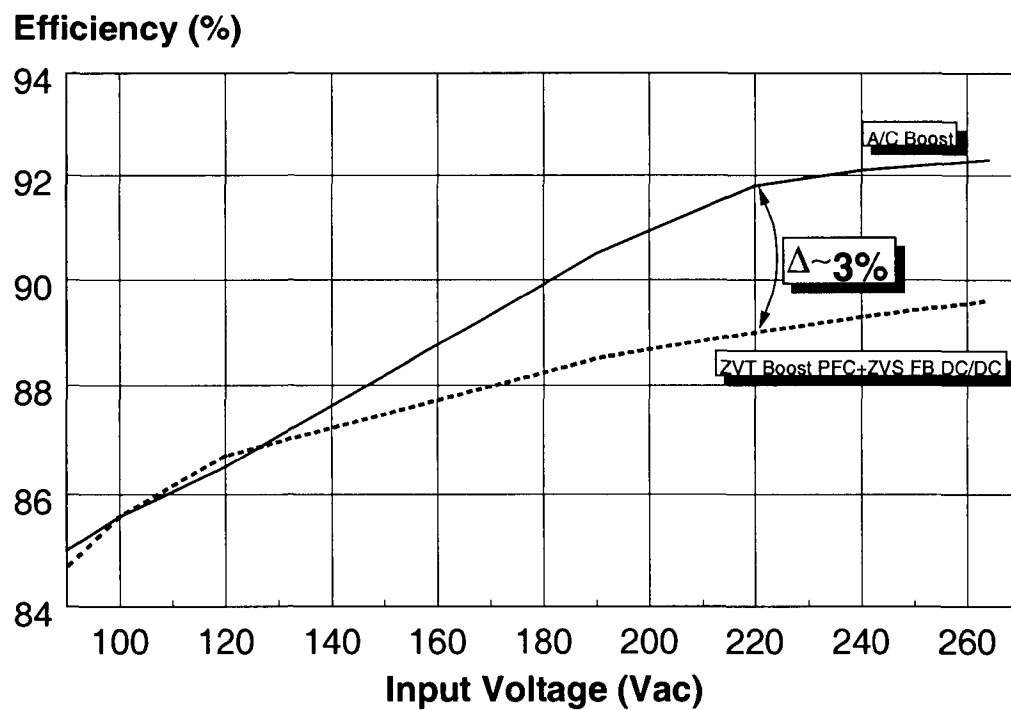
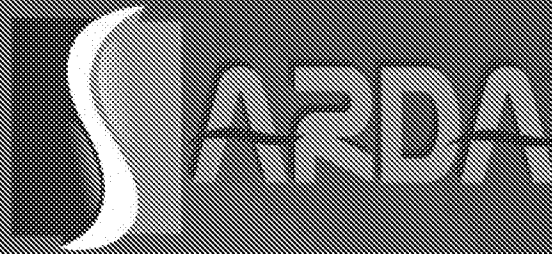


Fig. 4.14 Experimental efficiency comparison between 2 stage and single stage implementations. $P_o = 1$ kW.



12V PowerStage in Embedded Die System-in-Package

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Agenda

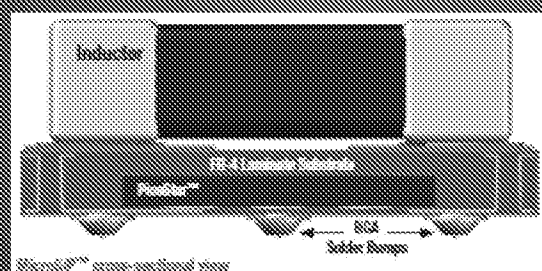
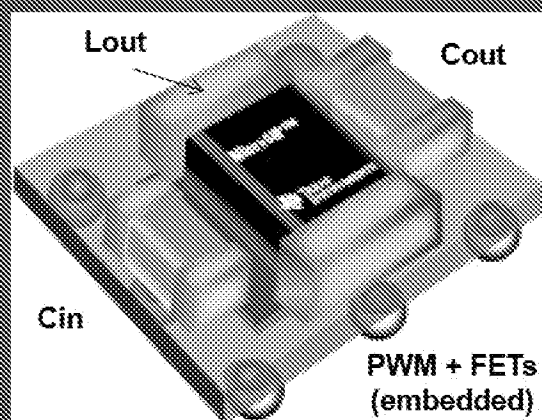
- Introduction: 3D Power Packaging
- State-of-the-Art 12V input POL Solutions
- What if you need a higher power density solution?
 - GaAs – an enabling high frequency FET
 - Multiphase – to shrink size of inductor
 - Integration – a requirement for multi-MHz switching
- Embedded Die System-in-Package approach
- Summary

3D Power Packaging is a Game Changer

using Embedded Die in Substrate Technology

In Production today

One Example: TI's MicroSiP™ Modules



Nice compact solution

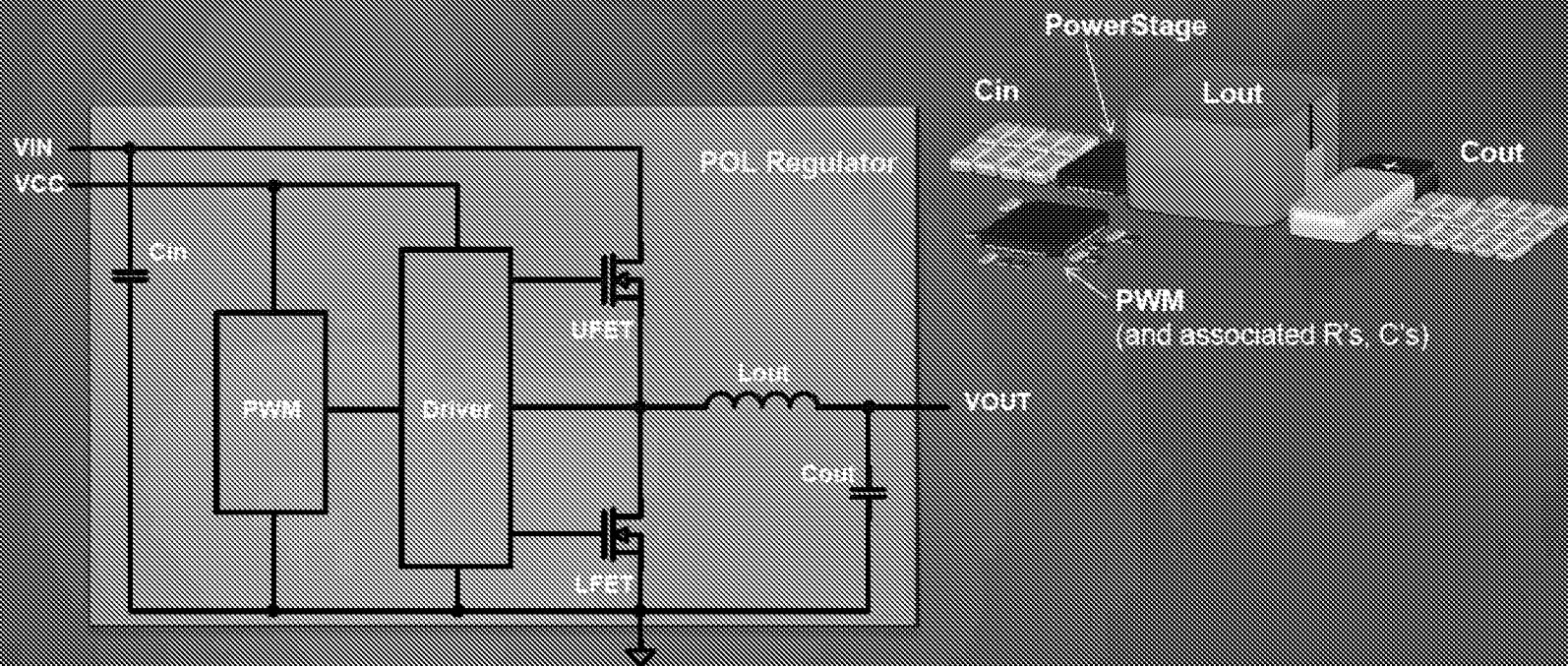
Low Input Voltage: 5V max

Low Output Current: ~2A max

*But what about higher voltages
and higher currents?*

*Does Embedded Die in Substrate
Technology add value there?*

12V Input Buck POL Regulator

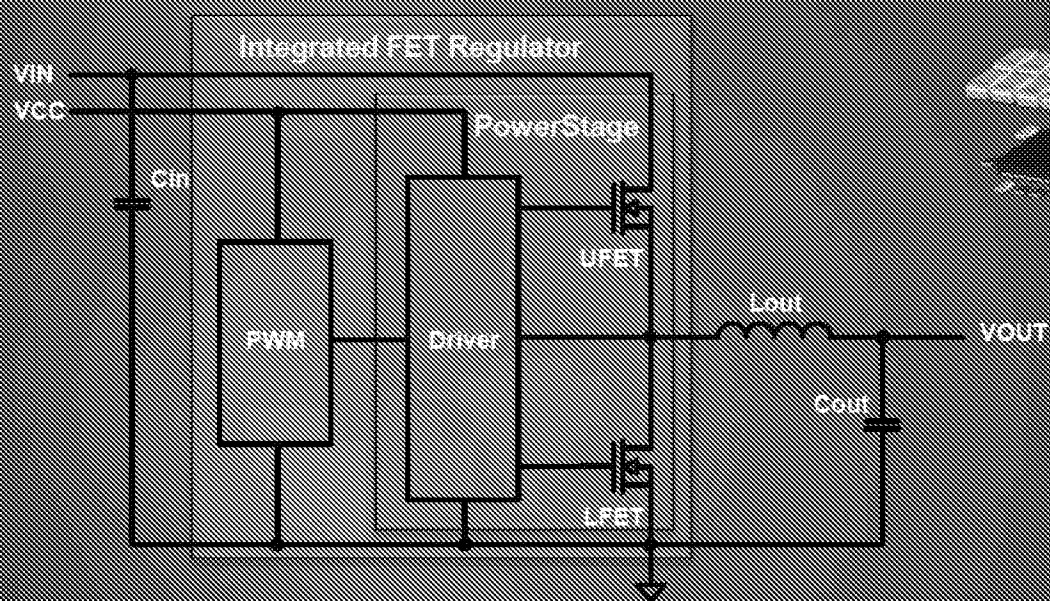


Can Embedded Die in Substrate Technology add value in 12V input, high Output Current applications?

*Answer: An emphatic Yes **

** In conjunction with Compound Semiconductor FETs, namely GaAs*

Common Integration Approaches



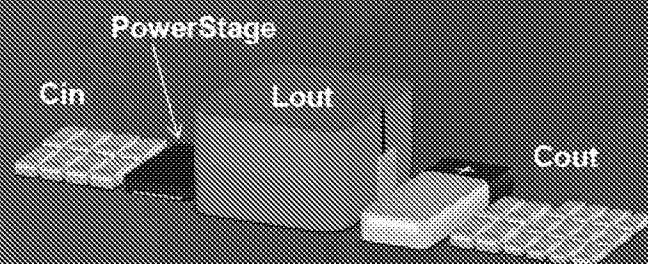
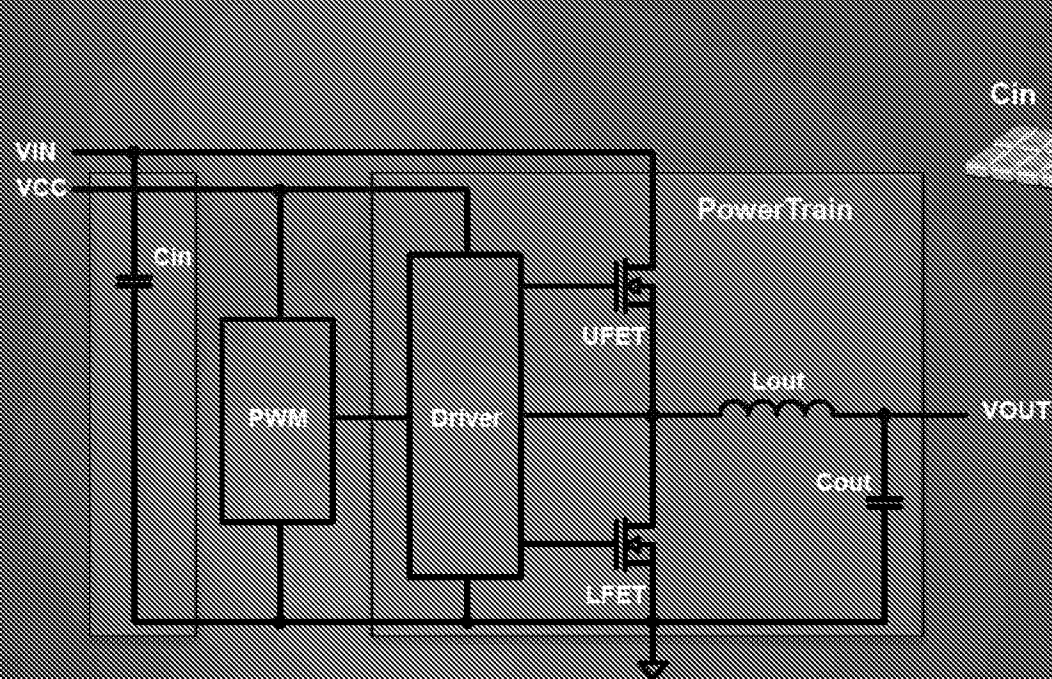
PowerStage

- Driver + FETs
- Typically multi-chip for high current applications ($>10A$)

Integrated FET Regulator

- PWM + Driver + FETs
- Typically monolithic for low current applications ($<10A$)
- Very common for low V_{in} ($\leq 5V$)

For Benchmarking Purposes – Examine PowerTrain

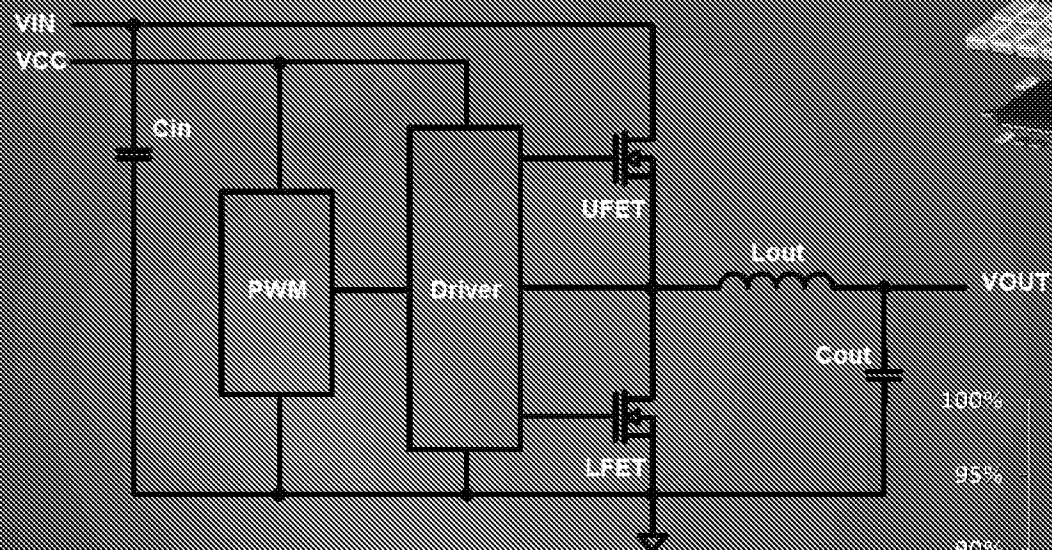


PowerTrain

- PowerStage + Output Filter + Input Cap
- Determines efficiency and majority of POL size

State-of-Art 12V input Buck POL Solution

30A output current example



Low Rdson MOSFETs

- 1mΩ LFET

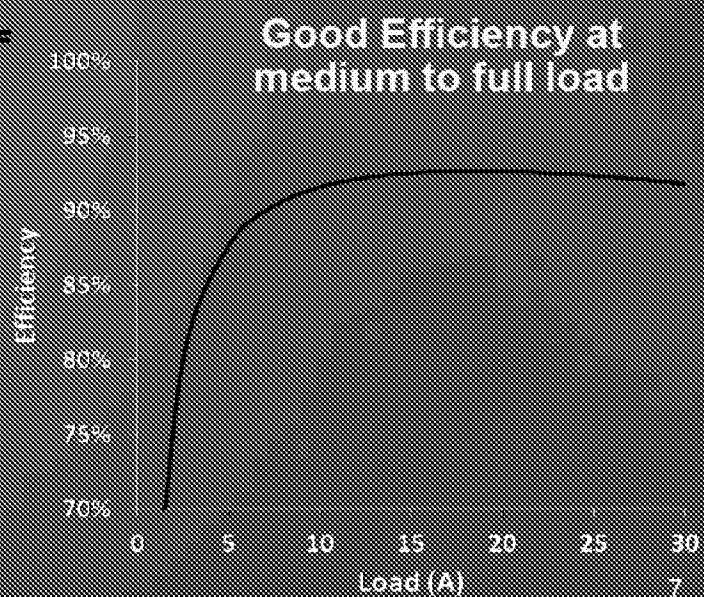
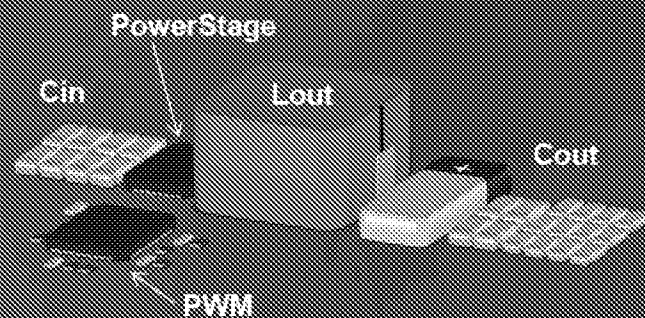
Low DCR Inductor

- 0.2mΩ

Low Switching Frequency

- 500-700kHz

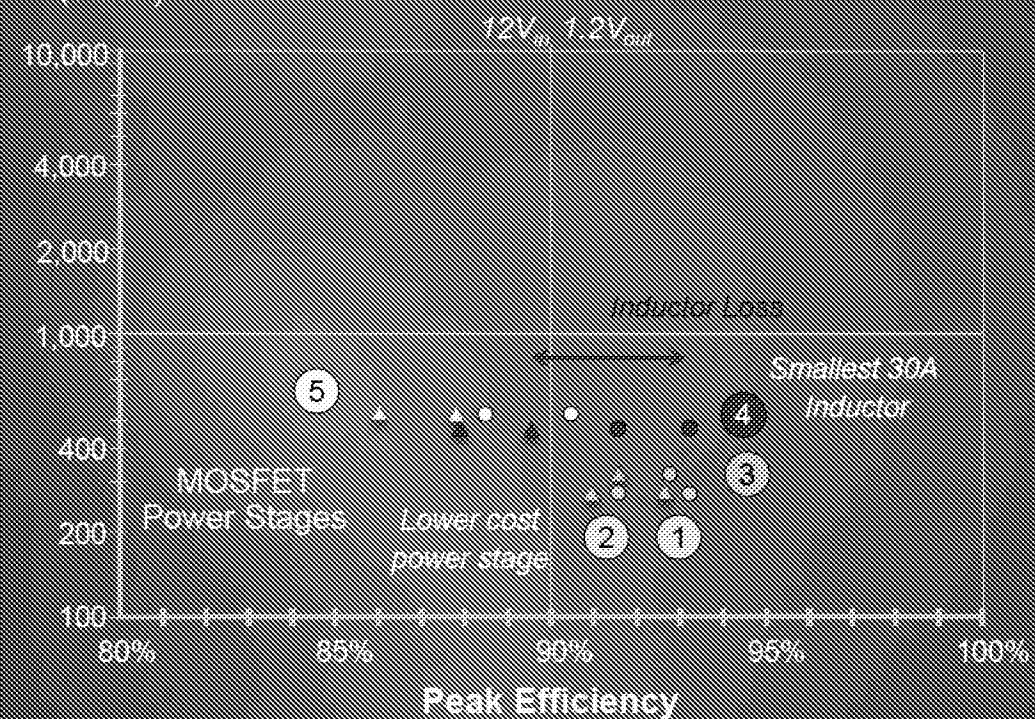
Bulky – but if you have the room for these solutions, by all means use them



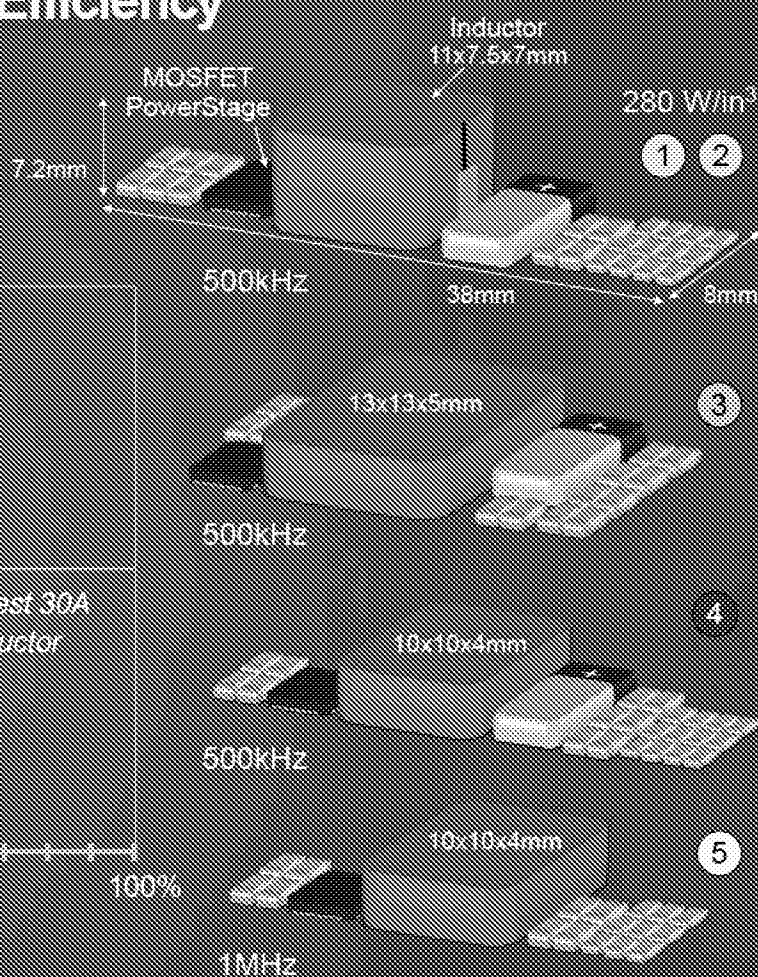
Key Metrics – Density (Size) and Efficiency

30A output current

Power Train
Density
(W/in³)



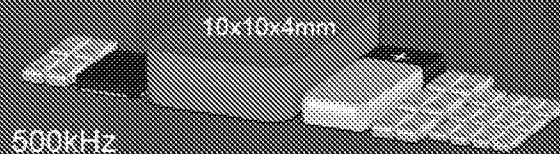
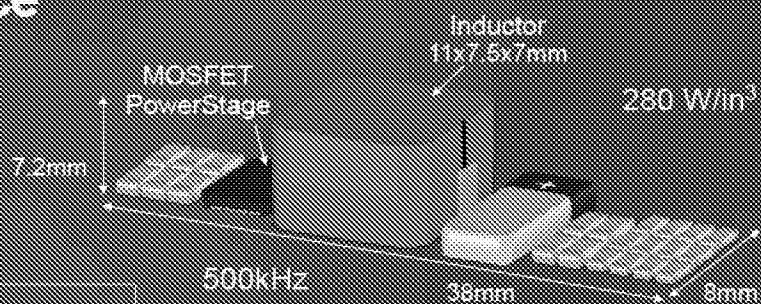
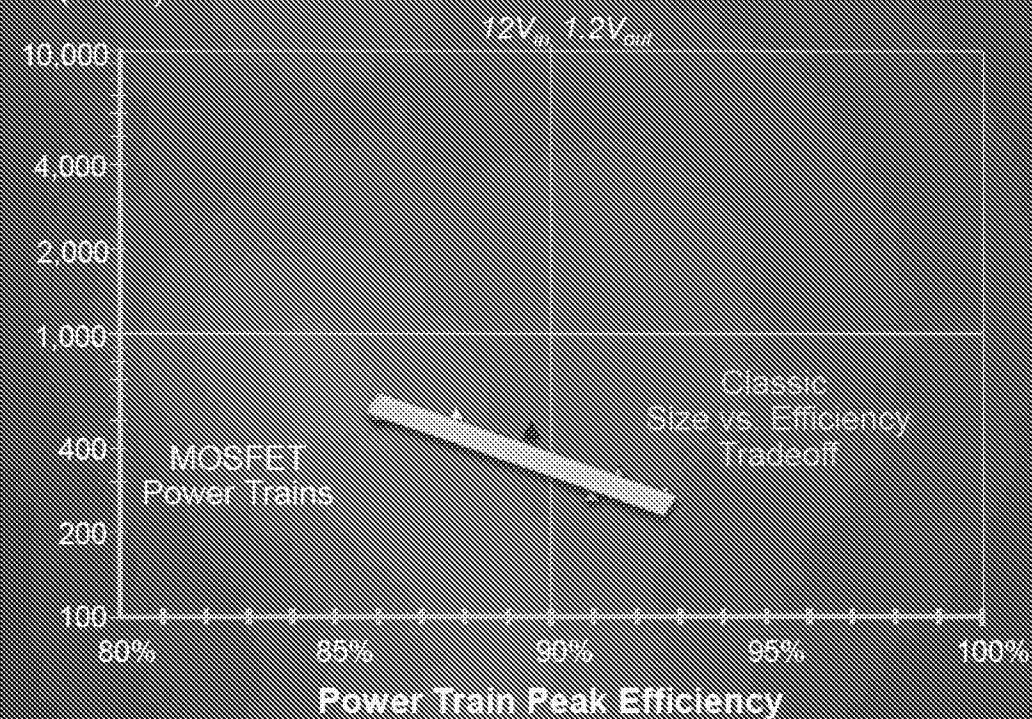
- Power Stage efficiency
- ▲ Power Train efficiency



MOSFET PowerTrain Performance

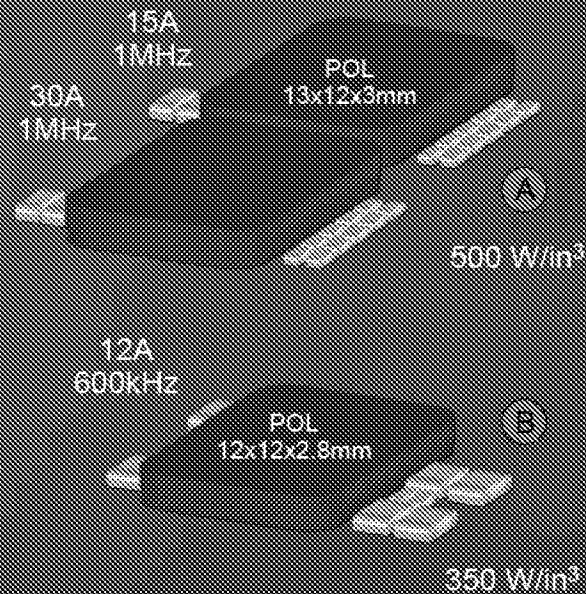
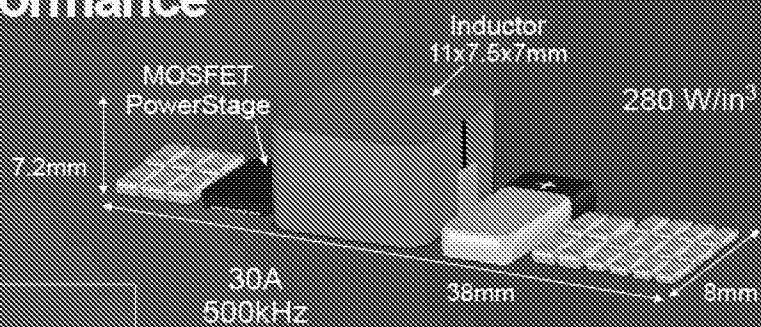
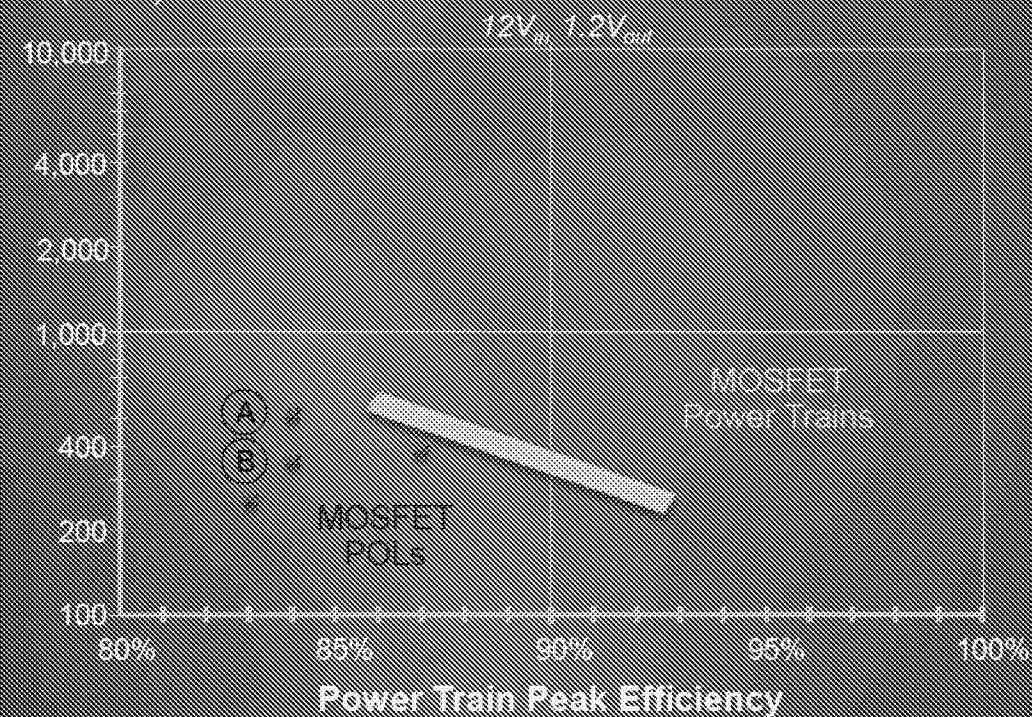
30A output current

Power Train
Density
(W/in³)



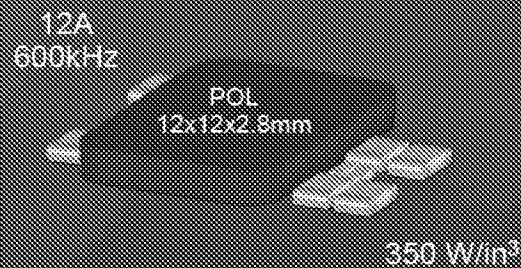
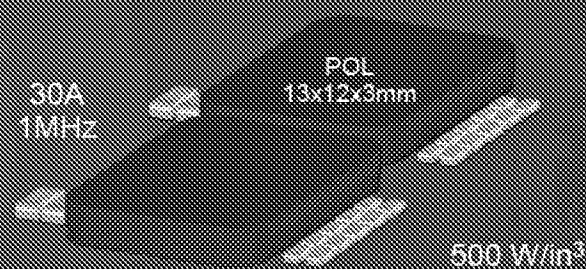
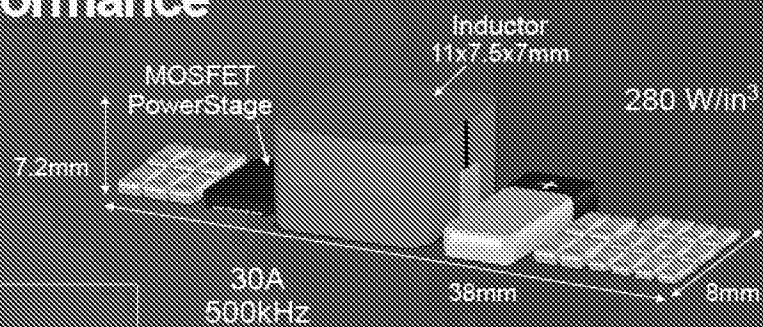
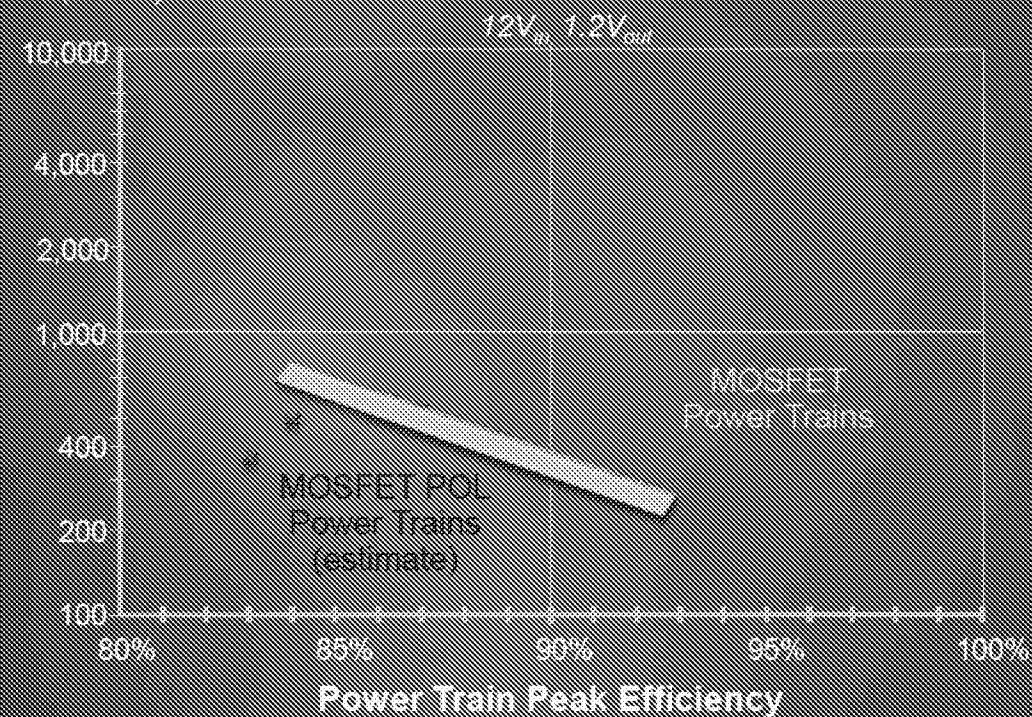
High Density POL Regulator Performance

Power Train
Density
(W/in³)



High Density POL Regulator Performance

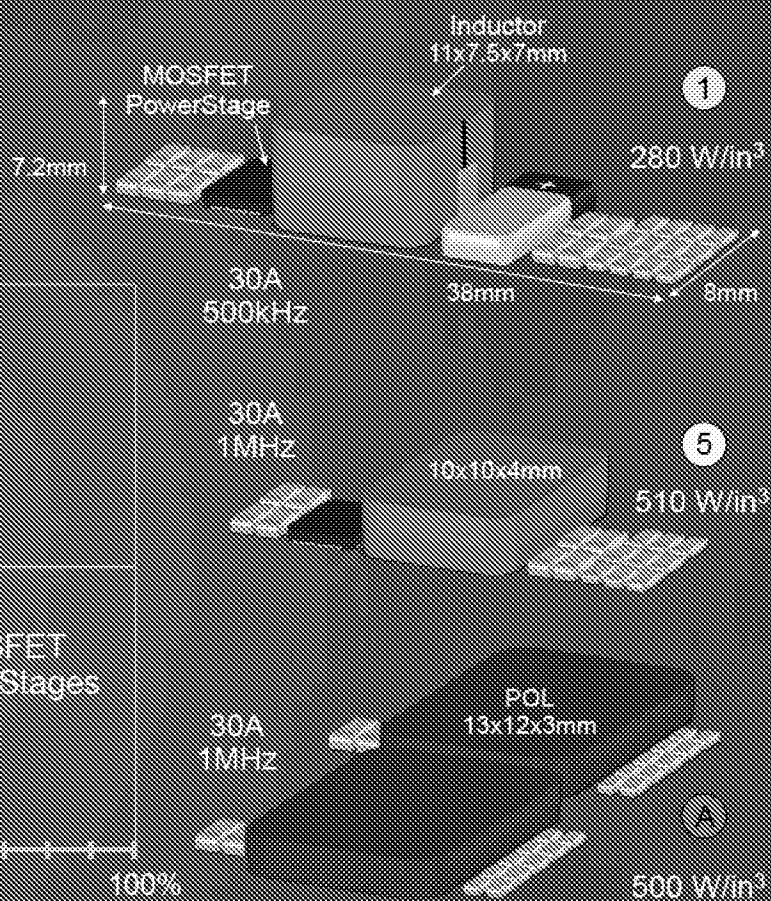
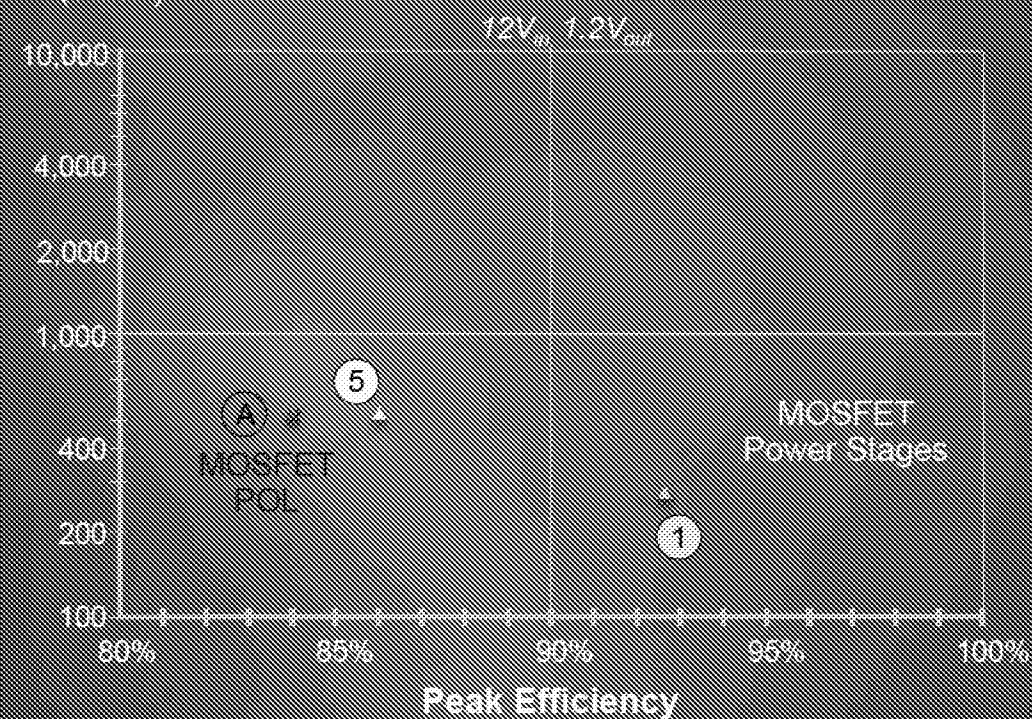
Power Train
Density
(W/in³)



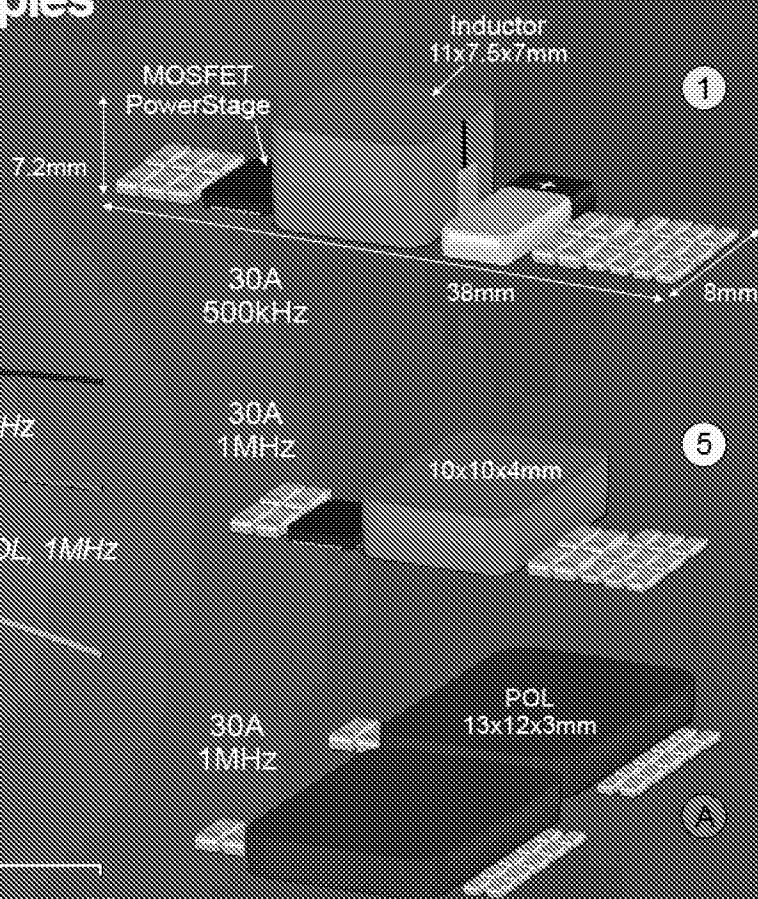
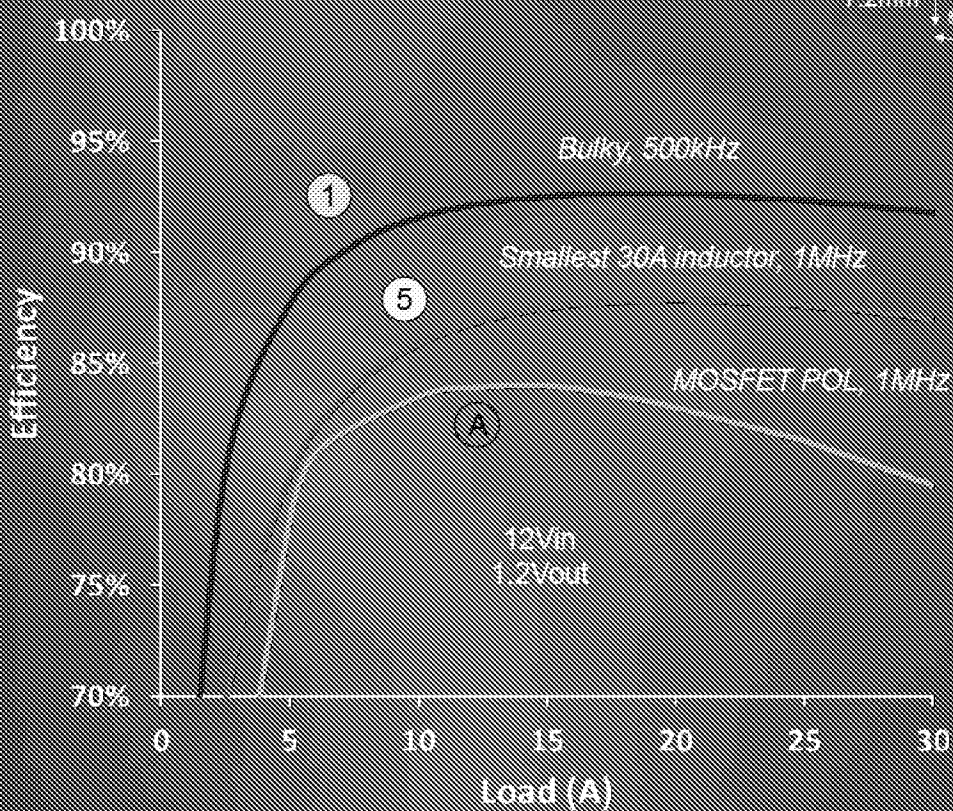
Closer Look at Three Examples

30A output current

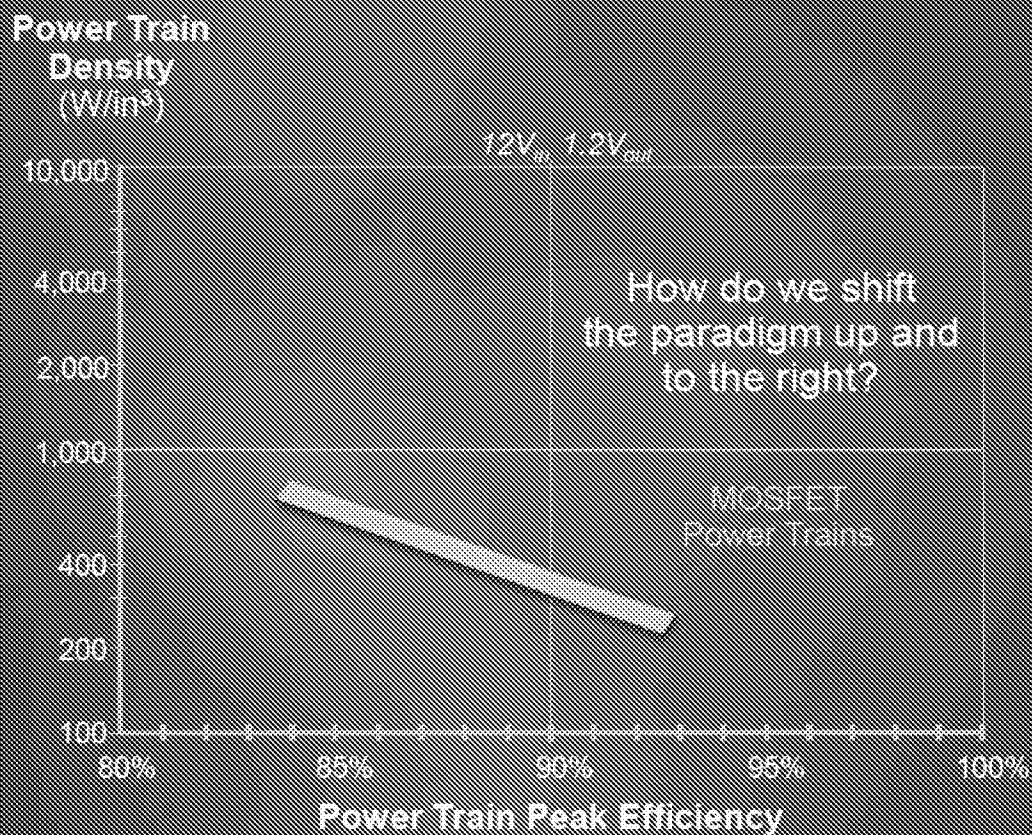
Power Train
Density
(W/in³)



Efficiency Curves of Three Examples

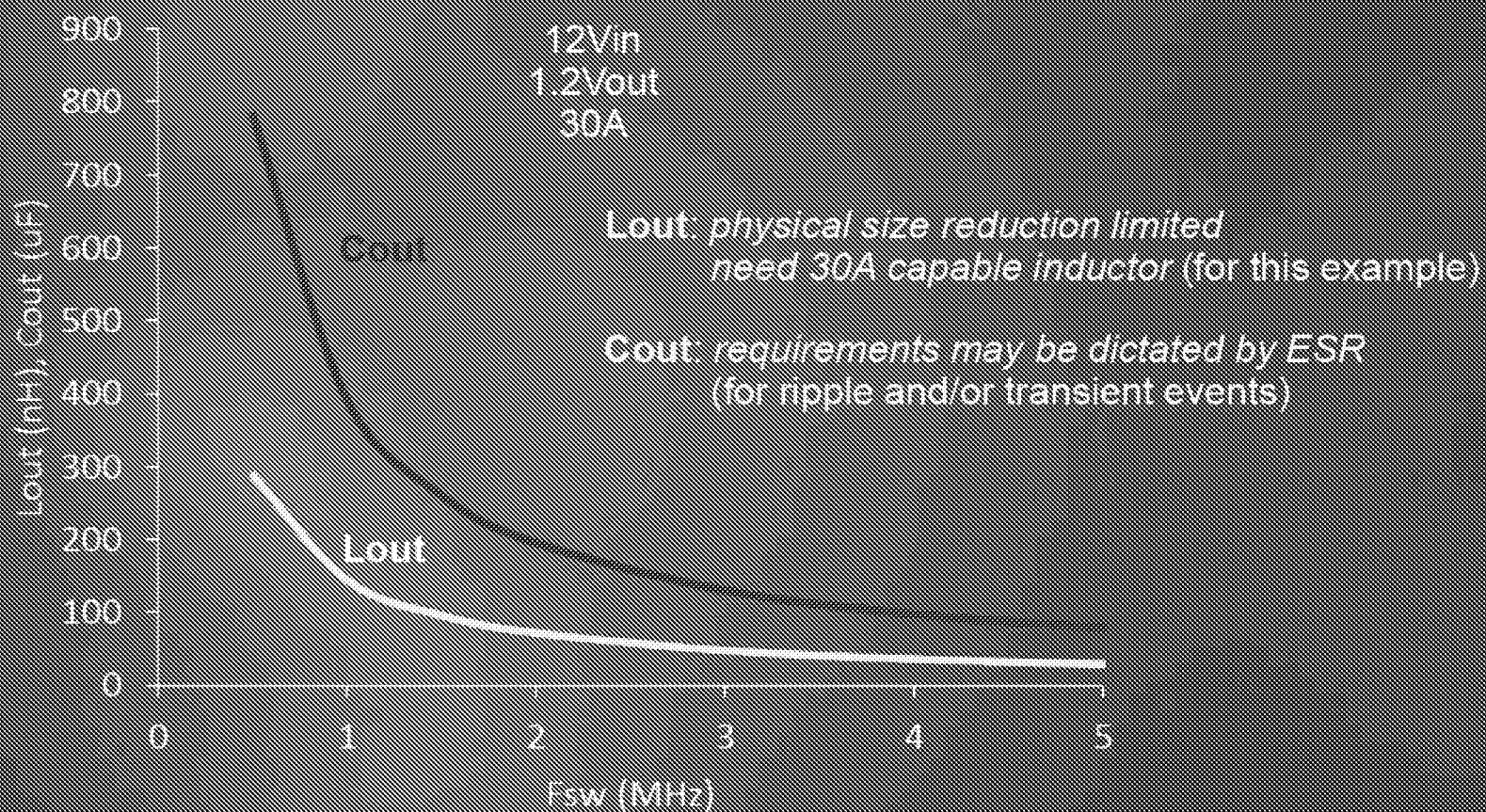


High Density POL Regulator Performance



We need higher Switching Frequency

to shrink the Output Filter

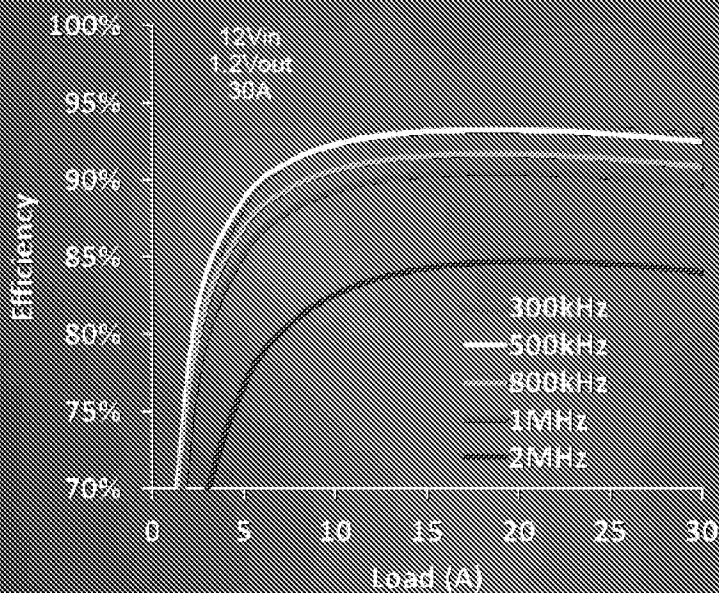


Challenges to Higher Density 12Vin POL solutions

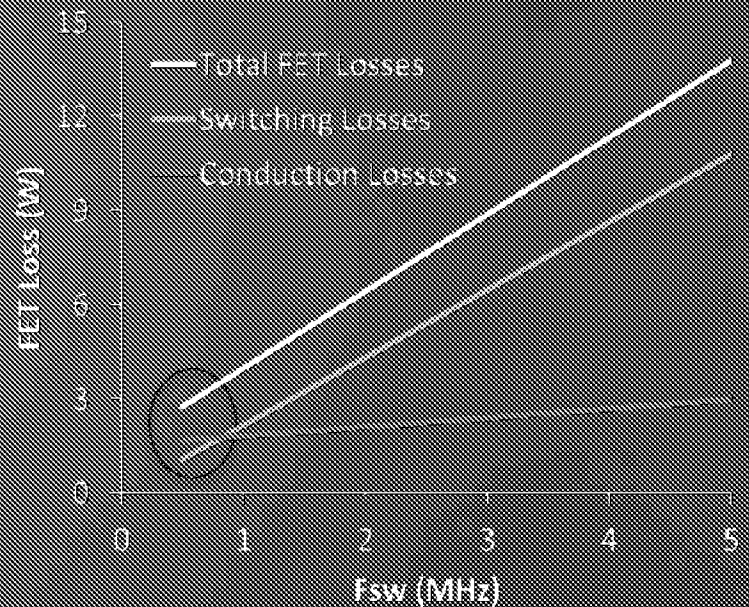
(1) FET Switching Losses

MOSFETs are not up to the challenge of multi-MHz operation

State of the art Powerstage Efficiency



State of the art Powerstage FET Losses



Limited to less than ~1MHz

GaAs FET Advantages for 12V Input POLs

	Vertical MOSFETs	GaN-on-Silicon FETs	GaAs FETs	Benefit
Electron Mobility (cm ² /Vs)	1,400	1,800	8,500	Fast Switching
R _{DS(on)} *Q _G (mΩ-nC)	~30	~25	Gen1 12 → Gen2 6	Reduced Switching loss (ability to go to higher F _{sw})
Body Diode	Yes	No	No	No Qrr loss
R _{DS(on)} temperature coefficient (% / °C)	0.4	0.2	0.2	Reduced Conduction loss
FET Structure	Vertical	Lateral	Lateral	Monolithic integration of UFET / LFET (reduced parasitics)

Green text: merit

Additional Information on Sarda's GaAs FET developments:

- * APEC 2013: "The **gFET™** Switch: A New Low Voltage High Speed GaAs HEMT for Switching Applications", Robert White, et al.
- * IMAPS 2013: "A Highly-Integrated GaAs-based Module for DC-DC Regulators", Greg Miller, et al.
- * APEC 2014: "Recent Developments In GaAs Power Switching Devices Including Device Modeling", Robert White, et al.

Challenges to Higher Density 12Vin POL solutions

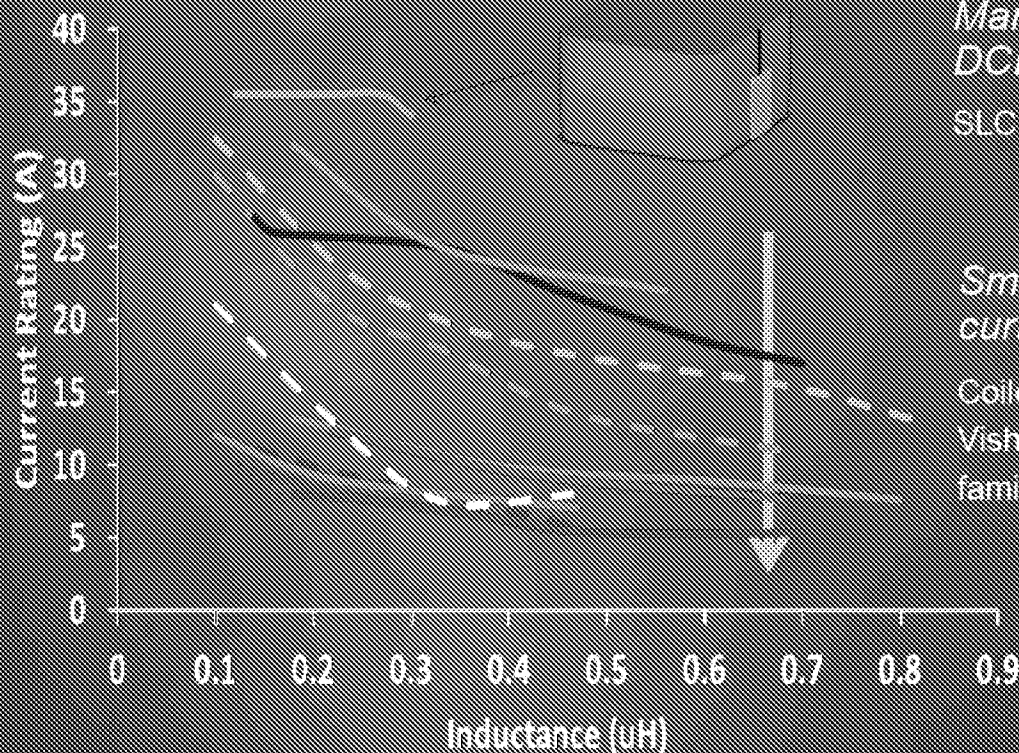
(1) FET Switching Losses → Sarda GaAs FET

(2) High current inductors are large

30A inductors are large

Many choices for low DCR, bulky Inductors

SLC1175 example



Smaller form-factor Inductors – current capability drops

Coilcraft XAL, XFL (solid lines),
Vishay IHLP (dashed lines)
families shown